AN AUDITED ASSEMBLY FOR THE CONVEX LARGE-N POLYGONAL PÓLYA–SZEGŐ THEOREM WITH A PATHWISE LCA FRONTIER

Abstract. We assemble the argument in logical order and separate proved interfaces from conditional ones. The disk endpoint, physical trace transfer, complete cubic tensor, structured collapsed-bubble commutator, material transport, exact fixed-normal support Hessian, fan-normal metric, strict side activation, and compact global implication are proved. Revision 24 separates the reduced disk form q_D^0 from the physical half-Hessian $\mathbf{q}_D = 2j_{0,1}^2 q_D^0$, corrects the physical corner exponent in the support-time specialization, and retracts the former near-regular three-periodic material transfer for the flux-matching clause JE–FM. The unproved separated JP–SC/JP–JE/JP–FS route is removed from the active DAG. The closed $D0_{\mathrm{ns}}$, JE–P, and JE–V rows form the disk endpoint JP–DE. A new honest fixed-fan scalar residual keeps the exact forcing and integrated support Hessian in the same Schur comparison. The exact completed-action row JC–ALG and the support-time logarithm cancellation JC–TL are proved. Revision 24 retains the complete regular block JC–AR for every character: it combines the genuine Schwarz–Christoffel mixed finite part, a rational completed-covector ratio certificate, and a uniform finite-N true-flux/reduced-resolvent bridge. The exact open cut inside the remaining pathwise payment JC–PL is now JC–SP+JC–CL, namely the completed-action Hessian modulus and coarse/far action localization. The jointly renormalized pure-support row, canonical super-near finite energy bridge, and exact completed-path activity are proved subrows. The former fixed-reserve RF–IM target is rigorously disproved by an active period-three family and replaced by the open relative-root interface RF–REL. The terminal-radius pure-power row is resummed exactly; the noninvariant DAR row is retracted, while the exact completed third-jet/QRCI estimate remains open. The regular coarse leading row is reduced to an unproved monotone-transport gap-simplex inequality. JC–PL implies the former one-sided comparison JP–JC and hence J0 by exact disk completion. JC–PL is the sole open mathematical input. Hence this file is a complete conditional assembly, not an unconditional certificate.

Contents

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1. Main theorem and assembly order

For $A > 0$, let $R_N(A)$ be the regular N-gon of area A.

Theorem 1.1 (Conditional convex large-N Pólya–Szegő theorem)**.** *Assume the pathwise LCA payment JC–PL, Contract 16.34. Then there exists N_0 such that, for every $N \geq N_0$ and every bounded convex planar polygon P of area A with exactly N genuine sides,*

$$\lambda_1(P) \geq \lambda_1(R_N(A)).$$

Equality holds if and only if P is congruent to $R_N(A)$.

The proof is organized by the following acyclic implication chain:

$$\mathrm{F51}+\mathrm{H0}+\mathrm{QI}\longrightarrow \mathrm{F65},\qquad \mathrm{F51}\longrightarrow \mathrm{F129},\qquad \mathrm{F51}+\mathrm{F65}+\mathrm{F129}\longrightarrow \mathrm{TR},$$
$$\mathrm{F51}\longrightarrow \mathrm{CT}_{\mathrm{core}},\qquad \mathrm{F51}+\mathrm{TR}+\mathrm{CT}_{\mathrm{core}}\longrightarrow \mathrm{QB},$$
$$\mathrm{F144}\longrightarrow \mathrm{F148}\longrightarrow \mathrm{DF0B},\qquad \mathrm{F144}\longrightarrow \mathrm{DF0A},$$
$$\mathrm{F144}+\mathrm{DF0A}+\mathrm{DF0B}\longrightarrow \mathrm{ZG}\longrightarrow \mathrm{DF},$$
$$\mathrm{F51}+\mathrm{CT}_{\mathrm{core}}\longrightarrow \mathrm{PT}\longrightarrow \mathrm{DM},$$
$$\mathrm{CT}_{\mathrm{core}}\longrightarrow \{\text{JP-EH},\text{JP-NS}\},$$
$$\text{JP-NS}+\mathrm{TR}+\mathrm{CT}_{\mathrm{core}}+\mathrm{QB}+\mathrm{DF}+\mathrm{DM}\longrightarrow \text{JP-DE}=\{D0_{\mathrm{ns}},\text{JE-P},\text{JE-V}\},$$
$$\text{JP-EH}+\text{JP-NS}+\text{JP-DE}\longrightarrow \{\text{JC-ALG},\text{JC-TL}\},$$
$$\text{JC-ALG}+\text{JC-TL}\longrightarrow \text{JC-PL}\longrightarrow \text{JP-JC},$$
$$\text{JP-DE}+\text{JP-JC}\longrightarrow \mathrm{NR}\longrightarrow \mathrm{J0},$$
$$\mathrm{F57}\longrightarrow \mathrm{A0},\qquad \mathrm{C0}+\mathrm{A0}+\mathrm{J0}\longrightarrow \mathrm{G0}\longrightarrow \text{Theorem 1.1}.$$

Every imported label is namespaced by its module identifier. The final ledger below distinguishes closed, retired, open, and dependent rows.

Part 1. Disk endpoint foundations and physical trace

2. Foundation F51: exact disk Hessian and Bessel–Schur normalization

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2.1. Fourier and energy conventions. For a real 2π-periodic function f, use

$$\widehat{f}_n=\frac{1}{2\pi}\int_0^{2\pi} f(\theta)e^{-in\theta}\,d\theta,\qquad f(\theta)=\sum_{n\in\mathbb{Z}}\widehat{f}_ne^{in\theta}. \tag{2.1}$$

Our normalized circle inner product is

$$\langle f,g\rangle_{L^2_*}:=\frac{1}{2\pi}\int_0^{2\pi} f(\theta)\overline{g(\theta)}\,d\theta=\sum_{n\in\mathbb{Z}}\widehat{f}_n\overline{\widehat{g}_n}. \tag{2.2}$$

Let

$$H^s_{\mathrm{ng}}:=\left\{f\in H^s(\mathbb{S}^1):\widehat{f}_0=\widehat{f}_1=\widehat{f}_{-1}=0\right\},\qquad \Pi_{\mathrm{ng}}f=\sum_{|n|\ge 2}\widehat{f}_ne^{in\theta}. \tag{2.3}$$

The normalized homogeneous and inhomogeneous half-order forms are

$$\|f\|^2_{\dot H^{1/2}_*}:=\sum_{|n|\ge 2}|n|\,|\widehat{f}_n|^2, \tag{2.4}$$

$$q_+(f,g):=\sum_{|n|\ge 2}(|n|+1)\widehat{f}_n\overline{\widehat{g}_n}. \tag{2.5}$$

If U is the decaying harmonic extension of f to the half-cylinder $\mathbb{S}^1\times(0,\infty)$, Parseval's identity gives

$$\int_{\mathbb{S}^1\times(0,\infty)}|\nabla U|^2\,d\theta\,dy=2\pi\|f\|^2_{\dot H^{1/2}_*}. \tag{2.6}$$

Thus a strip computation made in cylinder energy must be divided by 2π before it is inserted into (2.4). This is the source of the two constants

$$K_{\mathrm{cyl}} = \frac{\zeta(3)}{\pi^3}, \qquad K_* := \frac{K_{\mathrm{cyl}}}{2\pi} = \frac{\zeta(3)}{2\pi^4}. \tag{2.7}$$

They are not competing normalizations of the same displayed form.

2.2. **The exact fan/support affine class.** Let

$$\alpha_i > 0, \qquad \sum_{i=1}^{N} \alpha_i = 2\pi, \qquad \theta_{i+1} - \theta_i = \alpha_i \tag{2.8}$$

cyclically, and put $\nu_i = (\cos\theta_i, \sin\theta_i)$. On the edge cell

$$K_i = [-\alpha_{i-1}/2, \alpha_i/2] \tag{2.9}$$

write

$$a = \alpha_{i-1}, \quad b = \alpha_i, \quad s = \frac{x + a/2}{(a+b)/2}, \quad r_- = \frac{z_i - z_{i-1}}{a}, \quad r_+ = \frac{z_{i+1} - z_i}{b}. \tag{2.10}$$

Definition 2.1 (Exact trigonometric support trace)**.** For $z \in \mathbb{R}^N$, define $T_\alpha z$ on K_i by

$$(T_\alpha z)_i(x) = z_i \cos x + \big((1-s)a_i^- + s a_i^+\big) \sin x, \tag{2.11}$$

where

$$a_i^- = r_- \frac{a}{\sin a} - z_i \tan\frac{a}{2}, \qquad a_i^+ = r_+ \frac{b}{\sin b} + z_i \tan\frac{b}{2}. \tag{2.12}$$

The formulas on adjacent cells agree at their common endpoint, so $T_\alpha z$ is continuous and belongs to $H^1(\mathbb{S}^1)$.

Indeed, at the right endpoint of K_i, direct substitution gives

$$(T_\alpha z)_i(b/2) = \frac{z_i + z_{i+1}}{2\cos(b/2)} = (T_\alpha z)_{i+1}(-b/2). \tag{2.13}$$

The exact area row is

$$L_\alpha z = \sum_{i=1}^{N} \ell_i z_i, \qquad \ell_i = \tan\frac{\alpha_{i-1}}{2} + \tan\frac{\alpha_i}{2}, \tag{2.14}$$

and the admissible coefficient space is $Z_\alpha = \ker L_\alpha$. Let

$$V_\alpha := \Pi_{\mathrm{ng}} T_\alpha Z_\alpha \subset H^{1/2}_{\mathrm{ng}}. \tag{2.15}$$

Let $B^{\mathrm{ex}}_\alpha \in H^{1/2}(\mathbb{S}^1)$ be the exact canonical fan profile in this fixed material convention. The normalization of that profile is part of the definition; no factor is silently absorbed into it. Set

$$b_\alpha := \Pi_{\mathrm{ng}} B^{\mathrm{ex}}_\alpha, \qquad \mathcal{A}_\alpha = b_\alpha + V_\alpha. \tag{2.16}$$

All endpoint values below depend only on the affine subspace $\mathcal{A}_\alpha$, so replacing B^{ex}_α by $B^{\mathrm{ex}}_\alpha + T_\alpha z_0$ with $z_0 \in Z_\alpha$ changes no value.

2.2.1. *Translations.*

Lemma 2.2 (Exact translation identity)**.** *For $c \in \mathbb{R}^2$, put $z_i = c \cdot \nu_i$. Then $L_\alpha z = 0$ and*

$$T_\alpha z(\theta) = c \cdot (\cos\theta, \sin\theta). \tag{2.17}$$

Consequently $\Pi_{\mathrm{ng}} T_\alpha z = 0$.

Proof. Let $\tau_i = (-\sin\theta_i, \cos\theta_i)$. From

$$\nu_{i-1} = \nu_i \cos a - \tau_i \sin a, \qquad \nu_{i+1} = \nu_i \cos b + \tau_i \sin b \tag{2.18}$$

one obtains from (2.12)

$$a_i^- = a_i^+ = c \cdot \tau_i. \tag{2.19}$$

Equation (2.11) is therefore exactly

$$(c \cdot \nu_i)\cos x + (c \cdot \tau_i)\sin x = c \cdot \nu(\theta_i + x), \tag{2.20}$$

which proves (2.17). The area row vanishes because translations preserve area. Algebraically, it is the closure identity $\sum_i \ell_i \nu_i = 0$ for the tangential polygon with normal fan α. The last assertion follows from the Fourier support of $c \cdot \nu(\theta)$. □

Thus centering only chooses a representative of the translation orbit. It does not add a finite-rank loss to the nongeometric endpoint Schur value.

2.3. **The disk multiplier.** Let $j = j_{0,1}$ be the first positive zero of J_0. For $n \ge 2$, set

$$k_n = j\,\frac{J_{n+1}(j)}{J_n(j)}. \tag{2.21}$$

Lemma 2.3 (A rational enclosure for j)**.** *One has*

$$2.4 < j < 2.405, \qquad \frac{j^2}{4} < 1.447. \tag{2.22}$$

Proof. The series

$$J_0(x) = \sum_{m\ge 0} \frac{(-1)^m (x^2/4)^m}{(m!)^2} \tag{2.23}$$

is alternating with decreasing terms on the indicated interval after its first term. Rational partial sums through orders eight and nine give

$$0.0025076832943 < J_0(12/5) < 0.0025076834966,$$
$$-0.000090558004 < J_0(481/200) < -0.000090557793.$$

Moreover $J_0' = -J_1$, and

$$J_1(x) = \frac{x}{2} \sum_{m\ge 0} \frac{(-1)^m (x^2/4)^m}{m!(m+1)!} > 0 \qquad (0 < x \le 2.405), \tag{2.24}$$

because the alternating sum is at least $1-(2.405)^2/8>0$. Hence J_0 is strictly decreasing there and its first zero lies in the displayed interval. The final bound follows by squaring 2.405. $\square$

Theorem 2.4 (Positive order-(-1) splitting)**.** *For every $n\geq 2$,*

$$1+j\frac{J_n'(j)}{J_n(j)}=n+1-k_n, \tag{2.25}$$

and

$$0<k_n\leq\frac{2x}{n+1-x}<2,\qquad k_n<\frac{4}{n},\qquad x:=\frac{j^2}{4}<1.447. \tag{2.26}$$

In particular

$$\frac{1}{3}(n+1)\leq n+1-k_n\leq n+1\qquad(n\geq 2). \tag{2.27}$$

Proof. The Bessel recurrence

$$jJ_n'(j)=nJ_n(j)-jJ_{n+1}(j) \tag{2.28}$$

gives (2.25). Write

$$J_n(j)=\frac{(j/2)^n}{n!}F_n,\qquad F_n=\sum_{m\geq 0}\frac{(-1)^m x^m}{m!(n+1)_m}. \tag{2.29}$$

For $n\geq 2$, the ratio of consecutive absolute terms is at most $x/(n+1)<1/2$. Alternating-series bounds give

$$0<1-\frac{x}{n+1}\leq F_n\leq 1. \tag{2.30}$$

Consequently

$$0<k_n=\frac{2x}{n+1}\frac{F_{n+1}}{F_n}\leq\frac{2x}{n+1-x}. \tag{2.31}$$

The right-hand side is decreasing in n and is less than 1.864 at $n=2$. Direct cross multiplication, using $x<1.447$, also gives $2x/(n+1-x)<4/n$ for every $n\geq 2$. Hence $n+1-k_n\geq n-1\geq(n+1)/3$, proving (2.27). $\square$

Define the positive self-adjoint multiplier

$$K_{-1}e^{in\theta}=k_{|n|}e^{in\theta},\qquad |n|\geq 2, \tag{2.32}$$

and the reduced disk form

$$q^0_{\mathbb{D}}(f,g):=q_+(f,g)-k(f,g),\qquad k(f,g):=\langle f,K_{-1}g\rangle_{L^2_*}. \tag{2.33}$$

The physical disk quadratic Taylor form (equivalently, one half of the second variation of the scale-invariant deficit at the disk) is

$$\mathsf{Q}_D(f,g):=2j^2q^0_{\mathbb{D}}(f,g). \tag{2.34}$$

Thus the full Hessian is $2\mathsf{Q}_D=4j^2q^0_{\mathbb{D}}$. The factor $2j^2$ is not included in either q_+ or $q^0_{\mathbb{D}}$.

Corollary 2.5 (Coercivity and negative order)**.** *For $f \in H^{1/2}_{\mathrm{ng}}$,*

$$\frac{1}{3}q_+(f,f) \le q^0_{\mathbb{D}}(f,f) \le q_+(f,f). \tag{2.35}$$

Moreover

$$|k(f,g)| \le C_j \|f\|_{H^{-1/2}} \|g\|_{H^{-1/2}}. \tag{2.36}$$

Proof. Equation (2.35) is (2.27) summed over Fourier modes. By (2.26), $k_n \le C_j/n$; Cauchy–Schwarz with weight $1/n$ gives (2.36). ☐

2.4. **Exact terminal Schur identity.** For the affine class (2.16), define

$$\mathcal{P}^0_+(\alpha) := \inf_{v \in V_\alpha} q_+(b_\alpha + v, b_\alpha + v), \tag{2.37}$$

$$\widetilde{\mathcal{P}^{\mathrm{ex}}_D}(\alpha) := \inf_{v \in V_\alpha} q^0_{\mathbb{D}}(b_\alpha + v, b_\alpha + v), \tag{2.38}$$

$$\mathcal{P}^{\mathrm{ex}}_D(\alpha) := 2j^2 \widetilde{\mathcal{P}^{\mathrm{ex}}_D}(\alpha). \tag{2.39}$$

Let $f_\alpha \in \mathcal{A}_\alpha$ be the unique q_+-minimal residual. Thus

$$q_+(f_\alpha, v) = 0 \qquad (v \in V_\alpha). \tag{2.40}$$

On V_α, define

$$A_{D,\alpha}[v,w] = q^0_{\mathbb{D}}(v,w), \qquad g_\alpha(v) = k(f_\alpha, v). \tag{2.41}$$

Theorem 2.6 (Terminal Bessel–Schur formula)**.** *For every positive fan,*

$$\widetilde{\mathcal{P}^{\mathrm{ex}}_D}(\alpha) = \mathcal{P}^0_+(\alpha) - k(f_\alpha, f_\alpha) - \langle g_\alpha, A^{-1}_{D,\alpha} g_\alpha \rangle. \tag{2.42}$$

The inverse and dual pairing are taken on the quotient of V_α by its zero trace; by (2.35) they are well defined.

Proof. Every element of $\mathcal{A}_\alpha$ has a unique representation $f_\alpha + w$ modulo the zero trace, with $w \in V_\alpha$. Using (2.40),

$$\begin{aligned} q^0_{\mathbb{D}}(f_\alpha + w, f_\alpha + w) &= q_+(f_\alpha, f_\alpha) - k(f_\alpha, f_\alpha) \\ &\quad + A_{D,\alpha}[w,w] - 2g_\alpha(w). \end{aligned}$$

Completing the square in the coercive form $A_{D,\alpha}$ gives (2.42). ☐

2.5. **The uniform fan.** Let $\alpha_* = a\mathbf{1}$, $a = 2\pi/N$. Rotation by a acts on coefficient vectors by the cyclic shift and on traces by the unitary rotation $R_a f(\theta) = f(\theta - a)$.

Lemma 2.7 (Uniform Bessel forcing vanishes)**.** *Assume the canonical choice of $B^{\mathrm{ex}}_{\alpha_*}$ is cyclically equivariant. Then*

$$g_{\alpha_*} = 0. \tag{2.43}$$

Proof. The affine class, q_+, and the canonical datum are invariant under R_a, so uniqueness of the q_+ projection gives $R_a f_{\alpha*} = f_{\alpha*}$. The multiplier K_{-1} commutes with R_a. Hence for $z \in Z_{\alpha*}$,

$$k(f_{\alpha*}, \Pi_{\mathrm{ng}} T_{\alpha*} z) = k\left(f_{\alpha*}, \Pi_{\mathrm{ng}} T_{\alpha*} \frac{1}{N} \sum_{m=0}^{N-1} \sigma^m z\right), \tag{2.44}$$

where σ is the cyclic shift. At the uniform fan, $L_{\alpha*} z = 0$ is equivalent to $\sum_i z_i = 0$, so the averaged coefficient vector is zero. Therefore the displayed pairing vanishes for every element of $V_{\alpha*}$. □

Combining Theorem 2.6 and Lemma 2.7 gives the exact endpoint difference

$$\begin{aligned}\widetilde{\mathcal{P}_D^{\mathrm{ex}}}(\alpha) - \widetilde{\mathcal{P}_D^{\mathrm{ex}}}(\alpha_*) = \mathcal{P}_+^0(\alpha) - \mathcal{P}_+^0(\alpha_*) \\ - \left[k(f_\alpha, f_\alpha) - k(f_{\alpha*}, f_{\alpha*})\right] \\ - \langle g_\alpha, A_{D,\alpha}^{-1} g_\alpha \rangle.\end{aligned} \tag{2.45}$$

3. Node H0: the correct-source homogeneous gap

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3.1. Exact tangential-polygon geometry.

Let

$$\alpha_i > 0, \qquad \sum_{i=1}^{N} \alpha_i = 2\pi, \qquad \theta_{i+1} - \theta_i = \alpha_i \tag{3.1}$$

cyclically, and let $\nu_i = (\cos\theta_i, \sin\theta_i)$. Put

$$\phi_i = \theta_i + \frac{\alpha_i}{2}, \qquad K_i = [\phi_{i-1}, \phi_i]. \tag{3.2}$$

The polygon with all support numbers equal to one is

$$P(\alpha, \mathbf{1}) = \bigcap_i \{x : x \cdot \nu_i < 1\}. \tag{3.3}$$

Its area and its area-π scaling factor are

$$A_\alpha = \sum_i \tan\frac{\alpha_i}{2}, \qquad h_\alpha = \left(\frac{\pi}{A_\alpha}\right)^{1/2}. \tag{3.4}$$

On K_i use

$$x = \theta - \theta_i, \qquad -\frac{\alpha_{i-1}}{2} \le x \le \frac{\alpha_i}{2}. \tag{3.5}$$

The exact radial defect of the translated, area-normalized tangential polygon is

$$\rho_\alpha^{\tan}(\theta) = h_\alpha \sec x - 1, \qquad \theta \in K_i. \tag{3.6}$$

Translation of the polygon does not change its eigenvalue, so the auxiliary discrete centering translation is absent from (3.6).

Definition 3.1 (Correct principal source)**.** Define the continuous edge source

$$G_\alpha(\theta) = \frac{x^2}{2}, \qquad \theta \in K_i, \tag{3.7}$$

with x as in (3.5).

Lemma 3.2 (Exact source expansion)**.** *Let $S = \alpha_{\max}$. There is a constant c_α such that*

$$\rho_\alpha^{\tan} = c_\alpha + G_\alpha + R_\alpha, \tag{3.8}$$

where, on K_i,

$$|R_\alpha(\theta)| \le CS^2 x^2, \qquad |R'_\alpha(\theta)| \le CS^2|x|. \tag{3.9}$$

Consequently

$$\|R_\alpha\|_{H^{1/2}(\mathbb{S}^1)} \le CS\|G_\alpha\|_{H^{1/2}(\mathbb{S}^1)}. \tag{3.10}$$

The same estimate holds after removal of the modes $0, \pm 1$.

Proof. Taylor's formula and (3.1) give

$$A_\alpha = \pi + \frac{1}{24}\sum_i \alpha_i^3 + O(S^4), \qquad h_\alpha = 1 + O(S^2). \tag{3.11}$$

Take $c_\alpha = h_\alpha - 1$. Since

$$\sec x - 1 = \frac{x^2}{2} + O(x^4), \qquad (\sec x - 1)' = x + O(x^3), \tag{3.12}$$

uniformly for $|x| \le S$, equations (3.8)–(3.9) follow. The remainder is continuous at each ϕ_i, because the two adjacent endpoint values use the same gap α_i.

More intrinsically, write $R_\alpha = m_\alpha G_\alpha$, extending the quotient continuously at $x = 0$. The even analytic quotient

$$m_\alpha(x) = 2h_\alpha \frac{\sec x - 1}{x^2} - 1 \tag{3.13}$$

satisfies

$$\|m_\alpha\|_{L^\infty} \le CS^2, \qquad \|m'_\alpha\|_{L^\infty(K_i)} \le CS. \tag{3.14}$$

The endpoint values from the two incident edges agree, so m_α is a global Lipschitz multiplier. The standard Lipschitz multiplier estimate on $H^{1/2}(\mathbb{S}^1)$ now gives (3.10). A fixed Fourier projection is bounded on $H^{1/2}$. $\square$

3.2. **The old normal-cell source is not the polygonal source.** For comparison, define the normal-cell bubble

$$B_\alpha(\theta_i + y) = y(\alpha_i - y), \qquad 0 \le y \le \alpha_i. \tag{3.15}$$

Let $T_\alpha z$ be any vertex-compatible support trace obtained from a vector field affine along each straight edge. In particular, on K_i it has the exact trigonometric form

$$(T_\alpha z)(\theta) = z_i \cos x + (A_i + B_i x) \sin x. \tag{3.16}$$

The coefficients are constrained only by matching the common vertex velocities. Thus $T_\alpha z$ is smooth in the interior of every K_i.

Proposition 3.3 (Principal source obstruction)**.** *For every positive fan and every $c \neq 0$,*

$$G_\alpha \notin cB_\alpha + \{T_\alpha z : z \in \mathbb{R}^N\} + \operatorname{span}\{1, \cos\theta, \sin\theta\}. \tag{3.17}$$

The same assertion holds with the homogeneous support trace

$$T^0_\alpha z = z_i + x\big((1-s)r_{i-1} + sr_i\big) \quad \text{on } K_i. \tag{3.18}$$

Hence an effective polygonal source of the form $cB_\alpha + O_{\mathrm{scaled}}(S^3)$ cannot follow from exact polygon geometry and vertex-compatible support completion.

Proof. In the sense of periodic distributions,

$$G''_\alpha = 1 - \sum_i \alpha_i \delta_{\phi_i}, \tag{3.19}$$

$$B''_\alpha = -2 + \sum_i (\alpha_{i-1} + \alpha_i)\delta_{\theta_i}. \tag{3.20}$$

Indeed, $G'_\alpha = x$ on K_i and its jump at ϕ_i is $-\alpha_i$. Likewise, $B''_\alpha = -2$ between consecutive normals and the derivative jump at θ_i is $\alpha_{i-1} + \alpha_i$.

Every normal θ_i lies strictly inside K_i, whereas every ϕ_i is a polygonal vertex direction. By (3.16), the singular part of $(T_\alpha z)''$ is supported only on the set $\{\phi_i\}$. The same is true for (3.18); the three geometric modes are smooth. If (3.17) failed, comparison of the atoms at θ_i in the second distributional derivatives would give $c(\alpha_{i-1} + \alpha_i) = 0$ for every i, contrary to $c \neq 0$.

The two atom measures in (3.19)–(3.20) have coefficients of order S and their twice-integrated profiles have size S^2 on a normalized cell. Thus the discrepancy occurs at principal order, not in an endpoint-weighted $O(S^3)$ remainder. ☐

Remark 3.4. The bulk curvatures explain the formerly proposed constant: choosing $c = -1/2$ makes the absolutely continuous parts of G''_α and $(cB_\alpha)''$ agree. It does not move the atomic part from the normals to the vertices. Keeping only the bulk Taylor coefficient therefore loses exactly the geometric datum that the critical $H^{1/2}$ energy detects.

3.3. **The corrected homogeneous Schur problem.** Put

$$r_i = \frac{z_{i+1} - z_i}{\alpha_i}. \tag{3.21}$$

On K_i, let

$$T^0_\alpha z = z_i + x\big((1-s)r_{i-1} + s r_i\big), \qquad s = \frac{x + \alpha_{i-1}/2}{(\alpha_{i-1} + \alpha_i)/2}. \tag{3.22}$$

For

$$\widehat{f}_n = \frac{1}{2\pi}\int_0^{2\pi} f(\theta) e^{-in\theta}\, d\theta, \qquad \|f\|^2_{\dot H^{1/2}_*} = \sum_{n\neq 0} |n|\, |\widehat{f}_n|^2, \tag{3.23}$$

define

$$\mathcal{P}^0_{G,|D|}(\alpha) := \inf_{z\in Z^0_\alpha} \|G_\alpha + T^0_\alpha z\|^2_{\dot H^{1/2}_*}, \tag{3.24}$$

$$\mathcal{P}^0_{G,+}(\alpha) := \inf_{z\in Z^0_\alpha} \left(\|G_\alpha + T^0_\alpha z\|^2_{\dot H^{1/2}_*} + \|G_\alpha + T^0_\alpha z\|^2_{L^2_*} \right), \tag{3.25}$$

Here Z^0_α may be the full coefficient space or any homogeneous area/centering subspace for which $z = 0$ is admissible at the uniform fan. The lower bounds below are pointwise in z and are therefore unaffected by these rows.

Let $\mathcal{N}_i(f)$ denote the least Dirichlet energy in the half-strip $K_i \times (0,\infty)$ with bottom trace $f|_{K_i}$ and homogeneous Neumann data on the two vertical walls. Restriction of the periodic harmonic extension gives

$$2\pi \|f\|^2_{\dot H^{1/2}_*} \geq \sum_i \mathcal{N}_i(f). \tag{3.26}$$

3.4. **Exact one-edge energy and the rational Bellman inequality.** On one edge put

$$a = \alpha_{i-1}, \quad b = \alpha_i, \quad w = a+b, \quad t = \frac{a}{w}, \quad X = 1 - 2t, \tag{3.27}$$

and

$$\rho_- = \frac{r_{i-1}}{w}, \qquad \rho_+ = \frac{r_i}{w}, \qquad \delta = \rho_+ - \rho_-, \qquad q = t\rho_- + (1-t)\rho_+. \tag{3.28}$$

After subtracting the harmless constant z_i and using $y = s$, the bottom trace is $w^2 h(y)$, where

$$h(y) = \frac{1}{8}(y-t)^2 + \frac{1}{2}(y-t)\big((1-y)\rho_- + y\rho_+\big). \tag{3.29}$$

Lemma 3.5 (Exact edge energy)**.** *With*

$$K_{\mathrm{cyl}} = \frac{\zeta(3)}{\pi^3}, \tag{3.30}$$

the Neumann half-strip energy of (3.29) *is*

$$\mathcal{N}_i(G_\alpha + T^0_\alpha z) = \frac{K_{\mathrm{cyl}}}{4} w^4 \left[\left(\delta + \frac{1}{4}\right)^2 + 7\left(q + \frac{X}{4}\right)^2 \right]. \tag{3.31}$$

Proof. Let $u = h'(0)$ and $v = h'(1)$. Twice integrating the cosine coefficients by parts gives, for $k \geq 1$,

$$2\int_0^1 h(y)\cos(k\pi y)\,dy = \frac{2\big((-1)^k v - u\big)}{(k\pi)^2}. \tag{3.32}$$

The energy of this cosine mode in the unit half-strip is $k\pi/2$ times the square of its coefficient. Since

$$\sum_{k\ \text{even}} \frac{1}{k^3} = \frac{1}{8}\zeta(3), \qquad \sum_{k\ \text{odd}} \frac{1}{k^3} = \frac{7}{8}\zeta(3), \tag{3.33}$$

the total energy is

$$\frac{K_{\mathrm{cyl}}}{4} w^4\big((v-u)^2 + 7(v+u)^2\big). \tag{3.34}$$

Direct differentiation of (3.29) gives

$$v - u = \delta + \frac{1}{4}, \qquad v + u = q + \frac{X}{4}, \tag{3.35}$$

which proves (3.31). ∎

Lemma 3.6 (Rational edge Bellman inequality)**.** *For every* $0 \leq t \leq 1$ *and every* $\rho_-, \rho_+ \in \mathbb{R}$*,*

$$\begin{aligned} &\frac{1}{4}\left[\left(\delta + \frac{1}{4}\right)^2 + 7\left(q + \frac{X}{4}\right)^2\right] \\ &\quad \geq \frac{1}{10}\big(t^4 + (1-t)^4\big) + \frac{1}{320} + (1-t)^3\rho_+ - t^3\rho_-. \end{aligned} \tag{3.36}$$

Equality holds at $t = 1/2$ *and* $\rho_- = \rho_+ = 0$*.*

Proof. Put

$$s = t(1-t), \qquad d = s(1-2s), \qquad e = X(1-s). \tag{3.37}$$

The last two terms on the right of (3.36) that depend on the slopes satisfy

$$(1-t)^3\rho_+ - t^3\rho_- = d\delta + eq. \tag{3.38}$$

Completing the two squares shows that the left side of (3.36) minus its right side is

$$\begin{aligned} &\frac{1}{4}\left(\delta + \frac{1}{4} - 2d\right)^2 + \frac{7}{4}\left(q + \frac{X}{4} - \frac{2e}{7}\right)^2 \\ &\quad + X^2\left(\frac{9}{280} - \frac{X^2}{280} - \frac{X^4}{112} - \frac{X^6}{64}\right). \end{aligned} \tag{3.39}$$

The bracket is decreasing as a function of $X^2 \in [0,1]$ and its value at $X^2 = 1$ is $9/2240$. Hence every term in (3.39) is nonnegative. The stated equality is immediate. ∎

Restoring a, b, w in Lemma 3.6 and using Lemma 3.5 gives the pointwise inequality

$$\mathcal{N}_i(G_\alpha + T_\alpha^0 z) \geq K_{\mathrm{cyl}} \left\{ \frac{1}{10}(a^4 + b^4) + \frac{1}{320}(a+b)^4 + b^3 r_i - a^3 r_{i-1} \right\}. \tag{3.40}$$

3.5. **The replacement no-ratio gap.**

Theorem 3.7 (Corrected homogeneous $|D|$ gap)**.** *Let*

$$\bar{\alpha} = \frac{2\pi}{N}, \qquad \mathfrak{J}_4(\alpha) = \sum_i \alpha_i^4 - N\bar{\alpha}^4. \tag{3.41}$$

For every positive cyclic fan,

$$\boxed{\mathcal{P}^0_{G,|D|}(\alpha) - \mathcal{P}^0_{G,|D|}(\bar{\alpha}\mathbf{1}) \geq \frac{\zeta(3)}{10\pi^4}\mathfrak{J}_4(\alpha).} \tag{3.42}$$

The conclusion remains valid after imposing the homogeneous area and centering rows described after (3.25)*.*

Proof. Sum (3.40). The slope potential telescopes exactly:

$$\sum_i (\alpha_i^3 r_i - \alpha_{i-1}^3 r_{i-1}) = 0. \tag{3.43}$$

Together with (3.26), this gives, pointwise in z,

$$2\pi \|G_\alpha + T_\alpha^0 z\|^2_{\dot{H}^{1/2}_*} \geq K_{\mathrm{cyl}} \left[\frac{1}{5}\sum_i \alpha_i^4 + \frac{1}{320}\sum_i (\alpha_{i-1} + \alpha_i)^4 \right]. \tag{3.44}$$

For the uniform fan, $z = 0$ makes the periodic harmonic extension even across every vertical wall. Thus strip localization is an equality. Lemmas 3.5 and 3.6 are also equalities, and

$$2\pi \mathcal{P}^0_{G,|D|}(\bar{\alpha}\mathbf{1}) = \frac{K_{\mathrm{cyl}}}{4} N\bar{\alpha}^4. \tag{3.45}$$

Finally,

$$\sum_i (\alpha_{i-1} + \alpha_i)^4 \geq 16 N\bar{\alpha}^4 \tag{3.46}$$

by Jensen's inequality. Subtract (3.45) from (3.44), take the infimum over z, and use $K_{\mathrm{cyl}}/(2\pi) = \zeta(3)/(2\pi^4)$. This proves (3.42). □

3.6. **The identity multiplier has a nonnegative relative defect.**

For a function f on the circle, set

$$\|f\|^2_{L^2_*} = \frac{1}{2\pi}\int_{\mathbb{S}^1} |f - \operatorname{avg} f|^2. \tag{3.47}$$

On one edge the trace, after a cell constant is removed, is $w^2 h(y)$ with h given by (3.29) and $d\theta = (w/2)dy$. Write

$$\operatorname{Var}(h) = \int_0^1 h^2 - \left(\int_0^1 h\right)^2. \tag{3.48}$$

Lemma 3.8 (Exact rational variance inequality)**.** *For all* $0 \le t \le 1$ *and all* $\rho_-, \rho_+ \in \mathbb{R}$,

$$\frac{1}{2}\operatorname{Var}(h) \ge \frac{1}{23040} + \frac{1}{180}\left((1-t)^4\rho_+ - t^4\rho_-\right). \tag{3.49}$$

Equality holds at $t = 1/2$ *and* $\rho_- = \rho_+ = 0$.

Proof. Up to an irrelevant constant, write $h(y) = Ay^2 + By$, where

$$A = \frac{1}{8} + \frac{\delta}{2}, \qquad B = \frac{q}{2} - \frac{\delta}{2} - \frac{t}{4}. \tag{3.50}$$

The elementary moments of y and y^2 give

$$\frac{1}{2}\operatorname{Var}(h) = \frac{2A^2}{45} + \frac{B^2}{24} + \frac{AB}{12}. \tag{3.51}$$

Put

$$s = t(1-t), \quad X = 1-2t, \quad d_4 = s(1-3s), \quad e_4 = X(1-2s). \tag{3.52}$$

Then

$$(1-t)^4\rho_+ - t^4\rho_- = d_4\delta + e_4 q. \tag{3.53}$$

Define

$$\delta_* = 4d_4 - \frac{1}{4}, \qquad q_* = \frac{4}{15}e_4 - \frac{X}{4}. \tag{3.54}$$

Substitution in (3.51) and completion of the two independent squares gives the exact identity

$$\begin{aligned} &\frac{1}{2}\operatorname{Var}(h) - \frac{1}{23040} - \frac{1}{180}(d_4\delta + e_4 q) \\ &\quad = \frac{(\delta-\delta_*)^2}{1440} + \frac{(q-q_*)^2}{96} \\ &\qquad + \frac{X^2}{345600}\left(176 + 52X^2 + 116X^4 - 135X^6\right). \end{aligned} \tag{3.55}$$

For $|X| \le 1$, the last polynomial is at least $176 - 135 = 41$. This proves (3.49) and its equality statement. □

Restoring the physical widths in (3.49) gives

$$\min_{C\in\mathbb{R}} \int_{K_i} |G_\alpha + T^0_\alpha z + C|^2 \ge \frac{(a+b)^5}{23040} + \frac{1}{180}(b^4 r_i - a^4 r_{i-1}). \tag{3.56}$$

Summing removes the last potential. Since a single global mean is no better than independent cell constants,

$$2\pi\|G_\alpha + T^0_\alpha z\|^2_{L^2_*} \ge \frac{1}{23040}\sum_i(\alpha_{i-1} + \alpha_i)^5. \tag{3.57}$$

At the uniform fan, $z = 0$ gives equality in (3.57). Jensen's inequality therefore shows that the identity-multiplier relative defect is nonnegative.

Corollary 3.9 (Corrected homogeneous $|D|+I$ gap)**.** *For every positive cyclic fan,*

$$\boxed{\mathcal{P}^0_{G,+}(\alpha) - \mathcal{P}^0_{G,+}(\bar{\alpha}\mathbf{1}) \geq \frac{\zeta(3)}{10\pi^4}\mathfrak{J}_4(\alpha).} \tag{3.58}$$

The conclusion remains valid with the homogeneous area and centering rows.

Proof. Add (3.57) to the pointwise order-one estimate (3.44), subtract the two equalities at the uniform fan, and use

$$\sum_i (\alpha_{i-1}+\alpha_i)^5 \geq N(2\bar{\alpha})^5. \tag{3.59}$$

Taking the infimum over the same coefficient space proves (3.58). □

4. Foundation F65: the exact angular disk endpoint

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4.1. Fan, correct source, and the two support spaces.

Let

$$\alpha_i > 0, \qquad \sum_i \alpha_i = 2\pi, \qquad \theta_{i+1}-\theta_i = \alpha_i, \qquad \bar{\alpha} = \frac{2\pi}{N}, \tag{4.1}$$

and put

$$S = \alpha_{\max} + \bar{\alpha}, \qquad \mathfrak{J}_4(\alpha) = \sum_i \alpha_i^4 - N\bar{\alpha}^4. \tag{4.2}$$

Set $\phi_i = \theta_i + \alpha_i/2$ and $K_i = [\phi_{i-1}, \phi_i]$. On K_i write

$$a = \alpha_{i-1}, \quad b = \alpha_i, \quad x = \theta - \theta_i, \quad s = \frac{x+a/2}{(a+b)/2}, \tag{4.3}$$

and define the correct principal source

$$G_\alpha(\theta) = \frac{x^2}{2}. \tag{4.4}$$

For a coefficient vector z, put

$$r_- = \frac{z_i - z_{i-1}}{a}, \qquad r_+ = \frac{z_{i+1}-z_i}{b}. \tag{4.5}$$

The homogeneous and exact vertex-compatible traces are

$$(T^0_\alpha z)_i(x) = z_i + x\big((1-s)r_- + sr_+\big), \tag{4.6}$$

$$(T_\alpha z)_i(x) = z_i \cos x + \big((1-s)a_i^- + sa_i^+\big)\sin x, \tag{4.7}$$

where

$$a_i^- = r_-\frac{a}{\sin a} - z_i \tan\frac{a}{2}, \qquad a_i^+ = r_+\frac{b}{\sin b} + z_i\tan\frac{b}{2}. \tag{4.8}$$

Both traces are continuous at the vertices ϕ_i.

The homogeneous and exact area rows are

$$L^0_\alpha z = \frac{1}{2}\sum_i (\alpha_{i-1}+\alpha_i) z_i, \tag{4.9}$$

$$L_\alpha z = \sum_i \left(\tan\frac{\alpha_{i-1}}{2} + \tan\frac{\alpha_i}{2}\right) z_i. \tag{4.10}$$

Let $Z^0_\alpha = \ker L^0_\alpha$ and $Z_\alpha = \ker L_\alpha$. Centering rows may be added throughout; translations will be removed exactly in Section 4.4.

Work in $H = H^{1/2}(\mathbb{S}^1)/\mathbb{R}$ with

$$q_+(f) = \sum_{n\neq 0} (|n|+1)|\widehat{f}_n|^2, \qquad \widehat{f}_n = \frac{1}{2\pi}\int_0^{2\pi} f(\theta)e^{-in\theta}\,d\theta. \tag{4.11}$$

Set

$$V^0_\alpha = T^0_\alpha Z^0_\alpha, \qquad V_\alpha = T_\alpha Z_\alpha, \tag{4.12}$$

and

$$P^G_0(\alpha) = \inf_{v\in V^0_\alpha} q_+(G_\alpha + v), \tag{4.13}$$

$$P^G_{\mathrm{tr}}(\alpha) = \inf_{v\in V_\alpha} q_+(G_\alpha + v). \tag{4.14}$$

The corrected homogeneous theorem gives

$$P^G_0(\alpha) - P^G_0(\bar\alpha\mathbf{1}) \geq \frac{\zeta(3)}{10\pi^4}\mathfrak{J}_4(\alpha). \tag{4.15}$$

4.2. **No-ratio exact-trace graph.**

Lemma 4.1 (Local jet coercivity)**.** *There are absolute constants $0 < c < C$ such that, with $w = a + b$,*

$$cw(z_i^2 + w^2 r_-^2 + w^2 r_+^2) \leq \|T^0_\alpha z\|^2_{L^2(K_i)} \leq Cw(z_i^2 + w^2 r_-^2 + w^2 r_+^2). \tag{4.16}$$

The constants are independent of a/b.

Proof. After $x = (w/2)(y-t)$ with $t = a/w$, the three normalized polynomials are

$$1, \qquad (y-t)(1-y), \qquad (y-t)y, \qquad 0 \leq y \leq 1. \tag{4.17}$$

The matrix carrying the two slope variables to the linear and quadratic coefficients has determinant one and it and its inverse are uniformly bounded for $0 \leq t \leq 1$. Finite-dimensional norm equivalence proves (4.16). □

Lemma 4.2 (Exact-minus-homogeneous graph)**.** *For S sufficiently small there is a bijection $C_\alpha : Z^0_\alpha \to Z_\alpha$ for which*

$$J_\alpha(T^0_\alpha z) = T_\alpha C_\alpha z \tag{4.18}$$

satisfies

$$\|(J_\alpha - I)v\|_H \leq CS^{3/2}\|v\|_H, \qquad v \in V^0_\alpha. \tag{4.19}$$

Proof. Taylor's formula in (4.7)–(4.8) gives on K_i

$$\|T_\alpha z - T_\alpha^0 z\|_{L^\infty(K_i)} \le C\big(w^2|z_i| + w^3(|r_-| + |r_+|)\big),$$

$$\|\partial_\theta(T_\alpha z - T_\alpha^0 z)\|_{L^\infty(K_i)} \le C\big(w|z_i| + w^2(|r_-| + |r_+|)\big). \tag{4.20}$$

The error is continuous at every wall. Summing (4.20), using Lemma 4.1, cyclic Poincaré on the area kernel, and $\|f\|_{H^{1/2}}^2 \le C\|f\|_{L^2}\|f\|_{H^1}$ gives

$$\|T_\alpha z - T_\alpha^0 z\|_H \le CS^{3/2}\|T_\alpha^0 z\|_H. \tag{4.21}$$

For $z \in Z_\alpha^0$, define

$$C_\alpha z = z - \frac{L_\alpha z}{L_\alpha \mathbf{1}}\mathbf{1}. \tag{4.22}$$

Since

$$\left|\tan\frac{u}{2} - \frac{u}{2}\right| \le Cu^3, \tag{4.23}$$

the coefficient correction is $O(S^2)$ in the trace metric. The analogous formula with L_α^0 is the inverse map. Combining this row correction with (4.21) proves (4.19). □

Lemma 4.3 (Hilbert Schur perturbation)**.** *If $J : V_0 \to V_1$ is a bijection with $\|(J - I)v\| \le \eta\|v\|$, $\eta < 1/2$, then*

$$\left|\operatorname{dist}(b, V_1)^2 - \operatorname{dist}(b, V_0)^2\right| \le C\eta\|b\|^2. \tag{4.24}$$

Proof. Map the orthogonal projection of $-b$ onto V_0 through J and use it as a competitor in V_1. Apply the same argument to J^{-1}. □

The continuous edge quadratic satisfies the no-ratio spline bound

$$q_+(G_\alpha) \le C\sum_i \alpha_i^4 \le CS^3. \tag{4.25}$$

Indeed, affine normalization on each K_i controls the same-cell and adjacent-cell terms, while dyadic annuli control the separated-cell part. Lemma 4.3 therefore yields, for

$$R_{\mathrm{tr}}^G = P_{\mathrm{tr}}^G - P_0^G, \tag{4.26}$$

the absolute estimate

$$|R_{\mathrm{tr}}^G(\alpha)| \le CS^{9/2}. \tag{4.27}$$

4.3. **Relative trace comparison.**

Proposition 4.4 (Quasiuniform trace Hessian)**.** *Suppose*

$$(1-\eta_0)a \le \gamma_i \le (1+\eta_0)a, \qquad \sum_i \gamma_i = 2\pi, \tag{4.28}$$

where $0 < \eta_0 < 1/4$. If $\sum_i \delta_i = 0$, then

$$|D^2R_{\mathrm{tr}}^G(\gamma)[\delta,\delta]| \le Ca^{1/2}\sum_i \gamma_i^2\delta_i^2. \tag{4.29}$$

Proof. If $\delta \neq 0$, complexify $\gamma(z) = \gamma + z\delta$ in the disk

$$|z| < r, \qquad r = \frac{c_0 a}{\|\delta\|_{\ell^\infty}}. \tag{4.30}$$

For every cyclic arc I,

$$\left| z \sum_{i\in I} \delta_i \right| \leq c_0 a \# I \leq C c_0 \sum_{i \in I} \gamma_i. \tag{4.31}$$

Thus the cellwise affine material map changes every arc length by a fixed small relative amount. The pulled-back Gagliardo kernel, the L^2 row, the two traces, the area graph, and the polynomial source $G_{\gamma(z)}$ are holomorphic. Their Gram inverses stay uniformly bounded by Lemma 4.1. The proof of (4.27) applies to the complex bilinear Schur formula and gives

$$|R^G_{\mathrm{tr}}(\gamma(z))| \leq C a^{9/2}. \tag{4.32}$$

Cauchy's estimate gives

$$|D^2 R^G_{\mathrm{tr}}(\gamma)[\delta, \delta]| \leq C a^{5/2} \|\delta\|^2_{\ell^\infty}. \tag{4.33}$$

Use $\sum_i \gamma_i^2 \delta_i^2 \geq c a^2 \|\delta\|^2_{\ell^\infty}$ to obtain (4.29). □

Lemma 4.5 (Relative trace remainder)**.** *For every sufficiently fine positive fan,*

$$|R^G_{\mathrm{tr}}(\alpha) - R^G_{\mathrm{tr}}(\bar{\alpha}\mathbf{1})| \leq C(S^{1/4} + S^{1/2}) \mathfrak{J}_4(\alpha). \tag{4.34}$$

Proof. Put $a = \bar{\alpha}$ and $d_i = \alpha_i - a$. The exact identity

$$\mathfrak{J}_4(\alpha) = \sum_i d_i^2 (\alpha_i^2 + 2a\alpha_i + 3a^2) \tag{4.35}$$

implies $\max_i |d_i|^4 \leq C\mathfrak{J}_4(\alpha)$. If $\mathfrak{J}_4 \geq S^{17/4}$, use (4.27). Otherwise $\max |d_i|/S \leq CS^{1/16}$, so the segment $a\mathbf{1} + td$ is quasiuniform. Cyclic invariance makes the first derivative of R^G_{tr} vanish at $a\mathbf{1}$ on the zero-sum direction. Taylor's formula, Proposition 4.4, and (4.35) give the $CS^{1/2}\mathfrak{J}_4$ bound. The two cases prove (4.34). □

Theorem 4.6 (Correct-source exact-trace gap)**.** *There is* $S_{\mathrm{tr}} > 0$ *such that, for* $S \leq S_{\mathrm{tr}}$,

$$\boxed{P^G_{\mathrm{tr}}(\alpha) - P^G_{\mathrm{tr}}(\bar{\alpha}\mathbf{1}) \geq \frac{\zeta(3)}{20\pi^4} \mathfrak{J}_4(\alpha).} \tag{4.36}$$

Proof. Combine (4.15) and (4.34), decreasing S_{tr} until the latter coefficient is at most $\zeta(3)/(20\pi^4)$. □

4.4. **Exact translations.**

Lemma 4.7 (Translation quotient)**.** *For $c \in \mathbb{R}^2$, let $z_i = c \cdot (\cos\theta_i, \sin\theta_i)$. Then*

$$L_\alpha z = 0, \qquad T_\alpha z(\theta) = c \cdot (\cos\theta, \sin\theta). \tag{4.37}$$

Consequently

$$P^G_{\mathrm{tr}}(\alpha) = \inf_{v \in \Pi_{\mathrm{ng}} V_\alpha} q_+(\Pi_{\mathrm{ng}} G_\alpha + v), \tag{4.38}$$

where Π_{ng} removes the modes $0, \pm 1$.

Proof. With $\tau_i = (-\sin\theta_i, \cos\theta_i)$, the identities

$$\nu_{i-1} = \nu_i \cos a - \tau_i \sin a, \qquad \nu_{i+1} = \nu_i \cos b + \tau_i \sin b \tag{4.39}$$

give $a_i^- = a_i^+ = c \cdot \tau_i$ in (4.8). Hence (4.7) equals $c \cdot \nu(\theta)$. Translation invariance of area gives $L_\alpha z = 0$. Orthogonality of the Fourier modes proves (4.38). ☐

4.5. **Bessel order-minus-one reduction.**

Let $j = j_{0,1}$ and, for $n \geq 2$, set

$$k_n = j\,\frac{J_{n+1}(j)}{J_n(j)}. \tag{4.40}$$

On the range of Π_{ng}, define

$$k(f,g) = \sum_{|n|\geq 2} k_{|n|} \widehat{f}_n \widehat{g}_{-n},$$

$$q_D(f,g) = q_+(f,g) - k(f,g). \tag{4.41}$$

The Bessel recurrence and its alternating fixed-argument series give

$$0 < k_n < \frac{4}{n}, \qquad \frac{1}{3} q_+(f) \leq q_D(f) \leq q_+(f). \tag{4.42}$$

Put

$$b^G_\alpha = \Pi_{\mathrm{ng}} G_\alpha, \qquad W_\alpha = \Pi_{\mathrm{ng}} V_\alpha, \tag{4.43}$$

and

$$P^G_D(\alpha) = \inf_{v \in W_\alpha} q_D(b^G_\alpha + v). \tag{4.44}$$

Lemma 4.8 (No-ratio exact-trace quasi-interpolant)**.** *There is a linear operator*

$$I_\alpha : H^{3/2}_{\mathrm{ng}}(\mathbb{S}^1) \longrightarrow W_\alpha \tag{4.45}$$

such that, for every sufficiently fine positive fan,

$$\|\psi - I_\alpha \psi\|_{H^{1/2}} \leq CS \|\psi\|_{H^{3/2}}. \tag{4.46}$$

The constant is independent of the smallest cell and every adjacent mesh ratio.

Proof. It is enough to work first in the homogeneous trace space. Put $M = \lfloor S^{-1} \rfloor$ and let $\phi = P_{\le M}\psi$ be the Fourier projection onto $|n| \le M$. Directly from the Fourier norms,

$$\|\psi - \phi\|_{H^{1/2}} \le CS\|\psi\|_{H^{3/2}}, \qquad \|\phi''\|_{L^2} \le CS^{-1/2}\|\psi\|_{H^{3/2}}. \tag{4.47}$$

Take the nodal values $z_i = \phi(\theta_i)$ and set $Q_\alpha\phi = T^0_\alpha z$. We give the global estimate rather than invoke a mesh-regular interpolation theorem. On $K_i = [\phi_{i-1}, \phi_i]$, write $a = \alpha_{i-1}$, $b = \alpha_i$, $w = a + b$, and $\omega_i = [\theta_{i-1}, \theta_{i+1}]$. Since

$$r_- = \frac{1}{a}\int_{\theta_{i-1}}^{\theta_i} \phi'(u)\,du, \qquad r_+ = \frac{1}{b}\int_{\theta_i}^{\theta_{i+1}} \phi'(u)\,du,$$

Cauchy–Schwarz and the fundamental theorem of calculus give

$$|r_- - \phi'(\theta_i)| + |r_+ - \phi'(\theta_i)| \le Cw^{1/2}\|\phi''\|_{L^2(\omega_i)}. \tag{4.48}$$

The trace formula (4.6) reproduces every affine function. Using (4.48) in that formula, and using Taylor's integral formula for ϕ, yields for $x \in K_i$

$$|\phi(x) - Q_\alpha\phi(x)| \le Cw^{3/2}\|\phi''\|_{L^2(\omega_i)}, \tag{4.49}$$

$$|\phi'(x) - (Q_\alpha\phi)'(x)| \le Cw^{1/2}\|\phi''\|_{L^2(\omega_i)}. \tag{4.50}$$

Indeed, after subtracting the affine Taylor polynomial at θ_i, the first row consists of x times a convex combination of the two slope errors, and the differentiated row contains in addition $2x(r_+ - r_-)/w$; all its coefficients are bounded for $-a/2 \le x \le b/2$, including when a/b tends to zero or infinity.

The intervals ω_i have overlap at most two when their L^2-mass is assigned back to the two constituent normal cells. Since $w \le 2S$, integration of (4.49)–(4.50) and summation give

$$\|\phi - Q_\alpha\phi\|_{L^2} \le CS^2\|\phi''\|_{L^2}, \qquad \|\phi - Q_\alpha\phi\|_{H^1} \le CS\|\phi''\|_{L^2}. \tag{4.51}$$

The Fourier interpolation inequality $\|e\|^2_{H^{1/2}} \le \|e\|_{L^2}\|e\|_{H^1}$, (4.47), and (4.51) imply

$$\|\phi - Q_\alpha\phi\|_{H^{1/2}} \le CS^{3/2}\|\phi''\|_{L^2} \le CS\|\psi\|_{H^{3/2}}. \tag{4.52}$$

It remains only to impose the exact finite rows. Replace z by

$$z^0 = z - \frac{L^0_\alpha z}{L^0_\alpha \mathbf{1}}\mathbf{1}.$$

Because $T^0_\alpha \mathbf{1} = 1$, this changes the homogeneous trace only by a constant. Put $v^0 = T^0_\alpha z^0 \in V^0_\alpha$ and define

$$I_\alpha\psi = \Pi_{\mathrm{ng}} J_\alpha v^0 \in W_\alpha, \tag{4.53}$$

where J_α is the exact area graph of Lemma 4.2. The projection Π_{ng} is an orthogonal contraction in every Fourier Sobolev norm, $\psi = \Pi_{\mathrm{ng}}\psi$, and (4.19) gives

$$\|\Pi_{\mathrm{ng}}(J_\alpha - I)v^0\|_{H^{1/2}} \le CS^{3/2}\|v^0\|_{H^{1/2}} \le CS\|\psi\|_{H^{3/2}}.$$

Together with (4.47) and (4.52), this is (4.46). $\square$

Lemma 4.9 (Galerkin negative-norm gain)**.** *Let $f_\alpha = b^G_\alpha + v_\alpha$ be the q_+-minimizing residual. Then*

$$\|f_\alpha\|_{H^{-1/2}} \le CS\|f_\alpha\|_{H^{1/2}}. \tag{4.54}$$

Proof. For $g \in H^{1/2}_{\mathrm{ng}}$ solve $(|D| + I)\psi = g$. Galerkin orthogonality and Lemma 4.8 give

$$|\langle f_\alpha, g\rangle| = |q_+(f_\alpha, \psi - I_\alpha\psi)| \le CS\|f_\alpha\|_{H^{1/2}}\|g\|_{H^{1/2}}. \tag{4.55}$$

Taking the dual supremum proves (4.54). ☐

Let

$$R^G_B(\alpha) = P^G_{\mathrm{tr}}(\alpha) - P^G_D(\alpha). \tag{4.56}$$

Completing the disk square gives the exact identity

$$R^G_B = k(f_\alpha, f_\alpha) + \langle g_\alpha, A^{-1}_{D,\alpha} g_\alpha\rangle, \qquad g_\alpha(v) = k(f_\alpha, v), \tag{4.57}$$

where $A_{D,\alpha} = q_D|_{W_\alpha}$. Equations (4.42), (4.54), (4.25), and (4.57) imply

$$0 \le R^G_B(\alpha) \le CS^5. \tag{4.58}$$

Proposition 4.10 (Relative Bessel correction)**.** *For every sufficiently fine positive fan,*

$$|R^G_B(\alpha) - R^G_B(\bar\alpha\mathbf{1})| \le C(S^{1/2} + S)\mathfrak{J}_4(\alpha). \tag{4.59}$$

Proof. In a quasiuniform band, pull all functions to a fixed real fan. If $h = \|\delta\|_{\ell^\infty}/a$, the first two material derivatives of q_+ are bounded by Ch^r in $H^{1/2}$, while those of k are bounded by Ch^r in $H^{-1/2}$. The latter follows from

$$|\Delta^r k_n| \le C_r(1+n)^{-1-r}, \qquad 0 \le r \le 3, \tag{4.60}$$

and three discrete summations by parts in the pulled-back kernel. Differentiating the Galerkin equation and (4.54) twice gives

$$\|\partial_t^r f_t\|_{H^{1/2}} \le Ch^r a^{3/2}, \qquad \|\partial_t^r f_t\|_{H^{-1/2}} \le Ch^r a^{5/2}, \quad r = 0, 1, 2. \tag{4.61}$$

Twice differentiating (4.57) therefore yields

$$|D^2 R^G_B(\gamma)[\delta, \delta]| \le Ca \sum_i \gamma_i^2 \delta_i^2. \tag{4.62}$$

Use (4.58) when $\mathfrak{J}_4 \ge S^{9/2}$. In the complementary regime, (4.35) makes the radial segment from the uniform fan quasiuniform; cyclic invariance kills the first derivative there, and (4.62) with Taylor's formula gives $CS\mathfrak{J}_4$. This proves (4.59). ☐

Corollary 4.11 (Correct-source Bessel gap)**.** *There is $S_D > 0$ such that, for $S \le S_D$,*

$$\boxed{P^G_D(\alpha) - P^G_D(\bar\alpha\mathbf{1}) \ge \frac{\zeta(3)}{40\pi^4}\mathfrak{J}_4(\alpha).} \tag{4.63}$$

Proof. Subtract (4.59) from (4.36) and decrease S_D until the relative error is at most $\zeta(3)/(40\pi^4)$. □

4.6. **The exact secant source.** Let

$$A_\alpha = \sum_i \tan\frac{\alpha_i}{2}, \qquad h_\alpha = \left(\frac{\pi}{A_\alpha}\right)^{1/2}, \qquad \rho_\alpha^{\tan} = h_\alpha \sec x - 1 \quad \text{on } K_i. \tag{4.64}$$

Writing $c_\alpha = h_\alpha - 1$, one has

$$\rho_\alpha^{\tan} = c_\alpha + G_\alpha + E_\alpha, \qquad E_\alpha = h_\alpha(\sec x - 1) - \frac{x^2}{2}. \tag{4.65}$$

Lemma 4.12 (Scaled-spline secant remainder)**.** *For every sufficiently fine positive fan,*

$$\|E_\alpha\|_{H^{1/2}} \le CS^2 \|G_\alpha\|_{H^{1/2}}. \tag{4.66}$$

The estimate is independent of the smallest cell.

Proof. On K_i, normalized by $x = (a+b)(y-t)/2$, both G_α and E_α belong to a fixed compact family of even analytic profiles. Taylor's formula gives, through one normalized derivative,

$$|E_\alpha| + (a+b)|E'_\alpha| \le CS^2(a+b)^2. \tag{4.67}$$

The values from the two incident cells agree at each vertex.

For completeness, the mesh-independent assembly is as follows. Split the Slobodeckij integral into same-cell, adjacent-cell, and separated-cell pieces. Affine normalization and finite-dimensional norm equivalence apply to the first two pieces. For a separated pair K_i, K_j, subtract the cell means and use

$$\iint_{K_i\times K_j} \frac{d\theta\, d\varphi}{\operatorname{dist}_{\mathbb{S}^1}(\theta,\varphi)^2} \le C\frac{|K_i||K_j|}{d_{ij}^2}. \tag{4.68}$$

Summing in dyadic annuli bounds the mean-difference energy by the same-cell and adjacent-cell charges. Equation (4.67) makes every charge of E_α at most CS^2 times the corresponding charge of G_α. This proves (4.66) without a mesh-ratio assumption. □

Define the exact tangential-source disk value

$$P_D^{\tan}(\alpha) = \inf_{v\in W_\alpha} q_D\big(\Pi_{\mathrm{ng}}(\rho_\alpha^{\tan} - c_\alpha) + v\big) \tag{4.69}$$

and

$$R_{\sec}(\alpha) = P_D^{\tan}(\alpha) - P_D^G(\alpha). \tag{4.70}$$

Coercivity of (4.42), the Hilbert distance inequality, Lemma 4.12, and (4.25) give

$$|R_{\sec}(\alpha)| \le CS^5. \tag{4.71}$$

Lemma 4.13 (Relative secant correction)**.** *For every sufficiently fine positive fan,*

$$|R_{\rm sec}(\alpha) - R_{\rm sec}(\bar\alpha\mathbf{1})| \le C(S^{1/2}+S)\mathfrak{J}_4(\alpha). \tag{4.72}$$

Proof. Work in a real quasiuniform material segment $\gamma(t) = \gamma + t\delta$ and put $h = \|\delta\|_{\ell^\infty}/a$. Affine normalization of (4.64) and two differentiations of its elementary trigonometric coefficients give, for $r = 0, 1, 2$,

$$\|\partial_t^r G_{\gamma(t)}\|_{H^{1/2}} \le Ch^r a^{3/2}, \qquad \|\partial_t^r E_{\gamma(t)}\|_{H^{1/2}} \le Ch^r a^{7/2}. \tag{4.73}$$

The real-material q_D form and the exact trace/area graph have first and second derivatives bounded by Ch^r in their natural energy metrics; for the Bessel part this is the order-minus-one kernel estimate used in Proposition 4.10. Differentiating the two coercive Schur equations defining $P_D^{\tan}$ and P_D^G therefore shows that every second-derivative term in their difference is bounded by

$$Ch^2(a^{3/2})(a^{7/2}) = Ch^2a^5. \tag{4.74}$$

Thus

$$|D^2R_{\rm sec}(\gamma)[\delta,\delta]| \le Ca\sum_i \gamma_i^2\delta_i^2. \tag{4.75}$$

The two-regime argument used in Proposition 4.10, with threshold $\mathfrak{J}_4 = S^{9/2}$, proves (4.72). ☐

Theorem 4.14 (Exact tangential-polygon disk endpoint)**.** *There is $S_* > 0$ such that, whenever $S \le S_*$,*

$$\boxed{P_D^{\tan}(\alpha) - P_D^{\tan}(\bar\alpha\mathbf{1}) \ge \frac{\zeta(3)}{80\pi^4}\mathfrak{J}_4(\alpha).} \tag{4.76}$$

In particular, equality $P_D^{\tan}(\alpha) = P_D^{\tan}(\bar\alpha\mathbf{1})$ forces $\alpha_i = \bar\alpha$ for every i.

Proof. Combine (4.63) and (4.72), decreasing S_* until the latter coefficient is at most $\zeta(3)/(80\pi^4)$. Strict convexity of x^4 gives $\mathfrak{J}_4(\alpha) = 0$ only at the uniform fan. ☐

5. Foundation F129: the exact disk-source envelope

Audited source: `convex_largeN_polya_szego_stage129_EF_sharp_strip_gluing_closure.tex.`

5.1. Exact source and the three disk rows.

Let

$$\alpha_i > 0, \qquad \sum_i \alpha_i = 2\pi, \qquad a = \frac{2\pi}{N}, \qquad \mathfrak{J}_4(\alpha) = \sum_i \alpha_i^4 - Na^4. \tag{5.1}$$

Choose consecutive side normals θ_i, put $I_i = [\theta_i, \theta_{i+1}]$, and use $x = \theta - \theta_i$ on I_i. After removal of a constant, the unnormalized exact secant source is

$$B_\alpha(\theta_i + x) = F_{\alpha_i}(x) := \sec\bigl(\min\{x, \alpha_i - x\}\bigr) - 1, \qquad 0 \le x \le \alpha_i. \tag{5.2}$$

Both F_h and F_h' vanish at $0, h$, so the pieces form one continuous periodic source. The exact area-normalized source is

$$b_\alpha = h_\alpha \Pi_{\mathrm{ng}} B_\alpha, \qquad h_\alpha^2 = \frac{\pi}{A_\alpha}, \qquad A_\alpha = \sum_i \tan\frac{\alpha_i}{2}. \tag{5.3}$$

For $n \geq 2$, set

$$k_n = j_{0,1}\frac{J_{n+1}(j_{0,1})}{J_n(j_{0,1})}, \qquad m_n = n + 1 - k_n. \tag{5.4}$$

On nongeometric modes,

$$q_D(f) = q_{\dot H}(f) + q_0(f) - \mathcal{K}(f), \tag{5.5}$$

where

$$q_{\dot H}(f) = \sum_{|n|\geq 2} |n|\,|\widehat{f}_n|^2, \tag{5.6}$$

$$q_0(f) = \sum_{|n|\geq 2} |\widehat{f}_n|^2, \tag{5.7}$$

$$\mathcal{K}(f) = \sum_{|n|\geq 2} k_{|n|}|\widehat{f}_n|^2. \tag{5.8}$$

Stages 51 and 65 give, for $0 \leq r \leq 3$,

$$0 < k_n \leq \frac{C}{n}, \qquad |\Delta^r k_n| \leq C_r(1+n)^{-1-r}. \tag{5.9}$$

5.2. **The sharp broken-strip gluing lemma.** For $0 < h \leq h_0 < \pi$, let V_h be the Neumann harmonic extension of F_h to the half-strip $\Sigma_h = (0,h)\times(0,\infty)$:

$$\Delta V_h = 0, \qquad V_h(\cdot, 0) = F_h, \qquad \partial_x V_h(0,\cdot) = \partial_x V_h(h,\cdot) = 0. \tag{5.10}$$

Its limit at infinity is the cell mean

$$\mathfrak{m}(h) = \frac{1}{h}\int_0^h F_h(x)\,dx, \tag{5.11}$$

and put

$$\mathcal{N}(h) = \int_{\Sigma_h} |\nabla V_h|^2\,dx\,dy. \tag{5.12}$$

Lemma 5.1 (Scaled strip data)**.** *For $0 < h \leq h_0$,*

$$|\mathfrak{m}(h)| \leq Ch^2, \qquad 0 \leq \mathcal{N}(h) \leq Ch^4, \qquad |\mathcal{N}''(h)| \leq Ch^2. \tag{5.13}$$

If $w_h(y) = V_h(0,y) = V_h(h,y)$, then

$$w_h(y) = h^2 W_h(y/h), \tag{5.14}$$

where $h \mapsto W_h$ is bounded with two derivatives in $H^{1/2}(\mathbb{R}_+) \cap L^2(\mathbb{R}_+)$ after its constant limit is removed.

Proof. Write

$$F_h(hs) = h^2\Phi_h(s), \qquad \Phi_h(s) = \frac{\sec(h\min\{s, 1-s\}) - 1}{h^2}. \tag{5.15}$$

The family Φ_h, including its analytic value at $h = 0$, is bounded with two h-derivatives and four one-sided s-derivatives. Its value and first derivative vanish at $s = 0, 1$. Expand it in the Neumann cosine basis:

$$\Phi_h(s) = \overline{\Phi}_h + \sum_{\ell\geq 1} c_\ell(h)\cos(\ell\pi s). \tag{5.16}$$

The midpoint derivative jump gives $|\partial_h^r c_\ell(h)| \leq C_r(1+\ell)^{-2}$, $0 \leq r \leq 2$. Consequently

$$V_h(hs, hy) = h^2\left\{\overline{\Phi}_h + \sum_{\ell\geq 1} c_\ell(h)\cos(\ell\pi s)e^{-\ell\pi y}\right\}, \tag{5.17}$$

$$\mathcal{N}(h) = \frac{\pi h^4}{2}\sum_{\ell\geq 1}\ell\,|c_\ell(h)|^2. \tag{5.18}$$

Equations (5.13) and (5.14) now follow by termwise differentiation; the series are dominated by $\sum \ell^{-3}$. □

For $H > 0$, use the wall-trace norm

$$\|g\|_{X_H}^2 := \|g\|_{\dot H^{1/2}(\mathbb{R}_+)}^2 + H^{-1}\|g\|_{L^2(\mathbb{R}_+)}^2. \tag{5.19}$$

Functions below vanish at $y = 0$ and at infinity, so the homogeneous half-line norm has no constant ambiguity.

Lemma 5.2 (Wall equalization and lifting)**.** *Fix $a \leq h_0$. There are modified strip fields $\widetilde{V}_h$ with the same bottom value F_h, common limit $\mathfrak{m}(a)$ at infinity, and*

$$\int_{\Sigma_h} |\nabla\widetilde{V}_h|^2 \leq \mathcal{N}(h) + C\{\mathfrak{m}(h) - \mathfrak{m}(a)\}^2. \tag{5.20}$$

For two adjacent widths h, k, $H = \max\{h, k\}$, their wall jump $j_{h,k}$ satisfies

$$\|j_{h,k}\|_{X_H}^2 \leq C\{(h^2 - a^2)^2 + (k^2 - a^2)^2\}. \tag{5.21}$$

Moreover the jump can be lifted into the wider of the two adjacent strips, with zero bottom value, zero limit at infinity, and Dirichlet energy at most $C\|j_{h,k}\|_{X_H}^2$.

Proof. Choose a fixed smooth χ on $[0,\infty)$ with $\chi(0) = 0$ and $\chi = 1$ on $[1,\infty)$, and set

$$C_h(x, y) = -\{\mathfrak{m}(h) - \mathfrak{m}(a)\}\chi(y/h), \qquad \widetilde{V}_h = V_h + C_h. \tag{5.22}$$

The correction is independent of x. Since the horizontal mean of $\partial_y V_h$ is zero, its Dirichlet cross term with C_h vanishes. Direct scaling gives

$$\int_{\Sigma_h} |\nabla C_h|^2 = |\mathfrak{m}(h) - \mathfrak{m}(a)|^2\int_0^\infty |\chi'(s)|^2\,ds, \tag{5.23}$$

which proves (5.20).

The modified wall trace is zero at $y=0$ and tends to the common value $\mathfrak{m}(a)$. Subtract that common value. Equations (5.14) and (5.22), followed by scaling in (5.19), prove (5.21). For completeness, first suppose $h,k\in[a/2,2a]$. The fundamental theorem of calculus in the scale parameter gives

$$\|j_{h,k}\|_{X_H}^2\le C(h-k)^2(h+k)^2.$$

If at least one width lies outside $[a/2,2a]$, the triangle inequality and scaling of the two tail-equalized traces give

$$\|j_{h,k}\|_{X_H}^2\le C(h^4+k^4+a^4)\le C\{(h^2-a^2)^2+(k^2-a^2)^2\}.$$

When one width remains in $[a/2,2a]$, insert the nearer endpoint $a/2$ or $2a$ as an intermediate scale, use the first estimate on the short piece, and the coarse estimate on the other. This proves (5.21) in every ratio. The a^4 term is essential when both h and k are much smaller than a: it pays for changing their common strip tail to the regular-cell mean.

Finally, the Poisson extension in a quadrant is a right inverse of the vertical trace with energy $\|j\|^2_{\dot H^{1/2}(\mathbb{R}_+)}$. Multiply it by a horizontal cutoff supported before distance $H/3$. The cutoff row costs $CH^{-1}\|j\|_2^2$, proving the X_H estimate. Put the lifting in the wider adjacent strip. The two possible liftings assigned to one strip occupy its left and right thirds, so their overlap is uniformly bounded. □

Proposition 5.3 (Sharp no-ratio strip gluing)**.** *For every sufficiently fine positive fan,*

$$\sum_{n\neq0}|n|\,|\widehat B_{\alpha,n}|^2\le\frac{1}{2\pi}\sum_i\mathcal{N}(\alpha_i)+C\sum_i(\alpha_i^2-a^2)^2. \tag{5.24}$$

At the regular fan equality holds without the last term:

$$\sum_{n\neq0}|n|\,|\widehat B_{a\mathbf{1},n}|^2=\frac{N}{2\pi}\mathcal{N}(a). \tag{5.25}$$

Proof. Place $\widetilde V_{\alpha_i}$ over I_i. Lemma 5.2 gives wall jumps which vanish at the bottom and at infinity. Lift all jumps as in that lemma, assigning each lifting to the wider adjacent strip. The resulting field U belongs to $H^1(\mathbb{S}^1\times\mathbb{R}_+)$, has bottom trace B_α, and tends to the single constant $\mathfrak{m}(a)$.

The original V_{α_i} is harmonic and has zero normal derivative on its vertical walls. Hence it is Dirichlet-orthogonal to every jump lifting, which vanishes on the bottom and at infinity. It is also orthogonal to the constant-in-x tail correction. The only remaining cross term is between tail corrections and jump liftings; Cauchy–Schwarz absorbs it into their two energies. Lemma 5.2 therefore gives

$$\int|\nabla U|^2\le\sum_i\mathcal{N}(\alpha_i)+C\sum_i(\alpha_i^2-a^2)^2. \tag{5.26}$$

The harmonic-extension characterization of the periodic $\dot{H}^{1/2}$ norm proves (5.24).

For $\alpha_i = a$, all strip fields and their wall traces agree. No tail correction or lifting is present, and reflection of V_a across every vertical wall is exactly the periodic harmonic extension. This proves (5.25). □

5.3. **The order-one and identity source envelopes.** Recall the exact identity

$$\mathfrak{J}_4(\alpha) = \sum_i (\alpha_i - a)^2(\alpha_i^2 + 2a\alpha_i + 3a^2). \tag{5.27}$$

In particular,

$$\sum_i (\alpha_i^2 - a^2)^2 \le \mathfrak{J}_4(\alpha). \tag{5.28}$$

Theorem 5.4 (Sharp order-one source envelope)**.** *For every sufficiently fine positive fan,*

$$q_{\dot{H}}(\Pi_{\mathrm{ng}} B_\alpha) - q_{\dot{H}}(\Pi_{\mathrm{ng}} B_{a\mathbf{1}}) \le C\mathfrak{J}_4(\alpha). \tag{5.29}$$

Proof. Removing the modes ± 1 only decreases the left energy at α; they vanish at the regular fan by cyclic symmetry. Apply Proposition 5.3 and subtract (5.25). By Lemma 5.1 and Taylor's formula,

$$\begin{aligned}\sum_i \{\mathcal{N}(\alpha_i) - \mathcal{N}(a)\} &= \sum_i \{\mathcal{N}(\alpha_i) - \mathcal{N}(a) - \mathcal{N}'(a)(\alpha_i - a)\} \\ &\le C\sum_i (\alpha_i - a)^2(\alpha_i + a)^2 \le C\mathfrak{J}_4(\alpha).\end{aligned} \tag{5.30}$$

Use (5.28) for the gluing row. □

Lemma 5.5 (Identity source envelope)**.** *For every sufficiently fine positive fan,*

$$q_0(\Pi_{\mathrm{ng}} B_\alpha) - q_0(\Pi_{\mathrm{ng}} B_{a\mathbf{1}}) \le C\mathfrak{J}_4(\alpha). \tag{5.31}$$

Proof. Put

$$R(h) = \int_0^h F_h(x)^2\,dx, \qquad M(h) = \int_0^h F_h(x)\,dx = 2\int_0^{h/2} (\sec x - 1)\,dx. \tag{5.32}$$

Scaling gives $|R''(h)| \le Ch^3$. The linear terms cancel under $\sum_i (\alpha_i - a) = 0$, so

$$\sum_i \{R(\alpha_i) - R(a)\} \le Ch_0 \sum_i (\alpha_i - a)^2(\alpha_i + a)^2 \le C\mathfrak{J}_4(\alpha). \tag{5.33}$$

Moreover

$$M''(h) = \frac{1}{2}\sec\frac{h}{2}\tan\frac{h}{2} > 0. \tag{5.34}$$

Thus Jensen's inequality gives $\sum_i M(\alpha_i) \ge NM(a) > 0$; subtracting the squared global mean can only improve the relative upper bound. Finally, the regular source has no modes ± 1, while deleting those modes at α again decreases the energy. Parseval proves (5.31). □

5.4. **The smoother Bessel row.**

Lemma 5.6 (Regular Bessel baseline)**.** *For every sufficiently small a,*

$$0 \le \mathcal{K}(\Pi_{\mathrm{ng}} B_{a\mathbf{1}}) \le Ca^5. \tag{5.35}$$

Proof. Cyclic symmetry restricts the Fourier support to $N\mathbb{Z}$. The scaled cosine expansion in Lemma 5.1 gives $|\widehat{B}_{\ell N}| \le Ca^2(1+|\ell|)^{-2}$. Since $k_{\ell N} \le C/(|\ell|N) \le Ca/|\ell|$, summation proves (5.35). □

Lemma 5.7 (Quasiuniform Bessel material Hessian)**.** *Suppose*

$$\frac{a}{2} \le \gamma_i \le \frac{3a}{2}, \qquad \sum_i \gamma_i = 2\pi, \qquad \sum_i d_i = 0. \tag{5.36}$$

Then

$$\Big| D^2\{\mathcal{K}(\Pi_{\mathrm{ng}} B_\gamma)\}[d,d] \Big| \le Ca \sum_i \gamma_i^2 d_i^2. \tag{5.37}$$

Proof. Pull every normal interval affinely to the regular material interval. The profile (5.15) and its first two material derivatives are uniformly bounded after the factors γ_i^2, $\gamma_i d_i$, and d_i^2 are removed. The multiplier bounds (5.9) allow three cyclic Abel summations in the pulled-back kernel. The first removes the arbitrary root velocity, because consecutive midpoint velocities differ by $(d_i + d_{i+1})/2$; the next two give a summable $(1+|r|)^{-2}$ interaction between material cells at distance r. Cauchy–Schwarz after the full cyclic sum yields

$$|D^2\mathcal{K}| \le Ca \sum_i \gamma_i^2 d_i^2. \tag{5.38}$$

The factor a is the order-minus-one gain. This is the source-only specialization of the real-material Bessel Hessian estimate proved in stage 65; no Galerkin or support-space term is present here. □

Proposition 5.8 (Relative Bessel source row)**.** *For every sufficiently fine positive fan,*

$$-\{\mathcal{K}(\Pi_{\mathrm{ng}} B_\alpha) - \mathcal{K}(\Pi_{\mathrm{ng}} B_{a\mathbf{1}})\} \le C\mathfrak{J}_4(\alpha). \tag{5.39}$$

Proof. If $\mathfrak{J}_4(\alpha) \ge a^5$, positivity and Lemma 5.6 give

$$-\{\mathcal{K}(B_\alpha) - \mathcal{K}(B_{a\mathbf{1}})\} \le \mathcal{K}(B_{a\mathbf{1}}) \le C\mathfrak{J}_4(\alpha). \tag{5.40}$$

If $\mathfrak{J}_4(\alpha) < a^5$, (5.27) implies $\max_i |\alpha_i - a|^4 \le \mathfrak{J}_4 < a^5$. For small a, the whole affine segment $\gamma_i(t) = a + t(\alpha_i - a)$ lies in $[a/2, 3a/2]$. Cyclic invariance kills the first derivative at $a\mathbf{1}$. Lemma 5.7, Taylor's formula, and

$$\mathfrak{J}_4(\alpha) = 12 \int_0^1 (1-t) \sum_i \gamma_i(t)^2 (\alpha_i - a)^2 \, dt \tag{5.41}$$

prove the stronger bound $Ca\mathfrak{J}_4$. □

5.5. Exact source envelope and endpoint forcing.

Theorem 5.9 (Unnormalized exact disk-source envelope). *For every sufficiently fine positive fan,*

$$q_D(\Pi_{\mathrm{ng}} B_\alpha) - q_D(\Pi_{\mathrm{ng}} B_{a\mathbf{1}}) \le C\mathfrak{J}_4(\alpha). \tag{5.42}$$

Proof. Add Theorem 5.4, Lemma 5.5, and Proposition 5.8 according to the exact multiplier split (5.5). ☐

Theorem 5.10 (Area-normalized exact source envelope). *For every sufficiently fine positive fan,*

$$\boxed{q_D(b_\alpha) - q_D(b_{a\mathbf{1}}) \le C\mathfrak{J}_4(\alpha).} \tag{5.43}$$

Proof. Convexity of $\tan(x/2)$ and $\sum_i \alpha_i = Na$ give $A_\alpha \ge A_{a\mathbf{1}}$, hence $h_\alpha^2 \le h_{a\mathbf{1}}^2$. From (5.3),

$$\begin{aligned} q_D(b_\alpha) - q_D(b_{a\mathbf{1}}) = h_\alpha^2\{q_D(B_\alpha) - q_D(B_{a\mathbf{1}})\} \\ + (h_\alpha^2 - h_{a\mathbf{1}}^2) q_D(B_{a\mathbf{1}}). \end{aligned} \tag{5.44}$$

The second term is nonpositive. The factors h_α are uniformly bounded for fine fans, so Theorem 5.9 proves the claim. ☐

6. Node TR: physical trace and relative Schur transfer

Audited source: `TR_physical_trace.tex`.

6.1. The true straight-edge material trace.

Let $\alpha_i > 0$, $\sum_i \alpha_i = 2\pi$, and $\theta_{i+1} - \theta_i = \alpha_i$. Put

$$a = \alpha_{i-1}, \qquad b = \alpha_i, \qquad K_i = [\theta_i - a/2, \theta_i + b/2], \qquad x = \theta - \theta_i. \tag{6.1}$$

The area-normalized tangential polygon has common support

$$h_\alpha = \left(\frac{\pi}{\sum_j \tan(\alpha_j/2)} \right)^{1/2} \tag{6.2}$$

and radial boundary $h_\alpha \sec x$ on K_i.

For support increments $z = (z_i)$, the two endpoint displacement vectors of side i have components in the frame (ν_i, τ_i)

$$(z_i, a_i^-), \qquad (z_i, a_i^+), \tag{6.3}$$

where

$$a_i^- = \frac{z_i - z_{i-1}}{\sin a} - z_i \tan\frac{a}{2}, \qquad a_i^+ = \frac{z_{i+1} - z_i}{\sin b} + z_i \tan\frac{b}{2}. \tag{6.4}$$

Indeed these are obtained by differentiating the two adjacent support-line intersection equations.

The physical affine coordinate of the radial point $h_\alpha \nu_i + h_\alpha \tan x\, \tau_i$ is

$$t_i(x) = \frac{\tan x + \tan(a/2)}{\tan(a/2) + \tan(b/2)}. \tag{6.5}$$

Hence the genuine radial component of the straight-edge affine material velocity is

$$(\widetilde{T}_\alpha z)_i(x) = z_i \cos x + \big((1 - t_i(x))a_i^- + t_i(x)a_i^+\big) \sin x. \tag{6.6}$$

For comparison, the stage-65 angular trace is

$$(T_\alpha z)_i(x) = z_i \cos x + \big((1 - s_i(x))a_i^- + s_i(x)a_i^+\big) \sin x, \qquad s_i(x) = \frac{x + a/2}{(a+b)/2}. \tag{6.7}$$

We use the Fourier norm

$$\|f\|_{H^{1/2}}^2 = \sum_{n\in\mathbb{Z}} (1 + |n|)|\widehat{f}(n)|^2. \tag{6.8}$$

All quotient estimates below are unchanged if the equivalent disk form $q_{\mathbb{D}}$ is used.

Lemma 6.1 (Endpoint-uniform interpolation jets)**.** *Put $w = a+b$, $p = a/w$, $y = s_i(x)$, and assume $0 < w \le w_0 < \pi$. Then, uniformly for $0 \le p \le 1$,*

$$|t_i - s_i| \le Cw^2 s_i(1 - s_i), \tag{6.9}$$

$$w|t_i' - s_i'| \le Cw^2. \tag{6.10}$$

In particular, no endpoint factor occurs in (6.10).

Proof. In normalized variables,

$$\vartheta_{w,p}(y) = \frac{\tan(\frac{w}{2}(y-p)) + \tan(\frac{wp}{2})}{\tan(\frac{wp}{2}) + \tan(\frac{w(1-p)}{2})}, \qquad t_i(x) = \vartheta_{w,p}(y). \tag{6.11}$$

The denominator is between $w/2$ and $C_{w_0}w$. Direct differentiation gives

$$\partial_y^2 \vartheta_{w,p}(y) = \frac{w^2}{2} \frac{\sec^2(\frac{w}{2}(y-p)) \tan(\frac{w}{2}(y-p))}{\tan(\frac{wp}{2}) + \tan(\frac{w(1-p)}{2})}, \tag{6.12}$$

and hence $|\partial_y^2 \vartheta_{w,p}| \le Cw^2$. The function $f = \vartheta_{w,p} - y$ vanishes at 0 and 1. The Green function of $-\partial_y^2$ on $[0,1]$ therefore gives $|f(y)| \le Cw^2 y(1-y)$. Moreover $\int_0^1 f'(y)\, dy = 0$, so $|f'(y)| \le \|f''\|_\infty \le Cw^2$. Since $\partial_x = (2/w)\partial_y$, these are exactly (6.9)–(6.10). $\square$

For the next lemma, write $w_i = \alpha_{i-1} + \alpha_i$ and $d_i = a_i^+ - a_i^-$. The length of K_i is $w_i/2$.

Lemma 6.2 (Stable extraction of the internal quadratic coefficient)**.** *For S sufficiently small,*

$$w_i^2 |d_i|^2 \le C\left([T_\alpha z]_{H^{1/2}(K_i)}^2 + w_i^4 |z_i|^2\right). \tag{6.13}$$

On the exact area kernel, modulo exact support translations,

$$\sum_i w_i |z_i|^2 \le C \|\Pi_{\mathrm{ng}} T_\alpha z\|_{H^{1/2}(\mathbb{T})}^2. \tag{6.14}$$

Both constants are independent of adjacent ratios.

Proof. Pull K_i to $0 \le y \le 1$ and put

$$Z = z_i, \qquad A = w_i a_i^-, \qquad B = w_i a_i^+, \qquad \xi = \frac{w_i}{2}(y-p).$$

The normalized trace is

$$F_{w_i,p}(y) = Z\cos\xi + \{(1-y)A + yB\}\frac{\sin\xi}{w_i}. \tag{6.15}$$

At $w_i = 0$ its continuous limit is

$$F_{0,p}(y) = Z + \frac{1}{2}(y-p)\{(1-y)A + yB\}. \tag{6.16}$$

Modulo constants, the last two normalized polynomials have coefficient matrix

$$\begin{pmatrix} 1+p & -p \\ -1 & 1 \end{pmatrix},$$

whose determinant is one and whose inverse is uniformly bounded for $0 \le p \le 1$. Finite-dimensional norm equivalence therefore gives

$$|A|^2 + |B|^2 \le C[F_{0,p}]^2_{H^{1/2}(0,1)}. \tag{6.17}$$

Taylor's formula in (6.15), in the same finite-dimensional space, gives

$$[F_{w_i,p} - F_{0,p}]_{H^{1/2}(0,1)} \le C w_i^2(|Z| + |A| + |B|). \tag{6.18}$$

After decreasing the mesh threshold, the A, B terms are absorbed into (6.17). Since $B - A = w_i d_i$ and the $H^{1/2}$ seminorm is invariant under affine rescaling in one dimension, this proves (6.13).

We also need the constant and translation gauges. The full normalized L^2 three-jet matrix associated with (6.15) is uniformly invertible. Hence, for every constant c,

$$\sum_i w_i |z_i - c|^2 \le C\|T_\alpha z - c\|^2_{L^2(\mathbb{T})}. \tag{6.19}$$

Every first harmonic is exactly the angular trace of a support translation $z_i^{\mathrm{tr}} = q \cdot \nu_i$, and $L_\alpha z^{\mathrm{tr}} = 0$. Subtract the unique translation producing the $n = \pm 1$ part of $T_\alpha z$. For the resulting representative,

$$T_\alpha z = g + c, \qquad g = \Pi_{\mathrm{ng}} T_\alpha z.$$

The exact area row is

$$L_\alpha z = \sum_i \ell_i z_i = 0, \qquad \ell_i = \tan\frac{\alpha_{i-1}}{2} + \tan\frac{\alpha_i}{2} \asymp w_i. \tag{6.20}$$

By Cauchy–Schwarz and (6.19),

$$|c| L_\alpha \mathbf{1} = |L_\alpha(z - c\mathbf{1})| \le C\Big(\sum_i w_i |z_i - c|^2\Big)^{1/2},$$

where $L_\alpha \mathbf{1} \asymp 1$. Thus $|c| \le C\|g\|_{L^2}$. Applying (6.19) once more proves (6.14). Exact translation subtraction does not change d_i, because $a_i^- = a_i^+ = q \cdot \tau_i$ for a translation. $\square$

Lemma 6.3 (Stable zero-wall bubble decomposition)**.** *Let E equal E_i on K_i, where*

$$E_i = (t_i - s_i)d_i \sin x. \tag{6.21}$$

Then

$$\|E\|^2_{H^{1/2}(\mathbb{T})} \le C \sum_i w_i^6 |d_i|^2. \tag{6.22}$$

Proof. Lemma 6.1 gives, in normalized coordinates,

$$|E_i(y)| \le C w_i^3 |d_i| y(1-y), \qquad |\partial_y E_i(y)| \le C w_i^3 |d_i|. \tag{6.23}$$

In particular, E_i vanishes at both endpoints. The same-cell Gagliardo integral is at most $C w_i^6 |d_i|^2$. For the cross-cell integral use

$$|E_i(\theta) - E_j(\varphi)|^2 \le 2|E_i(\theta)|^2 + 2|E_j(\varphi)|^2.$$

For $\theta \in K_i$, integration of the circular kernel over $\mathbb{T} \setminus K_i$ is bounded by

$$C\left(1 + \operatorname{dist}(\theta, \partial K_i)^{-1}\right).$$

The endpoint factor in (6.23) makes the resulting Hardy integral at most $C w_i^6 |d_i|^2$. Summing over i, and adding the L^2 row, proves (6.22). This argument sums the whole complement of each cell at once, so it has no mesh-depth or adjacent-ratio constant. □

Lemma 6.4 (Physical versus angular interpolation)**.** *Let $S = \max_i \alpha_i + 2\pi/N$. On the exact area and translation quotient,*

$$\|\Pi_{\mathrm{ng}}(\widetilde{T}_\alpha - T_\alpha)z\|_{H^{1/2}(\mathbb{T})} \le C S^2 \|\Pi_{\mathrm{ng}} T_\alpha z\|_{H^{1/2}(\mathbb{T})}. \tag{6.24}$$

The constant is independent of every adjacent-cell ratio.

Proof. Projection is contractive in (6.8). By Lemmas 6.2 and 6.3,

$$\begin{aligned}
\|\Pi_{\mathrm{ng}} E\|^2_{H^{1/2}} &\le C \sum_i w_i^6 |d_i|^2 \\
&\le C S^4 \sum_i [T_\alpha z]^2_{H^{1/2}(K_i)} + C \sum_i w_i^8 |z_i|^2 \\
&\le C S^4 \|\Pi_{\mathrm{ng}} T_\alpha z\|^2_{H^{1/2}} + C S^7 \sum_i w_i |z_i|^2 \\
&\le C S^4 \|\Pi_{\mathrm{ng}} T_\alpha z\|^2_{H^{1/2}}.
\end{aligned}$$

Here we use the representative constructed in the proof of Lemma 6.2, for which $T_\alpha z = \Pi_{\mathrm{ng}} T_\alpha z + c$; consequently its local seminorms are those of $\Pi_{\mathrm{ng}} T_\alpha z$. Also $w_i \le 2S$, and the same-cell Gagliardo integrals are positive parts of the global seminorm. Taking square roots proves (6.24). □

Remark 6.5 (The repaired terminology). $T_\alpha z$ remains a useful trigonometric spline and the stage-65 disk gap for it is valid. Equation (6.6), not (6.7), is the trace of an actual affine map of a straight edge. The distinction is a small relative error, but it is not an exact identity and may not be suppressed in a value comparison.

6.2. **Transfer of D0 to the physical trace.** We first fix a normalization which was ambiguous in the historical notation. The reduced Fourier–Bessel form of F51 is denoted by $q^0_{\mathbb{D}}$; the physical half-Hessian form is

$$\boxed{\mathbf{q}_{\mathbb{D}}(f,g) := 2j_{0,1}^2 q^0_{\mathbb{D}}(f,g) = \frac{1}{2}D^2\mathcal{L}(0)[fe_r, ge_r].} \tag{6.25}$$

Every disk endpoint, metric, pairing, and distance below uses $\mathbf{q}_{\mathbb{D}}$. Thus the angular endpoint imported from F65 is first multiplied by $2j_{0,1}^2$; this changes only its positive universal gap constant, and changes neither its minimizer nor any relative trace comparison.

Let

$$b_\alpha = \Pi_{\mathrm{ng}}\{h_\alpha \sec x - 1\} \tag{6.26}$$

on the cells K_i. Define the angular and physical disk values

$$P_D(\alpha) = \inf_{z\in Z_\alpha} \mathbf{q}_{\mathbb{D}}(b_\alpha + \Pi_{\mathrm{ng}}T_\alpha z), \tag{6.27}$$

$$\widetilde{P}_D(\alpha) = \inf_{z\in Z_\alpha} \mathbf{q}_{\mathbb{D}}(b_\alpha + \Pi_{\mathrm{ng}}\widetilde{T}_\alpha z). \tag{6.28}$$

Here Z_α includes the exact area row and is quotiented by translations. Let

$$\widetilde{\mathfrak{G}}_\alpha(z) = \mathbf{q}_{\mathbb{D}}(\Pi_{\mathrm{ng}}\widetilde{T}_\alpha z), \tag{6.29}$$

and let $\tilde{z}^\sharp_\alpha$ be the corresponding Schur corrector.

Write $H_{\mathrm{ng}} = \Pi_{\mathrm{ng}}H^{1/2}(\mathbb{T})$ and let $\langle\cdot,\cdot\rangle_D$ be the bilinear form associated with $\mathbf{q}_{\mathbb{D}}$. The Fourier–Bessel bounds from the disk Hessian give

$$c\|f\|^2_{H^{1/2}} \le \mathbf{q}_{\mathbb{D}}(f) \le C\|f\|^2_{H^{1/2}}, \qquad f\in H_{\mathrm{ng}}. \tag{6.30}$$

Set

$$V_\alpha = \Pi_{\mathrm{ng}}T_\alpha Z_\alpha, \qquad \widetilde{V}_\alpha = \Pi_{\mathrm{ng}}\widetilde{T}_\alpha Z_\alpha. \tag{6.31}$$

Thus $P_D(\alpha)$ and $\widetilde{P}_D(\alpha)$ are the squared $\mathbf{q}_{\mathbb{D}}$-distances from $-b_\alpha$ to V_α and $\widetilde{V}_\alpha$, respectively.

Lemma 6.6 (Differentiated Hilbert projection calculus)**.** *Let q_t be uniformly coercive C^2 Hilbert forms on a fixed Hilbert space, let $P_t, \widetilde{P}_t$ be the corresponding orthogonal projections onto two C^2 families of closed subspaces, and put $\Delta P_t = \widetilde{P}_t - P_t$. Suppose, for $0\le r\le 2$,*

$$\|\partial_t^r q_t\| \le C_r h^r, \tag{6.32}$$

$$\|\partial_t^r P_t\| + \|\partial_t^r \widetilde{P}_t\| \le C_r h^r, \qquad \|\partial_t^r \Delta P_t\| \le C_r h^r\varepsilon, \tag{6.33}$$

$$\|\partial_t^r b_t\| \le C_r h^r B. \tag{6.34}$$

Then

$$\left|\partial_t^2\left\{\mathrm{dist}_{q_t}(-b_t,\widetilde{V}_t)^2 - \mathrm{dist}_{q_t}(-b_t,V_t)^2\right\}\right| \le Ch^2\varepsilon B^2. \tag{6.35}$$

Proof. Orthogonal completion of the square gives

$$\operatorname{dist}_{q_t}(-b_t, \widetilde{V}_t)^2 - \operatorname{dist}_{q_t}(-b_t, V_t)^2 = -q_t(\Delta P_t b_t, (\widetilde{P}_t + P_t)b_t). \tag{6.36}$$

Differentiate this identity twice. Every resulting term contains one of ΔP_t, $\partial_t \Delta P_t$, or $\partial_t^2 \Delta P_t$; the remaining factors are bounded by (6.32)–(6.34). Each term is therefore at most $Ch^2 \varepsilon B^2$. □

Lemma 6.7 (Uniform material calculus for the paired trace spaces)**.** *Let*

$$(1-\eta_0)a \le \gamma_i \le (1+\eta_0)a, \qquad \sum_i \gamma_i = 2\pi, \qquad 0 < \eta_0 < \frac{1}{4}, \tag{6.37}$$

and let $\sum_i \delta_i = 0$. *If*

$$R_{\mathrm{tr}}(\alpha) = \widetilde{P}_D(\alpha) - P_D(\alpha), \tag{6.38}$$

then

$$|D^2 R_{\mathrm{tr}}(\gamma)[\delta, \delta]| \le Ca \sum_i \gamma_i^2 \delta_i^2. \tag{6.39}$$

Proof. The assertion is trivial for $\delta = 0$. Otherwise put

$$h = \frac{\|\delta\|_{\ell^\infty}}{a}, \qquad \gamma(t) = \gamma + t\delta, \qquad |t| < c_0 h^{-1}. \tag{6.40}$$

Let Φ_t be the piecewise affine circle map carrying the normal nodes of γ to those of $\gamma(t)$, and pull all functions back by Φ_t. For every oriented cyclic arc I of length at most π,

$$\left| t \sum_{i \in I} \delta_i \right| \le c_0 a \# I \le C c_0 \sum_{i \in I} \gamma_i. \tag{6.41}$$

Consequently Φ_t' and every chord ratio are a fixed small relative perturbation of one. Differentiating the pulled Gagliardo kernel

$$\frac{\Phi_t'(\xi)\Phi_t'(\eta)}{\sin^2((\Phi_t(\xi) - \Phi_t(\eta))/2)} \tag{6.42}$$

zero, one, or two times proves

$$\|\partial_t^r q_t\| \le C_r h^r, \qquad 0 \le r \le 2. \tag{6.43}$$

Here the order-one part is exactly (6.42); the L^2 row is elementary. For the Bessel remainder use its established order-minus-one coefficients

$$|\Delta^r k_n| \le C_r (1 + |n|)^{-1-r}, \qquad 0 \le r \le 3, \tag{6.44}$$

and discrete summation by parts. It satisfies the same estimate, with one extra order of decay.

The normalized secant formula and two material differentiations give

$$\|\partial_t^r b_{\gamma(t)}\|_{H^{1/2}} \le C_r h^r a^{3/2}, \qquad 0 \le r \le 2. \tag{6.45}$$

Indeed a cell source has amplitude $O(a^2)$ and scale-invariant $H^{1/2}$ charge $O(a^4)$; summing $O(a^{-1})$ cells gives $O(a^3)$. Every normalized material derivative costs at most h. The global factor $h_{\gamma(t)}$ obeys the same bounds after differentiating its explicit tangent sum.

We next fix the coefficient space. Let

$$\tau_t^1 = (\cos\theta_i(t))_i, \qquad \tau_t^2 = (\sin\theta_i(t))_i$$

be the two exact support-translation vectors. They satisfy $L_{\gamma(t)}\tau_t^\mu = 0$. With the area weights $\ell_i(t)$ from (6.20), the 2×2 Gram matrix

$$M_t^{\mu\nu} = \sum_i \ell_i(t)\tau_{t,i}^\mu\tau_{t,i}^\nu$$

is uniformly comparable with the identity in the quasiuniform band. Define the explicit translation gauge

$$\mathcal{Q}_t z = z - \sum_{\mu,\nu=1}^{2}(M_t^{-1})_{\mu\nu}\left(\sum_i \ell_i(t) z_i \tau_{t,i}^\nu\right)\tau_t^\mu \tag{6.46}$$

and transport the area kernel by

$$C_t z = \mathcal{Q}_t\left(z - \frac{L_{\gamma(t)}z}{L_{\gamma(t)}\mathbf{1}}\mathbf{1}\right). \tag{6.47}$$

The coefficients of $L_{\gamma(t)}$ are comparable to a, and their first two normalized derivatives are bounded by $C_r h^r$. The same is true of τ_t^μ, M_t^{-1}, $\mathcal{Q}_t$, and C_t. The sampling estimate (6.14) makes the range of C_t uniformly transverse to the area and translation gauges. Thus, on the area/translation complement X at $t=0$, the pulled trace injections

$$B_t = \Pi_{\mathrm{ng}}(T_{\gamma(t)}C_t)\circ\Phi_t, \qquad \widetilde{B}_t = \Pi_{\mathrm{ng}}(\widetilde{T}_{\gamma(t)}C_t)\circ\Phi_t$$

are uniformly bounded below in the pulled q_t norm.

Apply the normalized calculation (6.11)–(6.23) on this quasiuniform material band. Its first two material derivatives give

$$|\partial_t^r\{(t_i - s_i)\sin x\}| \le C_r h^r a^3 y(1-y), \tag{6.48}$$

$$|\partial_y\partial_t^r\{(t_i - s_i)\sin x\}| \le C_r h^r a^3, \qquad 0\le r\le 2. \tag{6.49}$$

The derivative row uses the bound without an endpoint factor from (6.10). The coefficient-extraction and zero-wall proofs then apply after each material differentiation. Together with the derivative bounds for C_t, this gives

$$\|\partial_t^r B_t\| + \|\partial_t^r \widetilde{B}_t\| \le C_r h^r, \tag{6.50}$$

$$\|\partial_t^r(\widetilde{B}_t - B_t)\| \le C_r h^r a^2, \qquad 0\le r\le 2. \tag{6.51}$$

All operator norms use the fixed energy norm on X and the pulled q_t norm on the target. Notice that the endpoint factors were proved before differentiating, so no inverse minimum cell is present.

For completeness, the q_t-orthogonal projection onto $\operatorname{ran} B_t$ is

$$P_t = B_t G_t^{-1} B_t^{*t}, \qquad G_t = B_t^{*t} B_t, \tag{6.52}$$

where $*t$ is the adjoint in q_t. The corresponding formula with $\widetilde{B}_t$ defines $\widetilde{P}_t$. Uniform trace coercivity bounds both Gram inverses. Differentiating (6.52) and

$$\partial_t G_t^{-1} = -G_t^{-1}(\partial_t G_t)G_t^{-1}$$

twice, using (6.43), (6.50), and (6.51), yields

$$\begin{aligned} &\|\partial_t^r P_t\| + \|\partial_t^r \widetilde{P}_t\| \le C_r h^r, \\ &\|\partial_t^r(\widetilde{P}_t - P_t)\| \le C_r h^r a^2, \qquad 0 \le r \le 2. \end{aligned} \tag{6.53}$$

Lemma 6.6, with $\varepsilon = a^2$ and $B = a^{3/2}$, now gives

$$|D^2 R_{\mathrm{tr}}(\gamma)[\delta,\delta]| \le Ch^2 a^5 = Ca^3\|\delta\|_{\ell^\infty}^2. \tag{6.54}$$

Finally $\sum_i \gamma_i^2\delta_i^2 \ge ca^2\|\delta\|_{\ell^\infty}^2$, which proves (6.39). □

Lemma 6.8 (Relative physical-trace Schur transfer)**.** *There is a modulus $\omega_{\mathrm{tr}}(S) \to 0$ such that*

$$|\{\widetilde{P}_D(\alpha) - P_D(\alpha)\} - \{\widetilde{P}_D(\alpha_*) - P_D(\alpha_*)\}| \le \omega_{\mathrm{tr}}(S)\mathfrak{J}_4(\alpha). \tag{6.55}$$

Moreover $\widetilde{\mathfrak{G}}_\alpha$ and $\mathfrak{G}_\alpha$ are uniformly equivalent, with relative error $O(S^2)$.

Proof. The equivalence of the support metrics follows from Lemma 6.4, the Bessel coercivity $\mathbf{q}_{\mathbb{D}} \asymp H^{1/2}$, and polarization; more precisely,

$$|\widetilde{\mathfrak{G}}_\alpha(z) - \mathfrak{G}_\alpha(z)| \le CS^2\mathfrak{G}_\alpha(z). \tag{6.56}$$

Lemma 6.4 also gives a graph bijection $J_\alpha : V_\alpha \to \widetilde{V}_\alpha$ satisfying $\|J_\alpha - I\| \le CS^2$ in the $\mathbf{q}_{\mathbb{D}}$ norm. The elementary Hilbert distance comparison therefore gives

$$|R_{\mathrm{tr}}(\alpha)| \le CS^2\mathbf{q}_{\mathbb{D}}(b_\alpha) \le CS^5. \tag{6.57}$$

The final source estimate follows by normalized same/adjacent/separated Gagliardo summation for the continuous secant source: $\mathbf{q}_{\mathbb{D}}(b_\alpha) \le C\sum_i \alpha_i^4 \le CS^3$.

Put $a = 2\pi/N$, $d_i = \alpha_i - a$, and recall the exact identity

$$\mathfrak{J}_4(\alpha) = \sum_i d_i^2(\alpha_i^2 + 2a\alpha_i + 3a^2). \tag{6.58}$$

If $\mathfrak{J}_4(\alpha) \ge S^{9/2}$, apply (6.57) at α and at $\alpha_* = a\mathbf{1}$ to obtain

$$|R_{\mathrm{tr}}(\alpha) - R_{\mathrm{tr}}(\alpha_*)| \le CS^{1/2}\mathfrak{J}_4(\alpha). \tag{6.59}$$

Suppose instead that $\mathfrak{J}_4(\alpha) < S^{9/2}$. Then $\|d\|_{\ell^\infty}^4 \le C\mathfrak{J}_4(\alpha)$, so $\|d\|_{\ell^\infty}/S \le CS^{1/8}$. Since $S = a + \max_i \alpha_i$, the whole segment $\gamma(t) = a\mathbf{1} + td$ lies in a fixed quasiuniform band and $a \asymp S$. Cyclic invariance makes $DR_{\mathrm{tr}}(a\mathbf{1})[d] = 0$, because the gradient at the regular fan is constant and $\sum_i d_i = 0$. Taylor's formula and Lemma 6.7 give

$$\begin{aligned} |R_{\mathrm{tr}}(\alpha) - R_{\mathrm{tr}}(\alpha_*)| &\le Ca\int_0^1 (1-t)\sum_i \gamma_i(t)^2 d_i^2\,dt \\ &\le CS\mathfrak{J}_4(\alpha), \end{aligned} \tag{6.60}$$

where the last inequality follows from (6.58) and quasiuniformity. Combining (6.59) and (6.60) proves (6.55) with

$$\omega_{\mathrm{tr}}(S) = C(S^{1/2} + S). \tag{6.61}$$

□

Theorem 6.9 (Physical disk endpoint D0)**.** *For all sufficiently fine positive fans,*

$$\mathbf{q}_{\mathbb{D}}(b_\alpha + \Pi_{\mathrm{ng}}\widetilde{T}_\alpha z) - \widetilde{P}_D(\alpha_*) \geq c_f \mathfrak{J}_4(\alpha) + c_s \widetilde{\mathfrak{G}}_\alpha(z - \widetilde{z}^\sharp_\alpha), \tag{6.62}$$

where $c_f, c_s > 0$ are independent of all minimum cells.

Proof. Complete the physical support square exactly:

$$\mathbf{q}_{\mathbb{D}}(b_\alpha + \Pi_{\mathrm{ng}}\widetilde{T}_\alpha z) = \widetilde{P}_D(\alpha) + \widetilde{\mathfrak{G}}_\alpha(z - \widetilde{z}^\sharp_\alpha).$$

The angular endpoint gap of stages 65 and 129, together with Lemma 6.8, gives $\widetilde{P}_D(\alpha) - \widetilde{P}_D(\alpha_*) \geq c_f \mathfrak{J}_4$ after decreasing the fine-mesh threshold. Uniform coercivity gives the support constant c_s. □

6.3. **An exact fixed-disk scalar for the polygon.** Before the harmless area dilation, let $p_i(h)$ be the intersection of the support lines with normals ν_i, ν_{i+1}. It depends linearly on (h_i, h_{i+1}). On side i define

$$X_{\alpha,z}(\theta) = (1 - t_i(x))p_{i-1}(h_\alpha \mathbf{1} + z) + t_i(x)p_i(h_\alpha \mathbf{1} + z). \tag{6.63}$$

As θ traverses K_i, this traverses the actual straight side; hence its image is exactly $\partial P(\alpha, h_\alpha \mathbf{1} + z)$. Scale normalization of the eigenvalue makes the omitted final area dilation irrelevant.

Write

$$X_{\alpha,z}(\theta) = e_r(\theta) + W_{\alpha,z}(\theta). \tag{6.64}$$

At $z = 0$, (6.63) is the radial parametrization of the tangential polygon. Equations (6.3)– (6.6) therefore give the exact identity

$$\Pi_{\mathrm{ng}}(W_{\alpha,z} \cdot e_r) = b_\alpha + \Pi_{\mathrm{ng}}\widetilde{T}_\alpha z =: \widetilde{\tau}(\alpha, z). \tag{6.65}$$

Choose a fixed annular extension of W which equals W at $r = 1$ and vanishes for $r \leq 1/2$. For $\|W\|_{W^{1,\infty}}$ small, the map $F_W = \mathrm{id} + EW$ is bi-Lipschitz. Pulling stiffness and mass back by F_W defines the scale-normalized first eigenvalue

$$\mathcal{L}(W) = \frac{|F_W(\mathbb{D})|}{\pi}\lambda_1(F_W(\mathbb{D})). \tag{6.66}$$

Lemma 6.10 (Parametric fixed-chart analyticity)**.** *$\mathcal{L}$ is real analytic on a fixed small ball of $W^{1,\infty}(\mathbb{T};\mathbb{R}^2)$. It depends only on the boundary image, and*

$$D\mathcal{L}(0) = 0, \qquad \frac{1}{2}D^2\mathcal{L}(0)[W, W] = \mathbf{q}_{\mathbb{D}}(\Pi_{\mathrm{ng}}(W \cdot e_r)). \tag{6.67}$$

Proof. The pullback coefficients are rational analytic functions of DF_W and $\det DF_W$ on a small L^∞ ball. Uniform ellipticity, simplicity of the first eigenvalue, and the Riesz projection give operator-norm analyticity exactly as in the radial fixed chart. Two extensions with the same boundary image are related by a bi-Lipschitz change of variables and therefore give the same eigenvalue and area. The disk is critical for the scale-normalized functional. Reparametrization invariance removes the tangential component from its second differential; the normal component is the Fourier–Bessel disk Hessian. This proves (6.67). □

Define the nonlinear spectral remainder

$$\mathcal{N}(W) = \mathcal{L}(W) - \mathcal{L}(0) - \mathbf{q}_{\mathbb{D}}(\Pi_{\mathrm{ng}}(W \cdot e_r)). \tag{6.68}$$

Since (6.63) parametrizes the actual polygon,

$$\boxed{\lambda_1(P(\alpha, H_\alpha(z))) = \mathcal{L}(0) + \mathbf{q}_{\mathbb{D}}(\widetilde{\tau}(\alpha, z)) + \mathcal{N}(W_{\alpha,z}).} \tag{6.69}$$

Here the left side is area normalized; equivalently one may use the unnormalized support polygon because (6.66) already performs the scale normalization.

Part 2. Complete cubic tensor and quadratic-bubble reduction

7. Node CT: exact cubic tensor and retraction of CT-JM

Audited source: `CT_complete_tensor.tex`*.*

7.1. Two functional spaces and the physical support quotient.

The earlier working copies used the same symbol $\mathcal{X}$ for two different objects. We separate them here. For an arbitrary boundary vector field $W : \mathbb{T} \to \mathbb{R}^2$ let

$$\|W\|_{\mathcal{Y}}^2 := \sum_{n\in\mathbb{Z}} (1 + |n|)|\widehat{W}(n)|^2. \tag{7.1}$$

Thus $\mathcal{Y} = H^{1/2}(\mathbb{T}; \mathbb{R}^2)$ is the fixed space in which the disk Taylor tensors are estimated.

The support space is a different, fan-dependent quotient. Choose side normals $\nu_i = e_r(\theta_i)$, put $\tau_i = e_\theta(\theta_i)$, and write

$$\mathsf{Z}_\alpha := \left\{ z \in \mathbb{R}^N : \sum_i \left(\tan \frac{\alpha_{i-1}}{2} + \tan \frac{\alpha_i}{2} \right) z_i = 0 \right\} / \{(q \cdot \nu_i)_i : q \in \mathbb{R}^2\}. \tag{7.2}$$

For $x = \theta - \theta_i$ on $K_i = [\theta_i - \alpha_{i-1}/2, \theta_i + \alpha_i/2]$, set

$$\begin{aligned} a_i^- &= \frac{z_i - z_{i-1}}{\sin \alpha_{i-1}} - z_i \tan \frac{\alpha_{i-1}}{2}, \\ a_i^+ &= \frac{z_{i+1} - z_i}{\sin \alpha_i} + z_i \tan \frac{\alpha_i}{2}, \\ t_i(x) &= \frac{\tan x + \tan(\alpha_{i-1}/2)}{\tan(\alpha_{i-1}/2) + \tan(\alpha_i/2)}. \end{aligned} \tag{7.3}$$

The normal component of the genuine straight-side material velocity is

$$v_\alpha(z)(\theta) = z_i \cos x + \{(1 - t_i(x))a_i^- + t_i(x)a_i^+\} \sin x. \tag{7.4}$$

Its tangential component, needed only in the curvature row, is

$$w_\alpha(z)(\theta) = -z_i \sin x + \{(1 - t_i(x))a_i^- + t_i(x)a_i^+\} \cos x. \tag{7.5}$$

Let Π_{ng} delete the constant and first harmonics. The physical Schur form and the support Hilbert norm are, by definition,

$$\widetilde{\mathfrak{G}}_\alpha(z) := q_{\mathbb{D}}(\Pi_{\mathrm{ng}} v_\alpha(z)), \qquad \|z\|_{\mathcal{X}_\alpha} := \widetilde{\mathfrak{G}}_\alpha(z)^{1/2}. \tag{7.6}$$

The disk Bessel coercivity makes this a norm on the quotient Z_α. When a physical support slot is denoted by $Z_\alpha(z)$, we use the unambiguous convention

$$\|Z_\alpha(z)\|_{\mathcal{X}_\alpha} := \|z\|_{\mathcal{X}_\alpha}. \tag{7.7}$$

In particular, the comparison imported later by NR is not an additional theorem: with the current notation it is the identity

$$\boxed{\|Z_\alpha(z)\|^2_{\mathcal{X}_\alpha} = \widetilde{\mathfrak{G}}_\alpha(z).} \tag{7.8}$$

For later bookkeeping put $A_i(x) = (1-t_i(x))a_i^- + t_i(x)a_i^+$. Direct rotation gives the Cartesian identity

$$v_\alpha(z)e_r + w_\alpha(z)e_\theta = z_i\nu_i + A_i(x)\tau_i \qquad (x \in K_i). \tag{7.9}$$

The endpoint formulas (7.3) make the two Cartesian values at every common vertex identical. They do *not*, however, give a ratio-free reconstruction of the full vector trace from its normal component.

Proposition 7.1 (Failure of full-vector reconstruction)**.** *There is no constant independent of N for which*

$$\|v_\alpha(z)e_r + w_\alpha(z)e_\theta\|_{H^{1/2}} \le C\|\Pi_{\mathrm{ng}} v_\alpha(z)\|_{H^{1/2}}$$

holds on all exact support classes.

Proof. Take an even regular fan, $\alpha_i = a = 2\pi/N$, and the alternating support mode $z_i = (-1)^i$, which satisfies the area row and has no translation component. The endpoint formulas give

$$a_i^- = z_i \cot(a/2), \qquad a_i^+ = -z_i \cot(a/2).$$

Thus the normal trace has a fixed normalized-cell profile of amplitude $O(1)$, while the tangential coefficient A_i runs between values of size $\cot(a/2) \asymp a^{-1}$. Scale invariance of the cell $\dot{H}^{1/2}$ seminorm and summation over the N cells give

$$\|\Pi_{\mathrm{ng}} v_\alpha(z)\|^2_{H^{1/2}} \asymp N, \qquad \|v_\alpha(z)e_r + w_\alpha(z)e_\theta\|^2_{H^{1/2}} \asymp Na^{-2}.$$

The squared ratio therefore diverges like a^{-2}. ∎

Consequently tangential material velocities may enter only through the completed physical-coordinate chain rule in (7.27); they cannot be treated as an $H^{1/2}$ support slot. This is exactly the vertex-reparametrization issue isolated below.

7.2. **Fixed-domain coefficient ledger.** Let E be a fixed annular extension from boundary vector fields to $W^{1,\infty}(\mathbb{D};\mathbb{R}^2)$, vanishing on $\{|x| \le 1/2\}$. For three boundary directions V_i put

$$F_t(x) = x + \sum_{i=1}^{3} t_i EV_i(x), \qquad G_i = D(EV_i), \qquad G(t) = \sum_i t_i G_i.$$

Write

$$J(t) = \det(I + G(t)), \qquad P(t) = (I + G(t))^{-1}, \qquad C(t) = J(t)P(t)P(t)^T. \tag{7.10}$$

If I is an ordered list of distinct indices, a subscript I denotes $\partial_I|_{t=0}$, and $I_0 \dot\cup \cdots \dot\cup I_r = I$ denotes an ordered set partition, empty blocks being allowed.

All coefficient derivatives through cubic order are explicit. Namely,

$$\begin{aligned} J_i &= \operatorname{tr} G_i, \\ J_{ij} &= (\operatorname{tr} G_i)(\operatorname{tr} G_j) - \operatorname{tr}(G_i G_j), \qquad J_{ijk} = 0, \end{aligned} \tag{7.11}$$

$$\begin{aligned} P_i &= -G_i, \\ P_{ij} &= G_i G_j + G_j G_i, \\ P_{ijk} &= -\sum_{\pi \in S_3} G_{\pi(i)} G_{\pi(j)} G_{\pi(k)}, \end{aligned} \tag{7.12}$$

$$C_I = \sum_{I_0 \dot\cup I_1 \dot\cup I_2 = I} J_{I_0} P_{I_1} P_{I_2}^T. \tag{7.13}$$

Here $J_\varnothing = 1$ and $P_\varnothing = I$. Formula (7.13), rather than an ellipsis, is the metric/Jacobian ledger.

On $H_0^1(\mathbb{D})$ define

$$A_t(u,v) = \int_{\mathbb{D}} \nabla u \cdot C(t) \nabla v, \qquad M_t(u,v) = \int_{\mathbb{D}} J(t) uv, \tag{7.14}$$

$$s(t) = \pi^{-1} \int_{\mathbb{D}} J(t), \qquad \widehat{A}_t = s(t) A_t. \tag{7.15}$$

Thus the generalized eigenvalue of $(\widehat{A}_t, M_t)$ is the first Dirichlet eigenvalue after normalization to area π. Its scale rows are

$$\widehat{A}_I = \sum_{I_0 \dot\cup I_1 = I} s_{I_0} A_{I_1}, \qquad A_I(u,v) = \int \nabla u \cdot C_I \nabla v, \qquad M_I(u,v) = \int J_I uv. \tag{7.16}$$

No area constraint has yet been imposed in (7.16); in particular, the scale terms have not been silently deleted.

To use a fixed inner product, let $S(t)$ be multiplication by $J(t)^{-1/2}$ and set

$$H(t) = S(t)\widehat{A}_t S(t) \tag{7.17}$$

in the form sense on $L^2(\mathbb{D})$. The mass/Jacobian jets are

$$\begin{aligned} S_i &= -\tfrac{1}{2}J_i, \\ S_{ij} &= \tfrac{3}{4}J_iJ_j - \tfrac{1}{2}J_{ij}, \\ S_{ijk} &= -\tfrac{15}{8}J_iJ_jJ_k + \tfrac{3}{4}(J_{ij}J_k + J_{ik}J_j + J_{jk}J_i), \end{aligned} \tag{7.18}$$

and the completed operator jets are

$$H_I = \sum_{I_0 \dot\cup I_1 \dot\cup I_2 = I} S_{I_0}\widehat{A}_{I_1}S_{I_2}. \tag{7.19}$$

Equations (7.11)–(7.19) list every direct metric, Jacobian, mass, and scale monomial through third order.

7.3. **The full symmetric state tensor.** Let u_0 be the positive normalized disk state, $H_0u_0 = \lambda_0u_0$, and put

$$Q = I - |u_0\rangle\langle u_0|, \qquad R = Q(H_0 - \lambda_0)^{-1}Q, \qquad h_i = QH_iu_0. \tag{7.20}$$

The inverse in (7.20) is positive because λ_0 is the first simple eigenvalue.

Proposition 7.2 (Complete third reduced-resolvent formula)**.** *For arbitrary directions V_1, V_2, V_3, the full symmetric third derivative is*

$$\begin{aligned} \lambda_{123} = \langle u_0, H_{123}u_0\rangle & \\ &- \sum_{\mathrm{cyc(ij,k)}} \{\langle QH_{ij}u_0, Rh_k\rangle + \langle h_k, RQH_{ij}u_0\rangle\} \\ &+ \sum_{\pi\in S_3} \Big\langle h_{\pi(1)}, R(H_{\pi(2)} - \lambda_{\pi(2)}I)Rh_{\pi(3)}\Big\rangle. \end{aligned} \tag{7.21}$$

Here $\lambda_i = \langle u_0, H_iu_0\rangle$. At the scale-normalized disk $\lambda_i = 0$ for every boundary direction.

Proof of Proposition 7.2. Choose the analytic normalized eigenvector $u(t)$ with $\langle u(t), u(t)\rangle = 1$. The first differentiated state equation gives

$$Qu_i = -Rh_i, \qquad \langle u_i, u_0\rangle = 0. \tag{7.22}$$

Differentiate once more, project by Q, and use the differentiated normalization to eliminate the component along u_0. Substitution in $\lambda_i = \langle u, H_iu\rangle$ gives (7.21). For a repeated direction it reduces to

$$\lambda''' = \langle H'''\rangle - 6\langle QH''u_0, Rh\rangle + 6\langle h, R(H' - \lambda'I)Rh\rangle,$$

which also checks all multiplicities. Polarization proves the displayed symmetric formula. Criticality follows from dilation-normalized disk stationarity. ∎

Proposition 7.3 (Two energetic slots in the disk cubic tensor)**.** *The complete tensor of Proposition 7.2 extends to two $\mathcal{Y}$ slots and one Lipschitz coefficient slot, and*

$$|D^3\mathcal{L}(0)[U,V,W]| \leq C\sum_{\mathrm{cyc}} \|U\|_{\mathcal{Y}}\|V\|_{\mathcal{Y}}\|W\|_{W^{1,\infty}}. \tag{7.23}$$

The statement concerns the complete vector tensor; no radial reparametrization row has been separated from it.

Proof. For smooth traces use the exact physical chain rule (7.27). The normal tensor is the null form (7.34) plus the seven order-one rows (7.43). Its multipliers have total order at most two and obey the stated allocation by the dyadic argument of Lemma 7.6.

It remains to audit the curvature part. The disk Hessian has symbol $d^2(1+\mu_n)$ on $|n| \geq 2$, hence order one with the standard discrete differences. The three terms in Γ_{ij} have orders zero, one, and one. After symmetric polarization, their multipliers are finite permutations of

$$(1+\mu_{n_1}), \qquad n_2(1+\mu_{n_1}), \qquad n_1+n_2+n_3=0,$$

up to the harmless shifts by one caused by e_r, e_θ. They have total order at most two. On each dyadic block put the two largest half derivatives in L^2 and the remaining coefficient, together with its one allowed derivative, in L^∞. The finite-rank area and translation rows vanish by (7.28). Discrete Coifman–Meyer summation proves (7.23). Density defines the two energetic slots and preserves extension independence. ☐

The uncancelled rows in (7.21) are now literally:

row	exact source
third direct row	$\langle H_{123}u_0, u_0\rangle$ with (7.19);
second-state rows	the six H_{ij}–R–H_k pairings in the second line of (7.21);
three first-state rows	the six H_i–R–H_j–R–H_k pairings in the last line;
state normalization	the three $-\lambda_i H_j$ insertions in that same last line;
metric/Jacobian	the C_I terms of (7.13);
mass	the nonempty S_I terms of (7.19);
scale/area factor	the nonempty s_I terms of (7.16).

Thus the word "state rows" does not conceal a differentiated state, and the geometric rows are finite set-partition sums.

7.4. **Physical coordinates, curvature, area, and translations.** Write a boundary vector direction as

$$V_i = f_i e_r + w_i e_\theta. \tag{7.24}$$

The radial graph of the image of $e_r + \sum t_i V_i$ has first jet f_i and second jet

$$\Gamma_{ij} = w_i w_j - w_i f_j' - w_j f_i'. \tag{7.25}$$

Indeed, take the modulus and argument of $e^{i\theta}(1+\sum t_i f_i + i\sum t_i w_i)$ and then express the modulus in the new angular variable. This gives (7.25) before using any invariance.

The exact area-and-centering chart adds a unique χ_{ij} in the span of $1, \cos\theta, \sin\theta$ to the second radial jet. Its coefficients are

$$\begin{aligned} \chi_{ij} &= c_{ij}^0 + c_{ij}^1 \cos\theta + c_{ij}^2 \sin\theta, \\ c_{ij}^0 &= -\frac{1}{2\pi}\int_{\mathbb{T}} (\Gamma_{ij} + f_i f_j), \\ c_{ij}^1 &= -\frac{1}{\pi}\int_{\mathbb{T}} (\Gamma_{ij} + 2 f_i f_j)\cos\theta, \\ c_{ij}^2 &= -\frac{1}{\pi}\int_{\mathbb{T}} (\Gamma_{ij} + 2 f_i f_j)\sin\theta. \end{aligned} \tag{7.26}$$

These are obtained by twice differentiating

$$\tfrac{1}{2}\int R^2 = \pi, \qquad \tfrac{1}{3}\int R^3(\cos\theta, \sin\theta) = 0.$$

They are the area row and the two translation-projection rows; there is no unspecified finite-rank correction.

Let $\Lambda^{\mathrm{n}}(\rho)$ denote the scale-normalized scalar in the normal radial chart. Since $D\Lambda^{\mathrm{n}}(0) = 0$, the exact cubic chain rule is

$$\begin{aligned} D^3\mathcal{L}(0)[V_1, V_2, V_3] = D^3\Lambda^{\mathrm{n}}(0)[f_1, f_2, f_3] \\ + \sum_{\mathrm{cyc(ij,k)}} D^2\Lambda^{\mathrm{n}}(0)[f_k, \Gamma_{ij} + \chi_{ij}]. \end{aligned} \tag{7.27}$$

Scale and translation invariance imply

$$D^2\Lambda^{\mathrm{n}}(0)[g, \psi] = 0, \qquad \psi \in \operatorname{span}\{1, \cos\theta, \sin\theta\}. \tag{7.28}$$

The disk Hessian used throughout this node is the following literal Fourier–Bessel form. If $d = \partial_r u_0(1)$ for the positive L^2-normalized disk state and $\mathcal{D}_\circ$ is the multiplier in (7.36), set

$$q_{\mathbb{D}}(g,h) := d^2 \int_{\mathbb{T}} (\Pi_{\mathrm{ng}} g)(\mathcal{D}_\circ + 1)(\Pi_{\mathrm{ng}} h), \qquad q_{\mathbb{D}}(g) := q_{\mathbb{D}}(g,g). \tag{7.29}$$

Thus

$$\boxed{\frac{1}{2}D^2\Lambda^{\mathrm{n}}(0)[g,h] = q_{\mathbb{D}}(g,h)}, \qquad c\|\Pi_{\mathrm{ng}} g\|_{H^{1/2}}^2 \le q_{\mathbb{D}}(g) \le C\|\Pi_{\mathrm{ng}} g\|_{H^{1/2}}^2. \tag{7.30}$$

Indeed, for a mean-zero diagonal direction the second boundary row is $u_2| = d(f\mathcal{D}_\circ f + f^2/2)$. Green's identity gives the raw second Taylor coefficient $d^2 \int (f\mathcal{D}_\circ f + f^2/2)$. Multiplication by the area factor $1+\varepsilon^2(2\pi)^{-1}\int f^2$ adds $j^2(2\pi)^{-1}\int f^2 = (d^2/2)\int f^2$, because $d^2 = j^2/\pi$. This proves the first identity by polarization. Moreover, $\mu_{\pm 1} = -1$ and $\mu_n + 1 \asymp 1 + |n|$ for $|n| \geq 2$, which proves the coercivity statement and the geometric kernel (7.28).

Consequently all χ_{ij} rows cancel exactly. The surviving $D^2\Lambda^{\mathrm{n}}[f_k, \Gamma_{ij}]$ rows are the curvature rows. For $V_1 = V_2 = fe_r$ and $V_3 = we_\theta$, (7.27) becomes

$$D^3\mathcal{L}(0)[fe_r, fe_r, we_\theta] = -2D^2\Lambda^{\mathrm{n}}(0)[f, wf'], \tag{7.31}$$

or, equivalently, $-4q_{\mathbb{D}}(f, wf')$. After multiplication by the Taylor factor $1/2$ appropriate to the differentiated cubic term $3\mathcal{T}_3$, the same row is $-2q_{\mathbb{D}}(f, wf')$. This distinction fixes the multiplicity used downstream.

Proposition 7.4 (Retraction of the curvature-as-remainder estimate)**.** *Let f_α be the normal fan field and let $Z_\alpha(z) = v_\alpha(z)e_r + w_\alpha(z)e_\theta$ be a physical support slot. The formerly asserted estimate*

$$|4q_{\mathbb{D}}(f_\alpha, w_\alpha(z)f'_\alpha)| \leq CS^{5/2}\|Z_\alpha(z)\|_{\mathcal{X}_\alpha}. \tag{7.32}$$

is false uniformly as $N \to \infty$. Consequently the curvature row cannot be placed by itself in the $o(1)$ mixed remainder.

Proof. Take $N = 3M$, $a = 2\pi/N$, and the three-periodic fan $\alpha_{3j+r} = ap_r$ with $p = (1/6, 8/3, 1/6)$. Choose a three-periodic support vector satisfying the exact area row and the two translation rows; after scaling it tends to $(17, -2, 0)$. In the material coordinate $y = \theta/a$ one has

$$f_\alpha = a^2F_p + O(a^4), \qquad w_\alpha(z)f'_\alpha = G_{p,z} + O(a^2), \qquad \|Z_\alpha(z)\|^2_{\mathcal{X}_\alpha} \asymp a^{-1}.$$

The high-frequency expansion $\mathcal{D}_\circ + 1 = |D| + 1 + O(|D|^{-1})$ gives

$$q_{\mathbb{D}}(f_\alpha, w_\alpha(z)f'_\alpha) = 2\pi d^2 m_{p,z}a + O(a^2), \qquad m_{p,z} \neq 0.$$

Here $m_{p,z}$ is the mean Fourier pairing on the material period of length three. The factor 2π comes from its $2\pi/(3a)$ repetitions and the change $d\theta = a\,dy$. For a check independent of tail monotonicity, put

$$\widehat{H}(n) = \frac{1}{3}\int_0^3 H(y)e^{-2\pi i n y/3}\,dy$$

for every three-periodic material profile, and put

$$S_K = 2\sum_{n=1}^{K}\frac{2\pi n}{3}\,\mathrm{Re}\{\overline{\widehat{F}_p(n)}\widehat{G}_{p,z}(n)\}.$$

On each of the three cells the two functions are polynomials of degree at most two. For $omega = 2\pi n/3$ their integrals are evaluated by

$$I_0(l, u; \omega) = \frac{e^{-i\omega l} - e^{-i\omega u}}{i\omega}, \qquad I_j(l, u; \omega) = \left[-\frac{x^j e^{-i\omega x}}{i\omega}\right]_l^u + \frac{j}{i\omega}I_{j-1}(l, u; \omega).$$

Substitution of the rational cell endpoints and coefficients, with outward rational enclosures for π, sine, and cosine, gives at $K = 4096$

$$2.23372 < S_{4096} < 2.23374.$$

Moreover F_p' has total variation 6 and direct differentiation of the three quadratic pieces, including their endpoint jumps, gives $\mathrm{TV}(G_{p,z}) < 210$. Twice integrating by parts in $\widehat{F}_p$ and once in $\widehat{G}_{p,z}$ therefore yields

$$|m_{p,z} - S_K| \le \frac{\mathrm{TV}(F_p')\,\mathrm{TV}(G_{p,z})}{2\pi^2 K}, \qquad |m_{p,z} - S_{4096}| < 0.01559.$$

Thus $2.218 < m_{p,z} < 2.250$, in particular $m_{p,z} \neq 0$. The first ten modes give the additional regression value 2.159716, but are not used for the rigorous tail separation. The left side of (7.32) is therefore of order a, whereas its right side is $O(a^2)$. This contradicts a uniform constant. ☐

The distributional curvature pairing remains part of the exact identity

$$\begin{aligned} (7.33)\qquad & -4q_{\mathbb{D}}(f_\alpha, w_\alpha(z) f_\alpha') \\ &\qquad = D^3\mathcal{L}(0)[B_\alpha, B_\alpha, Z_\alpha(z)] - D^3\Lambda^{\mathrm{n}}(0)[f_\alpha, f_\alpha, v_\alpha(z)]. \end{aligned}$$

but Proposition 7.1 and Proposition 7.4 show that its two terms may not be estimated separately. In fact even their sum is not a small remainder, as proved below; it must be retained in a joint principal block.

7.5. **Extraction of the unique order-two symbol.** For real periodic functions define the same symmetric null form used by the quadratic-bubble node,

$$\mathcal{C}(f_1, f_2, f_3) = \frac{1}{3}\sum_{\mathrm{cyc}} \int_{\mathbb{T}} f_1\{(|D|f_2)(|D|f_3) - f_2' f_3'\}. \tag{7.34}$$

With $j = j_{0,1}$, the fixed nonzero disk solvability constant in our integral convention is

$$\boxed{\kappa_{\mathbb{D}} = -\frac{6j^2}{\pi}}. \tag{7.35}$$

Indeed, if $d = \partial_r u_0(1)$, then $\partial_\lambda u_0(1) = d/(2j^2)$; the coefficient of the cubic boundary solvability row is therefore $-2j^2$ times its angular mean. The minus sign comes from solving $(d/(2j^2))\,\delta\lambda + d\,R = 0$ for the spectral increment. Conversion from the Taylor coefficient to D^3 supplies the factor 6. This sign is fixed by the outward radial convention in (7.24); it is not absorbed into the definition of $\mathcal{C}$.

On the nongeometric subspace let

$$\begin{aligned} (7.36)\qquad & \mathcal{D}_\circ e^{in\theta} = \mu_n e^{in\theta} \quad (n \neq 0), \qquad\qquad \mathcal{D}_\circ 1 = 0, \\ & \qquad \mu_n = j\frac{J_{|n|}'(j)}{J_{|n|}(j)} = |n| + r_n, \qquad |\Delta^k r_n| \le C_k(1+|n|)^{-1-k}. \end{aligned}$$

The Bessel equation also gives the exact second normal derivative

$$\partial_r^2 \mathcal{E}_j g|_{r=1} = |D|^2 g + \mathcal{B}_1 g, \qquad \mathcal{B}_1 e^{in\theta} = (-\mu_n - j^2)e^{in\theta}, \tag{7.37}$$

where $\mathcal{B}_1$ has order one.

Proposition 7.5 (Exact seven-row boundary formula)**.** *Let f be real and $\Pi_0 f = f$, where Π_0 deletes only the constant mode. Then*

$$D^3\Lambda^{\mathrm{n}}(0)[f,f,f] = \kappa_{\mathbb{D}}\,\mathfrak{B}(f), \tag{7.38}$$

where

$$\mathfrak{B}(f) = \int_{\mathbb{T}} f\,\mathcal{D}_\circ\left(f\mathcal{D}_\circ f + \frac{1}{2}f^2\right) - \frac{1}{2}\int_{\mathbb{T}} f^2(|D|^2 + \mathcal{B}_1)f + \frac{2-j^2}{6}\int_{\mathbb{T}} f^3. \tag{7.39}$$

Consequently

$$\mathfrak{B}(f) = \mathcal{C}(f,f,f) + \sum_{a=1}^{7} \mathfrak{r}_a(f), \tag{7.40}$$

with the seven literal lower rows

$$\mathfrak{r}_1(f) = \int f|D|(f\,r(D)f), \qquad \mathfrak{r}_2(f) = \int f\,r(D)(f|D|f), \tag{7.41}$$

$$\mathfrak{r}_3(f) = \int f\,r(D)(f\,r(D)f), \qquad \mathfrak{r}_4(f) = \frac{1}{2}\int f|D|(f^2), \tag{7.42}$$

$$\mathfrak{r}_5(f) = \frac{1}{2}\int f\,r(D)(f^2), \qquad \mathfrak{r}_6(f) = -\frac{1}{2}\int f^2\mathcal{B}_1 f, \tag{7.43}$$

$$\mathfrak{r}_7(f) = \frac{2-j^2}{6}\int f^3. \tag{7.44}$$

Here $r(D) = \mathcal{D}_\circ - |D|$ on nonzero modes and is zero on the constant mode. The polarized formulas are obtained by symmetric polarization of (7.43)*.*

Proof. Write $u_0(r) = c_0 J_0(jr)$, $d = \partial_r u_0(1)$, and expand without factorials

$$u = u_0 + \varepsilon u_1 + \varepsilon^2 u_2 + \varepsilon^3 u_3 + O(\varepsilon^4), \qquad \lambda = j^2 + \varepsilon^2\lambda_2 + \varepsilon^3\lambda_3 + O(\varepsilon^4).$$

The linear term vanishes because f has zero mean. The first two Dirichlet rows are exactly

$$u_1|_{r=1} = -df, \qquad u_2|_{r=1} = d\left(f\mathcal{D}_\circ f + \frac{1}{2}f^2\right). \tag{7.45}$$

The nonzero angular modes of $\partial_r u_2$ are therefore $d\mathcal{D}_\circ(f\mathcal{D}_\circ f + f^2/2)$. Its constant mode contains the second spectral correction and the state-normalization correction, but it disappears after pairing with the mean-zero function f. The Bessel equation gives the exact identities

$$\partial_r^2 u_1|_{r=1} = -d(|D|^2 + \mathcal{B}_1)f, \qquad \partial_r^3 u_0(1) = (2-j^2)d. \tag{7.46}$$

The third Dirichlet row is

$$u_3|_{r=1} = -f\partial_r u_2 - \frac{1}{2}f^2\partial_r^2 u_1 - \frac{1}{6}f^3\partial_r^3 u_0.$$

Choose the state normalization with $\langle u_1, u_0\rangle = 0$. Green's identity then gives $\lambda_3 = d\int_{\mathbb{T}} u_3|_{r=1}$. Since the Rellich identity yields $d^2 = j^2/\pi$, multiplication by $3! = 6$ gives $-6d^2 = \kappa_{\mathbb{D}}$. Substitution of (7.45)–(7.46) proves (7.38)–(7.39).

Finally insert $\mathcal{D}_\circ = |D| + r(D)$. The two terms containing only $|D|$ are

$$\int f|D|(f|D|f) - \frac{1}{2}\int f^2|D|^2 f = \mathcal{C}(f,f,f).$$

The remaining terms are exactly (7.43). □

Lemma 7.6 (Seven-row multiplier audit)**.** *For the symmetric polarization $\mathfrak{r}_a(f_1,f_2,f_3)$ of every row in (7.43),*

$$|\mathfrak{r}_a(f_1,f_2,f_3)| \le C\sum_{p<q}\|f_p\|_{H^{1/2}}\|f_q\|_{H^{1/2}}\|f_\ell\|_{L^\infty}. \tag{7.47}$$

The constant is uniform over $a = 1,\dots,7$.

Proof. The symbols of $r(D)$ and $\mathcal{B}_1$ satisfy

$$|\Delta^k r_n| \le C_k(1+|n|)^{-1-k},$$
$$|\Delta^k(-\mu_n - j^2)| \le C_k(1+|n|)^{1-k}.$$

After polarization, the seven multipliers are finite permutations of

$$|n_2|r_{n_3},\quad r_{n_2}|n_3|,\quad r_{n_2}r_{n_3},\quad |n_2|,\quad r_{n_2},\quad -\mu_{n_3} - j^2,\quad 1,\qquad n_1+n_2+n_3=0. \tag{7.48}$$

Each multiplier has total order at most one and satisfies

$$|\Delta^\gamma m(n_1,n_2,n_3)| \le C_\gamma(1+\max_i|n_i|)^{1-|\gamma|}.$$

On a dyadic block the two largest frequencies are comparable. Put the corresponding half derivatives in L^2 and the remaining low-frequency factor in L^∞. Discrete Coifman–Meyer summation, followed by Cauchy–Schwarz in the common high index, proves (7.47). The finitely many modes with $|n| \le 2$ are absorbed by the inhomogeneous norms. □

Theorem 7.7 (Exact cubic split and literal remainder formula)**.** *For mean-zero normal directions, $\Pi_0 f_i = f_i$, one has the exact identity*

$$\boxed{D^3\Lambda^{\mathrm{n}}(0)[f_1,f_2,f_3] = \kappa_{\mathbb{D}}\mathcal{C}(f_1,f_2,f_3) + \mathcal{R}_1(f_1,f_2,f_3).} \tag{7.49}$$

The remainder is not defined by an order symbol. Its literal formula is

$$\begin{aligned}\mathcal{R}_1(f_1,f_2,f_3) = {}& \langle u_0, H_{123}u_0\rangle - \sum_{\mathrm{cyc(ij,k)}}\{\langle QH_{ij}u_0, Rh_k\rangle + \langle h_k, RQH_{ij}u_0\rangle\}\\ &+ \sum_{\pi\in S_3}\Big\langle h_{\pi(1)}, R(H_{\pi(2)} - \lambda_{\pi(2)}I)Rh_{\pi(3)}\Big\rangle - \kappa_{\mathbb{D}}\mathcal{C}(f_1,f_2,f_3),\end{aligned} \tag{7.50}$$

where every H_I is the finite set-partition expression (7.19), with (7.16), (7.13), and (7.11) substituted. Thus (7.50), unlike the old tag names alone, specifies every scalar monomial and its multiplicity. Equivalently, after the necessary direct/state cancellations have been performed, the same remainder is the seven-row grouped expression

$$\boxed{\mathcal{R}_1(f_1,f_2,f_3)=\kappa_{\mathbb{D}}\sum_{a=1}^{7}\mathfrak{r}_a(f_1,f_2,f_3),} \tag{7.51}$$

where $\mathfrak{r}_a(f_1,f_2,f_3)$ denotes symmetric polarization of the corresponding line of (7.43). Formula (7.51), rather than the uncancelled raw monomials, is the row decomposition used for analytic estimates.

For bookkeeping, expand (7.50) fully and assign each scalar monomial its first applicable tag in the ordered list

$$r(D),\quad \mathcal{B}_1,\quad C_I\ (I\neq\varnothing),\quad S_I\ (I\neq\varnothing),\quad s_I\ (I\neq\varnothing). \tag{7.52}$$

This gives the following disjoint ledger:

$$\mathcal{R}_1=R^r_{\mathrm{state}}+R_{rr}+R_{\mathrm{met}}+R_{J/M}+R_{\mathrm{scale}}. \tag{7.53}$$

Here:

1. *R^r_{state} consists of the state monomials in (7.21) containing at least one r_n from (7.36);*
2. *R_{rr} consists of those containing $\mathcal{B}_1$ from (7.37);*
3. *R_{met} consists of the remaining nonprincipal monomials tagged by a nonempty C_I in (7.13);*
4. *$R_{J/M}$ consists of the remaining monomials whose first tag is a nonempty S_I in (7.19);*
5. *R_{scale} consists of the remaining monomials whose first tag is a nonempty s_I in (7.16);*

The priority (7.52) makes the list disjoint, and the set-partition formulae make it exhaustive.

Proof of Theorem 7.7. Proposition 7.5 proves the diagonal identity (7.49) together with (7.51); symmetric polarization gives arbitrary nongeometric slots. Independently, Proposition 7.2 gives the complete reduced-resolvent expression. Subtracting the same principal null form from that expression gives (7.50). Hence the fixed-domain formula and the seven grouped boundary rows are two exact representations of the same trilinear remainder.

Expanding (7.50) and applying (7.52) records the origin of every raw monomial in (7.53). No estimate is taken before the raw metric and state rows have been recombined into the seven rows (7.51); this is the required order-two direct/resolvent cancellation. ☐

Corollary 7.8 (Order-one normal remainder)**.** *For mean-zero normal slots,*

$$|\mathcal{R}_1(f_1,f_2,f_3)|\le C\sum_{p<q}\|f_p\|_{H^{1/2}}\|f_q\|_{H^{1/2}}\|f_\ell\|_{L^\infty},\qquad \{\ell\}=\{1,2,3\}\setminus\{p,q\}. \tag{7.54}$$

Proof. Use (7.51) and sum (7.47) over the seven rows. □

For later fan inputs the constant scale mode must be retained explicitly. For arbitrary normal slots write

$$g_i = \Pi_0 f_i, \qquad \psi_i = (I - \Pi_0) f_i,$$

and, for $\varnothing \neq A \subset \{1,2,3\}$, let $\xi_i^A = \psi_i$ if $i \in A$ and $\xi_i^A = g_i$ otherwise. Define the finite seven-term geometric remainder by

$$\mathcal{R}_{\mathrm{geo}}(f_1, f_2, f_3) := \sum_{\varnothing \neq A \subset \{1,2,3\}} D^3\Lambda^{\mathrm{n}}(0)[\xi_1^A, \xi_2^A, \xi_3^A] \tag{7.55}$$

and set

$$\mathcal{R}_1^{\mathrm{full}}(f_1, f_2, f_3) := \mathcal{R}_1(g_1, g_2, g_3) + \mathcal{R}_{\mathrm{geo}}(f_1, f_2, f_3). \tag{7.56}$$

Multilinearity gives, with no omitted low mode,

$$\boxed{D^3\Lambda^{\mathrm{n}}(0)[f_1, f_2, f_3] = \kappa_{\mathbb{D}}\mathcal{C}(g_1, g_2, g_3) + \mathcal{R}_1^{\mathrm{full}}(f_1, f_2, f_3).} \tag{7.57}$$

Every term of (7.55) contains a constant. Apply Proposition 7.3 with that smooth finite-mode slot as the Lipschitz coefficient slot and the other two as energetic slots. Together with Corollary 7.8, this proves the same order-one bound for $\mathcal{R}_1^{\mathrm{full}}$ as in (7.54), with every f_i replaced by its full inhomogeneous norm. For the fan input,

$$\|(I - \Pi_0) f_\alpha\|_{W^{1,\infty}} \le CS^2, \tag{7.58}$$

so the geometric pure and mixed rows have the lower majorants CS^5 and $CS^{7/2}\|Z\|_{\mathcal{X}}$, respectively.

The physical curvature is a separate tensor, not a summand of the normal remainder. Define

$$\mathcal{K}_{\mathrm{curv}}(V_1, V_2, V_3) := \sum_{\mathrm{cyc(ij,k)}} D^2\Lambda^{\mathrm{n}}(0)[f_k, \Gamma_{ij}]. \tag{7.59}$$

Set

$$\mathcal{R}_1^{\mathrm{phys}}(V_1, V_2, V_3) := \mathcal{R}_1^{\mathrm{full}}(f_1, f_2, f_3) + \mathcal{K}_{\mathrm{curv}}(V_1, V_2, V_3). \tag{7.60}$$

Then (7.27) and the geometric kernel give the typed identity

$$\boxed{D^3\mathcal{L}(0)[V_1, V_2, V_3] = \kappa_{\mathbb{D}}\mathcal{C}(f_1, f_2, f_3) + \mathcal{R}_1^{\mathrm{phys}}(V_1, V_2, V_3).} \tag{7.61}$$

The χ_{ij} rows are absent from this formula because they vanish exactly, not because they have been assigned to $\mathcal{R}_1$.

For clarity, the cancellations used above are recorded separately:

family	cancellation or surviving order
two principal state derivatives	self-adjointness and one integration by parts combine them into (7.34);
mixed-sign order-two interactions	cancel inside $(\lvert D\rvert f_2)(\lvert D\rvert f_3) - f_2' f_3'$;
direct metric versus second state	their order-two pieces reconstruct the boundary acceleration in Proposition 7.5; after the combination only the seven order-one rows (7.43) remain;
state normalization	zero because $\lambda_i = 0$ at the normalized disk;
mass and scale	no tangential multiplier, hence order zero;
curvature	total order two in general; its ambient two-energetic-slot bound does not imply a radial support bound. Both the separate remainder estimate and the CT–JM small-sum estimate are false; the row is retained inside the honest JP–JC joint scalar audit;
area and translations	exact Hessian kernel (7.28).

We next define the scalar and support functionals used in the relative argument. Put $\alpha_* = a\mathbf{1}$, $\gamma(t) = \alpha_* + t(\alpha - \alpha_*)$, and write $f_t = f_{\gamma(t)}$. Fix at $t = 0$ a complement to the area and two translation rows. Let

$$C_t : \mathsf{Z}_{\alpha_*} \longrightarrow \mathsf{Z}_{\gamma(t)} \tag{7.62}$$

be the analytic constraint transport obtained by adding the unique linear combination of the scale and two translation vectors that satisfies the three rows at $\gamma(t)$. Its 3×3 Gram matrix is a uniform perturbation of the regular-fan matrix on the material polydisc, so C_t and C_t^{-1} are uniformly bounded. For a class z in the fixed complement define

$$F_1(t) := \mathcal{R}_1^{\mathrm{full}}(f_t, f_t, f_t), \tag{7.63}$$

$$\Psi_{1,n}(t)[z] := \mathcal{R}_1^{\mathrm{full}}(f_t, f_t, v_{\gamma(t)}(C_t z)), \tag{7.64}$$

$$\Psi_{1,t}(t)[z] := -4q_{\mathbb{D}}(f_t, w_{\gamma(t)}(C_t z) f_t'). \tag{7.65}$$

The last line is exactly the two curvature placements in (7.27); it is not included a second time in $\Psi_{1,n}$.

Lemma 7.9 (Material Cauchy bounds for the normal order-one rows)**.** *In the uniformly comparable material regime (7.78), F_1 and $\Psi_{1,n}$ extend holomorphically to the line polydisc of radius $c/\delta(t)$, where*

$$\delta(t) = \max_i \frac{\lvert\alpha_i - a\rvert}{\gamma_i(t)}.$$

Their scalar and dual majorants are

$$|F_1| \le CS^5, \qquad \|\Psi_{1,n}\|_{\mathcal{X}^*_{\gamma(t)}} \le CS^{7/2}. \tag{7.66}$$

Consequently, for $E(t) = \sum_i \gamma_i(t)^2(\alpha_i - a)^2$*,*

$$|F_1''(t)| \le CSE(t), \tag{7.67}$$

$$\|\Psi'_{1,n}(t)\|_{\mathcal{X}^*_{\gamma(t)}} \le CS^{3/2}E(t)^{1/2}. \tag{7.68}$$

Proof. Split every polar cell at its normal and pull both halves to $[0,1]$. The secant profile, the traces (7.4)–(7.5), and the constraint matrix in (7.62) are rational analytic functions of the incident widths. Their denominators stay in a fixed sector on the stated polydisc. The normal estimate (7.54) together with the finite geometric expansion (7.55) gives both bounds in (7.66). Same and adjacent cells use the normalized jets. On separated cells, subtracting both cell means makes the first-separating dyadic annuli summable. No bound contains the number of cells.

One-variable Cauchy on $\gamma(t) + \zeta(\alpha - \alpha_*)$ gives respectively $CS^5\delta(t)^2$ and $CS^{7/2}\delta(t)$. If k realizes $\delta(t)$, comparability gives

$$E(t) \ge \gamma_k(t)^2(\alpha_k - a)^2 \asymp S^4\delta(t)^2.$$

This proves (7.67)–(7.68). □

Lemma 7.10 (Relative closure of the pure cubic row)**.** *Use the material notation of Section 7.6. Let* f_α *be the exact normal fan input and let* $Z_\alpha(z)$ *be one physical support slot. On the exact scale/translation quotient there is a modulus* $\varepsilon_1 \to 0$ *such that*

$$|\mathcal{R}_1^{\text{full}}(f_\alpha, f_\alpha, f_\alpha) - \mathcal{R}_1^{\text{full}}(f_{a\mathbf{1}}, f_{a\mathbf{1}}, f_{a\mathbf{1}})| \le \varepsilon_1(S)\mathfrak{J}_4(\alpha). \tag{7.69}$$

Equivalently, in the typed physical notation (7.60)*, the left side is*

$$\left|\mathcal{R}_1^{\text{phys}}(B_\alpha, B_\alpha, B_\alpha) - \mathcal{R}_1^{\text{phys}}(B_{a\mathbf{1}}, B_{a\mathbf{1}}, B_{a\mathbf{1}})\right|. \tag{7.70}$$

Proof. The full remainder estimate following (7.56), (7.58), and the fan sizes give absolute majorants CS^5 for the pure row.

Fix $0 < \beta < 1$. If $\mathfrak{J}_4 \ge S^{5-\beta}$, these bounds divided by the right side of (7.69) gives CS^β. In the complementary regime the exact quartic Bregman identity (7.86) and the quantitative estimate following it give $\|\alpha - a\mathbf{1}\|_\infty/S \le 2S^{(1-\beta)/2}$, so for small S the entire segment is uniformly comparable and lies in the material polydisc (7.82). Lemma 7.9 gives the required material derivative bound. Cyclic symmetry kills the first pure derivative at the regular fan. Integrate with (7.85). This proves (7.69) with no cell-count factor. □

Proposition 7.11 (Retraction of CT–JM)**.** *There is no modulus* $\varepsilon_{\text{JM}}(S) \to 0$ *for which*

$$\left|\mathcal{R}_1^{\text{phys}}(B_\alpha, B_\alpha, Z_\alpha(z))\right| \le \varepsilon_{\text{JM}}(S)\mathfrak{J}_4(\alpha)^{1/2}\|Z_\alpha(z)\|_{\mathcal{X}_\alpha} \tag{7.71}$$

holds uniformly on the exact scale/translation quotient.

Proof. Use the three-periodic family from Proposition 7.4. Its seven normal rows satisfy, by the displayed order-one multiplier estimate,

$$\mathcal{R}_1^{\mathrm{full}}(f_\alpha, f_\alpha, v_\alpha(z)) = O(a^3),$$

whereas

$$-4q_{\mathbb{D}}(f_\alpha, w_\alpha(z)f'_\alpha) = -8\pi d^2 m_{p,z}a + O(a^2), \qquad m_{p,z} \neq 0.$$

For this family $S \asymp a$, $\mathfrak{J}_4(\alpha) \asymp a^3$, and $\|Z_\alpha(z)\|_{\mathcal{X}_\alpha} \asymp a^{-1/2}$; hence the product on the right of (7.71) is comparable to a. Dividing by that product gives a positive limiting lower bound, contradicting $\varepsilon_{\mathrm{JM}}(S) \to 0$. □

The surviving leading support forcing in the differentiated cubic Taylor row is therefore

$$2q_{\mathbb{D}}(f_\alpha, v_\alpha(z)) - 2q_{\mathbb{D}}(f_\alpha, w_\alpha(z)f'_\alpha) = 2q_{\mathbb{D}}\big(f_\alpha, v_\alpha(z) - w_\alpha(z)f'_\alpha\big). \tag{7.72}$$

For a polar graph R_α, the corresponding genuine polygon-normal trace is

$$v_{\alpha,\mathrm{eff}}(z) = \frac{v_\alpha(z) - (R'_\alpha/R_\alpha)w_\alpha(z)}{\sqrt{1 + (R'_\alpha/R_\alpha)^2}}. \tag{7.73}$$

Equations (7.72)– (7.73), together with all remaining cubic rows, are the exact input exported to the JP/JC joint scalar audit. No estimate in this node replaces that joint principal analysis.

Remark 7.12 (Sign and geometric tests). The Fourier symbol in (7.34) vanishes when the two differentiated frequencies have opposite signs and equals twice their product when they have the same sign. A constant in any slot gives zero by Parseval. The full tensor, including (7.27), vanishes on dilation and translation paths; the principal and lower rows need not vanish there separately.

7.6. **Uniform analytic tail on material cells.** Let

$$\alpha_i > 0, \qquad \sum_i \alpha_i = 2\pi, \qquad a = 2\pi/N, \qquad S = a + \max_i \alpha_i, \tag{7.74}$$

and let B_α be the physical fan boundary field.

Theorem 7.13 (Unconditional fixed-chart and natural-order facts)**.** *On a fixed sufficiently small real or complex $W^{1,\infty}$ ball, the pulled-back stiffness and mass forms, their compact solution operator, and the scale-normalized simple first-eigenvalue branch $\mathcal{L}(W)$ are analytic in operator norm. On that complexified chart its bilinear Hessian, and hence at every real near-disk active polygon the genuine polygon-preserving Hessian on continuous physical-side-affine fields, satisfies*

$$|D^2\mathcal{L}(W)[U,V]| \le C\|U\|_{H^{1/2}(\partial P)}\|V\|_{H^{1/2}(\partial P)}. \tag{7.75}$$

Here the trace norms on the complexified chart are transported to the underlying real boundary; these norms are uniformly equivalent throughout the fixed chart. The constant is independent of the number, the minimum size, and the ratios of the sides.

Proof. For the first assertion use the annular extension in the fixed-domain ledger. The coefficients J, P, C, S, and H in (7.10)–(7.17) are convergent rational power series in $D(EW)$ on a fixed L^∞ ball. Uniform ellipticity, the disk spectral gap, the resolvent identity, and the Riesz projection therefore give the operator-norm analytic simple branch. This applies directly to polygonal boundary traces and does not round their corners.

For the second assertion, extend a physical affine-side trace by a uniform right inverse from $H^{1/2}(\partial P;\mathbb{R}^2)$ to $H^1(P;\mathbb{R}^2)$. The radial bi-Lipschitz maps between the disk and the polygons give a uniform norm for this inverse. Convexity and the common inner and outer radii give uniform bounds for the eigenvalue, its spectral gap, the ground state, and its gradient. At a supporting line the barrier $s(D-s)$ gives $u(x) \le C\,\mathrm{dist}(x,\partial P)$; a scaled interior estimate then gives $\|\nabla u\|_\infty \le C$.

In the exact second-variation formula, the first coefficient jet is linear and the second is quadratic in the two extension gradients. The direct rows are bounded by the product of their H^1 norms; the two state forcings have the same H^{-1} bounds, and the reduced resolvent is uniformly coercive. On the smaller complex ball the real part of the pulled-back form remains uniformly coercive and the same spectral contour separates the analytic simple branch; the identical energy estimates therefore hold for the complex bilinear Hessian. Polarization proves (7.75). Approximation by Lipschitz extensions and invariance under changes with zero boundary trace justify the use of the energy extensions. ☐

Put

$$\mathcal{R}_4(W) = \mathcal{L}(W) - \mathcal{L}(0) - \frac{1}{2}D^2\mathcal{L}(0)[W,W] - \frac{1}{6}D^3\mathcal{L}(0)[W,W,W]. \tag{7.76}$$

Corollary 7.14 (Unconditional tame scalar fourth remainder)**.** *On a fixed smaller real or complex chart ball,*

$$\boxed{|\mathcal{R}_4(W)| \le C\|W\|^2_{H^{1/2}}\|W\|^2_{W^{1,\infty}}.} \tag{7.77}$$

The constant is independent of the number, minimum size, and ratios of the physical sides. This is a scalar statement and makes no assertion about the retired support-covector row of CT–M4.

Proof. The scale-normalized disk branch has $D\mathcal{L}(0) = 0$. For fixed W put

$$g(\zeta) = D^2\mathcal{L}(\zeta W)[W,W].$$

Theorem 7.13 makes g analytic for $|\zeta| < r_0/\|W\|_{W^{1,\infty}}$ and bounds it there by $C\|W\|^2_{H^{1/2}}$. Cauchy's coefficient estimate, after decreasing the chart so that this radius is greater than 2, gives for $0 \le t \le 1$

$$|g(t) - g(0) - tg'(0)| \le C\|W\|^2_{H^{1/2}}\|W\|^2_{W^{1,\infty}}t^2.$$

Twice integrating the scalar Taylor formula yields

$$\mathcal{R}_4(W) = \int_0^1 (1-t)\{g(t) - g(0) - tg'(0)\}\,dt,$$

and proves (7.77). $\square$

Call a material line $h_i(t) = a + td_i$ uniformly comparable when

$$\frac{a}{2} \le h_i(t) \le \frac{3a}{2} \qquad (1 \le i \le N,\ 0 \le t \le 1). \tag{7.78}$$

For such a line put

$$E(t) = \sum_i h_i(t)^2 d_i^2.$$

Remark 7.15 (Retired CT–M4 proposal: not an active obligation). Historically, this proposal assumed that CT–JP supplied a no-ratio Hilbert form $\mathfrak{G}_\alpha^{\mathrm{JP}}$ on the exact support quotient and wrote $\|z\|_{\mathcal{X}_\alpha^{\mathrm{JP}}} = \mathfrak{G}_\alpha^{\mathrm{JP}}(z)^{1/2}$. After splitting every physical cell at its normal and pulling the two halves to fixed unit intervals, prove the following four estimates with a constant independent of the number, minimum size, and ratios of the cells and moduli $\omega_4, \omega_{4,s} \downarrow 0$ at the origin:

$$\begin{aligned} |\mathcal{R}_4(B_\alpha)| &\le CS^5, \\ \|D\mathcal{R}_4(B_\alpha)\|_{(\mathcal{X}_\alpha^{\mathrm{JP}})^*} &\le CS^{7/2}, \end{aligned} \tag{7.79}$$

$$\begin{aligned} \left| \frac{d^2}{dt^2} \mathcal{R}_4(B_{h(t)}) \right| &\le \omega_4(S)E(t), \\ \left\| \frac{d}{dt} \left\{ D\mathcal{R}_4(B_{h(t)}) \circ \mathcal{T}_t \right\} \right\|_{(\mathcal{X}_\alpha^{\mathrm{JP}})^*} &\le \omega_{4,s}(S)E(t)^{1/2}, \end{aligned} \tag{7.80}$$

Here the varying support fibers are identified by the uniformly bounded constraint-preserving material transport

$$\mathcal{T}_t : \mathcal{X}_\alpha^{\mathrm{JP}} \longrightarrow \mathcal{X}_{h(t)}^{\mathrm{JP}}, \qquad \|\mathcal{T}_t\| + \|\mathcal{T}_t^{-1}\| \le C, \qquad \mathcal{T}_1 = I.$$

It is obtained by the fixed half-cell pullback followed by the exact area/translation projection. In the second estimate of (7.80), the derivative is the total derivative of the displayed fixed-fiber covector and therefore includes the variation of the physical support field as the fan changes. The estimates concern the completed direct-plus-state functional; its order-two rows may not be estimated separately.

Remark 7.16 (Why the old CT–M4 proposal was insufficient). Theorem 7.13 proves scalar analyticity and an absolute $H^{1/2}$ Hessian bound, but neither gives the vanishing moduli in (7.80). Those moduli express the cancellation between the direct metric row and the reduced-resolvent row at the natural boundary order. Historical rounded Calderón arguments estimated those rows separately and therefore do not prove this obligation.

Corollary 7.14 now closes the first, purely scalar bound in (7.79). It does not close the support-covector bound there, either relative path estimate in (7.80), or the moving support-fiber comparison. Thus it is sufficient for JP–BU but does not resurrect CT–M4.

The former fifth "support Hessian" row is not part of CT–M4. It used the wrong radial Schur metric and cannot control the physical support problem. Its correct replacement is the exact support Hessian in JP–EH, integrated without a post-cubic support remainder.

The former CT–AH hypothesis—bounded holomorphy of $W \mapsto D^2\mathcal{L}(W)$ in $\mathrm{Bil}(H^{1/2} \times H^{1/2})$ on a uniform complex $W^{1,\infty}$ ball—does not by itself imply CT–M4, because the ambient full-vector norm need not be uniformly comparable to the joint support metric. CT–AH is no longer a vertex of the active DAG.

The no-ratio physical sizes are

$$\|B_\alpha\|_{\mathcal{Y}}^2 \le CS^3, \qquad \|B_\alpha\|_{W^{1,\infty}} \le CS. \tag{7.81}$$

Split each material cell at its normal and pull both halves to $[0,1]$. The secant, support, Jacobian, and finite constraint coefficients are rational analytic functions of the incident widths. Hence, on a fan satisfying (7.78), they extend to

$$|\zeta_i - \alpha_i| < c\alpha_i, \qquad \sum_i \zeta_i = 2\pi, \tag{7.82}$$

with the same bounds as (7.81). The denominator sector and the chart radius are independent of N.

Lemma 7.17 (Archived consequences of the retired CT–M4 proposal)**.** *Assume the retired CT–M4 proposal 7.15. Let*

$$F_4(\alpha) = \mathcal{R}_4(B_\alpha),$$
$$\Psi_4(\alpha)[Z] = D\mathcal{R}_4(B_\alpha)[Z].$$

At the real fan α,

$$|F_4| \le CS^5, \qquad \|\Psi_4\|_{(\mathcal{X}_\alpha^{\mathrm{JP}})^*} \le CS^{7/2}. \tag{7.83}$$

If $h_i(t) = a + td_i$ is uniformly comparable in the sense of (7.78) and $E(t) = \sum_i h_i(t)^2 d_i^2$, then

$$|\partial_t^2 F_4(h(t))| \le \omega_4(S)E(t), \qquad \|\partial_t\{\Psi_4(h(t)) \circ \mathcal{T}_t\}\|_{(\mathcal{X}_\alpha^{\mathrm{JP}})^*} \le \omega_{4,s}(S)E(t)^{1/2}. \tag{7.84}$$

Proof. The two bounds in (7.83) are (7.79); the two bounds in (7.84) are (7.80). The identification of the varying dual spaces is the fixed material pullback required in CT–M4, so no derivative of a moving corner or hidden sum over material coordinates is introduced. ☐

The quartic action identities

$$\mathfrak{J}_4(\alpha) = 12\int_0^1 (1-t)E(t)\,dt, \qquad \int_0^1 E(t)\,dt \le C\mathfrak{J}_4(\alpha) \tag{7.85}$$

give the relative estimates as follows. At the regular fan, cyclic symmetry makes $DF_4(a\mathbf{1})$ a constant material covector, so it vanishes on $\sum_i d_i = 0$.

The same symmetry makes $\Psi_4(a\mathbf{1})$ a constant support covector, which vanishes on the area/translation quotient. Taylor's formula and (7.84) therefore give

$$|F_4(\alpha) - F_4(a\mathbf{1})| \le \omega_4(S) \int_0^1 (1-t)E(t)\,dt \le C\omega_4(S)\mathfrak{J}_4(\alpha).$$

Apply the fundamental theorem of calculus to the fixed-fiber covector $\Psi_4(h(t)) \circ \mathcal{T}_t$. The regular-fan value vanishes on the transported quotient, and Cauchy–Schwarz gives

$$|\Psi_4(\alpha)[Z]| \le \omega_{4,s}(S) \left(\int_0^1 E(t)\,dt\right)^{1/2} \|Z\|_{\mathcal{X}_\alpha^{\mathrm{JP}}} \le C\omega_{4,s}(S)\mathfrak{J}_4(\alpha)^{1/2}\|Z\|_{\mathcal{X}_\alpha^{\mathrm{JP}}}.$$

These are the pure and mixed tail bounds in the uniformly comparable regime. The following elementary dichotomy verifies that this regime and the absolute regime exhaust all positive fans. The exact Bregman identity is

$$\mathfrak{J}_4(\alpha) = \sum_i d_i^2(\alpha_i^2 + 2a\alpha_i + 3a^2). \tag{7.86}$$

Let $m = \max_i \alpha_i$, so $S = a + m$, and put $\rho = \mathfrak{J}_4(\alpha)^{1/2}/S^2$. Since $m \ge a$ and the summand at an index where $\alpha_i = m$ is at least $(m-a)^2 m^2$,

$$\frac{m-a}{S} \le 2\rho.$$

If $\rho \le 1/8$, then $a = (S - (m-a))/2 \ge 3S/8$; applying (7.86) at every index gives

$$\max_i \frac{|d_i|}{S} \le \frac{8}{3\sqrt{3}}\rho < 2\rho, \qquad \max_i \frac{|d_i|}{a} \le \frac{16}{3}\rho.$$

Fix $0 < \beta < 1$. If $\mathfrak{J}_4 < S^{5-\beta}$, then $\rho < S^{(1-\beta)/2}$; for all sufficiently small S the last bound is at most $1/2$, and the whole segment is uniformly comparable by convexity. In the complementary case $\mathfrak{J}_4 \ge S^{5-\beta}$, (7.83) gives respectively the relative factors CS^β and $CS^{1+\beta/2}$. Thus the two cases are exhaustive and both are $o(1)$ multiples of the required defects. Thus, under CT–M4, the quartic-and-higher Taylor tail has the uniform material bounds required by CT. This deduction uses no arbitrary-order tensor estimate.

7.7. **Literal fan-to-bubble dictionary.** Choose normal angles θ_i with $\theta_{i+1} - \theta_i = \alpha_i$ and set

$$K_i = [\theta_i - \alpha_{i-1}/2, \theta_i + \alpha_i/2]. \tag{7.87}$$

On K_i, $x = \theta - \theta_i$, the exact normal fan input is

$$f_\alpha(\theta) = h_\alpha \sec x - 1, \qquad h_\alpha = \left(\frac{\pi}{\sum_j \tan(\alpha_j/2)}\right)^{1/2}. \tag{7.88}$$

Define, on the same cell and with no change of nodes,

$$c_\alpha = h_\alpha - 1,$$

$$q_\alpha(\theta) = \tfrac{1}{2}x^2, \qquad r_\alpha(\theta) = (h_\alpha - 1)\tfrac{1}{2}x^2 + h_\alpha(\sec x - 1 - \tfrac{1}{2}x^2). \tag{7.89}$$

Then the coordinate dictionary is the literal equality

$$\boxed{f_\alpha = c_\alpha + q_\alpha + r_\alpha.} \tag{7.90}$$

The function q_α is continuous at the midpoint vertices because both adjacent values are $\alpha_i^2/8$, and

$$q_\alpha'' = 1 - \sum_i \alpha_i \delta_{\theta_i + \alpha_i/2}. \tag{7.91}$$

Taylor's formula on each normalized half-cell gives

$$\|r_\alpha\|_\infty \le CS^4, \qquad \|r_\alpha\|_{W^{1,\infty}} \le CS^3, \qquad \|r_\alpha\|^2_{H^{1/2}} \le CS^7. \tag{7.92}$$

Constants vanish in (7.34), including after polarization, because $\int(|D|g)(|D|h) = \int g'h'$. The tame estimate for the null form and (7.92) place every term containing r_α in the order-one/tail remainder already controlled above. The normal component of the physical support slot is exactly the trace $v_\alpha(z)$ used in PT. Therefore the two surviving principal rows of the third derivative are precisely

$$\kappa_{\mathbb{D}}\mathcal{C}(q_\alpha, q_\alpha, q_\alpha), \qquad \kappa_{\mathbb{D}}\mathcal{C}(q_\alpha, q_\alpha, v_\alpha(z)), \tag{7.93}$$

with the same nodes, normalization, and projection as the quadratic-bubble interface; the common fixed coefficient $\kappa_{\mathbb{D}}$ remains outside both terminal estimates.

For downstream Taylor assembly one must use $\mathcal{T}_3 = (3!)^{-1}D^3\mathcal{L}(0)$. Thus the corresponding pure Taylor row and differentiated mixed Taylor row are

$$\boxed{\frac{\kappa_{\mathbb{D}}}{6}\mathcal{C}(q_\alpha, q_\alpha, q_\alpha), \qquad \frac{\kappa_{\mathbb{D}}}{2}\mathcal{C}(q_\alpha, q_\alpha, v_\alpha(z)).} \tag{7.94}$$

The second coefficient is $3(\kappa_{\mathbb{D}}/6)$, because differentiating $\mathcal{T}_3(W, W, W)$ in one support slot produces three equal placements.

7.8. **CT reduction theorem and audit.**

Theorem 7.18 (Exact CT core)**.** *Unconditionally, the fixed-disk third derivative is the sum of $\kappa_{\mathbb{D}}$ times the null form* (7.34) *and the explicit normal order-one ledger* (7.53)*. The full physical-coordinate tensor is* (7.61)*; its area and translation rows vanish exactly. The radial null-form outputs are* (7.93)*, with Taylor factors* (7.94)*; the nonzero curvature support symbol* (7.72) *is retained separately and exported to the JP–JC audit. All unconditional stated constants are independent of the number and minimum size of active cells. The old material-tail proposal is not used by the active DAG.*

Proof. Combine Proposition 7.2, Theorem 7.7, Lemma 7.10, Propositions 7.4 and 7.11, and (7.90). The None of these statements imports an estimate for either terminal null-form row, and the active proof imports no premise from the archived calculation. ∎

audit item	disposition
state rows	all three families displayed in (7.21);
metric, mass, and scale	generated exhaustively by (7.11)–(7.19);
area and translation projection	explicit coefficients (7.26), then exact kernel cancellation;
curvature/tangential support	both the separate bound and CT–JM are retracted; (7.72) is retained for JP–JC;
principal symbol	same-sign null form (7.34);
all lower rows	disjoint ledger (7.53) and order-one estimate (7.54);
quartic pure/linear tail	old CT–M4 proposal and its consequences are retained only as an archive; the active exact-support route uses neither;
bubble output	literal equality (7.90), with identical cells and midpoint atoms.

8. Node QB: quadratic-bubble reduction

Audited source: `QB_bubble_reduction.tex`.

8.1. Quadratic polar spline. Let

$$\alpha_i > 0, \qquad \sum_i \alpha_i = 2\pi, \qquad a = \frac{2\pi}{N}, \qquad S = \max_i \alpha_i + a. \tag{8.1}$$

Choose normal angles θ_i with $\theta_{i+1} - \theta_i = \alpha_i$ and put

$$\varphi_i = \theta_i + \frac{\alpha_i}{2}. \tag{8.2}$$

The polar cell of side i is

$$K_i = [\varphi_{i-1}, \varphi_i] = [\theta_i - \alpha_{i-1}/2, \theta_i + \alpha_i/2]. \tag{8.3}$$

Define the continuous periodic quadratic spline

$$q_\alpha(\theta) = \frac{1}{2}(\theta - \theta_i)^2, \qquad \theta \in K_i. \tag{8.4}$$

Continuity follows because the two values at φ_i are both $\alpha_i^2/8$.

The exact radial fan graph is

$$f_\alpha(\theta) = h_\alpha \sec(\theta - \theta_i) - 1, \qquad \theta \in K_i, \tag{8.5}$$

with $h_\alpha = (\pi / \sum_j \tan(\alpha_j/2))^{1/2}$.

Lemma 8.1 (Secant/quadratic profile split)**.** *There is a constant c_α and a continuous remainder r_α such that*

$$f_\alpha = c_\alpha + q_\alpha + r_\alpha, \tag{8.6}$$

and, without a minimum-angle assumption,

$$\|r_\alpha\|_\infty \le CS^4, \qquad \|r_\alpha\|_{W^{1,\infty}} \le CS^3, \qquad \|r_\alpha\|_{H^{1/2}}^2 \le CS^7. \tag{8.7}$$

Also

$$\|q_\alpha\|_\infty \le CS^2, \qquad \|q_\alpha\|_{W^{1,\infty}} \le CS, \qquad \|q_\alpha\|_{H^{1/2}}^2 \le CS^3. \tag{8.8}$$

Proof. Take $c_\alpha = h_\alpha - 1$ and write, on a cell,

$$r_\alpha(x) = (h_\alpha - 1)\frac{x^2}{2} + h_\alpha\left(\sec x - 1 - \frac{x^2}{2}\right).$$

Since $h_\alpha - 1 = O(\sum_i \alpha_i^3) = O(S^2)$ and $\sec x - 1 - x^2/2 = O(x^4)$ through one normalized derivative, the first two estimates in (8.7) follow. Same and adjacent half-cells in the Slobodeckij seminorm cost $C\alpha_i^8$. On separated half-cells subtract both means and use the first-separating dyadic annulus. Thus

$$\|r_\alpha\|_{H^{1/2}}^2 \le C\sum_i \alpha_i^8 \le CS^4 \sum_i \alpha_i^4 \le CS^7.$$

The proof of (8.8) is identical with local amplitude α_i^2 and derivative amplitude α_i. □

8.2. **The higher secant profile is harmless.** Recall the symmetric null form of stage 138,

$$\mathcal{C}(g_1, g_2, g_3) = \frac{1}{3}\sum_{\text{cyc}} \int_{\mathbb{T}} g_1\{(|D|g_2)(|D|g_3) - g_2' g_3'\}\, d\theta. \tag{8.9}$$

It has total order two and the tame estimate

$$|\mathcal{C}(g_1, g_2, g_3)| \le C\sum_{p<q} \|g_p\|_{H^{1/2}}\|g_q\|_{H^{1/2}}\|g_\ell\|_{W^{1,\infty}}. \tag{8.10}$$

Lemma 8.2 (Constants disappear)**.** *For every g and every constant c,*

$$\mathcal{C}(g + c, g + c, g + c) = \mathcal{C}(g, g, g). \tag{8.11}$$

The same is true after polarization whenever a constant occupies any one slot.

Proof. Terms in which a differentiated slot is constant vanish. If the undifferentiated slot is constant, Parseval gives $\||D|g\|_2^2 = \|g'\|_2^2$. Polarization proves the last assertion. □

Theorem 8.3 (Secant-to-quadratic transfer)**.** *There are moduli $\varepsilon_q(S), \varepsilon_{q,s}(S) \to 0$ such that*

$$|\{\mathcal{C}(f_\alpha^3) - \mathcal{C}(q_\alpha^3)\} - \{\mathcal{C}(f_{a\mathbf{1}}^3) - \mathcal{C}(q_{a\mathbf{1}}^3)\}| \le \varepsilon_q(S)\mathfrak{J}_4(\alpha), \tag{8.12}$$

$$|\mathcal{C}(f_\alpha, f_\alpha, v_\alpha(z)) - \mathcal{C}(q_\alpha, q_\alpha, v_\alpha(z))| \le \varepsilon_{q,s}(S)\mathfrak{J}_4(\alpha)^{1/2}\widetilde{\mathfrak{G}}_\alpha(z)^{1/2}. \tag{8.13}$$

Here $v_\alpha(z)$ is the normal component of the physical support trace.

Proof. By Lemma 8.2, only r_α occurs in the difference. Equations (8.10)–(8.8) give the absolute estimates

$$|\mathcal{C}(f_\alpha^3) - \mathcal{C}(q_\alpha^3)| \le CS^6, \tag{8.14}$$

and

$$|\mathcal{C}(f_\alpha, f_\alpha, v) - \mathcal{C}(q_\alpha, q_\alpha, v)| \le CS^{9/2}\|v\|_{H^{1/2}}. \tag{8.15}$$

For example, an energetic (q, r) pair and a coefficient q cost $S^{3/2}S^{7/2}S = S^6$; the other placements are no larger.

Use the far threshold $\mathfrak{J}_4 \ge S^{6-\beta}$, $0 < \beta < 1$. Equations (8.14)– (8.15) give an $o(1)$ multiple of the required right sides. In the complementary regime,

$$\|\alpha - a\mathbf{1}\|_\infty / S \le CS^{(2-\beta)/4},$$

so the whole segment is in a fixed complex material polydisc. Pull every half-cell to $[0, 1]$ before differentiating. Cauchy's estimate applied to the two absolute majorants gives

$$|F_q''(t)| \le CS^2 E(t), \qquad \|\Psi_q'(t)\| \le CS^{5/2}E(t)^{1/2}.$$

Cyclic symmetry kills the regular first fan derivative and the regular support functional on the exact quotient. Integrate with the weighted and unweighted quartic actions. Physical trace equivalence replaces $\|v_\alpha(z)\|_{H^{1/2}}$ by $C\widetilde{\mathfrak{G}}_\alpha(z)^{1/2}$. $\square$

8.3. **Exact midpoint-atom formula.** Let

$$\mu_\alpha = \sum_i \alpha_i \delta_{\varphi_i}. \tag{8.16}$$

Lemma 8.4 (Distributional spline identity)**.** *One has*

$$q_\alpha'' = 1 - \mu_\alpha \tag{8.17}$$

on $\mathbb{T}$*. With Fourier convention* $\widehat{g}(n) = (2\pi)^{-1}\int_{\mathbb{T}} g(\theta)e^{-in\theta}\,d\theta$*, put*

$$M_\alpha(n) = \sum_i \alpha_i e^{-in\varphi_i}. \tag{8.18}$$

Then, for $n \ne 0$*,*

$$\widehat{q}_\alpha(n) = \frac{M_\alpha(n)}{2\pi n^2}. \tag{8.19}$$

Moreover the midpoint cancellation is the exact identity

$$M_\alpha(n) = \sum_i \alpha_i e^{-in\varphi_i}\left\{1 - \operatorname{sinc}\left(\frac{n\alpha_i}{2}\right)\right\}, \qquad n \ne 0. \tag{8.20}$$

Proof. Inside every open cell, $q_\alpha'' = 1$. At the vertex φ_i the one-sided derivatives are $\alpha_i/2$ and $-\alpha_i/2$, so the derivative jump is $-\alpha_i$. This proves (8.17); Fourier transformation gives (8.19).

The normal intervals $I_i = [\theta_i, \theta_{i+1}]$ partition the circle and have midpoint φ_i. Hence, for $n \neq 0$,

$$0 = \int_{\mathbb{T}} e^{-in\theta}\, d\theta = \sum_i \alpha_i e^{-in\varphi_i} \operatorname{sinc}\left(\frac{n\alpha_i}{2}\right).$$

Subtract this equality from (8.18). □

Theorem 8.5 (Exact discrete cubic scalar)**.** *The pure quadratic-bubble null form is*

$$\boxed{\mathcal{C}(q_\alpha, q_\alpha, q_\alpha) = \frac{1}{\pi^2} \operatorname{Re} \sum_{p,q\geq 1} \frac{M_\alpha(p) M_\alpha(q) \overline{M_\alpha(p+q)}}{pq(p+q)^2}.} \tag{8.21}$$

The series is interpreted by symmetric Abel limits and is absolutely convergent after the midpoint cancellation (8.20) *is inserted sourcewise.*

Proof. For a real g, Fourier expansion of (8.9) shows that only triples with two frequencies of the same sign survive. If $p, q > 0$ and the third frequency is $-(p+q)$, the unsymmetrized coefficient is $2pq$. The positive and negative triples are conjugate, so

$$\mathcal{C}(g, g, g) = 8\pi \operatorname{Re} \sum_{p,q\geq 1} pq\, \widehat{g}(p) \widehat{g}(q) \overline{\widehat{g}(p+q)}.$$

Insert (8.19); since $8\pi/(2\pi)^3 = 1/\pi^2$, this gives (8.21). Abel regularization r^{p+q} justifies rearrangement. Formula (8.20) gives two powers of $n\alpha_i$ at low frequency and the trivial bound at high frequency; dyadic source allocation then makes the Abel limit independent of the order of summation. □

8.4. **Final discrete obligations.**

Intermediate obligation 8.6 (DF: midpoint cubic Bregman estimate)**.** For the explicit scalar on the right of (8.21), prove

$$|\mathcal{C}(q_\alpha^3) - \mathcal{C}(q_{a\mathbf{1}}^3)| \leq CS\mathfrak{J}_4(\alpha). \tag{8.22}$$

Every cumulative phase produced by differentiating φ_i must first be summed cyclically; the midpoint factor in (8.20) is not to be differentiated row by row.

Intermediate obligation 8.7 (DM: polarized midpoint/support estimate)**.** For the polarization of (8.21) with one slot replaced by the physical normal trace $v_\alpha(z)$, prove

$$|\mathcal{C}(q_\alpha, q_\alpha, v_\alpha(z))| \leq \varepsilon_{\mathrm{DM}}(S) \mathfrak{J}_4(\alpha)^{1/2} \widetilde{\mathfrak{G}}_\alpha(z)^{1/2}, \qquad \varepsilon_{\mathrm{DM}}(S) \to 0. \tag{8.23}$$

Proposition 8.8 (Equivalence)**.** *DF and the secant-to-quadratic transfer imply the pure principal row NF. DF, DM, and the same transfer imply the mixed principal row NM. These are exactly the two terminal principal inputs exported by QB; the conversion to CF, CM, NR, and J0 is performed only in the current* `NR_assembly.tex` *node.*

Proof. Apply Theorem 8.3 to the pure and mixed null forms and absorb its moduli into the DF and DM moduli. No downstream tensor or value statement is imported here. □

Part 3. The pure midpoint row

9. Foundation F144: Hilbert stress and the exact midpoint identity

Audited source: `convex_largeN_polya_szego_stage144_NR_Hilbert_midpoint_quadrature_reduction.tex`.

9.1. A second exact form of the cubic null structure. Use the periodic Hilbert transform convention

$$\widehat{\mathcal{H}f}(n) = -i\,\mathrm{sgn}(n)\widehat{f}(n), \qquad |D| = \mathcal{H}\partial_\theta. \tag{9.1}$$

The symmetric null form is the one of stage 138; on the diagonal,

$$\mathcal{C}(f,f,f) = \int_{\mathbb{T}} f\{(|D|f)^2 - (f')^2\}. \tag{9.2}$$

Lemma 9.1 (Hilbert stress identity)**.** *For every smooth real periodic f,*

$$\boxed{\mathcal{C}(f,f,f) = \int_{\mathbb{T}} f''(\mathcal{H}f)^2.} \tag{9.3}$$

The identity extends by Abel regularization whenever the right side is a well-defined distributional pairing.

Proof. For a real function g the Hilbert product identity is

$$(\mathcal{H}g)^2 - g^2 = \mathcal{H}(2g\mathcal{H}g). \tag{9.4}$$

Indeed it follows first on trigonometric polynomials from $\mathcal{H}^2 = -I$ and then by density. Put $g = f'$. By skew-adjointness of $\mathcal{H}$ and commutation with ∂_θ,

$$\begin{aligned}\mathcal{C}(f,f,f) &= \int f\mathcal{H}(2f'\mathcal{H}f') = -2\int(\mathcal{H}f)f'(\mathcal{H}f)' \\ &= -\int f'\{(\mathcal{H}f)^2\}' = \int f''(\mathcal{H}f)^2.\end{aligned}$$

Abel regularization proves the distributional extension. □

Polarization of (9.3) also gives, for real f,g,v,

$$\mathcal{C}(f,g,v) = \frac{1}{3}\int_{\mathbb{T}}\{f''\mathcal{H}g\mathcal{H}v + g''\mathcal{H}f\mathcal{H}v + v''\mathcal{H}f\mathcal{H}g\}. \tag{9.5}$$

In particular,

$$3\mathcal{C}(q,q,v) = 2\int q''\mathcal{H}q\mathcal{H}v + \int v''(\mathcal{H}q)^2. \tag{9.6}$$

This will be the smaller starting point for the mixed row DM.

9.2. **The cubic scalar is a midpoint quadrature error.** Let $I_i = [\theta_i, \theta_{i+1}]$ have length $h_i > 0$ and midpoint c_i. On I_i put

$$q_\alpha(\theta_i + x) = \frac{1}{2} \min\{x^2, (h_i - x)^2\}. \tag{9.7}$$

Then, as in stage 139,

$$q''_\alpha = 1 - \mu_\alpha, \qquad \mu_\alpha = \sum_i h_i \delta_{c_i}. \tag{9.8}$$

Theorem 9.2 (Exact Hilbert midpoint formula). *Put $w_\alpha = \mathcal{H} q_\alpha$. Then*

$$\boxed{\mathcal{C}(q_\alpha, q_\alpha, q_\alpha) = \int_{\mathbb{T}} (q_\alpha - \overline{q}_\alpha)^2 - \sum_i h_i |w_\alpha(c_i)|^2.} \tag{9.9}$$

For the regular fan $a\mathbf{1}$, every sampled value is zero and

$$\mathcal{C}(q_{a\mathbf{1}}^3) = \frac{Na^5}{720} = \frac{\pi a^4}{360}. \tag{9.10}$$

Proof. Insert (9.8) in (9.3). The Hilbert transform is an L^2 isometry on mean-zero functions, so

$$\int w_\alpha^2 = \int (q_\alpha - \overline{q}_\alpha)^2.$$

This proves (9.9). At the regular fan, reflection symmetry about each midpoint makes q even and $\mathcal{H}q$ odd, hence $w(c_i) = 0$. Direct integration on one cell gives

$$\int_{I_i} q_\alpha = \frac{h_i^3}{24}, \qquad \int_{I_i} q_\alpha^2 = \frac{h_i^5}{320}. \tag{9.11}$$

Using $Na = 2\pi$ in the resulting variance gives (9.10). □

Thus the apparently cubic two-frequency obstruction is the difference of one explicit local scalar and one nonnegative one-frequency sampling row.

10. Foundation F148: the exact cumulative-free material matrix

Audited source: `convex_largeN_polya_szego_stage148_NR_quadrature_primitive_material_matrix.tex`.

10.1. **Clausen material derivative as a length gradient.** Let $I_j = [\theta_j, \theta_{j+1}]$ be a cyclic interval partition of $\mathbb{T}$, with length h_j and midpoint c_j. Put

$$\Phi(x) = \sum_{n \geq 1} \frac{\sin(nx)}{n^2}, \qquad K(x) = \Phi'(x) = -\log \left| 2 \sin \frac{x}{2} \right|. \tag{10.1}$$

The sampled conjugate of the quadratic fan spline is

$$w_i = \frac{1}{\pi} \sum_j h_j \Phi(c_i - c_j). \tag{10.2}$$

Along a positive affine path, let $h_j' = d_j$, $\sum_j d_j = 0$, and let $s_j = c_j'$. Stage 144 gives

$$W_i := \dot{w}_i = \frac{1}{\pi} \sum_{j \neq i} \{ d_j \Phi(c_i - c_j) + h_j (s_i - s_j) K(c_i - c_j) \}. \tag{10.3}$$

Fix the target i. Relabel cyclically and choose a partition endpoint θ_0 as a cut; write $\theta_0 + 2\pi$ for its second copy. The cut may be chosen at circular distance at least $\pi/3$ from c_i. This target-dependent relabelling changes neither (10.2) nor (10.3).

For $k < i$ define

$$\mathcal{E}_{ik}^- = \sum_{j<k} h_j K(c_i - c_j) + \frac{h_k}{2} K(c_i - c_k) - \int_{\theta_0}^{c_k} K(c_i - y)\, dy. \tag{10.4}$$

For $k > i$ define

$$\mathcal{E}_{ik}^+ = \int_{c_k}^{\theta_0 + 2\pi} K(c_i - y)\, dy - \frac{h_k}{2} K(c_i - c_k) - \sum_{j>k} h_j K(c_i - c_j). \tag{10.5}$$

The two arcs in these definitions do not contain c_i in their interiors. Finally put

$$\begin{aligned} E_i^- &= \sum_{j<i} h_j K(c_i - c_j) - \int_{\theta_0}^{\theta_i} K(c_i - y)\, dy, \\ E_i^+ &= \sum_{j>i} h_j K(c_i - c_j) - \int_{\theta_{i+1}}^{\theta_0 + 2\pi} K(c_i - y)\, dy, \end{aligned} \tag{10.6}$$

and define the matrix

$$\mathcal{E}_{ik} = \begin{cases} \mathcal{E}_{ik}^-, & k < i, \\ (E_i^- - E_i^+)/2, & k = i, \\ \mathcal{E}_{ik}^+, & k > i. \end{cases} \tag{10.7}$$

Theorem 10.1 (Exact cumulative-free material matrix)**.** *For every zero-sum length velocity d,*

$$\boxed{W_i = \frac{1}{\pi} \sum_k d_k \mathcal{E}_{ik}.} \tag{10.8}$$

Thus no node velocity s_j, root velocity, or cumulative sum of the d_j occurs in the final coefficient matrix.

Proof. Temporarily keep the cut fixed. Its length coordinates give

$$\frac{\partial c_j}{\partial h_k} = \begin{cases} 0, & j < k, \\ 1/2, & j = k, \\ 1, & j > k. \end{cases}$$

If $M_{ik} = \pi\, \partial w_i/\partial h_k$, differentiation of (10.2) gives

$$M_{ik} = \Phi(c_i - c_k) + \sum_j h_j \left(\frac{\partial c_i}{\partial h_k} - \frac{\partial c_j}{\partial h_k}\right) K(c_i - c_j), \tag{10.9}$$

where the self coefficient of $K(0)$ is zero before the Abel limit.

For $k < i$, equation (10.9) becomes

$$M_{ik} = \Phi(c_i - c_k) + \sum_{j<k} h_j K(c_i - c_j) + \frac{h_k}{2} K(c_i - c_k).$$

The fundamental theorem of calculus gives exactly

$$\int_{\theta_0}^{c_k} K(c_i - y)\, dy = \Phi(c_i - \theta_0) - \Phi(c_i - c_k).$$

After rotating the origin to the cut, the first term on the right is $\Phi(c_i)$. Hence $M_{ik} = \Phi(c_i) + \mathcal{E}^-_{ik}$. The same calculation on the right arc gives $M_{ik} = \Phi(c_i) + \mathcal{E}^+_{ik}$ for $k > i$.

For $k = i$, (10.9) reads

$$M_{ii} = \frac{1}{2} \sum_{j<i} h_j K(c_i - c_j) - \frac{1}{2} \sum_{j>i} h_j K(c_i - c_j).$$

The exact integrals over the two sides of I_i give $M_{ii} = \Phi(c_i) + (E^-_i - E^+_i)/2$. Therefore

$$M_{ik} = \Phi(c_i) + \mathcal{E}_{ik} \qquad \text{for every } k.$$

Since $\sum_k d_k = 0$, the common Clausen term disappears and (10.8) follows. This also proves that the result is independent of the chosen cut. □

10.2. **The matrix is a cotangent integral of a cellwise sawtooth.** On each cell define

$$\eta(y) = \begin{cases} -(y - \theta_j), & \theta_j < y < c_j, \\ \theta_{j+1} - y, & c_j < y < \theta_{j+1}. \end{cases} \tag{10.10}$$

Then, distributionally on I_j,

$$d\eta = h_j \delta_{c_j} - \mathbf{1}_{I_j}\, dy, \qquad |\eta| \le h_j/2, \qquad \int_{I_j} \eta = 0. \tag{10.11}$$

Moreover

$$K'(x) = -\frac{1}{2} \cot \frac{x}{2} = -\pi \kappa(x), \qquad \kappa(x) = \frac{1}{2\pi} \cot \frac{x}{2}. \tag{10.12}$$

Proposition 10.2 (Exact sawtooth representation)**.** *For $k < i$ and $k > i$, respectively,*

$$\mathcal{E}^-_{ik} = \int_{\theta_0}^{c_k} \eta(y) K'(c_i - y)\, dy, \tag{10.13}$$

$$\mathcal{E}^+_{ik} = -\int_{c_k}^{\theta_0 + 2\pi} \eta(y) K'(c_i - y)\, dy. \tag{10.14}$$

The terminal half-atom at c_k closes the cumulative mass to zero. Likewise E_i^- and E_i^+ are the corresponding full-cell integrals on the two sides of I_i. Hence every matrix entry is assembled from the same cellwise mean-zero sawtooth and one cotangent frequency.

Proof. On the left arc, the signed midpoint-quadrature measure consists of all full atoms before I_k, half of the atom at c_k, and minus Lebesgue measure on the arc. Its cumulative distribution is exactly η and vanishes at both endpoints. Stieltjes integration by parts therefore gives

$$\int K(c_i - y)\, d\eta(y) = -\int \eta(y) \frac{d}{dy} K(c_i - y)\, dy = \int \eta(y) K'(c_i - y)\, dy,$$

which is (10.13). The right arc has the opposite orientation, giving (10.14). The full-side statements are identical. □

Remark 10.3 (Both former cancellations are now identities)*.* The endpoint vanishing in stage 147 is encoded by the fact that η is zero at every cell endpoint. The cumulative-velocity cancellation is encoded by $\int_{I_j} \eta = 0$ and the disappearance of the common $\Phi(c_i)$ in Theorem 10.1. Thus neither cancellation needs to be recovered after a triangle inequality.

11. Node DF0A: intrinsic variance and the far stratum

Audited source: `DF0_variance.tex`*.*

11.1. **The intrinsic equality-stratum variance.** Let

$$h_i > 0, \qquad \sum_i h_i = 2\pi =: L, \qquad a = \frac{2\pi}{N}, \qquad S = \max_i h_i + a. \tag{11.1}$$

The quadratic cell bubble and its sampled Hilbert conjugate are

$$q_\alpha(\theta_i + x) = \frac{1}{2} \min\{x^2, (h_i - x)^2\}, \qquad 0 \le x \le h_i, \tag{11.2}$$

$$T(\alpha) = \sum_i h_i |\mathcal{H} q_\alpha(c_i)|^2, \qquad c_i = \theta_i + h_i/2. \tag{11.3}$$

Put $A_k = \sum_i h_i^k$ and define

$$\ell^2 := \frac{A_3}{L}, \qquad \boxed{\mathfrak{D}(\alpha) := \sum_i h_i (h_i^2 - \ell^2)^2 = A_5 - \frac{A_3^2}{L}.} \tag{11.4}$$

Lemma 11.1 (Pair-variance formula and equality strata)**.** *One has*

$$\mathfrak{D}(\alpha) = \frac{1}{2L} \sum_{i,j} h_i h_j (h_i^2 - h_j^2)^2. \tag{11.5}$$

Consequently $\mathfrak{D} \ge 0$, and it vanishes exactly when all positive cell lengths are equal. After zero-cell deletion, these are precisely the collapsed-uniform strata.

Proof. Expand the double sum and use $\sum_i h_i = L$ and $\sum_i h_i^3 = A_3$. Equality in a weighted variance holds precisely when h_i^2 is constant on its positive support. □

Recall the quartic Bregman charge

$$\mathfrak{J}_4(\alpha) = \sum_i (h_i - a)^2(h_i^2 + 2ah_i + 3a^2). \tag{11.6}$$

Proposition 11.2 (The intrinsic variance is paid by the DAG charge)**.** *For every positive fan,*

$$\boxed{\mathfrak{D}(\alpha) \le S\mathfrak{J}_4(\alpha).} \tag{11.7}$$

The estimate is independent of the minimum cell and all adjacent ratios.

Proof. The number $\ell^2 = A_3/L$ is the weighted least-squares minimizer of $m \mapsto \sum_i h_i(h_i^2 - m)^2$. Hence, using the competitor $m = a^2$,

$$\begin{aligned}
\mathfrak{D}(\alpha) &\le \sum_i h_i(h_i^2 - a^2)^2 \\
&= \sum_i h_i(h_i - a)^2(h_i + a)^2 \\
&\le S \sum_i (h_i - a)^2(h_i^2 + 2ah_i + 3a^2) = S\mathfrak{J}_4(\alpha).
\end{aligned}$$

□

11.2. **Exact relation with the cubic null form.** Stage 144 proves

$$\mathcal{C}(q_\alpha^3) = V(\alpha) - T(\alpha), \tag{11.8}$$

$$V(\alpha) = \frac{A_5}{320} - \frac{A_3^2}{1152\pi}. \tag{11.9}$$

Proposition 11.3 (Intrinsic defect identity)**.** *For every positive fan,*

$$\boxed{\mathcal{C}(q_\alpha^3) - \frac{1}{720}\sum_i h_i^5 = \frac{1}{576}\mathfrak{D}(\alpha) - T(\alpha).} \tag{11.10}$$

In particular the sharp inequality

$$\mathcal{C}(q_\alpha^3) \ge \frac{1}{720}\sum_i h_i^5 \tag{11.11}$$

is equivalent to $T \le \mathfrak{D}/576$.

Proof. Since $L = 2\pi$,

$$\frac{1}{320} - \frac{1}{720} = \frac{1}{576}, \qquad \frac{A_3^2}{1152\pi} = \frac{1}{576}\frac{A_3^2}{L}.$$

Insert these two equalities in (11.8)–(11.9). □

Remark 11.4 (Logical status of the sharp constant). Equation (11.10) is proved; inequality (11.11) is not used as an established fact. The unconditional proof only needs an unspecified universal constant in the following smaller endpoint leaf.

Intermediate obligation 11.5 (AVS: adaptive equality-stratum variance sampling)**.** Prove

$$\boxed{T(\alpha) \le C\mathfrak{D}(\alpha)} \tag{11.12}$$

for every positive fine fan. By Proposition 11.2, AVS implies HQ and hence DF. Unlike the deleted HMS/QPS route, AVS vanishes on every collapsed-uniform endpoint stratum and contains no path velocity.

11.3. **A global absolute sampling bound.** We use the following mesh-free sampling theorem in the precise special case needed here.

Lemma 11.6 (Adaptive truncated-Hilbert sampling)**.** *Let $\{I_i\}$ be any interval partition of the circle, with lengths h_i and midpoints c_i. If $f \in L^2(\mathbb{T})$ satisfies*

$$\mathrm{pv}\int_{I_i} \frac{1}{2\pi} \cot\frac{c_i - y}{2}\, f(y)\, dy = 0 \qquad \textit{for every } i, \tag{11.13}$$

then

$$\sum_i h_i |\mathcal{H}f(c_i)|^2 \le C\|f\|_2^2. \tag{11.14}$$

The constant is independent of all interval ratios.

Proof. The left side of (11.13) is the part removed by symmetric truncation at radius $h_i/2$. For every $x \in I_i$, the scale-restricted Cotlar inequality gives

$$|\mathcal{H}f(c_i)| \le C\{M(\mathcal{H}f)(x) + Mf(x)\}.$$

Indeed, split at radius h_i; moving the centre from c_i to x changes the far part by a Hardy–Littlewood average, and the intervening annulus is another such average. Square, integrate on I_i, sum the disjoint cells, and use the L^2 boundedness of M and $\mathcal{H}$. □

Theorem 11.7 (Unconditional absolute endpoint bound)**.** *For every positive fan,*

$$\boxed{T(\alpha) \le CA_5 = C\sum_i h_i^5.} \tag{11.15}$$

Proof. On its own cell, q_α is even about c_i, while the Hilbert kernel is odd. Thus (11.13) holds with $f = q_\alpha$. Moreover direct integration of the two half-bubbles gives

$$\|q_\alpha\|_2^2 = \sum_i \frac{h_i^5}{320}.$$

Apply Lemma 11.6. □

11.4. All fans away from the equality strata are closed.

Theorem 11.8 (Far-stratum AVS and HQ)**.** *Fix* $0 < \delta < 1$*. If*

$$\mathfrak{D}(\alpha) \geq \delta A_5, \tag{11.16}$$

then

$$T(\alpha) \leq C_\delta \mathfrak{D}(\alpha) \leq C_\delta S \mathfrak{J}_4(\alpha). \tag{11.17}$$

Thus AVS, HQ, and the pure cubic row DF are unconditional on the complete far-stratum region, including arbitrary minimum cells and ratios.

Proof. Theorem 11.7 and (11.16) give the first inequality; Proposition 11.2 gives the second. Stage 144 then gives HQ $\Rightarrow$ DF. □

11.5. Quantitative selection of the nearest collapsed-uniform stratum.

It remains to treat $\mathfrak{D} < \delta A_5$. This condition has a strong geometric consequence which we now make exact.

Lemma 11.9 (Concentration of physical angular mass)**.** *Assume* $0 < \delta < 1$ *and* $\mathfrak{D} < \delta A_5$*. Then*

$$\mathfrak{D} \leq \frac{\delta}{1-\delta} L\ell^4. \tag{11.18}$$

For $0 < \rho < 1$ *let*

$$\mathcal{B}_\rho = \{i : |h_i/\ell - 1| \geq \rho\}, \qquad B_\rho = \sum_{i \in \mathcal{B}_\rho} h_i. \tag{11.19}$$

Then

$$\boxed{\frac{B_\rho}{L} \leq \frac{\delta}{(1-\delta)\rho^2}.} \tag{11.20}$$

Proof. Equation (11.4) gives $A_5 = \mathfrak{D} + L\ell^4$; this proves (11.18). On a bad cell, $|h_i^2 - \ell^2| \geq \rho\ell^2$. Hence $\rho^2\ell^4 B_\rho \leq \mathfrak{D}$, and (11.20) follows. □

Let $\mathcal{G}_\rho$ be the complementary good set and $M = |\mathcal{G}_\rho|$. If $\delta < (1-\delta)\rho^2$, Lemma 11.9 ensures $M \geq 1$. Define the collapsed-uniform reference fan β by

$$\beta_i = \begin{cases} b := L/M, & i \in \mathcal{G}_\rho, \\ 0, & i \in \mathcal{B}_\rho. \end{cases} \tag{11.21}$$

After deleting its zero cells, β is the regular M-fan; therefore all its Hilbert midpoint samples vanish.

Lemma 11.10 (Quantitative distance to the selected stratum)**.** *Put* $\varepsilon = B_\rho/L$*. Then*

$$\frac{1-\rho}{1-\varepsilon} \leq \frac{b}{\ell} \leq \frac{1+\rho}{1-\varepsilon}. \tag{11.22}$$

If $\rho + \varepsilon \le 1/4$, then

$$\boxed{b^3 \sum_{i\in\mathcal{G}_\rho} (h_i - b)^2 + \sum_{i\in\mathcal{B}_\rho} h_i(h_i^2 - \ell^2)^2 \le C_\rho \mathfrak{D}(\alpha).} \tag{11.23}$$

Proof. Since every good length lies between $(1-\rho)\ell$ and $(1+\rho)\ell$,

$$M(1-\rho)\ell \le L - B_\rho \le M(1+\rho)\ell.$$

Using $b = L/M$ gives (11.22).

Write $e_i = h_i - \ell$ on the good set. There

$$h_i(h_i^2 - \ell^2)^2 \ge c_\rho \ell^3 e_i^2.$$

Also

$$M(b-\ell) = \sum_{i\in\mathcal{G}_\rho} e_i + B_\rho.$$

Therefore Cauchy–Schwarz, $M \asymp L/\ell$, and $\rho^2\ell^4 B_\rho \le \mathfrak{D}$ give

$$\begin{aligned} b^3 \sum_{\mathcal{G}_\rho} (h_i - b)^2 &\le C_\rho \ell^3 \sum_{\mathcal{G}_\rho} e_i^2 + C_\rho \frac{\ell^3 B_\rho^2}{M} \\ &\le C_\rho \mathfrak{D} + C_\rho \ell^4 \frac{B_\rho^2}{L} \le C_\rho \mathfrak{D}. \end{aligned}$$

The bad term in (11.23) is a sub-sum of $\mathfrak{D}$. ☐

The preceding lemmas reduce the near-stratum region to the following strictly localized endpoint estimate.

Intermediate obligation 11.11 (ELCA: equality-stratum first-separation square)**.** For the good/bad decomposition and collapsed-uniform reference (11.21), prove directly from the endpoint kernel-difference identity of stage 152 that

$$\begin{aligned} T(\alpha) & \\ &\le C_\rho \left\{ b^3 \sum_{i\in\mathcal{G}_\rho} (h_i - b)^2 + \sum_{i\in\mathcal{B}_\rho} h_i(h_i^2 - \ell^2)^2 \right\}. \end{aligned} \tag{11.24}$$

The regular M-fan is used only after zero-cell deletion. A maximal consecutive bad block is kept intact until its first separating scale; the two endpoint kernels may not be estimated separately inside that block. Good–good rectangles use the signed variable-grid discrete Calderón–Zygmund square, while every rectangle meeting a bad block is paid by the second term in (11.24). Descendants are squared before ancestors are summed.

Proposition 11.12 (ELCA is the only remaining pure endpoint row)**.** *ELCA implies AVS in the near-stratum region. Together with Theorem 11.8, it implies AVS, HQ, and DF for every positive fan.*

Proof. Lemma 11.10 bounds the right side of (11.24) by $C_\rho\mathfrak{D}$. This is AVS. The far and near regions exhaust all fans. Proposition 11.2 and stage 144 then give HQ and DF. ☐

12. Node DF0B: the comparable-grid discrete square

Audited source: `DF0_good_grid.tex`.

12.1. Quasiuniform data and the exact material matrix. Let $M \geq 2$, $b = 2\pi/M$, and let

$$h_i > 0, \qquad \sum_i h_i = 2\pi, \qquad (1-\rho)b \leq h_i \leq (1+\rho)b, \qquad 0 < \rho < \frac{1}{4}. \tag{12.1}$$

Put $d_i = h_i - b$ and $h_i(t) = b + td_i$. Then

$$(1-\rho)b \leq h_i(t) \leq (1+\rho)b, \qquad |d_i| \leq \rho b, \qquad \sum_i d_i = 0. \tag{12.2}$$

For a positive current partition $g = (g_i)$ with endpoints θ_i and midpoints c_i, the assembled foundation F148 gives the exact material derivative of

$$w_i = \mathcal{H}q_g(c_i) = \frac{1}{\pi}\sum_j g_j \Phi(c_i - c_j), \qquad \Phi'(x) = -\log\left|2\sin\frac{x}{2}\right|, \tag{12.3}$$

in the form

$$\dot{w}_i = \frac{1}{\pi}\sum_k d_k \mathcal{E}_{ik}(g). \tag{12.4}$$

For completeness, choose for each target i a cut θ_0 at circular distance at least $\pi/3$ from c_i. With $K = \Phi'$ and cyclic indices relabelled at that cut, the entries are

$$\begin{aligned}
\mathcal{E}^-_{ik} &= \sum_{j<k} g_j K(c_i - c_j) + \frac{g_k}{2}K(c_i - c_k) - \int_{\theta_0}^{c_k} K(c_i - y)\,dy, && k < i,\\
\mathcal{E}^+_{ik} &= \int_{c_k}^{\theta_0+2\pi} K(c_i - y)\,dy - \frac{g_k}{2}K(c_i - c_k) - \sum_{j>k} g_j K(c_i - c_j), && k > i,\\
\mathcal{E}_{ii} &= \frac{1}{2}\Bigg\{\sum_{j<i} g_j K(c_i - c_j) - \int_{\theta_0}^{\theta_i} K(c_i - y)\,dy \\
&\qquad\qquad - \sum_{j>i} g_j K(c_i - c_j) + \int_{\theta_{i+1}}^{\theta_0+2\pi} K(c_i - y)\,dy\Bigg\}.
\end{aligned} \tag{12.5}$$

Set $\mathcal{E}_{ik} = \mathcal{E}^-_{ik}$ for $k < i$ and $\mathcal{E}_{ik} = \mathcal{E}^+_{ik}$ for $k > i$. F148 proves (12.4) by differentiating the Clausen formula: every raw coefficient equals a common $\Phi(c_i)$ plus the displayed quadrature primitive, and the common term vanishes because $\sum_k d_k = 0$.

Every entry also has the following signed sawtooth representation. With

$$\eta(y)=\begin{cases}-(y-\theta_j), & \theta_j<y<c_j,\\ \theta_{j+1}-y, & c_j<y<\theta_{j+1},\end{cases} \tag{12.6}$$

one has on every complete cell

$$|\eta|\le g_j/2, \qquad \int_{I_j}\eta=0, \tag{12.7}$$

and, on the left or right arc selected by an antipodal cut,

$$\mathcal{E}_{ik}=\pm\int_{\operatorname{arc}(i,k)}\eta(y)K'(c_i-y)\,dy, \qquad K'(x)=-\frac{1}{2}\cot\frac{x}{2}. \tag{12.8}$$

The terminal half-cell is part of the integral; it is not separated from the complete-cell primitive.

12.2. **The variable-grid discrete CZ lemma.** Write $|i-k|_M$ for cyclic index distance. Since adding a constant to every entry of one row does not change (12.4) on $\sum d_k=0$, choose the representative $\widetilde{\mathcal{E}}$ whose one-sided primitives are centred at an antipodal cut. Thus the two terminal values at the cut are opposite (or zero when there is a single antipodal index). We do not subtract a row average, which could destroy the far-index size.

Lemma 12.1 (Signed primitive estimates on a comparable mesh)**.** *Assume*

$$\kappa b\le g_j\le\kappa^{-1}b \tag{12.9}$$

with fixed $0<\kappa<1$*. Then, after choosing the three usual shifted antipodal cuts, the normalized matrix* $A=b^{-1}\widetilde{\mathcal{E}}$ *satisfies*

$$|A_{ik}|\le\frac{C_\kappa}{1+|i-k|_M}, \tag{12.10}$$

$$|A_{i,k+1}-A_{ik}|+|A_{i+1,k}-A_{ik}|\le\frac{C_\kappa}{(1+|i-k|_M)^2} \tag{12.11}$$

away from the two harmless cut indices. At a cut the two one-sided values obey (12.10)*. Moreover, for every cyclic index interval* J*,*

$$\sum_{i\in J}\Big|\sum_{k\in J}A_{ik}\Big|^2 \quad+\quad \sum_{k\in J}\Big|\sum_{i\in J}A_{ik}\Big|^2 \quad\le\quad C_\kappa|J|. \tag{12.12}$$

All constants are independent of M *and of cumulative displacement of the variable grid from the regular grid.*

Proof. Quasiuniformity converts physical distance to index distance:

$$\operatorname{dist}(c_i,c_k)\asymp_\kappa b(1+|i-k|_M)$$

on the selected arc. In (12.8), a complete cell has both the endpoint cancellation and the zero mean in (12.7). Subtracting $K'(c_i-c_j)$ on that cell and using $|(K')'(x)| \le C/\operatorname{dist}(x, 2\pi\mathbb{Z})^2$ gives

$$\left|\int_{I_j} \eta(y) K'(c_i - y)\, dy\right| \le \frac{C_\kappa b}{(1 + |i-j|_M)^2}.$$

The terminal half-cell is estimated together with its endpoint atom and costs $C_\kappa b/(1 + |i - k|_M)$. Summing from the antipode proves (12.10). Advancing the terminal index by one cell leaves exactly one complete zero-mean cell, proving the k-difference in (12.11). Moving the target by one comparable cell and subtracting the old kernel value gives the i-difference. The cut values have physical distance comparable with one and satisfy the same size estimate directly.

For testing, retain the left/right signs from the antipodally centred primitive. Summation by parts with (12.11) shows that a target at index distance r from the nearer boundary of J contributes at most

$$C_\kappa \left(1 + \log \frac{|J| + 1}{r + 1}\right).$$

The square of this profile is summable:

$$\sum_{r=0}^{|J|-1} \left(1 + \log \frac{|J| + 1}{r + 1}\right)^2 \le C|J|.$$

This proves the first test. For the adjoint test, reverse the source and target roles in (12.8); the complete-cell zero mean and the terminal endpoint term are unchanged, so the same argument proves the second test. No absolute harmonic row sum is used. □

Lemma 12.2 (Finite cyclic discrete $T1$ square)**.** *Any cyclic matrix satisfying* (12.10)–(12.12) *obeys*

$$\|Ax\|_{\ell^2} \le C_\kappa \|x\|_{\ell^2}. \tag{12.13}$$

Proof. Use the three shifted dyadic grids on the cyclic index set and expand x and a test vector y in normalized Haar functions. If their support intervals are disjoint, subtract the kernel at the two interval centres. Equation (12.11) gives

$$|\langle A h_I, h_J\rangle| \le C_\kappa \frac{|I|^{1/2}|J|^{1/2} \min\{|I|, |J|\}}{(\operatorname{dist}(I, J) + \max\{|I|, |J|\})^2}.$$

The two-sided annular sum is Schur summable. For nested supports, split the larger Haar function into its constant child value plus a mean-zero remainder. The remainder has the same separated estimate; the constant parts form the two paraproducts. Their coefficients are Carleson because (12.12) is exactly the square testing condition for $A1_J$ and A^*1_J. The dyadic Carleson embedding theorem bounds both paraproducts. The three grids place every pair of cyclic intervals in one of these separated or nested configurations with comparable parent. Summing the Haar coefficients proves (12.13). □

Theorem 12.3 (Quasiuniform quadrature-primitive square)**.** *Under* (12.9)*, for every zero-sum direction d,*

$$\boxed{\sum_i \Big|\sum_k d_k \mathcal{E}_{ik}(g)\Big|^2 \le C_\kappa b^2 \sum_k d_k^2.} \tag{12.14}$$

Proof. The chosen row gauge does not change the action on d. Apply Lemmas 12.1–12.2 to $A = b^{-1}\widetilde{\mathcal{E}}$ and rescale. $\square$

12.3. **Endpoint integration inside one equality component.** Along (12.2), Theorem 12.3 and (12.4) give

$$\sum_i |\dot w_i(t)|^2 \le C_\rho b^2 \sum_k d_k^2. \tag{12.15}$$

At $t = 0$ the regular M-fan has $w_i(0) = 0$.

Theorem 12.4 (Complete quasiuniform AVS)**.** *Every fan satisfying* (12.1) *obeys*

$$\boxed{T(\alpha) = \sum_i h_i |\mathcal{H} q_\alpha(c_i)|^2 \le C_\rho b^3 \sum_i (h_i - b)^2 \le C_\rho \mathfrak{D}(\alpha).} \tag{12.16}$$

Hence HQ and DF are closed on the entire quasiuniform component, with no adjacent-ratio derivative or cumulative-position hypothesis.

Proof. Cauchy–Schwarz in t, (12.1), and (12.15) give

$$\begin{aligned} T(\alpha) &\le (1+\rho) b \int_0^1 \sum_i |\dot w_i(t)|^2 \, dt \\ &\le C_\rho b^3 \sum_i d_i^2. \end{aligned}$$

For the last inequality in (12.16), use the pair formula of stage 153. On (12.1),

$$h_i h_j (h_i^2 - h_j^2)^2 \asymp_\rho b^4 (h_i - h_j)^2.$$

Since $2\pi = Mb$ and $\sum_i d_i = 0$,

$$\mathfrak{D}(\alpha) \asymp_\rho \frac{b^4}{Mb} \sum_{i,j} (d_i - d_j)^2 = 2b^3 \sum_i d_i^2$$

up to constants depending only on ρ. Stage 153 gives AVS $\Rightarrow$ HQ $\Rightarrow$ DF. $\square$

The same square is needed at the collapsed locations at which a bad block will later be opened. Write $z_i = \theta_i$ for the cell endpoints.

Corollary 12.5 (Quasiuniform endpoint samples)**.** *Under* (12.1)*,*

$$\boxed{b \sum_i |\mathcal{H} q_\alpha(z_i)|^2 \le C_\rho b^3 \sum_i (h_i - b)^2.} \tag{12.17}$$

Proof. Repeat the material calculation with the target carried by the endpoint $z_i(t)$ instead of the midpoint $c_i(t)$. The exact length gradient is again a signed quadrature primitive. At the target endpoint the two incident terminal half-cells must be kept together; their quadratic profiles and first derivatives vanish at their common endpoint. Thus there is no boundary logarithm. Advancing either source or target by one cell leaves one complete mean-zero sawtooth, so (12.10)–(12.12) hold for the endpoint matrix with the same constants. Lemma 12.2 therefore gives the analogue of (12.15). At the regular fan the spline is even about every endpoint, so all endpoint Hilbert samples vanish. Integration in t proves (12.17). ☐

Remark 12.6 (Why stage 152 does not contradict this theorem). The false HMS/QPS family has half of its current cells of length εa while their fixed straight-path directions have size a. It leaves every fixed quasiuniform component. In Theorem 12.4, both the current lengths and the entire connecting segment remain comparable with b, and $|d_i| \le \rho b$.

12.4. **Effect on the ELCA ledger.** Use the good/bad selection of stage 153. On a maximal collection of good cells whose zero-cell deletion contains no intervening bad block, the current effective mesh is comparable with the selected stratum length b. The preceding theorem closes every good–good first-separation rectangle.

Corollary 12.7 (Only bad-block rectangles remain)**.** *In ELCA of stage 153, all rows whose source and target shadows avoid the bad set satisfy*

$$T_{\text{good-good}} \le C_\rho b^3 \sum_{i\in\mathcal{G}_\rho} (h_i - b)^2. \tag{12.18}$$

Thus the remaining pure endpoint ledger consists only of rectangles for which the source shadow or target shadow meets a maximal bad block. Such a block must be retained as a whole until its first separating scale and paid by

$$\sum_{i\in\mathcal{B}_\rho} h_i(h_i^2 - \ell^2)^2. \tag{12.19}$$

Proof. Apply Theorem 12.4 on each good component after the common collapsed-uniform reference has been fixed. At its two boundary cuts, keep the terminal half-cell with the adjacent bad block; those boundary rectangles are not included in (12.18). Disjoint good components have disjoint physical mass, so their squares sum. The reference-charge estimate of stage 153 then gives the displayed right side. Every omitted rectangle meets a bad block by construction. ☐

13. Node ZG: strong ZG-T1 retracted; structured ZG-COM closed

Audited source: `ZG_zero_gap.tex`.

13.1. **Intrinsic cut and source decomposition.** Let $I_i = [\theta_i, \theta_{i+1}]$ have length $h_i > 0$ and midpoint c_i. Put

$$q_h = \sum_i q_i, \qquad q_i(\theta_i + x) = \frac{1}{2}\min\{x^2, (h_i - x)^2\} \quad (0 \le x \le h_i), \tag{13.1}$$

where every q_i is extended by zero outside I_i. Define

$$T(h) = \sum_i h_i |\mathcal{H} q_h(c_i)|^2. \tag{13.2}$$

As in stage 153, let

$$\ell^2 = \frac{1}{2\pi}\sum_i h_i^3, \qquad \mathfrak{D}(h) = \sum_i h_i (h_i^2 - \ell^2)^2. \tag{13.3}$$

The far region $\mathfrak{D} \ge \delta \sum_i h_i^5$ is already closed by the absolute adaptive sampling estimate. In the complementary region choose $0 < \rho < 1/8$ and set

$$\mathcal{G} = \{i : |h_i/\ell - 1| < \rho\}, \qquad \mathcal{B} = \{1, \dots, N\} \setminus \mathcal{G}. \tag{13.4}$$

Put

$$M = |\mathcal{G}|, \qquad b = \frac{2\pi}{M}, \qquad E = \sum_{i \in \mathcal{B}} h_i. \tag{13.5}$$

The near-stratum selection gives $b \asymp_\rho \ell$ and E smaller than a fixed multiple of 2π. We fix the stratum threshold so that $E \le \pi/4$. If E is larger, the absolute sampling bound $T \le C \sum_i h_i^5$, together with (13.6) below and $b^4 \le C b^4 E$, already belongs to the closed far row. The following static budget is the only place where the definition of the bad set is used:

$$b^3 \sum_{i \in \mathcal{G}} (h_i - b)^2 + b^4 E + \sum_{i \in \mathcal{B}} h_i^5 \le C_\rho \mathfrak{D}(h). \tag{13.6}$$

Indeed, the good term is the reference-charge estimate of stage 153. On the bad set, with $r = h_i/\ell$,

$$|r - 1| \ge \rho, \qquad r^5 + r \le C_\rho r (r^2 - 1)^2.$$

Multiplication by ℓ^5, summation, and $b \asymp_\rho \ell$ give

$$b^4 E + \sum_{i \in \mathcal{B}} h_i^5 \le C_\rho \sum_{i \in \mathcal{B}} h_i (h_i^2 - \ell^2)^2,$$

which proves (13.6) without invoking the retracted block lemma of stage 155.

Split

$$q_{\mathrm{G}} = \sum_{i \in \mathcal{G}} q_i, \qquad q_{\mathrm{B}} = \sum_{i \in \mathcal{B}} q_i. \tag{13.7}$$

Then

$$T(h) \le 2T_{\mathrm{G}} + 2T_{\mathrm{B}}, \qquad T_X := \sum_i h_i |\mathcal{H} q_X(c_i)|^2. \tag{13.8}$$

13.2. **The bad source is an absolute square.** We recall the special adaptive sampling fact proved in stage 153. If an interval partition has midpoints c_i and $f \in L^2(\mathbb{T})$ satisfies

$$\operatorname{pv}\int_{I_i} \frac{1}{2\pi}\cot\frac{c_i-y}{2} f(y)\,dy = 0 \quad \text{for every } i, \tag{13.9}$$

then

$$\sum_i h_i |\mathcal{H}f(c_i)|^2 \le C\|f\|_2^2. \tag{13.10}$$

It follows by comparing the truncation at radius $h_i/2$ with the maximal Hilbert transform and integrating the Cotlar inequality over the disjoint cells.

Lemma 13.1 (Bad-source square)**.** *For the decomposition* (13.7)*,*

$$\boxed{T_{\mathrm{B}} \le C \sum_{j\in\mathcal{B}} h_j^5.} \tag{13.11}$$

Proof. If $i \in \mathcal{G}$, the function q_{B} vanishes on I_i. If $i \in \mathcal{B}$, its restriction to I_i is the even bubble q_i, while the cotangent kernel centred at c_i is odd. Thus (13.9) holds for $f = q_{\mathrm{B}}$ on every target cell. Equation (13.10) and the exact cell integral

$$\|q_{\mathrm{B}}\|_2^2 = \sum_{j\in\mathcal{B}} \int_{I_j} q_j^2 = \frac{1}{320}\sum_{j\in\mathcal{B}} h_j^5$$

prove (13.11). □

Remark 13.2. This single application of (13.10) is the decisive cleaning step. All bad midpoints are now inside the source q_{B} and are paid by their individual h_j^5 energies. A maximal bad block is never asked to pack its internal midpoint set at every dyadic scale.

13.3. **The good skeleton with zero-source gaps.** Delete the bad bubbles but retain their intervals as gaps. Thus q_{G} consists of comparable good bubbles, while a maximal consecutive bad block B_ν is a zero-source gap of length

$$e_\nu = \sum_{i\in B_\nu} h_i, \qquad \sum_\nu e_\nu = E. \tag{13.12}$$

Only the two endpoints ∂B_ν are singular locations for the good source.

For a target interval Q of physical length L_Q (either a member of one of the shifted grids or a node of the balanced target tree below), define the endpoint weight

$$\Omega_{\mathrm{end}}(B,Q) = \begin{cases} \dfrac{e}{L_Q}\left(1+\dfrac{\operatorname{dist}(B,Q)}{L_Q}\right)^{-3}, & e \le L_Q, \\[2ex] \dfrac{L_Q}{e}\left(1+\dfrac{\operatorname{dist}(\partial B,Q)}{L_Q}\right)^{-3}, & e > L_Q. \end{cases} \tag{13.13}$$

Lemma 13.3 (Genuine endpoint packing)**.** *For disjoint gaps, on each shifted dyadic grid and on every balanced target interval tree used below,*

$$\sup_Q \sum_\nu \Omega_{\mathrm{end}}(B_\nu, Q) + \sup_\nu \sum_Q \Omega_{\mathrm{end}}(B_\nu, Q) \le C. \tag{13.14}$$

Proof. For $e \le L_Q$, disjointness bounds the total gap length in the nth distance annulus by $C2^n L_Q$; the factor $L_Q^{-1}2^{-3n}$ is summable. For $e > L_Q$, sort gaps by $e \asymp 2^m L_Q$. The endpoints themselves need not be $2^m L_Q$-separated: two large gaps can be separated by one good cell. What is separated is the interior mass of the disjoint gaps. Hence an annulus of length $C2^n L_Q$ meets at most $C(1 + 2^{n-m})$ such gaps. Multiplication by $2^{-m}2^{-3n}$ is summable in m, n.

With a gap fixed, the coarse scale sum is $\sum_{L_Q \ge e} e/L_Q < \infty$. At fine scales there are only two singular points, so the annular sum is bounded at each scale and $\sum_{L_Q<e} L_Q/e < \infty$. This proves (13.14). The argument would fail with all internal bad midpoints in place of the two endpoints; Lemma 13.1 is what removed them. □

We next isolate the precise zero-gap extension of the variable-grid square. Only the good midpoint atoms occur in the opening operator:

$$d\tau_{\mathrm{G}} = \sum_{i\in\mathcal{G}} h_i \delta_{c_i}, \qquad d\mu = dx. \tag{13.15}$$

Thus the source and target measures are different. In particular the one-weight matrix theorem of stage 154 cannot be quoted here. We give the continuous-to-atomic operator and its square estimate explicitly.

The exact identity behind the next estimate is worth recording. With $\Phi(x) = \sum_{n\ge1} \sin(nx)/n^2$, twice integrating $q_j'' = \mathbf{1}_{I_j} - h_j\delta_{c_j}$ gives

$$\mathcal{H}q_{\mathrm{G}}(x) = \frac{1}{\pi}\sum_{j\in\mathcal{G}} \left\{ h_j \Phi(x - c_j) - \int_{I_j} \Phi(x-y)\,dy \right\}. \tag{13.16}$$

Each bracket has zero mass and zero first moment. Formula (13.16), rather than separate endpoint kernels, is used until the first source–target separating scale.

Put $B = \bigcup_\nu B_\nu$, $G^* = \bigcup_{j\in\mathcal{G}} I_j$, and

$$\gamma = \frac{E}{M}, \qquad \sigma = \mathbf{1}_B - \sum_{j\in\mathcal{G}} \frac{\gamma}{h_j}\mathbf{1}_{I_j}. \tag{13.17}$$

Then $\int_{\mathbb{T}} \sigma = 0$. Let u be the periodic primitive of σ with zero mean and set

$$R_t(x) = x - tu(x), \qquad 0 \le t \le 1. \tag{13.18}$$

On a good cell $R_t' = 1 + t\gamma/h_j$; on a gap $R_t' = 1 - t$. Consequently R_1 collapses every gap and maps I_j affinely onto an interval of length $h_j + \gamma$. In particular $R_1(c_j) = \widehat{c}_j$.

For $f \in L^2(\mu)$ put

$$f_0 = f - \frac{1}{2\pi}\int_{\mathbb{T}} f.$$

For an oriented cyclic arc from x to z, let $\epsilon_{x,z}$ be its signed indicator. If V_f is the periodic zero-mean primitive of f_0, then

$$V_f(z) - V_f(x) = \int_{\mathbb{T}} \epsilon_{x,z}(s) f_0(s)\, ds. \tag{13.19}$$

For $y \in I_j$, $j \in \mathcal{G}$, write $m(y) = c_j$ and $a_t(y) = R_t'(y)$. Define

(13.20)

$$\begin{aligned} P_t(x,y) &= \Phi(R_t x - R_t m(y)) - \Phi(R_t x - R_t y), \\ (\mathcal{T}_t f)(x) &= \frac{1}{\pi}\int_{G^*} \Big[-f_0(y) P_t(x,y) \\ &\qquad + a_t(y)\{\Phi'(R_t x - R_t m(y))[V_f(m(y)) - V_f(x)] \\ &\qquad\qquad - \Phi'(R_t x - R_t y)[V_f(y) - V_f(x)]\}\Big], dy. \end{aligned} \tag{13.21}$$

This is a linear map from $L^2(\mu)$ to functions on the good target points and annihilates constants. On the zero-mean subspace, (13.19) gives the raw kernel

$$\begin{aligned} K_t(x,s) = \frac{1}{\pi}\Big[&-\mathbf{1}_{G^*}(s) P_t(x,s) \\ &+ \int_{G^*} a_t(y)\{\Phi'(R_t x - R_t m(y))\epsilon_{x,m(y)}(s) \\ &\qquad - \Phi'(R_t x - R_t y)\epsilon_{x,y}(s)\}\, dy\Big]. \end{aligned} \tag{13.22}$$

For arbitrary f the actual kernel is the source-centred version

(13.23)

$$K_t^0(x,s) = K_t(x,s) - \frac{1}{2\pi}\int_{\mathbb{T}} K_t(x,r)\, dr, \qquad \mathcal{T}_t f(x) = \int_{\mathbb{T}} K_t^0(x,s) f(s)\, ds.$$

Below we suppress the superscript 0. The subtracted row constant has zero source difference. Integrating (13.22) in s, using $\int_{\mathbb{T}} \Phi = 0$, and pairing the two terminal half-cells gives

$$\left|\frac{1}{2\pi}\int_{\mathbb{T}} K_t(x,s)\, ds\right| + \left|\partial_x \frac{1}{2\pi}\int_{\mathbb{T}} K_t(x,s)\, ds\right| \le C_\rho b^2. \tag{13.24}$$

Thus the centred kernel obeys the same size, difference, and testing bounds (with a changed constant).

Proposition 13.4 (The strong ZG–$T1$ square is false)**.** *There is no constant independent of M for which*

$$\sum_{i \in \mathcal{G}} h_i |\mathcal{T}_0 f(c_i)|^2 \le C b^4 \|f\|^2_{L^2(\mathbb{T})} \tag{13.25}$$

holds for every f. In fact, on the uniform gap-free fan the norm of $\mathcal{T}_0 : L^2(dx) \to L^2(\tau_{\mathrm{G}})$ is at least a constant multiple of b.

Proof. Take $h_j = b = 2\pi/M$, $c_j = (j+\frac{1}{2})b$, and

$$f_M(\theta) = (2\pi)^{-1/2} e^{i(M-1)\theta}.$$

Put $n = M-1$ and $q = e^{-ib}$. Cellwise integration by parts in (13.21), followed by cyclic summation, gives for the unnormalised mode $e^{in\theta}$

$$\mathcal{T}_0(e^{in\cdot})(c_0) = \frac{B_M + C_M}{\pi i(M-1)}, \tag{13.26}$$

where

$$B_M = -(1-q)\sum_{j=0}^{M-1} q^j \Phi(jb), \tag{13.27}$$

$$C_M = be^{-ib/2}\sum_{j=1}^{M-1} \Phi'(jb)(1-q^j). \tag{13.28}$$

Here is the calculation before summing. Write $x = c_0$, $m = c_j$, $d = x - m = -jb$, $F(y) = e^{iny}$, and $P(r) = \Phi(d) - \Phi(d-r)$, and denote by $[\mathcal{T}_0 F(x)]_{I_j}$ the I_j-integral in (13.21). Integration by parts of $-FP$, using $F = (F/in)'$, cancels its interior $F\Phi'$ term against the last primitive term in (13.21) and gives

$$\begin{aligned}\pi[\mathcal{T}_0 F(x)]_{I_j} = \frac{1}{in}\Big\{ &-[F(m+r)P(r)]_{-b/2}^{b/2} \\ &+ F(x)[\Phi(d+b/2) - \Phi(d-b/2)] + b\Phi'(d)[F(m) - F(x)]\Big\}.\end{aligned} \tag{13.29}$$

For $j = 0$ the last product is understood by continuity and is zero. Now

$$F(c_j) = -e^{-ib/2}q^j, \qquad \Phi(-z) = -\Phi(z), \qquad \Phi'(-z) = \Phi'(z).$$

Substitution in (13.29), followed by one cyclic index shift in the endpoint terms, gives (13.27)–(13.28) and hence (13.26).

The Riemann sum in (13.27) satisfies

$$b\sum_{j=0}^{M-1} e^{-ijb}\Phi(jb) \longrightarrow \int_0^{2\pi} e^{-ix}\Phi(x)\,dx = -i\pi.$$

Since $1 - e^{-ib} = ib + O(b^2)$, it follows that $B_M \to -\pi$. Although Φ' has logarithmic endpoint singularities, the function

$$x \longmapsto \Phi'(x)(1 - e^{-ix})$$

extends continuously by zero at both endpoints. Hence

$$\begin{aligned} C_M &\longrightarrow \int_0^{2\pi} \Phi'(x)(1-e^{-ix})\,dx \\ &= -\int_0^{2\pi} \Phi'(x)e^{-ix}\,dx = -\pi; \end{aligned}$$

the last equality is the first cosine coefficient of $\Phi'(x) = \sum_{k\geq 1} \cos(kx)/k$. Thus

$$|\mathcal{T}_0(e^{in\cdot})(c_0)| \sim \frac{2}{M}. \tag{13.30}$$

Translation covariance gives the same modulus at all M target atoms. The normalisation of f_M cancels the total target mass 2π, and so

$$\|\mathcal{T}_0 f_M\|_{L^2(\tau_{\mathrm{G}})} = \frac{|B_M + C_M|}{\pi(M-1)} \sim \frac{2}{M}.$$

Since $b^2 = 4\pi^2/M^2$, the ratio of the two sides at the norm level is asymptotic to $M/(2\pi^2)$. This contradicts (13.25). If a nonempty deleted block is required, insert one gap of length $o(M^{-3})$ and distribute that length over the good cells. For each fixed M the finite cell integrals above depend continuously on those endpoints, so the same contradiction persists. □

Remark 13.5 (Consequence for the former proof route). The former ZG–$T1$ obligation asserted (13.25) for all f and all t. Proposition 13.4 shows that no completion of its proposed Haar paragraph can prove that claim. In particular, the previously stated kernel/testing package cannot have all the asserted uniform consequences. None of those claims is used below. The opening argument only needs the special density σ and the time integral appearing in the exact deformation identity.

Lemma 13.6 (Endpoint Calderón estimate for a small change of variable)**.** *Let $S : \mathbb{T} \to \mathbb{T}$ be an orientation-preserving, degree-one, piecewise affine map, and choose its lift in the form $S(x) = x + v(x)$. Suppose*

$$\|v'\|_\infty \leq \eta < 1. \tag{13.31}$$

For $f \in W^{1,\infty}(\mathbb{T})$ put

$$\mathfrak{C}_S f = (\mathcal{H} f) \circ S - \mathcal{H}(f \circ S).$$

Then

$$\|\mathfrak{C}_S f\|_2 \leq C_\eta \|S' - 1\|_2 \|f\|_\infty, \tag{13.32}$$

$$\|(\mathfrak{C}_S f)'\|_2 \leq C_\eta \|S' - 1\|_2 \|f'\|_\infty. \tag{13.33}$$

The same constants hold for limits of such maps with nonnegative derivative, provided each approximating map satisfies (13.31)*.*

Proof. We include the endpoint argument because replacing the right side of (13.32) by an arbitrary-input L^2 norm is precisely the false step retracted above. First smooth v and f; the estimates obtained below are uniform under this smoothing.

On a lifted interval, changing variables in the first Hilbert transform and writing

$$\Delta_v(x, y) = \frac{v(x) - v(y)}{x - y}$$

gives, for the singular part of the cotangent kernel,

$$\frac{1+v'(y)}{Sx-Sy}-\frac{1}{x-y}=\frac{v'(y)-\Delta_v(x,y)}{(x-y)(1+\Delta_v(x,y))}. \tag{13.34}$$

For the moment suppose $\|v'\|_\infty \le \eta_0$, where the universal $\eta_0 > 0$ will be chosen below. Expand the last denominator geometrically. It is therefore enough to estimate

$$A_m(v,g)(x)=\mathrm{pv}\int g(y)\frac{(v'(y)-\Delta_v(x,y))\Delta_v(x,y)^m}{x-y}\,dy, \qquad m\ge 0, \tag{13.35}$$

where in the application $g = f\circ S$. The following two endpoint multiplier bounds are the elementary Calderón estimate:

$$\|A_m(v,g)\|_2 \le C^{m+1}\|v'\|_2\|v'\|_\infty^m\|g\|_\infty, \tag{13.36}$$

$$\|\partial_x A_m(v,g)\|_2 \le C^{m+1}\|v'\|_2\|v'\|_\infty^m\|g'\|_\infty. \tag{13.37}$$

Here is a direct frequency verification. Use

$$\Delta_v(x,y)=\int_0^1 v'(y+s(x-y))\,ds$$

in (13.35), expand into Fourier series, and divide the frequencies into the regions in which one of the $m+1$ copies of v' has largest modulus. The difference $v'(y)-\Delta_v(x,y)$ makes the multiplier vanish unless such a v'-frequency dominates the frequency of g. On each region the multiplier and all its discrete differences are bounded by C^{m+1}. Smooth dyadic resolution consequently writes that piece as

$$\sum_r P_r v'\,T_r(P_{\le r+3}v',\dots,P_{\le r+3}v',P_{\le r+3}g),$$

up to a uniformly summable family of translates; the kernels of T_r have uniformly bounded L^1 norm. The square-function inequality for the sole high-frequency factor and the uniform L^∞ bounds for smooth low-frequency projections prove (13.36). After applying ∂_x, summation by parts in the Fourier frequency of g moves the output frequency onto g'; on the nonzero multiplier support its coefficient is bounded by the same largest v'-frequency. The identical decomposition proves (13.37). This also shows that the constants are independent of the number of affine pieces.

Choose η_0 so that $C\eta_0 < 1/2$. Summing (13.36)–(13.37) with the geometric coefficients gives the assertions for the flat singular kernel whenever $\|v'\|_\infty \le \eta_0$. The difference $\frac{1}{2}\cot(t/2)-1/t$ is smooth. Subtracting $g(x)$ (constants are annihilated) and integrating once by parts reduces its two estimates to convolution with an L^1 kernel, and gives the same right sides. Thus (13.32)–(13.33) hold in the small-distortion case. Notice that

$$\|g\|_\infty=\|f\|_\infty, \qquad \|g'\|_\infty \le (1+\eta_0)\|f'\|_\infty.$$

It remains to pass from η_0 to the stated fixed $\eta < 1$ without using the divergent series above. For $0 \le s \le 1$ define the degree-one map S_s by

$$S_s'(x) = \frac{S'(x)^s}{(2\pi)^{-1}\int_{\mathbb{T}} S'(y)^s\,dy}, \qquad S_0 = \mathrm{id}\,. \tag{13.38}$$

Up to a harmless rotation, $S_1 = S$. Partition $[0,1]$ into a number $L = L(\eta,\eta_0)$ of equal pieces and set $T_j = S_{(j+1)/L} \circ S_{j/L}^{-1}$. Since $1-\eta \le S' \le 1+\eta$, differentiating (13.38) in s gives

$$\|T_j' - 1\|_\infty \le \eta_0, \qquad \sum_{j=0}^{L-1} \|T_j' - 1\|_2 \le C_\eta \|S' - 1\|_2. \tag{13.39}$$

Indeed, on this compact interval the functions $\log t$ and $t-1$ are uniformly comparable, and change of variables by every S_s has Jacobian bounded above and below by constants depending only on η.

For degree-one maps the commutator has the exact cocycle

$$\mathfrak{C}_{S_2\circ S_1} f = \mathfrak{C}_{S_1}(f\circ S_2) + (\mathfrak{C}_{S_2} f)\circ S_1. \tag{13.40}$$

Iterate (13.40) over the T_j. All intermediate Jacobians and inverse Jacobians are bounded by C_η; composition preserves the L^∞ norm and multiplies the derivative norm by at most C_η. Apply the already proved small-distortion estimates to each summand and use (13.39). This proves (13.32)–(13.33) for every fixed $\eta < 1$.

Approximation and weak lower semicontinuity in H^1 remove the smoothing and prove the lemma. □

Put $R = R_1$ and $\widehat{h}_j = h_j + \gamma$ on the good indices. After deleting the gaps, the $\widehat{h}_j$ form a positive partition of the circle and remain comparable with b. Let $\widehat{q}$ be its complete quadratic spline. Set $p = \widehat{q} \circ R$ and define the monotone-pullback commutator

$$\mathfrak{C}_R \widehat{q}(x) := \mathcal{H}\widehat{q}(Rx) - \mathcal{H}p(x). \tag{13.41}$$

Changing variables $z = R(y)$ in the first Hilbert transform gives the entirely explicit kernel formula

$$\begin{aligned}\mathfrak{C}_R \widehat{q}(x) = \frac{1}{2\pi}\int_{\mathbb{T}} \widehat{q}(Ry)\Big\{R'(y)\cot\frac{Rx-Ry}{2}\\ - \cot\frac{x-y}{2}\Big\}\,dy,\end{aligned} \tag{13.42}$$

with principal values taken symmetrically. Here $R' = 0$ on every gap and $R' = 1 + \gamma/h_j$ on I_j.

Theorem 13.7 (Collapsed-bubble commutator square)**.** *One has*

$$\boxed{\sum_{i\in\mathcal{G}} h_i |\mathfrak{C}_R \widehat{q}(c_i)|^2 \le C_\rho b^4 E.} \tag{13.43}$$

The constant must be independent of the number, length, and subdivision of the gaps.

Proof. We resolve the collapse into geometrically summable, nondegenerate changes of variable. If B_ν is a gap of length e_ν, define the level-k geometry by

$$e_\nu^{(k)} = 2^{-k} e_\nu, \qquad h_j^{(k)} = h_j + (1 - 2^{-k})\gamma, \qquad k \ge 0. \tag{13.44}$$

These lengths sum to 2π. Let S_k be the affine map from the level-k circle to the level-$(k+1)$ circle which preserves the cyclic order of all cells. Thus

$$S_k' = \frac{1}{2} \quad \text{on the gaps,} \qquad S_k' = 1 + \frac{2^{-k-1}\gamma}{h_j^{(k)}} \quad \text{on the } j\text{th good cell.} \tag{13.45}$$

Put $R_0 = \mathrm{id}$ and $R_{k+1} = S_k \circ R_k$. Then R_k converges uniformly to the collapse R.

The good lengths are mutually comparable and their average is $g = (2\pi - E)/M$. Since $E \le \pi/4$ and $\rho < 1/8$, it follows directly from (13.45) that

$$\|S_k' - 1\|_\infty \le \frac{1}{2}, \qquad 1 \le R_k'|_{I_j} \le C_\rho. \tag{13.46}$$

Moreover, using $\sum_j (h_j^{(k)})^{-1} \le C_\rho M/b$ and $\gamma = E/M$, we have the exact layer computation

$$\|S_k' - 1\|_2^2 = \frac{1}{4}, 2^{-k}E + 2^{-2k-2}\gamma^2 \sum_{j \in \mathcal{G}} \frac{1}{h_j^{(k)}} \le C_\rho 2^{-k} E. \tag{13.47}$$

Let $R_{k,\infty}$ be the monotone tail collapse from the level-k geometry to the final complete fan, and put

$$p_k = \widehat{q} \circ R_{k,\infty}. \tag{13.48}$$

Then $p_0 = p$, $p_k = p_{k+1} \circ S_k$, and p_k vanishes on the level-k gaps. On a good cell $R_{k,\infty}$ is affine with derivative $\widehat{h}_j / h_j^{(k)}$. Since all these lengths are comparable with b, the explicit quadratic bubble gives, uniformly in k,

$$\|p_k\|_\infty \le C_\rho b^2, \qquad \|p_k'\|_\infty \le C_\rho b. \tag{13.49}$$

On the original good set define

$$G_k = (\mathcal{H} p_k) \circ R_k.$$

The identities $R_{k+1} = S_k \circ R_k$ and $p_k = p_{k+1} \circ S_k$ give the exact commutator cocycle

$$G_{k+1} - G_k = (\mathfrak{C}_{S_k} p_{k+1}) \circ R_k. \tag{13.50}$$

Lemma 13.6, (13.47), (13.49), and the change of variables on the good cells in (13.46) therefore imply

$$\|G_{k+1} - G_k\|_{L^2(G)} \le C_\rho b^2 2^{-k/2} E^{1/2}, \tag{13.51}$$

$$\|(G_{k+1} - G_k)'\|_{L^2(G)} \le C_\rho b 2^{-k/2} E^{1/2}. \tag{13.52}$$

Both series are summable. Since $G_0 = \mathcal{H}p$ and $G_k \to (\mathcal{H}\widehat{q}) \circ R$ in H^1 on the union of the original good cells, their sum yields

$$\|\mathfrak{C}_R\widehat{q}\|^2_{L^2(G)} + b^2\|(\mathfrak{C}_R\widehat{q})'\|^2_{L^2(G)} \le C_\rho b^4 E. \tag{13.53}$$

For $F \in H^1(I_i)$,

$$h_i|F(c_i)|^2 \le 2\int_{I_i} |F|^2 + h_i^2 \int_{I_i} |F'|^2.$$

Apply this to $F = \mathfrak{C}_R\widehat{q}$, sum over the disjoint good cells, and use $h_i \asymp_\rho b$ together with (13.53). This proves (13.43). ☐

Lemma 13.8 (Static zero-gap opening square)**.** *Let $\widehat{c}_j$ be the midpoints of the preceding collapsed partition and put*

$$U_i = \mathcal{H}\widehat{q}(\widehat{c}_i), \qquad i \in \mathcal{G}. \tag{13.54}$$

Then

$$\boxed{\sum_{i\in\mathcal{G}} h_i\,|\mathcal{H}q_{\mathrm{G}}(c_i) - U_i|^2 \le C_\rho b^4 E.} \tag{13.55}$$

The constant is independent of the number and subdivision of the gaps.

Proof. For $x = c_i$ define

$$W_i(t) = \frac{1}{\pi}\sum_{j\in\mathcal{G}} \int_{I_j} a_t(y)\{\Phi(R_tc_i - R_tc_j) - \Phi(R_tc_i - R_ty)\}\,dy. \tag{13.56}$$

Changing variables $z = R_t(y)$ shows that $W_i(t)$ is the Hilbert sample of the quadratic bubbles on the images $R_t(I_j)$, evaluated at their transported midpoint. Hence

$$W_i(0) = \mathcal{H}q_{\mathrm{G}}(c_i), \qquad W_i(1) = \mathcal{H}\widehat{q}(\widehat{c}_i) = U_i.$$

Differentiating (13.56) under the integral and using $u' = \sigma$ gives the exact identity

$$\dot{W}_i(t) = \mathcal{T}_t\sigma(c_i). \tag{13.57}$$

There is no omitted endpoint term: the gap motion is contained in the primitive differences in (13.21).

It remains to separate the part of the opening error already controlled by elementary Sobolev sampling. On a good cell put $a_j = 1 + \gamma/h_j = \widehat{h}_j/h_j$. The quadratic bubble scales exactly, so

$$p|_{I_j} = a_j^2 q_j, \qquad p = 0 \quad \text{on every gap.} \tag{13.58}$$

Consequently $r := q_{\mathrm{G}} - p$ belongs to $H^1(\mathbb{T})$ and, since $h_j \asymp_\rho b$ and the near stratum has bounded small E,

$$\|r\|_2^2 \le C_\rho E^2 \sum_{j\in\mathcal{G}} h_j^5 \le C_\rho b^4 E, \tag{13.59}$$

$$\|r'\|_2^2 \le C_\rho E^2 \sum_{j\in\mathcal{G}} h_j^3 \le C_\rho b^2 E. \tag{13.60}$$

Indeed $|a_j^2-1|\le C_\rho E$, while $Mb=2\pi$. For every $F\in H^1(I_j)$ the fundamental theorem of calculus and Cauchy–Schwarz give

$$h_j|F(c_j)|^2\le 2\int_{I_j}|F|^2+h_j^2\int_{I_j}|F'|^2.$$

Apply this with $F=\mathcal{H}r$, sum the good cells, and use the L^2 isometry of $\mathcal{H}$ and its commutation with differentiation. Equations (13.59)–(13.60) yield

$$\sum_{i\in\mathcal{G}}h_i|\mathcal{H}r(c_i)|^2\le C_\rho b^4E. \tag{13.61}$$

Finally, (13.41) gives the exact decomposition

$$\mathcal{H}q_{\mathrm{G}}(c_i)-\mathcal{H}\widehat{q}(Rc_i)=\mathcal{H}r(c_i)-\mathfrak{C}_R\widehat{q}(c_i).$$

Combine (13.61) with (13.43) to prove (13.55). Integrating (13.57) also shows that this static difference is $-\int_0^1\mathcal{T}_t\sigma(c_i)\,dt$; hence ZG–COM closes exactly the structured integrated opening-flow square, without a pointwise-in-t operator estimate.

□

The restriction $i\in\mathcal{G}$ in Lemma 13.8 is essential. On a bad midpoint the collapsed reference value is an endpoint trace, not zero. It must be retained in the endpoint layer, rather than incorrectly charged to the opening density alone.

Lemma 13.9 (Normalized-bubble bad-target trace)**.** *With the same notation,*

$$\boxed{\sum_{i\in\mathcal{B}}h_i|\mathcal{H}q_{\mathrm{G}}(c_i)|^2\le C_\rho\left\{b^3\sum_{j\in\mathcal{G}}(\widehat{h}_j-b)^2+b^4E\right\}.} \tag{13.62}$$

The constant is independent of the number and subdivision of the gaps.

Proof. Put

$$g:=\frac{2\pi-E}{M}=b-\gamma.$$

Thus $\widehat{h}_j-b=h_j-g$, and $g\asymp_\rho b$. On $0\le s\le 1$ write

$$\psi(s)=\frac{1}{2}\min\{s^2,(1-s)^2\},\qquad \overline{\psi}=\int_0^1\psi(s)\,ds=\frac{1}{24}.$$

For a good cell $I_j=[\theta_j,\theta_{j+1}]$ define the normalized bubble

$$p_j(\theta_j+h_js)=g^2\psi(s),\qquad p_{\mathrm{G}}=\sum_{j\in\mathcal{G}}p_j.$$

The amplitude error

$$f:=q_{\mathrm{G}}-p_{\mathrm{G}}=\sum_{j\in\mathcal{G}}(h_j^2-g^2)\psi\left(\frac{\cdot-\theta_j}{h_j}\right)\mathbf{1}_{I_j}$$

is even on every good target cell and is zero on every bad target cell. It therefore satisfies the self-zero hypothesis (13.9). Since $\int_0^1 \psi^2 = 1/320$ and $h_j, g \asymp_\rho b$,

$$\sum_i h_i |\mathcal{H} f(c_i)|^2 \le C\|f\|_2^2$$

$$= \frac{1}{320} \sum_{j\in\mathcal{G}} h_j (h_j^2 - g^2)^2 \le C_\rho b^3 \sum_{j\in\mathcal{G}} (h_j - g)^2. \tag{13.63}$$

It remains to estimate p_{G}. Decompose it as

$$p_{\mathrm{G}} = g^2 \overline{\psi}\, \mathbf{1}_{G^*} + r_{\mathrm{G}}, \qquad r_{\mathrm{G}}|_{I_j} = g^2 \left\{ \psi\left(\frac{\cdot - \theta_j}{h_j} \right) - \overline{\psi} \right\}. \tag{13.64}$$

Because $\mathcal{H}\mathbf{1} = 0$, $\mathcal{H}\mathbf{1}_{G^*} = -\mathcal{H}\mathbf{1}_B$. The function $\mathbf{1}_B$ is constant on every target cell, hence also satisfies (13.9). The adaptive sampling estimate gives

$$\sum_i h_i |\mathcal{H}(g^2 \overline{\psi}\, \mathbf{1}_{G^*})(c_i)|^2 \le C g^4 \|\mathbf{1}_B\|_2^2 = C g^4 E. \tag{13.65}$$

We now prove the same bound for r_{G} on the bad targets. Fix a maximal gap $B_\nu = (a, a+e)$ and $x \in B_\nu$. Put

$$r_- = x - a, \qquad r_+ = a + e - x.$$

The two good cells adjacent to a and $a+e$ have lengths comparable with g. On either adjacent cell the profile $\psi - \overline{\psi}$ has the common endpoint value $-\overline{\psi}$; its difference from that endpoint is ψ and vanishes quadratically. Since the periodic Hilbert kernel $K(t) = (2\pi)^{-1} \cot(t/2)$ is odd, pairing the two adjacent cells gives

$$\left| \int_{I_- \cup I_+} K(x-y) r_{\mathrm{G}}(y)\, dy \right| \le C_\rho g^2 \left(1 + \left| \log \frac{r_-}{r_+} \right| \right). \tag{13.66}$$

Indeed, the two constant endpoint pieces are

$$-g^2 \overline{\psi} \left\{ \int_0^{h_-} K(r_- + u)\, du - \int_0^{h_+} K(r_+ + u)\, du \right\}.$$

Lift the gap and its two adjacent cells to the real arc; this is unambiguous because $E \le \pi/4$ and the fan is sufficiently fine. Then

$$\int_0^h K(t+u)\, du = \frac{1}{\pi} \log \frac{\sin((t+h)/2)}{\sin(t/2)}$$

exactly. On this arc $\sin(v/2) \asymp v$, while $h_\pm \asymp_\rho g$. Moreover $u \mapsto \log(1+e^{-u})$ is 1-Lipschitz. Applying this with $u = \log(t/h)$ shows that the difference of the two logarithms is bounded by $C_\rho + |\log(r_-/r_+)|$, which proves the constant part of (13.66). For the two remaining pieces use $\psi(u/h) \le C \min\{(u/h)^2, ((h-u)/h)^2\}$; their integrals against $|K(t+u)|$ are uniformly bounded.

Every other good cell J is at distance at least $c_\rho g$ from x. On J the function r_{G} has zero mass and zero first moment about the midpoint. Two Taylor subtractions and $|K''(t)| \le C\,\mathrm{dist}(t, 2\pi\mathbb{Z})^{-3}$ give

$$\left|\int_J K(x-y) r_{\mathrm{G}}(y)\,dy\right| \le \frac{C_\rho g^5}{\mathrm{dist}(x,J)^3}.$$

The kth good cell in either cyclic direction is at distance at least $c_\rho k g$; summing k^{-3} and combining with (13.66) yields

$$|\mathcal{H} r_{\mathrm{G}}(x)| \le C_\rho g^2 \left(1 + \left|\log \frac{r_-}{r_+}\right|\right). \tag{13.67}$$

Finally, for an arbitrary subdivision of $(a, a+e)$ into target cells with midpoints c_i,

$$\sum_{I_i \subset B_\nu} h_i \left(1 + \left|\log \frac{c_i - a}{a+e-c_i}\right|\right)^2 \le Ce. \tag{13.68}$$

To see this, scale to $(0,1)$. On either half interval the displayed profile is monotone, so the value at a cell midpoint is bounded by twice its integral over the half of that cell nearer the corresponding endpoint. A cell crossing $1/2$ contributes $O(h_i)$. The remaining claim is the elementary finite integral

$$\int_0^1 \left(1 + \left|\log \frac{s}{1-s}\right|\right)^2 ds < \infty.$$

Equations (13.67)–(13.68), summed over the disjoint gaps, prove

$$\sum_{i \in \mathcal{B}} h_i |\mathcal{H} r_{\mathrm{G}}(c_i)|^2 \le C_\rho g^4 E. \tag{13.69}$$

Combining (13.63), (13.65), and (13.69), and using $g \asymp_\rho b$ and $h_j - g = \widehat{h}_j - b$, proves (13.62). □

We can now state the corrected replacement for the stage 155 opening square.

Lemma 13.10 (Zero-gap skeleton square)**.** *Let good source cells have lengths comparable with b, let disjoint zero-source gaps have total length E, and let the target measure put the actual cell mass at every good or bad midpoint. If the configuration is obtained from the collapsed regular M-fan by varying the good lengths to h_i and opening the gaps, then*

$$\boxed{\sum_i h_i |\mathcal{H} q_{\mathrm{G}}(c_i)|^2 \le C_\rho \left\{ b^3 \sum_{j \in \mathcal{G}} (h_j - b)^2 + b^4 E \right\}}. \tag{13.70}$$

The constant is independent of the number of gaps, the number of target cells inside a gap, and every gap ratio.

Proof. Let $\gamma = E/M$ and $\widehat{h}_j = h_j + \gamma$. Then $\sum_{j\in\mathcal{G}} \widehat{h}_j = 2\pi$ and

$$\sum_{j\in\mathcal{G}}(\widehat{h}_j - b)^2 = \sum_{j\in\mathcal{G}}(h_j - b)^2 - \frac{E^2}{M} \le \sum_{j\in\mathcal{G}}(h_j - b)^2.$$

The complete fan $\widehat{h}$ is comparable with b. The variable-grid $T1$ theorem of stage 154, applied on that complete fan, gives

$$\sum_{i\in\mathcal{G}} h_i|\mathcal{H}\widehat{q}(\widehat{c}_i)|^2 \le C_\rho b^3 \sum_{j\in\mathcal{G}}(h_j - b)^2,$$

because $h_i \asymp_\rho b$. This is precisely the square of the reference vector on the good target atoms. Lemma 13.8 bounds the good-target opening error by $C_\rho b^4 E$, and Lemma 13.9 bounds all bad target atoms. Since

$$\sum_{j\in\mathcal{G}}(\widehat{h}_j - b)^2 \le \sum_{j\in\mathcal{G}}(h_j - b)^2,$$

the inequality $|x+y|^2 \le 2|x|^2 + 2|y|^2$ proves (13.70). □

13.4. **AVS, HQ, and DF.**

Theorem 13.11 (Adaptive variance sampling)**.** *For every positive sufficiently fine fan,*

$$\boxed{T(h) \le C\mathfrak{D}(h) \le CS\mathfrak{J}_4(h).} \tag{13.71}$$

Proof. The far-stratum region is Theorem 4.1 of stage 153. In the near region, combine (13.8), Lemma 13.1, and Lemma 13.10, and then apply the static budget (13.6). This gives $T \le C_\rho\mathfrak{D}$. Stage 153 proves $\mathfrak{D} \le S\mathfrak{J}_4$. □

Corollary 13.12 (Pure midpoint row DF)**.** *The pure cubic Bregman row DF holds unconditionally.*

Proof. Stage 144 gives the exact identity

$$\mathcal{C}(q_h^3) - \mathcal{C}(q_{a\mathbf{1}}^3) = \frac{1}{720}\left(\sum_i h_i^5 - Na^5\right) + \frac{1}{576}\mathfrak{D}(h) - T(h), \qquad a = \frac{2\pi}{N}.$$

The linear part of the fifth-power Bregman difference cancels because $\sum_i (h_i - a) = 0$. Taylor's formula and $h_i, a \le S$ give

$$0 \le \sum_i h_i^5 - Na^5 \le CS\sum_i (h_i - a)^2(h_i^2 + 2ah_i + 3a^2) = CS\mathfrak{J}_4(h).$$

Also $\mathfrak{D} \le S\mathfrak{J}_4$ by stage 153, and $0 \le T \le C\mathfrak{D}$ by Theorem 13.11. Taking absolute values in the exact identity proves DF with no omitted sign case. □

Part 4. The mixed support row and exact support-Hessian assembly

14. Node PT: physical material transport and the mixed row DM

Audited source: `PT_material_transport.tex`.

14.1. The row and the two regimes.

Let

$$h_i > 0, \qquad \sum_i h_i = 2\pi, \qquad a = \frac{2\pi}{N}, \qquad S = a + \max_i h_i, \tag{14.1}$$

and put

$$\mathfrak{J}_4(h) = \sum_i h_i^4 - Na^4. \tag{14.2}$$

On $I_i = [\theta_i, \theta_{i+1}]$ define

$$q_h(\theta_i + x) = \frac{1}{2}\min\{x^2, (h_i - x)^2\}, \qquad 0 \le x \le h_i. \tag{14.3}$$

Let Z_h be the exact area tangent space of support increments in a fixed translation gauge, and let $v_h(z)$ be the physical-affine normal trace. On

$$K_i = [\theta_i - h_{i-1}/2, \theta_i + h_i/2], \qquad x = \theta - \theta_i,$$

it is

$$\begin{aligned} v_h(z)(\theta) &= \Pi_{\mathrm{ng}}\left[z_i \cos x + \{(1 - t_i(x))A_i^- + t_i(x)A_i^+\}\sin x\right], \\ t_i(x) &= \frac{\tan x + \tan(h_{i-1}/2)}{\tan(h_{i-1}/2) + \tan(h_i/2)}, \\ A_i^- &= \frac{z_i - z_{i-1}}{\sin h_{i-1}} - z_i \tan\frac{h_{i-1}}{2}, \qquad A_i^+ = \frac{z_{i+1} - z_i}{\sin h_i} + z_i \tan\frac{h_i}{2}. \end{aligned} \tag{14.4}$$

The exact support metric satisfies

$$c\|v_h(z)\|_{H^{1/2}}^2 \le \widetilde{\mathfrak{G}}_h(z) \le C\|v_h(z)\|_{H^{1/2}}^2. \tag{14.5}$$

The symmetric null form is

$$\mathcal{C}(f_1, f_2, f_3) = \frac{1}{3}\sum_{\mathrm{cyc}} \int_{\mathbb{T}} f_1\{(|D|f_2)(|D|f_3) - f_2' f_3'\}\, d\theta. \tag{14.6}$$

The row DM is

$$|\mathcal{C}(q_h, q_h, v_h(z))| \le \varepsilon_{\mathrm{DM}}(S)\mathfrak{J}_4(h)^{1/2}\widetilde{\mathfrak{G}}_h(z)^{1/2}, \qquad \varepsilon_{\mathrm{DM}}(S) \longrightarrow 0. \tag{14.7}$$

Fix $\beta = 1/2$. As in stage 161, split into

$$\mathfrak{J}_4(h) \ge S^{5-\beta} \qquad \text{(far)}, \tag{14.8}$$

$$\mathfrak{J}_4(h) < S^{5-\beta} \qquad \text{(near)}. \tag{14.9}$$

The fully polarized one-Lipschitz estimate

$$|\mathcal{C}(f, g, w)| \le C\|f'\|_\infty \|g\|_{\dot{H}^{1/2}} \|w\|_{\dot{H}^{1/2}} \tag{14.10}$$

follows from the same-sign Fourier formula by putting the low derivative block in L^∞; when the Lipschitz slot is high, the remaining separation factor is $2^{-|j-k|/2}$. Since

$$\|q_h'\|_\infty \le S, \qquad \|q_h\|_{\dot H^{1/2}} \le CS^{3/2}, \tag{14.11}$$

the far regime gives

$$|\mathcal{C}(q_h, q_h, v_h(z))| \le CS^{1/4}\mathfrak{J}_4(h)^{1/2}\widetilde{\mathfrak{G}}_h(z)^{1/2}. \tag{14.12}$$

In the near regime set

$$d_i = h_i - a, \qquad h_i(t) = a + td_i. \tag{14.13}$$

The exact identity

$$\mathfrak{J}_4(h) = \sum_i d_i^2(h_i^2 + 2ah_i + 3a^2) \tag{14.14}$$

gives $|d_i|^4 \le C\mathfrak{J}_4(h)$. In the near regime, $\max_i |d_i| \le CS^{(5-\beta)/4} = CS^{9/8} = o(S)$ as $S \to 0$. Since $S = a + \max_i h_i$ and $|h_i - a| \le \max_j |d_j|$, both a and every $h_i(t)$ therefore lie between fixed positive multiples of S, for all sufficiently small S. Hence

$$cS \le a \le h_i(t) \le CS, \qquad \delta(t) := \max_i \frac{|d_i|}{h_i(t)} \le C\frac{\max_i |d_i|}{S}. \tag{14.15}$$

14.2. **Lipschitz covariance of the same-sign null form.** For a periodic Lipschitz function u, put $F_\varepsilon(x) = x + \varepsilon u(x)$ and

$$\mathfrak{D}_u\mathcal{C}(f, g, w) = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} \mathcal{C}(f \circ F_\varepsilon^{-1}, g \circ F_\varepsilon^{-1}, w \circ F_\varepsilon^{-1}). \tag{14.16}$$

Lemma 14.1 (Critical Lipschitz covariance)**.** *For real $f \in W^{1,\infty}(\mathbb{T})$ and real $g, w \in H^{1/2}(\mathbb{T})$,*

$$\boxed{|\mathfrak{D}_u\mathcal{C}(f, g, w)| \le C\|u'\|_\infty\|f'\|_\infty\|g\|_{H^{1/2}}\|w\|_{H^{1/2}}.} \tag{14.17}$$

The polarized estimate is uniform for piecewise-affine u.

Proof. First take trigonometric polynomials. The same-sign representation of (14.6) is a sum, over $p, q > 0$, of the three placements of

$$pq\, \widehat{f}_p\widehat{g}_q\overline{\widehat{w}_{p+q}} \tag{14.18}$$

and their conjugates. Since

$$\frac{d}{d\varepsilon}\bigg|_0 (f \circ F_\varepsilon^{-1}) = -uf',$$

differentiating the three slots in (14.18) produces a quadrilinear multiplier with the deformation frequency ℓ. At $\ell = 0$ the three contributions add to zero: this is exactly invariance under a common translation. Therefore the summed multiplier is divisible by ℓ, and $\ell\widehat{u}_\ell$ is a Fourier coefficient of u'.

For completeness, this divisibility has a pointwise symbol bound. Put

$$\tau(r,s,t)=\sum_{\mathrm{cyc}}(|s||t|+st),\qquad r+s+t=0.$$

Thus τ is twice the product of the two same-sign frequencies. If $\ell+k+m+n=0$, direct substitution in the six possible sign orders gives

$$\begin{aligned}\big|k\tau(k+\ell,m,n)+m\tau(k,m+\ell,n)\\ +n\tau(k,m,n+\ell)\big|\le 3|\ell|\,|k|\,|m|^{1/2}|n|^{1/2}.\end{aligned}\tag{14.19}$$

For example, on a region with $m,n>0$ the only changes occur when $k+\ell$, $m+\ell$, or $n+\ell$ crosses zero; on each resulting interval the left side is affine in ℓ after its zero value at $\ell=0$ is removed, and its two endpoint slopes are bounded by the right side. The other sign orders are permutations. This proves (14.19) without a smoothness assumption on u.

We spell out the endpoint summation, since the pointwise bound alone is not an $L^\infty\times L^\infty\times L^2\times L^2$ theorem. Let P_j be cyclic Littlewood–Paley projections and write

$$U_j=P_j(u'),\quad F_j=P_j(f'),\quad G_j=|D|^{1/2}P_jg,\quad W_j=|D|^{1/2}P_jw.$$

Split each dyadic block further at the six sign boundaries

$$k=0,\quad m=0,\quad n=0,\quad k+\ell=0,\quad m+\ell=0,\quad n+\ell=0.$$

On every resulting interval the divided symbol

$$\frac{k\tau(k+\ell,m,n)+m\tau(k,m+\ell,n)+n\tau(k,m,n+\ell)}{\ell k|m|^{1/2}|n|^{1/2}}\tag{14.20}$$

and each discrete difference, multiplied by the corresponding dyadic scale, are bounded. At a boundary the one-sided limits agree with the formula in the adjacent sign chamber; if one of k,m,n is zero, the undivided symbol is zero. Thus no boundary atom or zero-frequency row is lost.

Let J be the largest dyadic index and recall that the frequency sum forces the two largest indices to differ by at most 3. Up to exchanging g,w and u',f', all flag rows have the following bounds: on the chambers where one of G,W is r generations below a high U or F block, direct substitution in the same-sign product gives an extra $2^{-r/2}$; when both G,W are lower by r,s and U,F are high, the divided symbol is bounded by $C2^{-(r+s)/2}$. For example, if $k>0$, $m,n>0$, and $\ell<0$ are in the latter configuration, then $k+\ell=-(m+n)$ and the numerator in (14.20) is $6kmn$, so division gives $C|mn|^{1/2}/|\ell|$. The other five sign orders are identical after a permutation. Thus all flag rows have the following bounds:

two high blocks	bound after the order-zero kernel is summed
G_J,W_J	$C\|U_{\le J}\|_\infty\|F_{\le J}\|_\infty\|G_J\|_2\|W_{J+O(1)}\|_2$
U_J,G_J	$C\sum_{r\ge0}2^{-r/2}\|U_J\|_\infty\|F_{\le J}\|_\infty\|G_{J+O(1)}\|_2\|W_{J-r}\|_2$
F_J,G_J	$C\sum_{r\ge0}2^{-r/2}\|U_{\le J}\|_\infty\|F_J\|_\infty\|G_{J+O(1)}\|_2\|W_{J-r}\|_2$
U_J,F_J	$C\sum_{r,s\ge0}2^{-(r+s)/2}\|U_J\|_\infty\|F_{J+O(1)}\|_\infty\|G_{J-r}\|_2\|W_{J-s}\|_2.$

The factors $2^{-r/2}$ are the unused half derivative in (14.19). The same table holds when a shifted frequency touches a sign boundary, using the one-sided divided symbol above.

Each Littlewood–Paley block is bounded in L^∞ by the full L^∞ norm. Sum first in r, s and then in J. Young's inequality for the geometric sequences and Cauchy–Schwarz for the G, W square functions give

$$C\|u'\|_\infty\|f'\|_\infty\left(\sum_j\|G_j\|_2^2\right)^{1/2}\left(\sum_j\|W_j\|_2^2\right)^{1/2}.$$

The finitely many zero and unit frequencies are bounded by the inhomogeneous $H^{1/2}$ norms. This is (14.17).

Approximate a Lipschitz u by smooth functions with uniformly bounded derivative. The estimate just proved is stable under this approximation, so it applies in particular to piecewise-affine velocities. □

Remark 14.2. Lemma 14.1 measures u' and not u. Consequently a large cumulative displacement of grid nodes and a common rotation do not enter the estimate. This is the precise replacement for the invalid bare-kernel order used in stage 161.

14.3. **Exact material fan identity and support transport.** Refine the fan mesh by the nodes $\theta_i(t)$ and the midpoints $c_i(t) = (\theta_i(t) + \theta_{i+1}(t))/2$. Let $F_t : \mathbb{T}_0 \to \mathbb{T}_t$ be the unique orientation-preserving map which is affine on every reference half-cell and sends $\theta_i(0), c_i(0)$ to $\theta_i(t), c_i(t)$. Set

$$G_{t,\varepsilon} = F_{t+\varepsilon} \circ F_t^{-1}, \qquad u_t = \partial_\varepsilon G_{t,\varepsilon}\big|_{\varepsilon=0}. \tag{14.21}$$

On the two halves of $I_i(t)$ one has

$$u_t' = \frac{d_i}{h_i(t)}, \qquad \|u_t'\|_\infty = \delta(t). \tag{14.22}$$

Lemma 14.3 (Exact fan transport)**.** *On both halves of $I_i(t)$,*

$$q_{h(t+\varepsilon)} \circ G_{t,\varepsilon} = \left(1 + \varepsilon\frac{d_i}{h_i(t)}\right)^2 q_{h(t)}. \tag{14.23}$$

Consequently, if a dot denotes the derivative of the pulled profile,

$$\|\dot{q}_t\|_{H^{1/2}} \le C\delta(t)S^{3/2}. \tag{14.24}$$

Proof. On either half-cell, q_h is one half of the squared distance from its nearest endpoint. The affine map multiplies that distance by $1 + \varepsilon d_i/h_i(t)$, proving (14.23).

Thus $\dot{q}_t = 2(d_i/h_i(t))q_{h(t)}$ on $I_i(t)$. Each cell piece vanishes quadratically at its outer endpoints. Its zero extension has $H^{1/2}$ norm squared at most $Ch_i(t)^4|d_i/h_i(t)|^2$ by scaling the fixed unit-cell profile. To sum the pieces, use comparability from (14.15). The $H^{1/2}$ pairing of two nonadjacent cell profiles at cyclic index distance r is bounded by $CS^4|a_ia_j|/(1+r)^2$, where

$a_i = d_i/h_i(t)$; adjacent pairs are bounded by the two-cell unit-profile norm. Schur summation of $(1+r)^{-2}$ gives

$$\Big\|\sum_i 2a_i q_i\Big\|_{H^{1/2}}^2 \le C\sum_i S^4|a_i|^2.$$

Hence

$$\|\dot q_t\|_{H^{1/2}}^2 \le C\delta(t)^2\sum_i h_i(t)^4 \le C\delta(t)^2 S^3,$$

because $\sum_i h_i(t) = 2\pi$ and $h_i(t) \le CS$. □

We next record the only finite-element statement about the physical support trace. The estimate is on its natural support space, not on an arbitrary ambient target.

Lemma 14.4 (Stable physical material transport)**.** *Along every comparable path* (14.15) *there is a linear transport $z_1 \mapsto z_t \in Z_{h(t)}$, preserving the exact area and translation quotient, such that, with*

$$\widetilde v_t = v_{h(t)}(z_t)\circ F_t, \tag{14.25}$$

interpreted infinitesimally on the common refined grid,

$$\sup_{0\le t\le 1}\|v_{h(t)}(z_t)\|_{H^{1/2}} \le C\|v_h(z_1)\|_{H^{1/2}}, \tag{14.26}$$

$$\|\dot{\widetilde v}_t\|_{H^{1/2}} \le C\delta(t)\|v_{h(t)}(z_t)\|_{H^{1/2}}. \tag{14.27}$$

Proof. We include the discrete endpoint estimate which prevents a hidden mesh-frequency loss. On a comparable cyclic mesh define

$$\|z\|_{\mathfrak h_S^{1/2}}^2 = S\sum_i |z_i|^2 + \sum_{i\ne j}\frac{|z_i - z_j|^2}{(1+|i-j|_{\mathrm{cyc}})^2}. \tag{14.28}$$

First let $I_h z$ be the affine interpolant with nodal values z_i. For non-neighbouring cells I_i, I_j at cyclic distance r, comparability gives

$$|x-y| \asymp S(1+r), \qquad |I_i| \asymp |I_j| \asymp S.$$

The corresponding Gagliardo rectangle is therefore comparable with $|\bar z_i - \bar z_j|^2/(1+r)^2$, where $\bar z_i$ is the cell average. The two-cell rectangles give $|z_{i+1} - z_i|^2$. Since

$$|\bar z_i - z_i| \le \tfrac12 |z_{i+1} - z_i|,$$

the triangle inequality converts nodal differences to average differences and conversely, with an error paid by these neighbouring rectangles. The summability of $(1+r)^{-2}$ pays each such error a bounded number of times. Together with $\|I_h z\|_2^2 \asymp S\sum_i |z_i|^2$, summing the rectangles in both directions proves

$$\|I_h z\|_{H^{1/2}}^2 \asymp \|z\|_{\mathfrak h_S^{1/2}}^2. \tag{14.29}$$

Now pull every patch K_i in (14.4) to $[0,1]$. With $w_i = (h_{i-1}+h_i)/2$, $p_i = h_{i-1}/(h_{i-1}+h_i)$, and

$$Z = z_i, \qquad A = w_i A_i^-, \qquad B = w_i A_i^+,$$

the normalized trace is

$$Z\cos\xi + \{(1-y)A + yB\}\frac{\sin\xi}{w_i}, \qquad \xi = w_i(y-p_i),$$

where replacing t_i by y changes the coefficient functions by $O(S^2)$ in C^1. Comparability places p_i in a fixed compact subinterval of $(0,1)$. As $w_i \to 0$, the local three-node map tends to

$$(z_{i-1}, Z, z_{i+1}) \longmapsto Z + (y-p_i)\left\{(1-y)\frac{Z - z_{i-1}}{2p_i} + y\frac{z_{i+1} - Z}{2(1-p_i)}\right\}. \tag{14.30}$$

This limiting map is injective: evaluation at $y = p_i$ first gives $Z = 0$, and division by $y - p_i$ then gives an affine function whose two endpoint coefficients force $z_{i-1} = z_{i+1} = 0$. Finite-dimensional norm equivalence, compactness in p_i, and an $O(S^2)$ perturbation therefore give uniform upper and lower bounds for the local map and its inverse. The physical trace and the affine interpolant are consequently related by uniformly invertible three-point finite-element maps. Applying the preceding rectangle argument to these local maps gives

$$\|v_h(z)\|^2_{H^{1/2}} \asymp \|z\|^2_{\mathfrak{h}^{1/2}_S} \tag{14.31}$$

on the exact quotient. Indeed, the area row controls the constant mode, while support translations are exactly the first two harmonics and are removed by Π_{ng}. Hence no low-mode kernel remains in either direction.

We next differentiate the actual coefficients. Put

$$b_i^- = \frac{h_{i-1}}{2}, \quad b_i^+ = \frac{h_i}{2}, \quad D_i = \tan b_i^- + \tan b_i^+, \quad N_i(x) = \tan x + \tan b_i^-.$$

For the pulled local coordinate $x = x(t)$, direct differentiation gives

$$\dot t_i(x) = \frac{\{\sec^2 x\,\dot x + \frac12\sec^2 b_i^-\,d_{i-1}\}D_i - N_i(x)\{\frac12\sec^2 b_i^-\,d_{i-1} + \frac12\sec^2 b_i^+\,d_i\}}{D_i^2}, \tag{14.32}$$

$$\dot A_i^- = \frac{\dot z_i - \dot z_{i-1}}{\sin h_{i-1}} - \frac{(z_i - z_{i-1})\cos h_{i-1}}{\sin^2 h_{i-1}}d_{i-1} \tag{14.33}$$

$$- \dot z_i \tan\frac{h_{i-1}}{2} - \frac12 z_i\sec^2\frac{h_{i-1}}{2}d_{i-1}, \tag{14.34}$$

$$\dot A_i^+ = \frac{\dot z_{i+1} - \dot z_i}{\sin h_i} - \frac{(z_{i+1} - z_i)\cos h_i}{\sin^2 h_i}d_i \tag{14.35}$$

$$+ \dot z_i \tan\frac{h_i}{2} + \frac12 z_i\sec^2\frac{h_i}{2}d_i. \tag{14.36}$$

Moreover

$$\partial_t \cos x = -\dot{x} \sin x, \qquad \partial_t \sin x = \dot{x} \cos x, \tag{14.37}$$

$$|\dot{x}| \le C\delta(t)S, \qquad |\dot{t}_i| \le C\delta(t). \tag{14.38}$$

Substitution of (14.32)–(14.37) in (14.4) shows that every pulled trace derivative is a bounded three-point finite-element combination of $\dot{z}$ and

$$Sa_i z_i, \qquad a_i(z_{i+1} - z_i), \qquad a_{i-1}(z_i - z_{i-1}), \qquad |a_i| \le \delta(t), \tag{14.39}$$

and their one-cell shifts. There is no unweighted term $a_i z_i$: the trigonometric frame contributes $d_i = Sa_i$, while every derivative of a normalized slope is paired with a nodal difference. The two one-sided wall terms cancel because the physical affine vector field is continuous. Differentiating Π_{ng} adds only the modes $1, \cos\theta, \sin\theta$ and their pulled derivatives $u_t \sin\theta, u_t \cos\theta$. After subtracting the irrelevant common rotation, periodic Poincaré gives $\|u_t\|_\infty \le C\|u_t'\|_\infty$; these fixed-dimensional rows obey the same bound as (14.39) and create no factor of N.

The multiplier bound can be proved without a Fourier endpoint theorem. Write $Dz_i = z_{i+1} - z_i$ and let $K(i,j) = (1 + |i-j|_{\mathrm{cyc}})^{-2}$. For $w_i = Sa_i z_i$,

$$S\sum_i |w_i|^2 \le C\|a\|_\infty^2 S \sum_i |z_i|^2,$$

$$\sum_{i\ne j} K(i,j)|w_i - w_j|^2 \le 2S^2\|a\|_\infty^2 \sum_{i \ne j} K(i,j)|z_i - z_j|^2 + 8S^2\|a\|_\infty^2 \sum_j |z_j|^2 \sum_i K(i,j).$$

Since $\sup_j \sum_i K(i,j) < \infty$ and S is bounded, this is controlled by (14.28). For $w_i = a_i Dz_i$ use

$$S\sum_i |w_i|^2 \le S\|a\|_\infty^2 \sum_i |Dz_i|^2 \le C\|a\|_\infty^2 \|z\|_{\mathfrak{h}_S^{1/2}}^2$$

and

$$|w_i - w_j|^2 \le 2\|a\|_\infty^2 |Dz_i - Dz_j|^2 + 8\|a\|_\infty^2 |Dz_j|^2$$

and

$$|Dz_i - Dz_j|^2 \le 2|z_{i+1} - z_{j+1}|^2 + 2|z_i - z_j|^2.$$

The last row is controlled by the $|i-j|_{\mathrm{cyc}} = 1$ part of (14.28). These estimates prove

$$\|Saz + a\Delta z\|_{\mathfrak{h}_S^{1/2}} \le C\|a\|_{\ell^\infty} \|z\|_{\mathfrak{h}_S^{1/2}}. \tag{14.40}$$

It remains to define, rather than merely invoke, the constraint transport. Let

$$L_t z = \sum_i \ell_i(t) z_i, \qquad \ell_i(t) = \tan\frac{h_{i-1}(t)}{2} + \tan\frac{h_i(t)}{2}, \tag{14.41}$$

and let

$$\tau_t^1 = (\cos\theta_i(t))_i, \qquad \tau_t^2 = (\sin\theta_i(t))_i.$$

These are the exact two support-translation vectors and satisfy $L_t\tau_t^\mu = 0$. The matrix

$$M_t^{\mu\nu} = \sum_i \ell_i(t)\tau_{t,i}^\mu \tau_{t,i}^\nu \tag{14.42}$$

is uniformly comparable with the identity. Define

$$\mathcal{Q}_t z = z - \sum_{\mu,\nu=1}^{2} (M_t^{-1})_{\mu\nu} \left(\sum_i \ell_i(t) z_i \tau_{t,i}^\nu \right) \tau_t^\mu, \tag{14.43}$$

$$C_t z = \mathcal{Q}_t \left(z - \frac{L_t z}{L_t \mathbf{1}} \mathbf{1} \right). \tag{14.44}$$

If z_1 is in the chosen exact gauge, set

$$z_t = C_t z_1. \tag{14.45}$$

Then $L_t z_t = 0$, z_t is orthogonal to both translation rows in the $\ell(t)$-weighted Gram product, and $C_1 z_1 = z_1$.

The explicit rows give

$$\ell_i \asymp S, \qquad |\dot{\ell}_i| \le C\delta(t) S, \tag{14.46}$$

$$\|\dot{\tau}_t^\mu\|_{\mathfrak{h}_S^{1/2}} \le C\delta(t), \qquad |\dot{M}_t| + |\partial_t M_t^{-1}| \le C\delta(t). \tag{14.47}$$

Indeed, $|\dot{\theta}_i| = |\sum_{k<i} d_k| \le C\delta(t)$ and adjacent differences of $\dot{\theta}_i$ equal d_i. Differentiating (14.43)–(14.44), and using the bounded area and translation functionals furnished by (14.31), first yields

$$\|C_t\| \le C, \qquad \|\dot{C}_t z_1\|_{\mathfrak{h}_S^{1/2}} \le C\delta(t)\|z_1\|_{\mathfrak{h}_S^{1/2}}. \tag{14.48}$$

On $Z_{h(1)}$ the terminal map C_1 is the identity. Hence

$$\|(C_t - C_1)|_{Z_{h(1)}}\| \le C \int_t^1 \delta(s)\, ds \le CS^{1/8}.$$

For sufficiently small S this norm is at most $1/2$. The Neumann series therefore proves that $C_t|_{Z_{h(1)}}$ is injective onto its image and that its inverse is uniformly bounded. Combining this inverse bound with (14.48) gives

$$\|C_t\| + \|(C_t|_{Z_{h(1)}})^{-1}\| \le C, \qquad \|\dot{C}_t z_1\|_{\mathfrak{h}_S^{1/2}} \le C\delta(t)\|C_t z_1\|_{\mathfrak{h}_S^{1/2}}. \tag{14.49}$$

Equations (14.31), (14.40), and (14.49) now give

$$\|\dot{\tilde{v}}_t\|_{H^{1/2}} \le C\delta(t)\|v_{h(t)}(z_t)\|_{H^{1/2}}.$$

The finite maps F_t and F_t^{-1} have uniformly bounded Lipschitz constants by (14.15), so their pullbacks are uniformly bounded on $H^{1/2}$. Finally,

$$\int_0^1 \delta(t)\, dt \le C \frac{\max_i |d_i|}{S} \le CS^{1/8} \tag{14.50}$$

in the near regime. Gronwall's inequality proves (14.26), and the displayed derivative estimate is (14.27). □

14.4. **The near material estimate.**

Proposition 14.5 (Comparable Lipschitz material bound)**.** *Along the near path* (14.13) *and the transport of Lemma 14.4,*

$$\boxed{\left|\frac{d}{dt}\mathcal{C}(q_{h(t)}, q_{h(t)}, v_{h(t)}(z_t))\right| \le CS^{5/2}\delta(t)\widetilde{\mathfrak{G}}_h(z_1)^{1/2}.} \tag{14.51}$$

Proof. Pull all three functions by the common refined-grid map. The derivative splits into the common-coordinate row and three profile rows. Lemma 14.1, (14.11), (14.22), and (14.26) bound the coordinate row by the right side of (14.51).

For either fan-profile row use the polarized estimate (14.10), placing the Lipschitz derivative on the unchanged fan slot, and then use (14.24):

$$|\mathcal{C}(q_t, \dot{q}_t, v_t)| \le CS\,(\delta(t)S^{3/2})\|v_t\|_{H^{1/2}}.$$

For the physical-profile row, use (14.10), (14.11), and (14.27):

$$|\mathcal{C}(q_t, q_t, \dot{\tilde{v}}_t)| \le CS\,S^{3/2}\delta(t)\|v_t\|_{H^{1/2}}.$$

Metric equivalence (14.5) completes the proof. □

Lemma 14.6 (Regular quotient cancellation)**.** *For the uniform fan and every exact support class,*

$$\mathcal{C}(q_{a\mathbf{1}}, q_{a\mathbf{1}}, v_{a\mathbf{1}}(z)) = 0. \tag{14.52}$$

Proof. The fan spline and the physical support construction commute with the cyclic shift. Hence the left side is a cyclic-invariant linear functional of the support coefficients, and therefore a multiple of $\sum_i z_i$. At the regular tangential polygon the exact area tangent condition is $\sum_i z_i = 0$; translations have already been removed. □

Theorem 14.7 (Unconditional mixed row DM)**.** *For every sufficiently fine positive fan and every exact support class,*

$$\boxed{|\mathcal{C}(q_h, q_h, v_h(z))| \le C(S^{1/4} + S^{1/2})\mathfrak{J}_4(h)^{1/2}\widetilde{\mathfrak{G}}_h(z)^{1/2}.} \tag{14.53}$$

In particular (14.7) *holds independently of every minimum cell and adjacent ratio.*

Proof. Use (14.12) in the far regime. In the near regime transport the final support class to the regular quotient and integrate (14.51). Equations (14.15) and (14.14) give

$$\mathfrak{J}_4(h)^{1/2} \ge cS\left(\sum_i d_i^2\right)^{1/2} \ge cS\max_i |d_i|.$$

Therefore

$$\begin{aligned} |\mathcal{C}(q_h, q_h, v_h(z))| &\le CS^{5/2}\frac{\max_i |d_i|}{S}\widetilde{\mathfrak{G}}_h(z)^{1/2} \\ &\le CS^{1/2}\mathfrak{J}_4(h)^{1/2}\widetilde{\mathfrak{G}}_h(z)^{1/2}, \end{aligned}$$

where Lemma 14.6 supplies the zero initial value. Combining the regimes proves (14.53). ☐

Corollary 14.8 (Handoff to the current NR assembly)**.** *Theorem 14.7 supplies the mixed terminal input DM. The exact flux audit shows, however, that DM does not imply the former JE–FM clause, which remains unavailable; its former period-three material falsification is retracted after the polygonal corner finite-part audit. NR and active-chart J0 are assembled conditionally in `JC_joint_comparison.tex` and `NR_assembly.tex`. At the current audit DF, the CT core, and JP–DE are closed, while JC–PL is the unique open premise and JP–JC is its dependent exact output. No historical CT proposition is imported in that assembly, and none of these conclusions uses a minimum fan, polar, or conformal cell.*

15. Node JP: exact support Hessian and the closed disk endpoint

Audited source: `JP_joint_principal.tex`.

15.1. **Fixed-normal support chart.** Fix cyclic unit normals ν_i with exterior angles $\alpha_i \in (0, \pi)$ and $\sum_i \alpha_i = 2\pi$. For a support vector $h = (h_i)$ put

(15.1)

$$P_\alpha(h) = \{x \in \mathbb{R}^2 : x \cdot \nu_i < h_i \text{ for all } i\}, \qquad \Phi_\alpha(h) = \frac{|P_\alpha(h)|}{\pi} \lambda_1(P_\alpha(h)).$$

We work in an active star on which all sides persist. Translation vectors $(b \cdot \nu_i)_i$ are quotiented exactly. Although Φ_α is also dilation invariant, at a nonstationary point its Hessian does not descend to an abstract dilation quotient. We therefore fix an affine scale section below.

Let $p_i(h)$ be the intersection of the two support lines with normals ν_i, ν_{i+1}. Solving the two linear equations shows that

$$p_i(h + tz) = p_i(h) + t p_i(z). \tag{15.2}$$

Consequently the physical-affine material parametrization of every side is affine in t. Its normal velocity on side $S_i(t)$ is exactly

$$V(t) \cdot \nu_i = z_i, \qquad \partial_\tau (V \cdot \nu_i) = 0. \tag{15.3}$$

This is the regular finite-energy replacement for the discontinuous piecewise-constant function obtained by writing z_i in the radial angular coordinate. That raw angular function is not itself an $H^{1/2}$ trace.

The area $A_\alpha(h) = |P_\alpha(h)|$ is a quadratic polynomial in h. Indeed, by (15.2),

$$
\begin{aligned}
A_\alpha(h) &= \frac{1}{2}\sum_i \det(p_i(h), p_{i+1}(h)),\\
A'_\alpha[h;z] &= \frac{1}{2}\sum_i \{\det(p_i(z), p_{i+1}(h)) + \det(p_i(h), p_{i+1}(z))\},\\
A''_\alpha[z,z] &= \sum_i \det(p_i(z), p_{i+1}(z)).
\end{aligned}
\tag{15.4}
$$

Equivalently, direct differentiation of the side lengths gives the ratio-sensitive but exact cyclic formula

$$
\begin{aligned}
A''_\alpha[z,z] = \sum_i z_i\{z_{i-1}\csc\alpha_{i-1} + z_{i+1}\csc\alpha_i \\
- z_i(\cot\alpha_{i-1} + \cot\alpha_i)\}.
\end{aligned}
\tag{15.5}
$$

Equivalently,

$$
A''_\alpha[z,z] = \frac{1}{2}\sum_i \left\{\tan\frac{\alpha_i}{2}(z_i + z_{i+1})^2 - \cot\frac{\alpha_i}{2}(z_{i+1} - z_i)^2\right\}. \tag{15.6}
$$

Thus the area row contains a negative jump channel; it may not be bounded from below independently of the endpoint part of $\mathfrak{D}^{\mathrm{red}}$.

Let $h_0 = h_\alpha\mathbf{1}$ be the canonical tangential support vector, $A_0 = A_\alpha(h_0)$, and define the linear scale functional

$$
s_\alpha(h) = \frac{A'_\alpha[h_0;h]}{2A_0}. \tag{15.7}
$$

Euler's identity for the quadratic area gives $s_\alpha(h_0) = 1$, while translation invariance of area makes s_α vanish on translation vectors. Fix once and for all a translation complement Y_α which contains h_0, and put

$$
\mathsf{X}_\alpha = \{z \in \mathsf{Y}_\alpha : s_\alpha(z) = 0\}. \tag{15.8}
$$

Then $h_0 + tz$ stays in the affine section $s_\alpha = 1$ for every $z \in \mathsf{X}_\alpha$. All support variables, Hessians, dual norms, and Riesz maps used in JP–SC/JP–JE/JP–FS refer to this fixed section.

To match the global wrapper, let $\mathcal{U}_{S_0,\rho_0}$ denote the normalized radial endpoint chart: $S = a + \max_i \alpha_i \le S_0$, all sides are active, and, after the fixed scale/translation normalization, the radial graph of $P_\alpha(h)$ is within ρ_0 of the unit circle in $W^{1,\infty}$. The radii S_0, ρ_0 are universal and may be decreased. No smallness of the affine-side parametrization relative to $P_\alpha(h_0)$ is imposed: such a condition would incorrectly exclude a positive side whose length is much smaller than its normal gap. The ratio-free compatibility between the radial endpoint chart, the whole fixed-support segment, and the support metric is proved below as JP–NS.

Proposition 15.1 (The radial chart does not imply a labelled affine-side chart)**.** *There are active area-π convex N-gons Q_N with positive normal fans such that*

$$\max_i \alpha_i + \frac{2\pi}{N} \longrightarrow 0, \qquad \|r_{Q_N} - 1\|_{W^{1,\infty}(\mathbb{S}^1)} \longrightarrow 0,$$

whereas the derivative of the labelled physical-affine displacement from the tangential polygon with the same fan tends to infinity. Thus no ratio-free radial-to-affine-side chart implication is available.

Proof. Put $a = 2\pi/N$, $\varepsilon = a^4$, $b = (2\pi - 2\varepsilon)/(N-2)$, and $t = a^2$. Choose consecutive normals $-\varepsilon, 0, \varepsilon$ and make all remaining gaps equal to b. For

$$c = \left\{ \frac{\pi}{2\tan(\varepsilon/2) + (N-2)\tan(b/2)} \right\}^{1/2} = 1 + O(a^2),$$

the constant support vector $c\mathbf{1}$ gives the area-π tangential polygon. Keep the central support equal to c, change the two adjacent supports to $c\cos\varepsilon + t\sin\varepsilon$, and leave all other supports equal to c. The central side then has endpoints $(c, -t)$ and (c, t) and length $2t$, whereas its tangential reference length is $2c\tan(\varepsilon/2)$.

Writing

$$\delta = \sin\varepsilon \left(t - c\tan\frac{\varepsilon}{2} \right) = t\varepsilon(1 + o(1)),$$

the two incident side lengths change by $-\delta(\cot\varepsilon + \cot b) = O(a^2)$ from positive lengths asymptotic to a, the next two change by $\delta\csc b$, and all others are unchanged. Hence all sides remain active. The affine scale-section factor is $1+O(bt\varepsilon)$, so scale and area normalization do not alter the following asymptotics. On the central labelled side the tangential derivative of the displacement is at least

$$\frac{t}{(1+o(1))c\tan(\varepsilon/2)} - 1 \sim \frac{2t}{\varepsilon} = \frac{2}{a^2} \longrightarrow \infty.$$

Every changed vertex is nevertheless displaced by $O(t)$; the two central displacements are exactly $t - c\tan(\varepsilon/2)$, and the outer displacements are $O(\delta/\sin b)$. On a side with normal θ_i the radial graph is $r(\theta) = h_i/\cos(\theta - \theta_i)$ and $|r'| = r|\tan(\theta - \theta_i)|$. Thus, after the harmless area normalization, $\|r_{Q_N} - 1\|_{W^{1,\infty}} \le Ca \to 0$. This proves the claim. $\square$

15.2. **Exact eigenvalue and scale-normalized Hessians.** Let $u_t > 0$ be the L^2-normalized first eigenfunction of $P_\alpha(h + tz)$, let λ_t be its eigenvalue, and put $p_t = -\partial_{\nu_t} u_t$ on the open sides. If g is boundary data, write $\mathfrak{D}^{\mathrm{red}}_{P_t}(g)$ for the reduced Helmholtz energy: solve the Eulerian state equation with trace $-g$, impose L^2 orthogonality to u_t, and evaluate $\int_{P_t} |\nabla\psi|^2 - \lambda_t \int_{P_t} \psi^2$. This is the same reduced form as in the polygon-preserving volume Hessian.

Theorem 15.2 (JP–EH: exact fixed-normal support Hessian)**.** *For every active support line $h(t) = h + tz$,*

$$\lambda_t' = -\sum_i z_i \int_{S_i(t)} p_t^2\, ds, \tag{15.9}$$

$$\lambda_t'' = 2\mathfrak{D}^{\mathrm{red}}_{P_t}(z_i \partial_{\nu_t} u_t). \tag{15.10}$$

Hence, with $A_t = A_\alpha(h(t))$ and $F_t(z) = \sum_i z_i \int_{S_i(t)} p_t^2\, ds$, one has the exact identity

$$\boxed{\frac{d^2}{dt^2}\Phi_\alpha(h(t)) = \frac{1}{\pi}\Big\{2A_t\mathfrak{D}^{\mathrm{red}}_{P_t}(z_i\partial_{\nu_t}u_t) - 2A_t'[z]F_t(z) + \lambda_t A_\alpha''[z,z]\Big\}.} \tag{15.11}$$

No stationarity, minimum side, adjacent-ratio, moving radial knot, or Taylor truncation is used in this formula.

Proof. Hadamard's formula and (15.3) give (15.9). Differentiate it on the moving polygon. If ψ_t is the Eulerian eigenfunction derivative, the material derivative of the boundary flux is

$$\dot p_t = -\partial_{\nu_t}\psi_t + V_\tau \partial_\tau p_t.$$

The differentiated arclength, normal, and transport rows combine, after one sidewise integration by parts, into

$$\lambda_t'' = 2\mathfrak{D}^{\mathrm{red}}_{P_t}((V\cdot\nu_t)\partial_{\nu_t}u_t) + 2\sum_i \int_{S_i(t)} p_t^2 V_\tau \partial_\tau (V\cdot\nu_i)\, ds.$$

The endpoint terms vanish by the convex-corner expansion of the ground state. The second row is zero by (15.3), proving (15.10). Finally differentiate $\Phi_\alpha = A_t\lambda_t/\pi$ twice, use $\lambda_t' = -F_t(z)$ and (15.4), and obtain (15.11). ☐

15.3. **Why the disk cubic truncation cannot be the global metric.** For reference, let

$$H_\alpha^{(3)}(z,z) = D^2\mathcal{L}(0)[Z_\alpha, Z_\alpha] + D^3\mathcal{L}(0)[B_\alpha, Z_\alpha, Z_\alpha] \tag{15.12}$$

be the formerly proposed disk-based support Hessian. The complete physical chain rule is understood in (15.12); in particular the two curvature terms are not omitted.

Proposition 15.3 (Alternating-mesh obstruction to the cubic metric)**.** *There are even cyclic fans with fixed adjacent ratio and admissible translation/area-normalized support directions for which $H_\alpha^{(3)}(z,z) < 0$. Consequently no positive no-ratio metric can be obtained by proving coercivity of* (15.12)*.*

Proof. Let $N = 2M$, $a = 2\pi/N$, take alternating exterior angles $a/10, 19a/10$, and set $z_{2j} = 1$, $z_{2j+1} = -1$. The alternating symmetry gives the two translation constraints and the exact area row. After the material rescaling $y = \theta/a$, one period has length 2. Put $p = (1/10, 19/10)$, $c_0 = 0$, $c_1 = 1/10$, and on the cell $x = y - c_r \in [-p_{r-1}/2, p_r/2]$ define

$$F_r = \frac{x^2}{2} - \frac{p_0^3 + p_1^3}{48}, \qquad W_0 = -\frac{360}{19} - \frac{400}{19}x, \qquad W_1 = -\frac{360}{19} + \frac{400}{19}x,$$

$$V_r = z_r + xW_r, \qquad U_r = xW_r, \qquad K_r = W_r^2 - 2W_rV_r'.$$

For period-two Fourier coefficients $\widehat{h}(n) = \frac{1}{2}\int_0^2 h(y)e^{-\pi i n y}\,dy$, write $\langle g, h\rangle_2 = \sum_{n\neq 0} \pi|n|\overline{\widehat{g}(n)}\widehat{h}(n)$. The small-cell expansions, in the physical angular coordinate, are

$$f = a^2F + O(a^4), \quad w = a^{-1}W + O(a), \quad v = V + O(a^2), \quad f' = aF' + O(a^3),$$

$$w^2 - 2wv' = a^{-2}K + O(1), \qquad wf' = U + O(a^2).$$

The exact physical-chain identity is

$$D^3\mathcal{L}(0)[B, Z, Z] = D^3\Lambda^n[f, v, v] + D^2\Lambda^n[f, w^2 - 2wv'] + 2D^2\Lambda^n[v, -wf'],$$

and $D^2\Lambda^n = 2q_{\mathbb{D}}$. The normal cubic is $O(1)$; rescaling the other three pairings therefore gives

$$H_\alpha^{(3)}(z, z) = \frac{4\pi d^2}{a}E + O(1), \qquad E = \langle V, V\rangle_2 + \langle F, K\rangle_2 - 2\langle V, U\rangle_2,$$

where $d > 0$ is the disk normalization constant. Exact rational-polynomial Fourier integration through mode 4096, with outward intervals of radius 10^{-12} including the trigonometric Taylor and rounding remainders, gives

$$A_{4096} = 46.5557803349116641,$$
$$B_{4096} = 22.9387363887007744,$$
$$C_{4096} = 35.1293696664596509,$$

and hence $E_{4096} = A_{4096} + B_{4096} - 2C_{4096} = -0.7642226093068633$. Integration by parts in each Fourier coefficient and Cauchy–Schwarz give the explicit tail bound

$$|E - E_K| \le \frac{\mathrm{TV}(V')^2}{4\pi^3K^2} + \frac{\mathrm{TV}(F')\,\mathrm{TV}(K)}{2\pi^2K} + \frac{\mathrm{TV}(V')\,\mathrm{TV}(U)}{\pi^2K}.$$

Here

$$\mathrm{TV}(F') = 4, \qquad \mathrm{TV}(V') = 1600/19,$$
$$\mathrm{TV}(U) = 476/19, \qquad \mathrm{TV}(K) = 523200/361.$$

The reproducible certificate is `tests/check_jp_cubic_counterexample.py`. At $K = 4096$ the right-hand side is less than 0.123892271, so $E < -0.640330338$. Therefore (15.12) is negative for all sufficiently small a. Reflection through a common cell vertex makes B even and Z odd, so the old linear forcing in this direction and $D^3\mathcal{L}(0)[Z, Z, Z]$ vanish. The negative sign is therefore intrinsic to the candidate Hessian. This is an obstruction to the truncation, not to the exact Hessian (15.11). □

Corollary 15.4 (Stationary specialization)**.** *If h is stationary for Φ_α under the fixed-normal support variations, then at $t=0$*

$$D^2\Phi_\alpha(h)[z,z] = \frac{2A}{\pi}\mathfrak{D}_P^{\mathrm{red}}(z_i\partial_\nu u) + \frac{\lambda}{\pi}\left\{A''_\alpha[z,z] - \frac{2A'_\alpha[z]^2}{A}\right\}. \tag{15.13}$$

On the area tangent row $A'_\alpha[z]=0$, the last square vanishes.

Proof. Stationarity and (15.9) give $F_0(z) = \lambda A'_\alpha[z]/A$. Substitute this in (15.11). ☐

15.4. **Exact value integration and the reduced frontier.** Let $h_\alpha\mathbf{1}$ denote the canonical tangential support vector and let $\alpha_* = a\mathbf{1}$, $a = 2\pi/N$. Put

$$\ell_\alpha = D_h\Phi_\alpha(h_\alpha\mathbf{1}) \tag{15.14}$$

restricted to X_α. Ordinary one-variable Taylor integration, using the genuine Hessian rather than its disk cubic truncation, gives the identity

$$\boxed{\Phi_\alpha(h_\alpha\mathbf{1}+z) - \Phi_\alpha(h_\alpha\mathbf{1}) = \ell_\alpha[z] + \int_0^1 (1-t)D_h^2\Phi_\alpha(h_\alpha\mathbf{1}+tz)[z,z]\,dt.} \tag{15.15}$$

The exact Hadamard variable is the constant side-normal speed z_i, not the radial component of the full material vertex field. Let Π_{ng} delete the constant and first Fourier harmonics. We import from the CT core the literal disk Bessel form

$$q_D(f) = d^2\int_{\mathbb{T}}(\Pi_{\mathrm{ng}}f)(\mathcal{D}_\circ+1)(\Pi_{\mathrm{ng}}f), \qquad q_D(f) \asymp \|\Pi_{\mathrm{ng}}f\|^2_{H^{1/2}},$$

where $d = \partial_r u_0(1)$ and $\mathcal{D}_\circ e^{in\theta} = jJ'_{|n|}(j)J_{|n|}(j)^{-1}e^{in\theta}$ for $n\neq 0$. Let $\theta_{i+1}-\theta_i = \alpha_i$ and define the periodic fan-normal sine interpolant by

$$(\mathcal{I}_\alpha z)(\theta) = \frac{\sin(\theta_{i+1}-\theta)z_i + \sin(\theta-\theta_i)z_{i+1}}{\sin\alpha_i}, \qquad \theta_i\le\theta\le\theta_{i+1}. \tag{15.16}$$

Put

$$\mathfrak{G}^{\mathrm{ns}}_\alpha(z) := q_D(\Pi_{\mathrm{ng}}\mathcal{I}_\alpha z), \qquad z\in\mathsf{X}_\alpha. \tag{15.17}$$

This is the metric interface JP-NS.

Proposition 15.5 (Closed chart entry for the normal-speed metric)**.** *The form $\mathfrak{G}^{\mathrm{ns}}_\alpha$ is a Hilbert form on X_α. There is a universal C such that*

$$P_\alpha(h_0+z)\in\mathcal{U}_{S_0,\rho_0} \implies h_0+[0,1]z \text{ is active}, \qquad \mathfrak{G}^{\mathrm{ns}}_\alpha(z)^{1/2}\le C(S_0+\rho_0), \tag{15.18}$$

and every polygon on the segment has a radial graph $1+\rho_t$ satisfying

$$\sup_{0\le t\le 1}\|\rho_t\|_{W^{1,\infty}(\mathbb{S}^1)} \le C(S_0+\rho_0). \tag{15.19}$$

No minimum angle or adjacent-ratio bound is used.

Proof. If $\mathfrak{G}^{\mathrm{ns}}_\alpha(z)=0$, then $\mathcal{I}_\alpha z$ contains only the constant and first Fourier modes. On every open normal-angle interval it satisfies $(\mathcal{I}_\alpha z)''+\mathcal{I}_\alpha z=0$. Applying this equation to a function of the form $c+b\cdot e_r(\theta)$ forces $c=0$. Hence $z_i=b\cdot\nu_i$ is an exact support translation, which is zero in the chosen complement X_α. This proves positive definiteness. It also shows that (15.17) descends exactly through the translation quotient, rather than merely after choosing the complement.

Let $P_i(z)$ be the displacement of the vertex shared by sides i and $i+1$ from its position in $P_\alpha(h_0)$. If a star-shaped boundary is $r(\varphi)e_r(\varphi)$, its one-sided outer normal is parallel to $re_r-r'e_\varphi$. Hence $\|r-1\|_{W^{1,\infty}}\le\rho_0<1/2$ puts the polar direction of each endpoint vertex within $C\rho_0$ of its adjacent normal cone. The tangential reference has $\|r_0-1\|_{W^{1,\infty}}\le CS_0$, and the cone width is at most S_0. The two vertex radii are bounded above and below universally. Therefore

$$\max_i|P_i(z)|\le C(S_0+\rho_0), \tag{15.20}$$

with no inverse angle or side length. At that vertex the constant-normal-speed identity gives

$$z_i=P_i(z)\cdot\nu_i,\qquad z_{i+1}=P_i(z)\cdot\nu_{i+1}.$$

Solving these two projection equations gives the exact identity

$$(\mathcal{I}_\alpha z)(\theta)=P_i(z)\cdot e_r(\theta),\qquad \theta_i\le\theta\le\theta_{i+1}.$$

Therefore

$$\int_{\theta_i}^{\theta_{i+1}}\left\{|\mathcal{I}_\alpha z|^2+|\partial_\theta\mathcal{I}_\alpha z|^2\right\}d\theta=\alpha_i|P_i(z)|^2.$$

Summation gives $\|\mathcal{I}_\alpha z\|_{H^1}\le C(S_0+\rho_0)$, and the Bessel equivalence $q_D\asymp H^{1/2}$ proves the metric bound. For fixed normals every side length is a linear function of the support numbers. It is positive at both h_0 and h_0+z; therefore it remains positive on their whole affine segment. This proves activity of every labelled side.

Every intermediate vertex is the affine combination of its endpoint vertices. Its radial and tangential coordinates relative to either incident normal therefore obey the same $C(S_0+\rho_0)$ bounds. On each side the exact formulas

$$r_t(\theta)=\frac{h_i(t)}{\cos(\theta-\theta_i)},\qquad r_t'(\theta)=r_t(\theta)\tan(\theta-\theta_i)$$

then give (15.19), uniformly in t. In view of Proposition 15.1, this radial/metric path statement, not a labelled affine-side derivative bound, is the ratio-free interface used downstream. $\square$

Proposition 15.6 (Exact area identity in the normal-speed chart)**.** *For $f=\mathcal{I}_\alpha z$ one has*

$$\boxed{A''_\alpha[z,z]=\int_{\mathbb{T}}(f^2-|f'|^2)\,d\theta.} \tag{15.21}$$

Moreover, on each interval $I_i = [\theta_i, \theta_{i+1}]$,

$$2\int_{I_i} f^2\,d\theta = \frac{\alpha_i - \sin\alpha_i}{4\sin^2(\alpha_i/2)}(z_{i+1}-z_i)^2 + \frac{\alpha_i + \sin\alpha_i}{4\cos^2(\alpha_i/2)}(z_{i+1}+z_i)^2. \tag{15.22}$$

Thus the two displayed coefficients are exactly $2\int_{I_i} f^2$. After the Hessian factor λ_α/π, this is the polygonal local L^2 row converging to the L^2 part of $2q_D$. The low-mode projection and the nonlocal order-one row of q_D remain global.

Proof. Write $s = \theta - (\theta_i + \theta_{i+1})/2$ and

$$X_i = \frac{z_i + z_{i+1}}{2\cos(\alpha_i/2)}, \qquad Y_i = \frac{z_{i+1} - z_i}{2\sin(\alpha_i/2)}.$$

The sine interpolation formula is exactly $f = X_i \cos s + Y_i \sin s$. Orthogonality of the even and odd summands on $[-\alpha_i/2, \alpha_i/2]$ gives (15.22). The same calculation gives

$$\int_{I_i} (f^2 - |f'|^2)\,d\theta = 2z_i z_{i+1}\csc\alpha_i - (z_i^2 + z_{i+1}^2)\cot\alpha_i.$$

Summing in i is precisely (15.5), which proves (15.21). $\square$

Henceforth the support Hilbert form used by JP–SC, JP–FS, and NR is fixed to be

$$\mathfrak{G}_\alpha^{\mathrm{JP}} := \mathfrak{G}_\alpha^{\mathrm{ns}}. \tag{15.23}$$

Remark 15.7 (Archived stronger JP–SC route). For every fixed $\delta \in (0,1)$ there are $S_0, \rho_0, \eta_0 > 0$, independent of the number, minimum size, and ratios of the sides, such that the following holds for every fan with $S = a + \max_i \alpha_i \le S_0$. For all $y, \xi \in \mathsf{X}_\alpha$ with the entire segment $h_0 + [0,1]y$ in the physical active support chart and $\mathfrak{G}_\alpha^{\mathrm{ns}}(y)^{1/2} \le \eta_0$,

$$D_h^2\Phi_\alpha(h_\alpha\mathbf{1} + y)[\xi,\xi] \ge (2-\delta)\mathfrak{G}_\alpha^{\mathrm{ns}}(\xi). \tag{15.24}$$

The exported Hilbert constant is $c = 2 - \delta$; it may not be silently decreased before JP–FS. The proof must use the complete expression (15.11). A lower bound for its reduced-energy diagonal alone is insufficient unless the area and mixed-flux rows have first been Schur-completed.

Remark 15.8 (Why stationary O–H does not discharge JP–SC). At the base point $h_\alpha\mathbf{1}$, the definition of the fixed scale section gives $A'_\alpha[\xi] = 0$, so the exact base Hessian is

$$D_h^2\Phi_\alpha(h_\alpha\mathbf{1})[\xi,\xi] = \frac{1}{\pi}\left\{2A_\alpha\mathfrak{D}^{\mathrm{red}}_{P_\alpha^0}(\xi_i\partial_\nu u_\alpha) + \lambda_\alpha A''_\alpha[\xi,\xi]\right\}. \tag{15.25}$$

The stationary O–H comparison is neither stated at this generally nonstationary tangential polygon nor relative to the scalar support metric. Its available error has the form $\omega(S)\|V\|^2_{H^{1/2}(\mathbb{T};\mathbb{R}^2)}$. This full-vector norm is not uniformly controlled by (15.17): the vertex solution contains the tangential coefficient

$$\frac{\xi_{i+1} - \xi_i \cos \alpha_i}{\sin \alpha_i},$$

which has an inverse-angle amplification that the scale-critical fan-normal interpolation does not. A rate-free $\omega(S) \to 0$ therefore cannot be absorbed into $\mathfrak{G}_\alpha^{\mathrm{ns}}$. A proof of JP–SC needs a support-compatible, endpoint-resummed relative lower bound for the complete form (15.25), followed by control of the mixed-flux row away from the base. Invoking stationary O–H or its full-vector norm is not such a bound.

Remark 15.9 (Why a pointwise corner-amplitude estimate is insufficient). The terminal jump–cotangent cancellation in the stationary O–H ledger uses sidewise stationary flux normalization. At the generally nonstationary tangential polygon one has only the global Rellich average, not $\int_{S_i} p^2 = (\lambda/\pi)|S_i|$ for each side. If β_i denotes the normalized leading flux amplitude at a corner of exterior angle α_i, separating that corner before its descendants leaves a schematic residual

$$(\beta_i^2 - 1)\alpha_i^{-1}(z_{i+1} - z_i)^2.$$

The corresponding sharp transition in the JP–NS form costs only $O((z_{i+1} - z_i)^2(1 + |\log \alpha_i|))$. Consequently even a uniform $\beta_i = 1 + o(1)$ is not an absorbable pointwise error. The amplitude loss must be retained with all descendant annuli and LCA pairs until their positive energy is recovered. This is why (15.27) is a completed operator inequality rather than a cornerwise estimate.

The fan-normal candidate reduces JP–SC to two precise estimates. In the pure two-ray wedge model, combining the endpoint reduced energy with the area row gives exactly the local integral identity (15.22). Both coefficients are nonnegative, and the common coefficient is positive. What is not supplied by this local calculation is the ratio-free global normalization of all terminal rays, annuli, and LCA pairs.

Remark 15.10 (Archived JP–S0 sufficient estimate). Prove a modulus $\omega_0(S) \to 0$, independent of all minimum angles and adjacent ratios, such that

$$\boxed{D_h^2\Phi_\alpha(h_0)[z, z] \geq (2 - \omega_0(S))\mathfrak{G}_\alpha^{\mathrm{ns}}(z), \qquad z \in \mathsf{X}_\alpha.} \tag{15.26}$$

The complete two-ray, annular, and LCA ledger must be retained, and its amplified jump must be combined with $\lambda_0 A_\alpha''[z, z]$ before either term is estimated.

At the base scale section, (15.21) rewrites this obligation without any hidden geometric row as

(15.27)
$$\frac{1}{\pi}\left\{2A_\alpha \mathfrak{D}^{\mathrm{red}}_{P^0_\alpha}(z_i\partial_\nu u_\alpha) + \lambda_\alpha \int_{\mathbb{T}} (f^2 - |f'|^2)\right\} \geq (2-\omega_0(S))q_D(\Pi_{\mathrm{ng}} f), \qquad f = \mathcal{I}_\alpha z.$$

The same/adjacent two-ray part is (15.22); the genuinely missing part is the no-ratio nonlocal terminal-ray, annular, and LCA normalization in (15.27). None of H0, QI, DF, or DM states this polygonal reduced-DtN operator inequality.

Remark 15.11 (Archived JP–ST sufficient estimate). Along every active fixed-scale support segment in the physical chart, prove a modulus $\omega_1(\rho, S) \to 0$ such that

(15.28)
$$D_h^2\Phi_\alpha(h_0+y)[\xi,\xi] \geq \min\left\{D_h^2\Phi_\alpha(h_0)[\xi,\xi], 2\mathfrak{G}^{\mathrm{ns}}_\alpha(\xi)\right\} - \omega_1(\rho,S)\mathfrak{G}^{\mathrm{ns}}_\alpha(\xi).$$

All reduced-energy, mixed-flux, and area rows in (15.11) are included. No two-sided absolute Hessian transport is required: a large positive endpoint coefficient need only remain above the disk threshold, not be approximated to additive $\mathfrak{G}^{\mathrm{ns}}_\alpha$ accuracy.

Proposition 15.12 (Typed sufficient cut for JP–SC)**.** *JP–S0 and JP–ST, together with Proposition 15.5, imply JP–SC with* $\mathfrak{G}^{\mathrm{JP}}_\alpha = \mathfrak{G}^{\mathrm{ns}}_\alpha$ *and* $c = 2-\delta$ *after decreasing* S_0 *and* ρ_0 *so that* $\omega_0(S) + \omega_1(\rho, S) \leq \delta$.

Proof. Choose the thresholds so that $\omega_0(S) + \omega_1(\rho, S) \leq \delta$. Then (15.26) makes the minimum in (15.28) at least $(2-\omega_0(S))\mathfrak{G}^{\mathrm{ns}}_\alpha(\xi)$. Hence (15.26)–(15.28) give $D_h^2\Phi_\alpha(h_0+y)[\xi,\xi] \geq (2-\delta)\mathfrak{G}^{\mathrm{ns}}_\alpha(\xi)$. The remaining chart and activity requirements are exactly (15.18). □

Remark 15.13 (Numerical warning is not a counterexample). Finite-element prechecks on alternating fans have not found a negative exact support Hessian. The quotient relative to the older physical-radial trace deteriorates as one angle ratio decreases at fixed modest N; that calculation may also underresolve the nearly flat corner exponent and is not a certified asymptotic. On the same samples, the quotient relative to $\mathfrak{G}^{\mathrm{ns}}_\alpha$ does not deteriorate. These are design diagnostics only: (15.26) still requires a ratio-free proof for the complete endpoint ledger, and no node status follows from the precheck.

Remark 15.14 (Status of the candidate cut). JP–S0 and JP–ST are typed sufficient subcontracts; they are not additional active DAG vertices. A replacement proof using another no-ratio support metric would have to replace JP–NS and the JP–SC interface together. Conversely, a rigorous degeneration of JP–S0 would reject only this particular completion route, not the target theorem.

Let $\|\ell_\alpha\|_{(\mathfrak{G}_\alpha^{\mathrm{JP}})^*}$ be the dual norm. We first remove the part of the former JP–BR contract which is already a consequence of the closed pure-fan nodes.

Theorem 15.15 (JP–BU: exact tangential-fan baseline upper bound)**.** *There are universal $C, S_0 > 0$ such that, for every positive fan with $S = a + \max_i \alpha_i \le S_0$,*

$$\boxed{\Phi_\alpha(h_\alpha \mathbf{1}) - \Phi_{\alpha_*}(h_{\alpha_*}\mathbf{1}) \le C\mathfrak{J}_4(\alpha).} \tag{15.29}$$

The constant is independent of the smallest angle and all adjacent ratios.

Proof. The exact disk-normalized expansion furnished by the closed TR, CT core, QB, and DF nodes splits the pure tangential-fan value into its quadratic disk-source row, its complete cubic row, and a fourth-and-higher scalar tail F_4; explicitly, if Δ_2 and Δ_3 denote the complete degree-two and degree-three scalar rows, then

$$\begin{aligned}\Phi_\alpha(h_\alpha \mathbf{1}) - \Phi_{\alpha_*}(h_{\alpha_*}\mathbf{1}) = [\Delta_2(\alpha) - \Delta_2(\alpha_*)] + [\Delta_3(\alpha) - \Delta_3(\alpha_*)] \\ + [F_4(\alpha) - F_4(\alpha_*)].\end{aligned}$$

The relative source transfer gives the required one-sided quadratic Bregman estimate, while the closed pure cubic row is absolute:

$$\begin{aligned}\Delta_2(\alpha) - \Delta_2(\alpha_*) &\le C\mathfrak{J}_4(\alpha),\\ |\Delta_3(\alpha) - \Delta_3(\alpha_*)| &\le C\mathfrak{J}_4(\alpha).\end{aligned}$$

It remains only to record why the scalar tail needs no support-direction CT–M4 estimate.

Put $d_i = \alpha_i - a$ and $\gamma_i(t) = a + td_i$. The unconditional tame scalar fourth-remainder corollary of the closed CT core (obtained by applying Cauchy's coefficient estimate to the complex natural-order Hessian) gives, on the real chart and on its material complex polydisc,

$$|F_4(\zeta)| \le C\|B_\zeta\|_{H^{1/2}}^2 \|B_\zeta\|_{W^{1,\infty}}^2 \le CS^5 \le CS^4. \tag{15.30}$$

Choose a fixed sufficiently small $\varepsilon > 0$. If $\mathfrak{J}_4(\alpha) \ge \varepsilon S^4$, applying (15.30) at the two endpoints gives the desired bound with constant $2C/\varepsilon$. If $\mathfrak{J}_4(\alpha) < \varepsilon S^4$, the exact identity

$$\mathfrak{J}_4(\alpha) = 12\int_0^1 (1-t)E(t)\,dt, \qquad E(t) = \sum_i \gamma_i(t)^2 d_i^2, \tag{15.31}$$

and $\mathfrak{J}_4 = \sum_i d_i^2(\alpha_i^2 + 2a\alpha_i + 3a^2)$ show, after decreasing ε, that $a \asymp S$ and the whole segment is uniformly comparable. The material complex radius in the direction d is at least $ca/\|d\|_{\ell^\infty}$. Cauchy's estimate and (15.30) therefore give

$$|F_4''(t)| \le CS^4 \frac{\|d\|_{\ell^\infty}^2}{a^2} \le CE(t).$$

Cyclic symmetry makes $F_4'(0)$ constant in the fan coordinates, hence $F_4'(0)[d] = 0$ because $\sum_i d_i = 0$. Taylor integration and (15.31) yield

$$|F_4(\alpha) - F_4(\alpha_*)| \le C \int_0^1 (1-t)E(t)\,dt \le C\mathfrak{J}_4(\alpha).$$

Adding the three rows proves (15.29). Only the pure scalar tail was used here; none of the retired support-linear CT–M4 estimates is being reintroduced. □

The remaining forcing admits a coordinate-free finite-dimensional description. Let

$$P_\alpha^0 = P_\alpha(h_\alpha \mathbf{1}), \qquad L_i = |S_i|, \qquad m_i = \int_{S_i} (-\partial_\nu u_\alpha)^2\,ds, \tag{15.32}$$

where $|P_\alpha^0| = \pi$ and u_α is L^2-normalized.

Proposition 15.16 (Exact flux representation of the support forcing)**.** *On the fixed affine scale section, equivalently $\sum_i L_i z_i = 0$ at $h_\alpha \mathbf{1}$, one has*

$$\boxed{\ell_\alpha[z] = -\sum_i m_i z_i = -\sum_i \left(m_i - \frac{\lambda_\alpha}{\pi} L_i\right) z_i.} \tag{15.33}$$

Moreover, with

$$r_{\alpha,i} = m_i - \frac{\lambda_\alpha}{\pi} L_i, \tag{15.34}$$

one has

$$\sum_i r_{\alpha,i} = 0, \qquad \sum_i r_{\alpha,i}\nu_i = 0. \tag{15.35}$$

Thus r_α is an exact covector on the scale/translation quotient.

Proof. Hadamard's formula and the first area variation give

$$D_h\Phi_\alpha(h_\alpha \mathbf{1})[z] = \frac{1}{\pi}\left\{\lambda_\alpha \sum_i L_i z_i - \pi \sum_i m_i z_i\right\}.$$

The first term vanishes on the fixed scale section. Rellich's identity on the tangential polygon and tangentiality of its boundary give

$$h_\alpha \sum_i m_i = 2\lambda_\alpha, \qquad \sum_i L_i = \frac{2\pi}{h_\alpha}.$$

These identities prove the first equation in (15.35) and allow the centered form in (15.33). Translation invariance in Hadamard's formula gives $\sum_i m_i \nu_i = 0$, while the boundary normal-balance identity is $\sum_i L_i \nu_i = 0$. Their difference proves the vector equation in (15.35). □

We now isolate the analytic input which is actually missing from the old TR/DM route. On the side-normal cell $K_i = [\theta_i - \alpha_{i-1}/2, \theta_i + \alpha_i/2]$, put

$$B_\alpha = \Pi_{\mathrm{ng}}\{h_\alpha \sec(\theta - \theta_i) - 1\}, \qquad K_\alpha z = \Pi_{\mathrm{ng}} \mathcal{I}_\alpha z. \tag{15.36}$$

Let $\langle\cdot,\cdot\rangle_D$ be the bilinear disk Bessel form and define

$$p_\alpha[z] = 2\langle B_\alpha, K_\alpha z\rangle_D, \qquad \mathfrak{G}^{\mathrm{JP}}_\alpha(z) = q_D(K_\alpha z), \tag{15.37}$$

$$P^{\mathrm{ns}}_D(\alpha) = \inf_{z\in \mathsf{X}_\alpha} q_D(B_\alpha + K_\alpha z), \qquad \Delta_\alpha = \Phi_\alpha(h_\alpha \mathbf{1}) - \Phi_{\alpha_*}(h_{\alpha_*}\mathbf{1}). \tag{15.38}$$

Cyclic symmetry gives $p_{\alpha_*} = 0$ on the regular scale/translation quotient.

We first close the disk part of the endpoint. This is not a transfer from the old physical-radial space: the normal-cell partition admits its own exact half-strip square.

For $I_i = [\theta_i, \theta_{i+1}]$, let

$$(\mathcal{I}^0_\alpha z)(\theta_i + y) = z_i + \frac{y}{\alpha_i}(z_{i+1} - z_i), \qquad 0 \le y \le \alpha_i, \tag{15.39}$$

and let G^{e}_α be the edge-quadratic source

$$G^{\mathrm{e}}_\alpha(\theta_i + y) = \frac{1}{2}\min\{y, \alpha_i - y\}^2. \tag{15.40}$$

This is precisely $x^2/2$ on the two side cells incident to the polygonal vertex direction $\theta_i + \alpha_i/2$. Put

$$q_+(f) = \sum_{n\neq 0}(|n|+1)|\widehat{f}_n|^2, \qquad L^0_\alpha z = \frac{1}{2}\sum_i(\alpha_{i-1} + \alpha_i)z_i, \tag{15.41}$$

and

$$P^{\mathrm{ns}}_{+,0}(\alpha) = \inf_{L^0_\alpha z = 0} q_+(G^{\mathrm{e}}_\alpha + \mathcal{I}^0_\alpha z). \tag{15.42}$$

The missing constant in (15.41) is understood as the global mean minimization.

Lemma 15.17 (Exact normal-cell half-strip and variance squares)**.** *Let $\mathcal{N}_i$ be the least Dirichlet energy in the Neumann half-strip $I_i \times (0,\infty)$ with prescribed bottom trace. Then*

$$\boxed{\mathcal{N}_i(G^{\mathrm{e}}_\alpha + \mathcal{I}^0_\alpha z) = \frac{\zeta(3)}{\pi^3}\left\{\frac{\alpha_i^4}{4} + 7(z_{i+1} - z_i)^2\right\}.} \tag{15.43}$$

Moreover, after allowing an independent local mean C_i,

$$\boxed{\inf_{C_i\in\mathbb{R}}\int_{I_i}|G^{\mathrm{e}}_\alpha + \mathcal{I}^0_\alpha z - C_i|^2 = \frac{\alpha_i^5}{720} + \frac{\alpha_i}{12}(z_{i+1} - z_i)^2.} \tag{15.44}$$

Consequently

$$P^{\mathrm{ns}}_{+,0}(\alpha) - P^{\mathrm{ns}}_{+,0}(\alpha_*) \ge \frac{\zeta(3)}{8\pi^4}\mathfrak{I}_4(\alpha). \tag{15.45}$$

The estimate is pointwise in the coefficient vector before imposing the area and translation rows.

Proof. Write $w = \alpha_i$, $t = y/w$, and

$$g(t) = \frac{1}{2}\min\{t, 1-t\}^2, \qquad \rho = \frac{z_{i+1} - z_i}{w^2}.$$

Modulo a constant, the normalized bottom trace is $w^2 h(t)$ with $h(t) = g(t) + \rho t$. Its distributional second derivative is $h'' = 1 - \delta_{1/2}$. Twice integrating the cosine coefficient gives, for $k \geq 1$,

$$2\int_0^1 h(t)\cos(k\pi t)\,dt = \frac{2\{\rho((-1)^k - 1) + \cos(k\pi/2)\}}{(k\pi)^2}.$$

Thus the even modes contain only the source and the odd modes only the slope. Since

$$\sum_{k \text{ even}} k^{-3} = \frac{1}{8}\zeta(3), \qquad \sum_{k \text{ odd}} k^{-3} = \frac{7}{8}\zeta(3),$$

conformal rescaling of the half-strip gives (15.43).

The function g is symmetric about $1/2$, so it is orthogonal in variance to $t - 1/2$. Direct integration gives

$$\int_0^1 g = \frac{1}{24}, \qquad \int_0^1 g^2 = \frac{1}{320}, \qquad \operatorname{Var}(g) = \frac{1}{720}, \qquad \operatorname{Var}(t) = \frac{1}{12}.$$

This proves (15.44).

Restriction of the periodic harmonic extension yields

$$2\pi \sum_{n \neq 0} |n||\widehat{f}_n|^2 \geq \sum_i \mathcal{N}_i(f).$$

For the L^2 row, independent cell means only decrease the infimum relative to one global mean. At the regular fan with $z = 0$, reflection at every normal line makes both restrictions equalities and all local means agree. Subtracting the regular value, using (15.43)– (15.44), and applying convexity of x^5 gives

$$P^{\mathrm{ns}}_{+,0}(\alpha) - P^{\mathrm{ns}}_{+,0}(\alpha_*) \geq \frac{\zeta(3)}{8\pi^4}\left(\sum_i \alpha_i^4 - Na^4\right) + \frac{1}{1440\pi}\left(\sum_i \alpha_i^5 - Na^5\right),$$

which proves (15.45). ☐

Let $Z_\alpha = \ker L_\alpha$, where

$$L_\alpha z = \sum_i \left\{\tan\frac{\alpha_{i-1}}{2} + \tan\frac{\alpha_i}{2}\right\} z_i, \tag{15.46}$$

and define the exact-sine, edge-source value

$$P^{\mathrm{e}}_{+,\sin}(\alpha) = \inf_{z \in Z_\alpha} q_+(G^{\mathrm{e}}_\alpha + \mathcal{I}_\alpha z). \tag{15.47}$$

Translations may be left in Z_α: the sine interpolant sends them exactly to the first harmonics, so the minimizer cancels those modes. This is equivalent to applying Π_{ng} and then taking the fixed translation complement.

Lemma 15.18 (Relative linear-to-sine area graph)**.** *There is a modulus $\omega_{\sin}(S) \to 0$, independent of all minimum angles and adjacent ratios, such that*

$$\left|\{P^{\mathrm{e}}_{+,\sin}(\alpha) - P^{\mathrm{ns}}_{+,0}(\alpha)\} - \{P^{\mathrm{e}}_{+,\sin}(\alpha_*) - P^{\mathrm{ns}}_{+,0}(\alpha_*)\}\right| \le \omega_{\sin}(S)\mathfrak{J}_4(\alpha). \tag{15.48}$$

One may take $\omega_{\sin}(S) = C(S^{1/4} + S^{1/2})$.

Proof. On I_i, Taylor's formula for the two endpoint basis functions gives

$$\|\mathcal{I}_\alpha z - \mathcal{I}^0_\alpha z\|_{L^\infty(I_i)} \le C\alpha_i^2(|z_i| + |z_{i+1}|),$$
$$\|\partial_\theta(\mathcal{I}_\alpha z - \mathcal{I}^0_\alpha z)\|_{L^\infty(I_i)} \le C\alpha_i(|z_i| + |z_{i+1}|).$$

For every constant c, direct integration of a linear function gives the ratio-free nodal sampling estimate

$$\sum_i (\alpha_{i-1} + \alpha_i)|z_i - c|^2 \le C\|\mathcal{I}^0_\alpha z - c\|^2_{L^2}.$$

If $L^0_\alpha z = 0$, then $\int_{\mathbb{T}} \mathcal{I}^0_\alpha z = L^0_\alpha z = 0$, so take $c = 0$. Summing the displayed jets, applying the sampling estimate, and using $\|e\|^2_{H^{1/2}} \le C\|e\|_{L^2}\|e\|_{H^1}$ gives

$$\|\mathcal{I}_\alpha z - \mathcal{I}^0_\alpha z\|_{H^{1/2}} \le CS^{3/2}\|\mathcal{I}^0_\alpha z\|_{H^{1/2}}. \tag{15.49}$$

Indeed, the squared L^2 and H^1 errors are bounded respectively by $CS^4\|\mathcal{I}^0_\alpha z\|^2_{L^2}$ and $CS^2\|\mathcal{I}^0_\alpha z\|^2_{L^2}$.

For $z \in \ker L^0_\alpha$, put

$$C_\alpha z = z - \frac{L_\alpha z}{L_\alpha \mathbf{1}}\mathbf{1}.$$

Then $C_\alpha z \in Z_\alpha$, and the analogous formula using L^0_α is its inverse. Since

$$L_\alpha z - L^0_\alpha z = \sum_i \left(\tan\frac{\alpha_i}{2} - \frac{\alpha_i}{2}\right)(z_i + z_{i+1}),$$

the sampling estimate and $|\tan(u/2) - u/2| \le Cu^3$ show that the scalar correction in C_α has size at most $CS^2\|\mathcal{I}^0_\alpha z\|_{H^{1/2}}$. The sine trace of the constant nodal vector has uniformly bounded $H^{1/2}$ norm. Combining this fact with (15.49) yields

$$\|\mathcal{I}_\alpha C_\alpha z - \mathcal{I}^0_\alpha z\|_{H^{1/2}} \le CS^{3/2}\|\mathcal{I}^0_\alpha z\|_{H^{1/2}}, \tag{15.50}$$

with constants independent of the smallest interval and all adjacent ratios.

The source bound $q_+(G^{\mathrm{e}}_\alpha) \le C\sum_i \alpha_i^4 \le CS^3$ follows by splitting the Slobodeckij integral into same-cell, adjacent-cell, and dyadically separated-cell pieces. The elementary Hilbert-space fact

$$\left|\mathrm{dist}(b, V_1)^2 - \mathrm{dist}(b, V_0)^2\right| \le C\eta\|b\|^2$$

whenever V_1 is the graph of a bijection $J : V_0 \to V_1$ with $\|J - I\| \le \eta < 1/2$, applied to (15.50), gives

$$|R_{\sin}(\alpha)| \le CS^{9/2}, \qquad R_{\sin} = P^{\mathrm{e}}_{+,\sin} - P^{\mathrm{ns}}_{+,0}. \tag{15.51}$$

It remains to make the comparison relative to $\mathfrak{J}_4$. In a quasiuniform material band $(1-\eta_0)a \le \gamma_i \le (1+\eta_0)a$, pull every interval affinely to the real fan and complexify $\gamma(t) = \gamma + t\delta$ for $|t| < ca/\|\delta\|_{\ell^\infty}$. The pulled-back Gagliardo kernel, the L^2 row, both interpolation graphs, the area row, and the quadratic source are holomorphic. The sampling coercivity above keeps the complex Gram inverse uniformly bounded after the numerical constant c is decreased. Thus the proof of (15.51) applies to the complex bilinear Schur formula and gives $|R_{\sin}(\gamma(t))| \le Ca^{9/2}$. Cauchy's estimate and $\sum_i \gamma_i^2\delta_i^2 \ge ca^2\|\delta\|_{\ell^\infty}^2$ give

$$|D^2R_{\sin}(\gamma)[\delta,\delta]| \le Ca^{1/2}\sum_i \gamma_i^2\delta_i^2,$$

Put $d_i = \alpha_i - a$. The exact identities

$$\mathfrak{J}_4 = \sum_i d_i^2(\alpha_i^2 + 2a\alpha_i + 3a^2) = 12\int_0^1 (1-t)\sum_i (a+td_i)^2 d_i^2\,dt$$

imply $\max_i |d_i|^4 \le C\mathfrak{J}_4$. If $\mathfrak{J}_4 \ge S^{17/4}$, subtract the two bounds (15.51). Otherwise, $a \asymp S$ and $\max_i |d_i|/a \le CS^{1/16}$, so the whole radial segment is quasiuniform. Cyclic symmetry kills $DR_{\sin}(a\mathbf{1})[d]$ because $\sum_i d_i = 0$. Taylor integration and the Hessian bound give the $CS^{1/2}\mathfrak{J}_4$ estimate in the near regime; the far regime gives $CS^{1/4}\mathfrak{J}_4$. This proves (15.48). ☐

Lemma 15.19 (Normal-sine Bessel and secant transfer). *With $\widehat{P}_D^{\mathrm{ns}} = (2\pi d^2)^{-1}P_D^{\mathrm{ns}}$, there is a modulus $\omega_D(S) \to 0$ such that*

$$\left|\{\widehat{P}_D^{\mathrm{ns}}(\alpha) - P^{\mathrm{e}}_{+,\sin}(\alpha)\} - \{\widehat{P}_D^{\mathrm{ns}}(\alpha_*) - P^{\mathrm{e}}_{+,\sin}(\alpha_*)\}\right| \le \omega_D(S)\mathfrak{J}_4(\alpha). \tag{15.52}$$

Proof. Put

$$W_\alpha = \Pi_{\mathrm{ng}}\mathcal{I}_\alpha Z_\alpha, \qquad \widehat{q}_D = (2\pi d^2)^{-1}q_D.$$

Because exact support translations belong to Z_α and are carried by $\mathcal{I}_\alpha$ to the first harmonics, minimizing q_+ over the unprojected sine space is the same as minimizing it over W_α after applying Π_{ng}.

We first construct the required no-ratio quasi-interpolant. For $\psi \in H^{3/2}_{\mathrm{ng}} := \Pi_{\mathrm{ng}}H^{3/2}(\mathbb{T})$, let $\phi = P_{\le M}\psi$, $M = \lfloor S^{-1}\rfloor$, sample *phi* at the normal angles, and use the linear interpolant (15.39). Fourier truncation gives

$$\|\psi - \phi\|_{H^{1/2}} \le CS\|\psi\|_{H^{3/2}}, \qquad \|\phi''\|_{L^2} \le CS^{-1/2}\|\psi\|_{H^{3/2}}. \tag{15.53}$$

Taylor's integral formula on each interval, followed by summation, gives

$$\|\phi - \mathcal{I}^0_\alpha\phi|_{\{\theta_i\}}\|_{L^2} \le CS^2\|\phi''\|_{L^2}, \qquad \|\phi - \mathcal{I}^0_\alpha\phi|_{\{\theta_i\}}\|_{H^1} \le CS\|\phi''\|_{L^2}.$$

The constants are independent of cell ratios because each point belongs to one normal interval and the estimate is charged to that same interval. Interpolation between L^2 and H^1, the tail (15.53), the exact mean correction in the homogeneous area row, and the graph (15.50) therefore produce

$Q_\alpha\psi \in \Pi_{\mathrm{ng}}\mathcal{I}_\alpha Z_\alpha$ with

$$\|\psi - Q_\alpha\psi\|_{H^{1/2}} \le CS\|\psi\|_{H^{3/2}}. \tag{15.54}$$

No overlap or ratio constant occurs.

Write the normalized disk multiplier as

$$\widehat{q}_D(f) = q_+(f) - k(f), \qquad k(f) = \sum_{|n|\ge 2} k_{|n|}|\widehat{f}_n|^2, \qquad 0 < k_n < \frac{4}{n}.$$

Let f_α be the q_+-minimizing residual of $\Pi_{\mathrm{ng}}G^{\mathrm{e}}_\alpha + W_\alpha$. If $g \in H^{1/2}_{\mathrm{ng}}$ and $(|D|+1)\psi = g$, Galerkin orthogonality and (15.54) give

$$|\langle f_\alpha, g\rangle| = |q_+(f_\alpha, \psi - Q_\alpha\psi)| \le CS\|f_\alpha\|_{H^{1/2}}\|g\|_{H^{1/2}}.$$

Taking the dual supremum proves the negative-norm gain

$$\|f_\alpha\|_{H^{-1/2}} \le CS\|f_\alpha\|_{H^{1/2}}.$$

Since $q_+(f_\alpha) \le q_+(G^{\mathrm{e}}_\alpha) \le CS^3$, the exact Hilbert Schur identity

$$\begin{aligned} R_B(\alpha) &:= P^{\mathrm{e}}_{+,\sin}(\alpha) - P^{\mathrm{e}}_D(\alpha) \\ &= k(f_\alpha, f_\alpha) + \langle g_\alpha, (\widehat{q}_D|_{W_\alpha})^{-1} g_\alpha\rangle, \qquad g_\alpha(v) = k(f_\alpha, v), \end{aligned}$$

where $P^{\mathrm{e}}_D = \inf_{v\in W_\alpha} \widehat{q}_D(\Pi_{\mathrm{ng}}G^{\mathrm{e}}_\alpha + v)$, gives

$$0 \le R_B(\alpha) \le CS^5. \tag{15.55}$$

For the source replacement, the exact decomposition on every side-normal cell is

$$h_\alpha \sec x - 1 = (h_\alpha - 1) + G^{\mathrm{e}}_\alpha + E_\alpha, \qquad E_\alpha = h_\alpha(\sec x - 1) - \frac{x^2}{2}.$$

Here x is measured from the side normal, so $G^{\mathrm{e}}_\alpha = x^2/2$ on both incident half-cells. Taylor's formula through one normalized derivative gives

$$|E_\alpha| + w|E'_\alpha| \le CS^2w^2$$

on a side cell of total angular width w. Splitting the Gagliardo integral into same, adjacent, and dyadically separated cells yields the ratio-free estimate

$$\|E_\alpha\|_{H^{1/2}} \le CS^2\|G^{\mathrm{e}}_\alpha\|_{H^{1/2}}. \tag{15.56}$$

The constant term $h_\alpha - 1$ disappears under Π_{ng}. Coercivity of $\widehat{q}_D$ and the Hilbert distance inequality therefore give

$$|R_{\sec}(\alpha)| \le CS^5, \qquad R_{\sec} = \widehat{P}^{\mathrm{ns}}_D - P^{\mathrm{e}}_D. \tag{15.57}$$

We finally turn both absolute corrections into relative ones. In a quasiuniform material band, pull all intervals to a fixed real fan and put $h = \|\delta\|_{\ell^\infty}/a$. The first two material derivatives of q_+ are bounded by Ch^r in $H^{1/2}$, while those of k are bounded by Ch^r in $H^{-1/2}$; the latter follows from $|\Delta^r k_n| \le C_r(1+n)^{-1-r}$ and discrete summation by parts. Differentiating the Galerkin equation and the negative-norm estimate gives, for $r = 0, 1, 2$,

$$\|\partial_t^r f_t\|_{H^{1/2}} \le Ch^r a^{3/2}, \qquad \|\partial_t^r f_t\|_{H^{-1/2}} \le Ch^r a^{5/2}.$$

Twice differentiating the displayed Schur identity gives

$$|D^2R_B(\gamma)[\delta,\delta]| \le Ca\sum_i \gamma_i^2\delta_i^2.$$

For R_{sec}, affine normalization of the two source profiles gives

$$\|\partial_t^r G^{\mathrm{e}}_{\gamma(t)}\|_{H^{1/2}} \le Ch^r a^{3/2}, \qquad \|\partial_t^r E_{\gamma(t)}\|_{H^{1/2}} \le Ch^r a^{7/2}, \qquad r=0,1,2.$$

Differentiating the two coercive Schur equations in (15.57) consequently gives the same Hessian bound for R_{sec}. Hence their sum $R_D = \widehat{P}^{\mathrm{ns}}_D - P^{\mathrm{e}}_{+,\sin}$ satisfies

$$|D^2R_D(\gamma)[\delta,\delta]| \le Ca\sum_i \gamma_i^2\delta_i^2. \tag{15.58}$$

If $\mathfrak{J}_4 \ge S^{9/2}$, subtract the endpoint bounds (15.55) and (15.57); this gives $CS^{1/2}\mathfrak{J}_4$. Otherwise the exact fourth-defect identity used in the preceding lemma makes the radial segment quasiuniform. Cyclic symmetry removes the first derivative of R_D at the regular fan, and Taylor's formula with (15.58) gives $CS\mathfrak{J}_4$. This proves (15.52) with $\omega_D(S) = C(S^{1/2}+S)$. □

Theorem 15.20 (Closed fan-normal disk endpoint and principal forcing)**.** *There are universal $S_D, \kappa, C_p > 0$ such that every positive fan with $S \le S_D$ satisfies*

$$P^{\mathrm{ns}}_D(\alpha) - P^{\mathrm{ns}}_D(\alpha_*) \ge 2\kappa\mathfrak{J}_4(\alpha), \tag{15.59}$$

$$\|p_\alpha\|_{(\mathfrak{G}^{\mathrm{JP}}_\alpha)^*} \le C_p\mathfrak{J}_4(\alpha)^{1/2}. \tag{15.60}$$

Both constants are independent of all minimum angles and adjacent ratios.

Proof. Combine (15.45), (15.48), and (15.52), then decrease S_D so that the two moduli consume at most half of the homogeneous margin. Multiplication by $2\pi d^2$ gives (15.59) for a universal $\kappa > 0$.

The closed exact source envelope from F129 gives

$$q_D(B_\alpha) - q_D(B_{\alpha_*}) \le C\mathfrak{J}_4(\alpha).$$

Hilbert completion and $p_{\alpha_*} = 0$ give

$$\frac{1}{4}\|p_\alpha\|^2_{(\mathfrak{G}^{\mathrm{JP}}_\alpha)^*} = \{q_D(B_\alpha) - q_D(B_{\alpha_*})\} - \{P^{\mathrm{ns}}_D(\alpha) - P^{\mathrm{ns}}_D(\alpha_*)\}.$$

Drop the favorable second brace and take square roots to obtain (15.60). □

Theorem 15.21 (Closed exact fan-normal value matching)**.** *There is a modulus $\varepsilon_V(S) \to 0$, independent of all minimum angles and adjacent ratios, such that every sufficiently fine positive fan satisfies*

$$\boxed{|\Delta_\alpha - \{q_D(B_\alpha) - q_D(B_{\alpha_*})\}| \le \varepsilon_V(S)\mathfrak{J}_4(\alpha).} \tag{15.61}$$

Proof. Let $W_\alpha = B_\alpha e_r$ be the radial physical boundary field of the area-normalized tangential polygon; its normal scalar trace at the disk is the source B_α in (15.36). The exact disk-normalized scalar expansion used in

JP–BU may be retained at its natural orders rather than weakened to a one-sided bound. It is

$$\Delta_\alpha = \{q_D(B_\alpha) - q_D(B_{\alpha_*})\} + \{\Delta_3(\alpha) - \Delta_3(\alpha_*)\} + \{F_4(\alpha) - F_4(\alpha_*)\}, \tag{15.62}$$

where Δ_3 is the complete scalar cubic row, including its factor $1/6$, and F_4 is the exact fourth-and-higher scalar remainder.

The closed CT cubic identity splits Δ_3 into its principal null form and the complete normal/geometric order-one remainder. CT's relative closure theorem controls the latter by $o(\mathfrak{J}_4)$. The QB secant-to-quadratic transfer and the closed DF theorem control the principal null form by the same scale. Consequently there is a modulus $\varepsilon_3(S) \to 0$ with

$$|\Delta_3(\alpha) - \Delta_3(\alpha_*)| \le \varepsilon_3(S)\mathfrak{J}_4(\alpha). \tag{15.63}$$

This is the pure scalar row, so no support-metric comparison is involved.

It remains to sharpen the tail estimate already used in JP–BU. On the real chart and its material complex disk, the tame scalar remainder gives

$$|F_4(\zeta)| \le C\|W_\zeta\|^2_{H^{1/2}}\|W_\zeta\|^2_{W^{1,\infty}} \le CS^5. \tag{15.64}$$

Put $d_i = \alpha_i - a$ and $\gamma_i(t) = a + td_i$. If $\mathfrak{J}_4 \ge S^{9/2}$, applying (15.64) at the two endpoints gives

$$|F_4(\alpha) - F_4(\alpha_*)| \le CS^{1/2}\mathfrak{J}_4(\alpha).$$

If $\mathfrak{J}_4 < S^{9/2}$, the exact fourth-defect identity implies $a \asymp S$ and keeps the whole radial segment quasiuniform. The material complex radius in direction d is at least $ca/\|d\|_{\ell^\infty}$. Cauchy's estimate and (15.64) give, with $E(t) = \sum_i \gamma_i(t)^2 d_i^2$,

$$|F_4''(t)| \le CS^5 \frac{\|d\|^2_{\ell^\infty}}{a^2} \le CSE(t).$$

Cyclic symmetry makes $F_4'(0)$ constant in the fan coordinates and hence $F_4'(0)[d] = 0$. Taylor integration and $\mathfrak{J}_4 = 12\int_0^1 (1-t)E(t)\,dt$ give the near estimate $CS\mathfrak{J}_4$. Together with (15.63) and (15.62), this proves (15.61) with $\varepsilon_V(S) = \varepsilon_3(S) + C(S^{1/2} + S)$. □

Proposition 15.22 (Exact forcing-residual covector)**.** *On the polar side cell K_i, put*

$$f_\alpha(\theta) = h_\alpha \sec(\theta - \theta_i) - 1$$

and let $Z_\alpha(z) = v_\alpha(z)e_r + w_\alpha(z)e_\theta$ be the continuous physical-affine support displacement. In the notation of the complete CT tensor, let $\kappa_{\mathbb{D}}\mathcal{C} + \mathcal{R}_{\mathrm{full}}$ be the normal third-derivative split and set

$$\Psi_{4,\alpha}[z] := D\mathcal{R}_4(f_\alpha e_r)[Z_\alpha(z)].$$

Then the exact residual $e_\alpha = \ell_\alpha - p_\alpha$ *is*

$$\boxed{\begin{aligned} e_\alpha[z] &= 2\langle f_\alpha, v_\alpha(z) - \mathcal{I}_\alpha z\rangle_D \\ &\quad + \frac{\kappa_{\mathbb{D}}}{2}\mathcal{C}(f_\alpha, f_\alpha, v_\alpha(z)) + \frac{1}{2}\mathcal{R}_{\mathrm{full}}(f_\alpha, f_\alpha, v_\alpha(z)) \\ &\quad - 2\langle f_\alpha, w_\alpha(z) f'_\alpha\rangle_D + \Psi_{4,\alpha}[z]. \end{aligned}} \tag{15.65}$$

Every term is evaluated after the exact area/translation projection. The formula is an identity of covectors on X_α*; in particular, it contains no norm comparison and no asymptotic remainder notation.*

Proof. The boundary of the tangential polygon is the disk boundary displaced by $W_\alpha = f_\alpha e_r$. Because vertices and side parametrizations are affine in the support numbers, differentiation in the support direction is the single physical slot $Z_\alpha(z)$. The exact scalar Taylor identity therefore gives

$$\ell_\alpha[z] = D^2\mathcal{L}(0)[W_\alpha, Z_\alpha(z)] + \frac{1}{2}D^3\mathcal{L}(0)[W_\alpha, W_\alpha, Z_\alpha(z)] + \Psi_{4,\alpha}[z].$$

The disk Hessian is $D^2\mathcal{L}(0)[W_\alpha, Z_\alpha(z)] = 2\langle f_\alpha, v_\alpha(z)\rangle_D$. The complete physical-coordinate CT identity gives

$$\begin{aligned} \frac{1}{2}D^3\mathcal{L}(0)[W_\alpha, W_\alpha, Z_\alpha(z)] &= \frac{\kappa_{\mathbb{D}}}{2}\mathcal{C}(f_\alpha, f_\alpha, v_\alpha(z)) \\ &\qquad + \frac{1}{2}\mathcal{R}_{\mathrm{full}}(f_\alpha, f_\alpha, v_\alpha(z)) - 2\langle f_\alpha, w_\alpha(z) f'_\alpha\rangle_D. \end{aligned}$$

The last coefficient is the curvature multiplicity after differentiating the cubic Taylor coefficient: the raw third derivative contributes $-4\langle f_\alpha, w_\alpha(z) f'_\alpha\rangle_D$ and is multiplied by $1/2$. Finally $p_\alpha[z] = 2\langle f_\alpha, \mathcal{I}_\alpha z\rangle_D$ because the disk form annihilates the constant and first harmonics. Subtraction proves (15.65). ☐

Remark 15.23 (Retraction of the period-three material forcing transfer). The former proposition which identified the three-cell material constant R_3 with the genuine polygonal covector $e_\alpha = \ell_\alpha - p_\alpha$ is withdrawn. The revision-22 Schwarz–Christoffel audit finds a same-order corner finite part already in the support diagonal, so the material-to-polygon transfer used below does not justify the mixed forcing coefficient either. The displayed material series remains a useful diagnostic, but it proves neither truth nor falsity of the former JE–FM estimate

$$\|e_\alpha\|_{(\mathfrak{G}^{\mathrm{ns}}_\alpha)^*} \le \varepsilon(S)\mathfrak{J}_4(\alpha)^{1/2}, \qquad \varepsilon(S) \longrightarrow 0. \tag{15.66}$$

JE–FM is absent from the active DAG, and no active implication uses this archived calculation.

The closed disk Schur, forcing, and value rows are retained as JP–DE. The unproved separated flux interface is absent from the active DAG; this does not remove the fan-normal metric. The replacement is the fixed-loss joint scalar comparison in `JC_joint_comparison.tex`.

Remark 15.24 (Retraction of the period-three joint-loss transfer). The former period-three joint-loss proposition is also withdrawn. Its coefficient E_3 is the material cell value, while the true polygonal support finite part is $P(1/3)/2 = 9\log 3/(4\pi)$. Moreover the material constant R_3 has not been transferred through the mixed polygonal corner layer. Consequently the limit formerly labelled "joint-period3-loss" is not a theorem about the genuine polygonal residual. The active JC–PL formulation already uses a fixed absorbable loss and does not depend on this archived diagnostic.

Remark 15.25 (Why the closed TR/DM metric cannot be transferred). TR completes its square in $\widetilde{G}_\alpha(z) = q_D(\Pi_{\mathrm{ng}}\widetilde{T}_\alpha z)$, not in $\mathfrak{G}_\alpha^{\mathrm{JP}}$. These metrics have no no-ratio comparison in the needed direction. Indeed, take adjacent angles $\varepsilon \ll a$ and support data $z_0 = 0$, $z_1 = z_2 = 1$, return to zero across ordinary cells, and remove the scale and translation modes. The sine trace has only one sharp transition, so

$$\mathfrak{G}_\alpha^{\mathrm{JP}}(z) \le C\{1 + \log(a/\varepsilon)\}.$$

On the adjacent physical cell, however, $a_1^- = (\sin\varepsilon)^{-1} - \tan(\varepsilon/2) \asymp \varepsilon^{-1}$. On a subinterval of length comparable to a this gives $|\widetilde{T}_\alpha z| \ge ca/\varepsilon$. Removing three low modes does not remove this localized mass, and hence

$$\widetilde{G}_\alpha(z) \ge c\frac{a^3}{\varepsilon^2}, \qquad \frac{\widetilde{G}_\alpha(z)}{\mathfrak{G}_\alpha^{\mathrm{JP}}(z)} \longrightarrow \infty.$$

Thus the closed DM estimate in the old physical-radial metric cannot imply the fan-normal dual estimate (15.66). This is a strict interface mismatch, not a missing citation.

Remark 15.26 (Diagnostic check of the new Schur sign). The reproducible script `tests/check_jp_joint_endpoint.py` samples the exact finite-dimensional Schur complement in (15.38) on the area/translation quotient. For regular-nearby, alternating-tiny, one-tiny-angle, and fixed-seed random fans with $N = 8, 12, 16, 24$, the sampled ratio

$$\frac{P_D^{\mathrm{ns}}(\alpha) - P_D^{\mathrm{ns}}(\alpha_*)}{\mathfrak{J}_4(\alpha)}$$

lies between 0.0189 and 0.0333; the sampled metric remains positive on the quotient. This independently checks the sign and normalization used in Theorem 15.20; the script itself is not a proof and its finite resolution does not test the mixed polygonal corner finite part isolated in the JC node.

Proposition 15.27 (Archived conditional JP–FS algebra)**.** *Assume JP–SC with $c = 2 - \delta$, $0 < \delta \le 1/2$, and JP–JE. After decreasing the common fan and chart thresholds,*

$$\boxed{\Delta_\alpha - \frac{1}{2c}\|\ell_\alpha\|^2_{(\mathfrak{G}_\alpha^{\mathrm{JP}})^*} \ge \kappa\mathfrak{J}_4(\alpha).} \tag{15.67}$$

Equivalently, with $\mathcal{Q}_\alpha = c\mathfrak{G}_\alpha^{\mathrm{JP}}$, the dual term is $\frac{1}{2}\sup_{z\ne 0}\frac{(\sum_i r_{\alpha,i}z_i)^2}{\mathcal{Q}_\alpha(z)}$.

Proof. Write $\|\cdot\|_* = \|\cdot\|_{(\mathfrak{G}_\alpha^{\mathrm{JP}})^*}$ and $e_\alpha = \ell_\alpha - p_\alpha$, and enlarge the two moduli to the common one

$$\varepsilon(S) := \max\{\varepsilon_V(S), \varepsilon_{\mathrm{JE}}(S)\} \longrightarrow 0.$$

Hilbert completion and $p_{\alpha_*} = 0$ give

$$P_D^{\mathrm{ns}}(\alpha) - P_D^{\mathrm{ns}}(\alpha_*) = q_D(B_\alpha) - q_D(B_{\alpha_*}) - \frac{1}{4}\|p_\alpha\|_*^2.$$

For $c = 2 - \delta$ and $\delta \le 1/2$,

$$\frac{1}{2c}\|\ell_\alpha\|_*^2 - \frac{1}{4}\|p_\alpha\|_*^2 \le \left\{\frac{\delta C_p^2}{6} + \frac{2C_p\varepsilon(S) + \varepsilon(S)^2}{3}\right\}\mathfrak{J}_4(\alpha).$$

Indeed, $1/(2c) - 1/4 \le \delta/6$, $1/(2c) \le 1/3$, and then use (15.60)–(15.66) and Cauchy–Schwarz. Combining this with (15.59) and (15.61) yields

$$\Delta_\alpha - \frac{1}{2c}\|\ell_\alpha\|_*^2 \ge \Big[2\kappa - \varepsilon(S) - \frac{\delta C_p^2}{6} - \frac{2C_p\varepsilon(S) + \varepsilon(S)^2}{3}\Big]\mathfrak{J}_4(\alpha).$$

Choose δ and then S_0, ρ_0 so that the total subtracted bracket is at most κ. This proves (15.67). Finally (15.33) gives the displayed supremum identity. □

Proposition 15.28 (Archived exact reduction of JP–BR)**.** *JP–BU and JP–FS give the former JP–BR interface, namely* (15.29) *and*

$$\Phi_\alpha(h_\alpha\mathbf{1}) - \Phi_{\alpha_*}(h_{\alpha_*}\mathbf{1}) - \frac{1}{2c}\|\ell_\alpha\|_{(\mathfrak{G}_\alpha^{\mathrm{JP}})^*}^2 \ge c_0\mathfrak{J}_4(\alpha). \tag{15.68}$$

Proof. Equation (15.33) and $\mathcal{Q}_\alpha = c\mathfrak{G}_\alpha^{\mathrm{JP}}$ give the exact identity

$$\frac{1}{2c}\|\ell_\alpha\|_{(\mathfrak{G}_\alpha^{\mathrm{JP}})^*}^2 = \frac{1}{2}\sup_{z\ne 0}\frac{(\sum_i r_{\alpha,i}z_i)^2}{\mathcal{Q}_\alpha(z)}.$$

Now apply JP–BU and JP–FS. □

Remark 15.29 (Why the closed DM row did not save JE–FM). In disk coordinates the first nonzero part of the exact covector is

$$2q_D(f_\alpha, v_\alpha - w_\alpha f_\alpha').$$

DM controls the null-form contribution in the v_α slot, but not the curvature placement $-2q_D(f_\alpha, w_\alpha f_\alpha')$. The certified three-periodic CT family makes this curvature placement comparable to a, while $\mathfrak{J}_4(\alpha)^{1/2}\mathfrak{G}_\alpha^{\mathrm{JP}}(z)^{1/2}$ is also comparable to a. Thus the archived material model does not behave as an $o(1)$ perturbation of the disk Schur margin. The old stationary O–H determinant cannot be inserted at the nonstationary tangential polygon either. After the corner-transfer retraction in Remark 15.23, this is only a diagnostic, not a proof that the genuine separate covector estimate is false. The fixed-loss scalar JP–JC, rather than an unproved citation to TR or DM, is the active replacement.

Theorem 15.30 (Archived separated completion)**.** *Assume JP–SC and JP–JE. Let $\mathcal{R}_\alpha$ be the Riesz map of $\mathfrak{G}_\alpha^{\mathrm{JP}}$ and set*

$$z_\alpha^\sharp = -c^{-1}\mathcal{R}_\alpha^{-1}\ell_\alpha. \tag{15.69}$$

Then, for every normalized $P_\alpha(h_0 + z) \in \mathcal{U}_{S_0,\rho_0}$,

$$\mathfrak{G}_\alpha^{\mathrm{JP}}(z_\alpha^\sharp) \le C\mathfrak{J}_4(\alpha), \tag{15.70}$$

$$\Phi_\alpha(h_\alpha\mathbf{1} + z) - \Phi_{\alpha_*}(h_{\alpha_*}\mathbf{1}) \ge c_0\mathfrak{J}_4(\alpha) + \frac{c}{2}\mathfrak{G}_\alpha^{\mathrm{JP}}(z - z_\alpha^\sharp). \tag{15.71}$$

Thus JP–SC plus JP–JE implies the active-chart coercive node directly; there is no support-direction CT–M4 or post-cubic remainder.

Proof. Proposition 15.27 supplies JP–FS from the two hypotheses; together with the closed JP–BU theorem, Proposition 15.28 supplies the baseline Schur margin. By JP–SC and (15.15),

$$\Phi_\alpha(h_\alpha\mathbf{1} + z) - \Phi_\alpha(h_\alpha\mathbf{1}) \ge \ell_\alpha[z] + \frac{c}{2}\mathfrak{G}_\alpha^{\mathrm{JP}}(z).$$

Hilbert completion gives

$$\ell_\alpha[z] + \frac{c}{2}\mathfrak{G}_\alpha^{\mathrm{JP}}(z) = \frac{c}{2}\mathfrak{G}_\alpha^{\mathrm{JP}}(z - z_\alpha^\sharp) - \frac{1}{2c}\|\ell_\alpha\|^2_{(\mathfrak{G}_\alpha^{\mathrm{JP}})^*}.$$

Proposition 15.28 supplies (15.68); add it to obtain (15.71). Equations (15.29) and (15.68) imply $\|\ell_\alpha\|^2_{(\mathfrak{G}_\alpha^{\mathrm{JP}})^*} \le C\mathfrak{J}_4(\alpha)$; use (15.69) to obtain (15.70). $\square$

15.5. **Audit disposition.**

item	disposition
fixed-normal vertex/support path	exact affine identity (15.2);
genuine side-normal velocity	constant z_i, equation (15.3);
JP–EH	closed by Theorem 15.2;
JP–NS	closed by Proposition 15.5; the downstream metric is fixed by (15.23);
raw q_D norm of the piecewise-constant true normal	rejected as ill-typed because vertex jumps have logarithmically divergent $H^{1/2}$ energy;
disk cubic candidate	rigorously noncoercive on the alternating family of Proposition 15.3;
JP–SC	archived stronger route; estimates (15.26) and (15.28) are not active premises;
JP–BU	closed by Theorem 15.15;
exact support forcing	centered side-flux covector (15.33)–(15.35);

$D0_{\mathrm{ns}}$ and JE–P	closed by Theorem 15.20;
JE–V	closed by Theorem 15.21;
JE–FM	unavailable; its former period-three material falsification is retracted by Remark 15.23, and the row is absent from the active DAG;
vanishing joint comparison	former material counterexample retracted by Remark 15.24; the active route uses a fixed absorbable loss in JP–JC and does not assume this statement;
JP–FS and JP–BR	archived conditional algebra; unnecessary on the JP–JC route;
support post-cubic and support-linear fourth tails	removed from the DAG by the exact value identity (15.15).

16. Node JC: honest joint endpoint comparison

Audited source: `JC_joint_comparison.tex`.

16.1. **Two honest scalars on one fixed fan.** Fix a positive normal fan α, let $h_\alpha \mathbf{1}$ be its area-π tangential support vector, and work on the fixed affine scale/translation section X_α of the JP node. Put

$$\Phi_\alpha(h) = \frac{|P_\alpha(h)|}{\pi}\lambda_1(P_\alpha(h)). \tag{16.1}$$

The closed JP–NS and disk-endpoint rows define

$$B_\alpha = \Pi_{\mathrm{ng}}\{h_\alpha \sec(\theta-\theta_i) - 1\}, \qquad K_\alpha z = \Pi_{\mathrm{ng}}\mathcal{I}_\alpha z, \tag{16.2}$$

and

$$\mathfrak{G}_\alpha^{\mathrm{JP}}(z) = q_D(K_\alpha z), \qquad p_\alpha[z] = 2\langle B_\alpha, K_\alpha z\rangle_D. \tag{16.3}$$

Let $\mathcal{R}_\alpha$ be the Riesz map of $\mathfrak{G}_\alpha^{\mathrm{JP}}$ and set

$$z_\alpha^D = -\frac{1}{2}\mathcal{R}_\alpha^{-1}p_\alpha. \tag{16.4}$$

Proposition 16.1 (Exact disk completion)**.** *On X_α one has*

$$q_D(B_\alpha + K_\alpha z) = P_D^{\mathrm{ns}}(\alpha) + \mathfrak{G}_\alpha^{\mathrm{JP}}(z - z_\alpha^D), \tag{16.5}$$

$$P_D^{\mathrm{ns}}(\alpha) = q_D(B_\alpha) - \frac{1}{4}\|p_\alpha\|^2_{(\mathfrak{G}_\alpha^{\mathrm{JP}})^*}, \tag{16.6}$$

$$\mathfrak{G}_\alpha^{\mathrm{JP}}(z_\alpha^D) = \frac{1}{4}\|p_\alpha\|^2_{(\mathfrak{G}_\alpha^{\mathrm{JP}})^*}. \tag{16.7}$$

At the regular fan α_*, cyclic symmetry gives $p_{\alpha_*} = 0$ and $z_{\alpha_*}^D = 0$.

Proof. Expand the fixed quadratic form:

$$q_D(B_\alpha + K_\alpha z) = q_D(B_\alpha) + p_\alpha[z] + \mathfrak{G}_\alpha^{\mathrm{JP}}(z).$$

The definition (16.4) is exactly $2\mathcal{R}_\alpha z_\alpha^D = -p_\alpha$. Completing the Hilbert square proves all three identities. The regular assertion follows because the regular source is cyclic while every vector in the quotient is orthogonal to the cyclic mode. ☐

Define the fixed-fan joint residual

$$\boxed{\mathcal{R}_\alpha(z) := \Phi_\alpha(h_\alpha \mathbf{1} + z) - \Phi_\alpha(h_\alpha \mathbf{1}) - \{q_D(B_\alpha + K_\alpha z) - q_D(B_\alpha)\}.} \tag{16.8}$$

Both terms in (16.8) are genuine scalar functions of z. The disk term is the fixed Fourier–Bessel quadratic form; it is not the eigenvalue of a moving disk domain.

Proposition 16.2 (Exact integrated form)**.** *Let $\ell_\alpha = D_h\Phi_\alpha(h_\alpha \mathbf{1})$ and $e_\alpha = \ell_\alpha - p_\alpha$. Whenever the fixed-normal support segment is active,*

$$\mathcal{R}_\alpha(z) = e_\alpha[z] + \int_0^1 (1-t)\{D_h^2\Phi_\alpha(h_\alpha \mathbf{1} + tz)[z,z] - 2\mathfrak{G}_\alpha^{\mathrm{JP}}(z)\}\,dt. \tag{16.9}$$

Proof. Apply the fundamental theorem of calculus twice to the polygonal scalar. For the disk scalar use $q_D(B + Kz) - q_D(B) = p[z] + \mathfrak{G}^{\mathrm{JP}}(z)$. Subtraction gives (16.9) exactly. ☐

16.2. **Exact Schur reduction and support-time cancellation.** Let $g_\alpha(\cdot,\cdot)$ be the bilinear form associated with $\mathfrak{G}_\alpha^{\mathrm{JP}}$, put $d_\alpha = z_\alpha^D$, and abbreviate

$$H_{\alpha,y} = D_h^2\Phi_\alpha(h_\alpha \mathbf{1} + y). \tag{16.10}$$

Equation (16.4) is equivalently

$$p_\alpha[\xi] = -2g_\alpha(d_\alpha, \xi). \tag{16.11}$$

Define the completed covector

$$b_\alpha[\xi] := \ell_\alpha[\xi] + g_\alpha(d_\alpha, \xi) = \ell_\alpha[\xi] - \frac{1}{2}p_\alpha[\xi] \tag{16.12}$$

and the pathwise Schur action

(16.13)

$$\mathcal{A}_\alpha(z) := b_\alpha[z] + \int_0^1 (1-t)\{H_{\alpha,tz}[z,z] - g_\alpha(z,z)\}\,dt + \frac{1}{2}g_\alpha(d_\alpha, d_\alpha) + \frac{\kappa}{2}\mathfrak{J}_4(\alpha).$$

Proposition 16.3 (JC–ALG: exact completed-action equivalence)**.** *For every active fixed-fan segment,*

$$\boxed{\mathcal{A}_\alpha(z) = \mathcal{R}_\alpha(z) + \frac{1}{2}\mathfrak{G}_\alpha^{\mathrm{JP}}(z - z_\alpha^D) + \frac{\kappa}{2}\mathfrak{J}_4(\alpha).} \tag{16.14}$$

Consequently the former JP–JC inequality is equivalent, without a loss of constants, to $\mathcal{A}_\alpha(z) \geq 0$.

Proof. Insert the integrated identity (16.9) and expand

$$\frac{1}{2}g_\alpha(z-d_\alpha, z-d_\alpha) = \frac{1}{2}g_\alpha(z,z) - g_\alpha(z,d_\alpha) + \frac{1}{2}g_\alpha(d_\alpha,d_\alpha).$$

By (16.11), $-g_\alpha(z,d_\alpha) = p_\alpha[z]/2$. Hence the linear row is $(\ell_\alpha - p_\alpha)[z] + p_\alpha[z]/2 = b_\alpha[z]$. Moreover

$$\int_0^1 (1-t)(-2g_\alpha(z,z))\,dt + \frac{1}{2}g_\alpha(z,z) = -\frac{1}{2}g_\alpha(z,z),$$

which is exactly the integral of $-g_\alpha(z,z)$ with weight $1-t$. The constant rows agree with (16.13), proving (16.14). □

The same identity has the following equivalent form, which displays the positive disk corrector square before any estimate:
(16.15)

$$\boxed{\mathcal{A}_\alpha(z) = \frac{1}{2}\mathfrak{G}_\alpha^{\mathrm{JP}}(z-d_\alpha) + e_\alpha[z] + \int_0^1 (1-t)\{H_{\alpha,tz}[z,z] - 2\mathfrak{G}_\alpha^{\mathrm{JP}}(z)\}\,dt + \frac{\kappa}{2}\mathfrak{J}_4(\alpha).}$$

Indeed, $b_\alpha = e_\alpha - g_\alpha(d_\alpha,\cdot)$ and $\int_0^1 (1-t)g_\alpha(z,z)\,dt = \mathfrak{G}_\alpha^{\mathrm{JP}}(z)/2$. Thus the period-three component of e_α is not to be bounded as an isolated covector: it must be paid jointly by the jump surplus in $H_{\alpha,tz} - 2\mathfrak{G}_\alpha^{\mathrm{JP}}$ and the fan row.

Lemma 16.4 (JC–TL: support-time logarithm cancellation)**.** *Let* $\beta > 0$, $0 < r \le 1$, *and*

$$I_\beta(r) := \int_0^1 (1-t)\left\{1 - ((1-t)+tr)^{2\beta}\right\}dt. \tag{16.16}$$

Then

$$0 \le I_\beta(r) \le \frac{\beta}{3}\log\frac{1}{r}, \tag{16.17}$$

$$I_\beta(r) \le \frac{\beta}{2(1+\beta)} \le \frac{\beta}{2}. \tag{16.18}$$

In particular,

$$\boxed{I_\beta(r) \le \frac{\beta}{3}\min\left\{\log\frac{1}{r}, \frac{3}{2}\right\}.} \tag{16.19}$$

All constants are independent of r.

Proof. For $0 < x \le 1$, $1 - x^{2\beta} \le 2\beta(-\log x)$. Weighted arithmetic–geometric mean gives $(1-t)+tr \ge r^t$ and hence

$$-\log((1-t)+tr) \le t\log(1/r).$$

Since $\int_0^1 t(1-t)\,dt = 1/6$, these two inequalities prove (16.17). Also $(1-t)+tr \ge 1-t$, so monotonicity of the power gives

$$I_\beta(r) \le I_\beta(0) = \frac{1}{2} - \frac{1}{2+2\beta} = \frac{\beta}{2(1+\beta)}.$$

This proves (16.18); taking the smaller bound gives (16.19). □

Lemma 16.5 (JC–TL$^{\min}$: switches of the shortest incident side)**.** *Let L_- and L_+ be positive affine functions on $[0,1]$, put $m(t) := \min\{L_-(t), L_+(t)\}$, and set $r := m(1)/m(0)$. Then, for every $\beta > 0$,*

$$\int_0^1 (1-t)\left\{1 - \left(\frac{m(t)}{m(0)}\right)^{2\beta}\right\} dt \le \begin{cases} I_\beta(r), & 0 < r \le 1, \\ 0, & r \ge 1. \end{cases} \tag{16.20}$$

Consequently every upper *bound in Lemma 16.4 remains valid for the positive part of the loss when a single affine length is replaced by the shortest of the two incident physical lengths. No switch-point boundary term is present.*

Proof. The pointwise minimum of affine functions is concave. If $r \le 1$, concavity gives

$$\frac{m(t)}{m(0)} \ge (1-t) + tr.$$

The function $x \mapsto 1 - x^{2\beta}$ is decreasing, so integration with the nonnegative weight $1-t$ proves the first line of (16.20). If $r \ge 1$, the chord joining the endpoint values lies above $m(0)$, and concavity gives $m(t) \ge m(0)$ throughout; the integrand is therefore nonpositive. The argument differentiates neither m nor $m^{2\beta}$, hence a change of the minimizing affine side creates no distributional boundary contribution. □

Remark 16.6 (Exact terminal normalization). For an actual physical side length with the affine form $L(t) = L_0\{(1-t)+tr\}$, the squared corner-flux power uses

$$\beta_i^{\mathrm{ph}} = \frac{\alpha_i}{\pi - \alpha_i}, \tag{16.21}$$

not α_i/π. Put $\Delta_i = a_i^- - a_i^+ = 2\sin(\alpha_i/2)\tau_i$. The integrated terminal amplitude loss obeys

(16.22)
$$\alpha_i \tau_i^2 I_{\beta_i^{\mathrm{ph}}}(r) \le |\Delta_i|^2 \min\left\{\frac{\alpha_i^2}{12(\pi-\alpha_i)\sin^2(\alpha_i/2)}\log\frac{1}{r}, \frac{\alpha_i^2}{8\pi\sin^2(\alpha_i/2)}\right\}.$$

Thus the logarithmic coefficient is $1/(3\pi) + O(\alpha_i)$, while the exact absolute cap is

$$\alpha_i \tau_i^2 I_{\beta_i^{\mathrm{ph}}}(r) \le c_T(S)|\Delta_i|^2, \qquad c_T(S) := \sup_{0<\alpha\le S} \frac{\alpha^2}{8\pi\sin^2(\alpha/2)}, \tag{16.23}$$

and $c_T(S) \to 1/(2\pi)$ as $S \downarrow 0$. This is the sharp support-time cap for a terminal length tending to zero. A growing terminal length has the favorable sign and is separated before the absolute estimate. The exponent α_i/π belongs instead to a conformal-prevertex power. No retained node proves that a conformal prevertex interval is affine in support time, so that substitution is not allowed. If the terminal cutoff is the smaller of two incident physical lengths, Lemma 16.5 applies globally: concavity of the minimum dominates the affine endpoint chord, so neither subdivision nor a switch correction is needed. (Restarting the Taylor weight after a subdivision would in fact change the ledger and is not the argument used here.) The right side is the descendant-annulus/terminal-cap jump charge on the same physical scale. Thus taking a side-ratio supremum before the support-time integral destroys a real cancellation and is inadmissible.

Formula (16.9) is the reason the two old obligations must not be estimated independently. In the three-periodic obstruction e_α has a non-vanishing normalized leading term; the surplus in the integrated Hessian is available in (16.9) to complete the corresponding Schur square.

Proposition 16.7 (JC–RS: exact quadratic root certificate)**.** *Put*

$$Q_\alpha[\xi,\xi] := H_{\alpha,0}[\xi,\xi] - g_\alpha(\xi,\xi). \tag{16.24}$$

Define the frozen quadratic root model

$$\mathcal{A}^{(2)}_\alpha(\xi) := \frac{1}{2}g_\alpha(d_\alpha,d_\alpha) + \frac{\kappa}{2}\mathfrak{J}_4(\alpha) + b_\alpha[\xi] + \frac{1}{2}Q_\alpha[\xi,\xi]. \tag{16.25}$$

Then $\mathcal{A}^{(2)}_\alpha(\xi) \ge 0$ for every ξ if and only if

$$\boxed{Q_\alpha \ge 0, \qquad \|b_\alpha\|^2_{Q^*_\alpha} \le g_\alpha(d_\alpha,d_\alpha) + \kappa\mathfrak{J}_4(\alpha).} \tag{16.26}$$

For a semidefinite Q_α, the second statement includes that b_α annihilates its kernel and uses the quotient dual norm.

Proof. This is Hilbert completion of the quadratic form (16.25). If it is nonnegative on the whole vector space, its quadratic part is semidefinite and its linear part annihilates the kernel. Its infimum is

$$\frac{1}{2}g_\alpha(d_\alpha,d_\alpha) + \frac{\kappa}{2}\mathfrak{J}_4(\alpha) - \frac{1}{2}\|b_\alpha\|^2_{Q^*_\alpha},$$

which is nonnegative exactly under (16.26). The reverse implication follows from the same completed square. □

Proposition 16.7 is an exact certificate for the frozen quadratic root model, not a necessary consequence of local nonnegativity of the full action: its positive constant term prevents that inference. To use it for JC–PL one must additionally compare the full integrated Hessian to this frozen model on the whole admissible amplitude range. The closed JP–DE estimates control p_α and hence d_α, but do not prove either that comparison or the displayed sharp dual bound for $b_\alpha = \ell_\alpha - p_\alpha/2$. The period-three computation passes the quadratic certificate with a sizeable numerical margin; a uniform no-ratio proof over all profiles is still required.

Proposition 16.8 (JC–RC: exact root/path sufficient reduction)**.** *Suppose a positive semidefinite form L_α satisfies, throughout the radial/metric path chart,*

$$\int_0^1 (1-t)\{H_{\alpha,tz} - H_{\alpha,0}\}[z,z]\,dt \ge -L_\alpha[z,z]. \tag{16.27}$$

Define

$$\widehat{Q}_\alpha := H_{\alpha,0} - g_\alpha - 2L_\alpha. \tag{16.28}$$

If

$$\widehat{Q}_\alpha \geq 0, \qquad \|b_\alpha\|^2_{\widehat{Q}^*_\alpha} \leq g_\alpha(d_\alpha, d_\alpha) + \kappa\mathfrak{J}_4(\alpha), \tag{16.29}$$

then JC–PL holds.

Proof. Split the Hessian integral in (16.13) at $H_{\alpha,0}$. The Bregman weight has mass $1/2$, so (16.27) gives

$$\mathcal{A}_\alpha(z) \geq \frac{1}{2}\{g_\alpha(d_\alpha, d_\alpha) + \kappa\mathfrak{J}_4(\alpha)\} + b_\alpha[z] + \frac{1}{2}\widehat{Q}_\alpha[z,z].$$

The right side is nonnegative by the same Hilbert completion as Proposition 16.7. □

This reduction records the required slack correctly. Terminal and nonterminal estimates must first be assembled into a genuine form L_α; one may not subtract their scalar upper bounds twice or replace (16.27) by a pointwise support-time supremum.

Remark 16.9 (Why the tempting exact Gram shortcut does not close the root). Let $\mathcal{H}_D$ be the disk boundary Hilbert space, let K be the fan-normal sine trace, T the genuine material/kinematic trace at the tangential endpoint, and B the fan source. Put

$$G = K^*K, \qquad p = 2K^*B, \qquad d = -G^{-1}p/2, \qquad c = -Kd = P_{K\mathsf{X}_\alpha}B, \qquad E = T - K.$$

For the pure quadratic disk principal block, where $\ell = 2T^*B$ and $H_0 = 2T^*T$, direct expansion gives, with $V = 2T - K$,

$$b[z] = \langle c, Vz\rangle_D + 2\langle B - c, Ez\rangle_D, \tag{16.30}$$

$$H_0 - G = V^*V - 2E^*E, \tag{16.31}$$

$$\frac{1}{2}G(d) + b[z] + \frac{1}{2}(H_0 - G)[z,z] = \frac{1}{2}\|c + Kz + 2Ez\|^2_D - \|Ez\|^2_D + 2\langle B - c, Ez\rangle_D. \tag{16.32}$$

Thus choosing $U = c$ and $V = 2T - K$ leaves the negative form $-2E^*E$ and an orthogonal-source cross term; they do not vanish merely because c is the projection of B onto $K\mathsf{X}_\alpha$.

For the genuine polygonal block write $\Gamma = H_{\alpha,0} - 2T^*T$ and let r be the difference between b_α and the first pairing in (16.30). A valid Gram repair must, at minimum, pay the block

$$\begin{pmatrix} \kappa\mathfrak{J}_4(\alpha) & r^* \\ r & \Gamma - 2E^*E \end{pmatrix} \geq 0$$

after adding any explicitly retained fan square and path-loss slack. No closed TR or JP–DE row proves this block. Equations (16.30)–(16.32) explain why the numerically favorable root ratio cannot be promoted by a bare Cauchy–Schwarz citation.

Remark 16.10 (Audited physical-trace branch: exact algebra, failed chart). There is an algebraically cleaner inactive branch. Let $\mathbf{q}_D = 2j_{0,1}^2 q_D^0$ be the physical disk half-Hessian, put

$$\widetilde{K}_\alpha z = \Pi_{\mathrm{ng}}\widetilde{T}_\alpha z, \qquad \widetilde{G}_\alpha(z) = \mathbf{q}_D(\widetilde{K}_\alpha z), \qquad \widetilde{p}_\alpha[z] = 2\langle B_\alpha, \widetilde{K}_\alpha z\rangle_D,$$

and let $\widetilde{d}_\alpha = -\frac{1}{2}\widetilde{\mathcal{R}}_\alpha^{-1}\widetilde{p}_\alpha$. The normalized TR endpoint gives the exact square

$$\mathbf{q}_D(B_\alpha + \widetilde{K}_\alpha z) = \widetilde{P}_D(\alpha) + \widetilde{G}_\alpha(z - \widetilde{d}_\alpha).$$

The corresponding physical residual and action satisfy the same exact Schur identity as Proposition 16.3; in its pure disk block the root is exactly

$$\frac{1}{2}\widetilde{G}_\alpha(z - \widetilde{d}_\alpha) + \frac{\kappa}{2}\mathfrak{J}_4(\alpha) \geq 0.$$

Thus the $E = T - K$ defect in (16.31) disappears from the disk block.

This does not close the polygonal action. The large component of $\widetilde{T}_\alpha$ produced by a short angle is a tangential slide of a straight side viewed in the disk radial frame; it can be nonzero even when the true side-normal speed is zero. It is cancelled at the same order by the nonlinear reparametrization remainder $\mathcal{N}$:

$$D_z^2\widetilde{\mathcal{R}}_\alpha(0)[\xi,\xi] = H_{\alpha,0}[\xi,\xi] - 2\widetilde{G}_\alpha(\xi).$$

By the exact value split in TR, the difference on the right is precisely the full second z-derivative of $\mathcal{N}(W_{\alpha,z})$ at $z = 0$; this formulation includes all chain-rule terms at the vector-field base point $W_{\alpha,0}$. The cancellation cannot be estimated from the radial image alone. For example, if $X(\theta) = e_r(\theta + \delta\sin 2\theta)$ and $W = X - e_r$, the boundary image is the unit circle, so $\mathcal{L}(W) = \mathcal{L}(0)$, whereas

$$f = W \cdot e_r = \cos(\delta \sin 2\theta) - 1, \qquad \mathcal{N}(W) = -\mathbf{q}_D(\Pi_{\mathrm{ng}} f).$$

Hence no image-small argument can yield $\mathcal{N}(W) \geq -\vartheta\mathbf{q}_D(\Pi_{\mathrm{ng}} f)$ with a universal $\vartheta < 1$. The radial-not-affine example in JP further shows that the full material $W^{1,\infty}$ norm can diverge on the radial chart. Consequently the old physical CF+CM→NR support-tail argument cannot be reactivated. The physical branch replaces JC–PL by an equivalent open full-radial-chart action; it is recorded as an inactive alternative, not as a second proof of J0.

Remark 16.11 (Audited Minkowski shortcut: two concavities do not order the ratio). For a fixed positive normal fan, support interpolation is Minkowski interpolation. Consequently

$$U(h) = |P(h)|^{1/2}, \qquad V(h) = \lambda_1(P(h))^{-1/2}$$

are positive, one-homogeneous concave functions, while $\Phi = (U/V)^2$. At a stationary point,

$$\frac{D^2\Phi}{\Phi} = 2\left(\frac{D^2U}{U} - \frac{D^2V}{V}\right)$$

on a nonradial direction. Both displayed curvatures are nonpositive, but their difference has no determined sign.

This is a genuine logical obstruction, not only a missing inequality. On the Lorentz cone $\{(x,y) : x > |y|\}$ put, with $c = 11/20$,

$$U(x,y) = \sqrt{x^2 - y^2}, \qquad V(x,y) = x - c\frac{y^2}{x}.$$

Both functions are positive, one-homogeneous and concave, since

$$D^2U = -\frac{1}{(x^2-y^2)^{3/2}}\begin{pmatrix} y^2 & -xy \\ -xy & x^2 \end{pmatrix}, \quad D^2V = -\frac{2c}{x^3}\begin{pmatrix} y^2 & -xy \\ -xy & x^2 \end{pmatrix}.$$

Moreover $U^2 = x^2 - y^2$ is exactly quadratic. At $(1,0)$ the ratio $(U/V)^2$ is stationary and has second derivative $1/5 > 0$ in the y direction. Nevertheless the same-area point $(5/3, 4/3)$ satisfies

$$U(5/3,4/3) = 1, \qquad \left(\frac{U}{V}\right)^2(5/3,4/3) = \frac{625}{729} < 1.$$

Thus area and eigenvalue Brunn–Minkowski concavity, even with exact quadratic area and strict local support minimality, do not give a global support comparison. The missing relative curvature comparison is again part of JC–PL.

Remark 16.12 (Audited stationary-endpoint bypass: why it is insufficient). The independent stationary O–H endpoint theorem cannot by itself replace JC–PL. Indeed, put the regular polygon and a hypothetical active global minimizer in one fixed-area joint chart at 0 and x. Since both are stationary, the exact identity is

$$0 = \langle \nabla\Phi_N(x) - \nabla\Phi_N(0), x\rangle = \int_0^1 D^2\Phi_N(tx)[x,x]\,dt. \tag{16.33}$$

The interior points tx are not stationary, so a Hessian theorem whose hypothesis is stationarity does not control this integral.

Endpoint coercivity alone is logically insufficient even at the relevant super-near scale. For $a > 0$ and $0 < c < 1/3$ consider

$$F_{a,c}(t,y) = at^2(a-t)^2 - ca^5\{3(t/a)^2 - 2(t/a)^3\} + |y|^2, \qquad E_a(u,v) = a^3u^2 + |v|^2.$$

The points $(0,0)$ and $(a,0)$ are stationary, while $F_{a,c}(a,0) = -ca^5 < F_{a,c}(0,0) = 0$. Both endpoint Hessians are strictly positive: their t-eigenvalues are respectively

$$(2-6c)a^3, \qquad (2+6c)a^3,$$

and the y-block is $2I$. Moreover, with $u = t/a$,

$$\frac{F_{a,c}(t,0)}{a^5} + c = (1-u)^2\{(u+c)^2 + c - c^2\} \ge 0,$$

so $(a,0)$ is the global minimum. Nevertheless $D^2_{tt}F_{a,c}(a/2,0) = -a^3$ and, for $x = (a,0)$,

$$\int_0^1 D^2F_{a,c}(sx)[x,x]\,ds = 0, \qquad E_a(x) = a^5.$$

Thus positive endpoint spectra, including uniform spectra in the naturally scaled norm, do not prove the path monotonicity needed in (16.33).

The minimal stationary-route replacement is a super-near joint statement, not an endpoint citation. After gauges, it would have to prove uniformly

$$\int_0^1 D^2\Phi_N(tx)[x,x]\,dt \geq \mu\left\{6a^2\sum_i d_i^2 + \mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(z - Z_N d)\right\} \tag{16.34}$$

for nonzero $x=(d,z)$ with $\mathfrak{J}_4(\alpha_*+d)+\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(z) \lesssim a^5$. At the regular polygon this requires the full cyclic fan/support Schur symbol, including its nonzero period-three mixed block; it must then be propagated diagonally along the super-near path. No retained node proves either the regular Schur complement or this relative path stability. Consequently the stationary route only narrows JC–PL; it does not close the main theorem.

16.3. **The unique open pathwise payment.** The closed disk endpoint and value rows provide universal $S_D, \kappa, C_p > 0$ and a modulus $\varepsilon_V(S) \to 0$ such that, for $S \leq S_D$,

$$P_D^{\mathrm{ns}}(\alpha) - P_D^{\mathrm{ns}}(\alpha_*) \geq 2\kappa\mathfrak{J}_4(\alpha), \tag{16.35}$$

$$\|p_\alpha\|_{(\mathfrak{G}^{\mathrm{JP}}_\alpha)^*} \leq C_p\mathfrak{J}_4(\alpha)^{1/2}, \tag{16.36}$$

$$|\Phi_\alpha(h_\alpha\mathbf{1}) - \Phi_{\alpha_*}(h_*\mathbf{1}) - \{q_D(B_\alpha) - q_D(B_{\alpha_*})\}| \leq \varepsilon_V(S)\mathfrak{J}_4(\alpha). \tag{16.37}$$

Proposition 16.13 (Regular disk-joint Schur symbol). *Let $\alpha_* = a\mathbf{1}$ with $2a < S_D$. Choose the canonical analytic scale/translation trivialization constructed in Proposition 16.27,*

$$J_\alpha : \mathsf{X}_{\alpha_*} \longrightarrow \mathsf{X}_\alpha, \qquad J_{\alpha_*} = I,$$

and put

$$F_D^\sharp(\alpha, z) := q_D(B_\alpha + K_\alpha J_\alpha z), \qquad \bar{z}^D_\alpha := J_\alpha^{-1} z^D_\alpha, \qquad Z_N\delta := D_\alpha \bar{z}^D_\alpha\big|_{\alpha_*}[\delta].$$

For every $\sum_i \delta_i = 0$ and every regular-quotient support vector ζ,

$$\begin{aligned}
\mathcal{D}_N(\delta,\zeta) &:= \frac{1}{2}D^2F_D^\sharp(\alpha_*,0)[(\delta,\zeta),(\delta,\zeta)] \qquad (16.38)\\
&= \frac{1}{2}D^2P_D^{\mathrm{ns}}(\alpha_*)[\delta,\delta] + \mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(\zeta - Z_N\delta)\\
&\geq 12\kappa a^2\sum_i \delta_i^2 + \mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(\zeta - Z_N\delta). \qquad (16.39)
\end{aligned}$$

Thus every cyclic Fourier block of the regular normal-sine disk-joint *symbol has a uniform completed Schur gap. This statement concerns the fixed disk scalar; it is not a lower bound for the polygonal joint Hessian.*

Proof. The endpoint data and J_α are analytic in the fixed quotient chart. If $\bar{\mathfrak{G}}^{\mathrm{JP}}{}_\alpha(w) := \mathfrak{G}^{\mathrm{JP}}_\alpha(J_\alpha w)$, exact disk completion gives

$$F_D^\sharp(\alpha,z) = P_D^{\mathrm{ns}}(\alpha) + \bar{\mathfrak{G}}^{\mathrm{JP}}{}_\alpha(z - \bar{z}^D_\alpha), \qquad \bar{z}^D_{\alpha_*} = 0.$$

Along $\alpha(t)=\alpha_*+t\delta$, $z(t)=t\zeta$, one has $\bar z^D_{\alpha(t)}=tZ_N\delta+O(t^2)$, while variation of $\bar{\mathfrak{G}}^{\mathrm{JP}}{}_{\alpha(t)}$ contributes only from order t^3 onward. Taking the coefficient of t^2 proves (16.38). Moreover

$$\mathfrak{J}_4(\alpha_*+t\delta)=6a^2t^2\sum_i\delta_i^2+4at^3\sum_i\delta_i^3+t^4\sum_i\delta_i^4.$$

Apply the closed estimate $P^{\mathrm{ns}}_D(\alpha)-P^{\mathrm{ns}}_D(\alpha_*)\ge 2\kappa\mathfrak{J}_4(\alpha)$. For fixed δ, both signs of sufficiently small t keep $\alpha_*+t\delta$ in the positive-fan region $S\le S_D$. Dividing the two signed inequalities first by t shows $DP^{\mathrm{ns}}_D(\alpha_*)[\delta]=0$; subtracting that linear term and then dividing by t^2 gives $\frac12 D^2P^{\mathrm{ns}}_D(\alpha_*)[\delta,\delta]\ge 12\kappa a^2\sum_i\delta_i^2$, proving (16.39). □

Proof contract 16.14 (JC–AR: regular completed-action Schur block)**.** In the trivialization of Proposition 16.13, put

$$F^\sharp_P(\alpha,z):=\Phi_\alpha(h_\alpha\mathbf{1}+J_\alpha z),\qquad \mathcal{E}_N(d,\zeta):=\frac12D^2(F^\sharp_P-F^\sharp_D)(\alpha_*,0)[(d,\zeta),(d,\zeta)].$$

Write $\bar z^D_\alpha=J_\alpha^{-1}z^D_\alpha$, so that $D\bar z^D_{\alpha_*}[d]=Z_Nd$, and use the exact completed coordinates

$$\mathbf{A}_N(d,w):=\mathcal{A}_{\alpha_*+d}(J_{\alpha_*+d}(\bar z^D_{\alpha_*+d}+w)).$$

Then $\mathbf{A}_N(0,0)=0$, $D\mathbf{A}_N(0,0)=0$, and the exact regular quadratic root is

(16.40)

$$\boxed{\mathcal{Q}_N(d,w):=\frac12D^2\mathbf{A}_N(0,0)[(d,w),(d,w)]=\mathcal{E}_N(d,w+Z_Nd)-\mathcal{E}_N(d,0)+\frac12\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w)+3\kappa a^2\sum_i d_i^2.}$$

The subtraction $\mathcal{E}_N(d,0)$ and the coefficients $1/2$ and 3κ are forced respectively by the base-value subtraction in $\mathcal{R}_\alpha$, the completed disk square, and $\frac{\kappa}{2}\mathfrak{J}_4$. They may not be replaced by the larger margins of the full disk-joint Hessian.

Let

$$\mathcal{P}_N(d,\zeta):=\frac12D^2F^\sharp_P(\alpha_*,0)[(d,\zeta),(d,\zeta)]$$

be the genuine polygonal joint half-Hessian. Exact disk completion gives the following sharper algebraic reduction:

(16.41)

$$\boxed{\mathcal{Q}_N(d,w)=\mathcal{P}_N(d,w+Z_Nd)-\mathcal{P}_N(d,0)+\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(Z_Nd)-\frac12\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w)+3\kappa a^2\sum_i d_i^2.}$$

Indeed Proposition 16.13 yields

$$\frac12D^2F^\sharp_D(\alpha_*,0)[(d,\zeta)]^2=\frac12D^2P^{\mathrm{ns}}_D(\alpha_*)[d,d]+\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(\zeta-Z_Nd).$$

Subtracting its values at $\zeta=w+Z_Nd$ and $\zeta=0$ gives $\mathfrak{G}^{\mathrm{JP}}(w)-\mathfrak{G}^{\mathrm{JP}}(Z_Nd)$, which proves (16.41). In particular the pure polygonal fan block cancels exactly. JC–AR therefore asks only for the polygonal support block and its fan–support coupling after the corrector is inserted; estimating the entire polygonal joint Hessian is unnecessary.

For $d=0$, equation (16.41) becomes

$$\mathcal{Q}_N(0,w)=\frac{1}{2}D_h^2\Phi_{\alpha_*}(h_*\mathbf{1})[w,w]-\frac{1}{2}\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w). \tag{16.42}$$

Consequently the archived regular specialization of JP–S0,

$$D_h^2\Phi_{\alpha_*}(h_*\mathbf{1})[w,w]\geq(2-\omega_0(a))\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w),\qquad \omega_0(a)\to 0,$$

would close the pure-support row of JC–AR with any fixed $\mu_s<1/2$ for all sufficiently large N. The remaining part is then the modewise Schur payment of the fan–support coupling in (16.41).

The open regular action target is to prove constants

$$\mu_f>0,\qquad \mu_s>0, \tag{16.43}$$

independent of N, such that

$$\boxed{\mathcal{Q}_N(d,w)\geq\mu_f a^2\sum_i d_i^2+\mu_s\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w).} \tag{16.44}$$

This is not a consequence of Proposition 16.13, which controls the full disk scalar rather than the completed action.

More explicitly, cyclic equivariance diagonalizes every surviving quotient mode k. With the normalization

$$\begin{aligned}\mathcal{Q}_{N,k}(d_k,w_k)&=q^{ff}_{N,k}|d_k|^2+2\operatorname{Re}(\overline{q^{fs}_{N,k}}d_k\overline{w_k})+q^{ss}_{N,k}|w_k|^2,\\ \mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w_k)&=g_{N,k}|w_k|^2,\qquad g_{N,k}>0,\end{aligned}$$

write the polygonal half-Hessian on the same character as

$$\mathcal{P}_{N,k}(d_k,\zeta_k)=p^{ff}_{N,k}|d_k|^2+2\operatorname{Re}(\overline{p^{fs}_{N,k}}d_k\overline{\zeta_k})+p^{ss}_{N,k}|\zeta_k|^2,\qquad (Z_Nd)_k=Z_{N,k}d_k.$$

Expanding the already proved polygonal reduction (16.41) gives the exact coefficients

$$\boxed{\begin{aligned}q^{ss}_{N,k}&=p^{ss}_{N,k}-\frac{1}{2}g_{N,k},\\ q^{fs}_{N,k}&=p^{fs}_{N,k}+p^{ss}_{N,k}\overline{Z_{N,k}},\\ q^{ff}_{N,k}&=3\kappa a^2+2\operatorname{Re}(p^{fs}_{N,k}Z_{N,k})+(p^{ss}_{N,k}+g_{N,k})|Z_{N,k}|^2.\end{aligned}} \tag{16.45}$$

Thus the pure polygonal fan coefficient $p^{ff}_{N,k}$ cancels. Before positive margins are inserted, the remaining Schur complement is

$$3\kappa a^2+\frac{1}{2}g_{N,k}|Z_{N,k}|^2-\frac{|p^{fs}_{N,k}+\frac{1}{2}g_{N,k}\overline{Z_{N,k}}|^2}{p^{ss}_{N,k}-\frac{1}{2}g_{N,k}}.$$

This is an algebraic simplification, not a positivity assertion: the polygonal support coefficient and the fan–support covector remain to be estimated.

The same cancellation remains exact after margins are inserted. Put

$$A_{N,k}:=p^{fs}_{N,k}+\frac{1}{2}g_{N,k}\overline{Z_{N,k}},\qquad r_{N,k}:=p^{ss}_{N,k}-\left(\frac{1}{2}+\mu_s\right)g_{N,k}.$$

Here $A_{N,k}$ is precisely the regular fan derivative of the completed covector $b_\alpha = \ell_\alpha - p_\alpha/2$ in the same orthonormal Fourier normalization. When $r_{N,k} > 0$, the full modal Schur row is equivalent to

(16.46)
$$\boxed{(3\kappa - \mu_f)a^2 + \left(\frac{1}{2} - \mu_s\right) g_{N,k}|Z_{N,k}|^2 - \frac{|A_{N,k} + \mu_s g_{N,k}\overline{Z_{N,k}}|^2}{r_{N,k}} \geq 0.}$$

Thus the mixed analytic blocker is a sharp all-character estimate for $D_\alpha(\ell_\alpha - p_\alpha/2)$, not for the discarded covector $\ell_\alpha - p_\alpha$.

Equation (16.44) is equivalent, mode by mode, to

$$\boxed{\begin{pmatrix} q^{ff}_{N,k} - \mu_f a^2 & q^{fs}_{N,k} \\ \overline{q^{fs}_{N,k}} & q^{ss}_{N,k} - \mu_s g_{N,k} \end{pmatrix} \succeq 0.} \tag{16.47}$$

Equivalently, with the usual quotient interpretation when the lower-right entry vanishes,

$$q^{ss}_{N,k} - \mu_s g_{N,k} \geq 0, \tag{16.48}$$

$$q^{ff}_{N,k} - \mu_f a^2 \geq \frac{|q^{fs}_{N,k}|^2}{q^{ss}_{N,k} - \mu_s g_{N,k}}. \tag{16.49}$$

In particular, the pure support specialization already asks for

$$D^2_h \Phi_{\alpha_*}(h_* \mathbf{1})[z, z] \geq (1 + 2\mu_s)\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(z). \tag{16.50}$$

The exact fixed-fan support Hessian identifies the left side with the regular-polygon reduced Dirichlet-to-Neumann block plus its geometric area row. More precisely, for the orthonormal character $z_i = N^{-1/2}e^{\mathrm{i}kai}$ let $f_{N,k} = \mathcal{I}_{\alpha_*}z$, $g_{N,k} = q_D(f_{N,k})$, let $p_N = -\partial_\nu u_N$, and let $\psi_{N,k}$ be the character-k reduced Helmholtz solution on the area-π regular polygon with boundary value $z_i p_N$ on side i. Writing

$$\mathcal{D}_{N,k} = \int_{R_N} |\nabla \psi_{N,k}|^2 - \lambda_N \int_{R_N} |\psi_{N,k}|^2,$$

the exact support Hessian and area symbol give

$$\boxed{p^{ss}_{N,k} = \mathcal{D}_{N,k} + \frac{\lambda_N}{\pi}\frac{\cos(ka) - \cos a}{\sin a}, \qquad q^{ss}_{N,k} = p^{ss}_{N,k} - \frac{1}{2}g_{N,k}.} \tag{16.51}$$

Hence the pure-support row is exactly the uniform sector inequality

$$\mathcal{D}_{N,k} + \frac{\lambda_N}{\pi}\frac{\cos(ka) - \cos a}{\sin a} \geq \left(\frac{1}{2} + \mu_s\right) g_{N,k}, \qquad 2 \leq k \leq N - 2.$$

All terms except the regular-polygon reduced Helmholtz DtN energy are explicit. No closed terminal/LCA theorem proves this inequality, even before the mixed Schur row (16.49) is considered.

The exact Schwarz–Christoffel representation also shows why the DtN and area rows must be estimated together. After a rotation,

$$F_N'(z) = c_N(1 - z^N)^{-2/N}, \qquad J_N = |F_N'|,$$

and, for the unnormalized character $z_i = e^{ikai}$, let $\sigma_{N,k}$ be the step function on $\mathbb{T}$ which equals e^{ikai} on the preimage arc of side i (its endpoint values are irrelevant). The pulled-back boundary trace is

$$h_{N,k} = \sigma_{N,k} c_N^{-1} |1 - e^{iN\theta}|^{2/N} q_N, \qquad q_N = -\partial_r(u_N \circ F_N)|_{\mathbb{T}}.$$

Uniform disk $W^{2,p}$ estimates give $q_N \to d_\circ$ uniformly, including at the prevertices. Keeping only the cross-prevertex rectangles in the Gagliardo norm gives the harmonic-extension lower bound $H(h_{N,k}) \geq cN^2\{1 - \cos(ka)\} - C$. To pass to the reduced Helmholtz energy, use the uniform boundary Friedrichs inequality

$$\|\phi\|^2_{L^2(R_N)} \leq \varepsilon \|\nabla \phi\|^2_{L^2(R_N)} + C_\varepsilon \|h_{N,k}\|^2_{L^2(\partial R_N)}.$$

Choose $\varepsilon < (2 \sup_N \lambda_N)^{-1}$. Rellich's identity gives $\|h_{N,k}\|^2_{L^2(\partial R_N)} = \int_{\partial R_N} p_N^2 = 2\lambda_N / h_N \leq C$ in the unnormalized character. Hence every admissible reduced competitor has energy at least $\frac{1}{2} H(h_{N,k}) - C$; the ground-state orthogonality only decreases the competitor set. This proves the rigorous DtN-only lower bound

$$\mathcal{D}^{\mathrm{un}}_{N,k} \geq cN^2\{1 - \cos(ka)\} - C. \tag{16.52}$$

This does *not* close the support row. Formula (16.51) uses the orthonormal character; in the unnormalized convention its area term must also be multiplied by N and is of order $-N^2\{1 - \cos(ka)\}$. The two leading corner-scale terms cancel in the disk limit. More precisely, the uncancelled coefficient contains

$$\frac{2}{a^2}\{\pi c_N^{-2} q_{N,*}^2 - \lambda_N\}\{1 - \cos(ka)\},$$

for the prevertex endpoint amplitude $q_{N,*}$. Rate-free convergence alone does not control this coefficient after multiplication by a^{-2}. One must consequently estimate the jointly renormalized remainder, rather than combine (16.52) with an area symbol written in the other normalization. The next proposition supplies precisely that expansion, uniformly through the mesoscopic regime $ka \downarrow 0$.

Lemma 16.15 (Fixed-character polygon–disk support limit)**.** *For every fixed integer $k \geq 2$,*

$$\frac{p^{ss}_{N,k}}{g_{N,k}} \longrightarrow 1 \qquad (N \to \infty). \tag{16.53}$$

Proof. Use the unnormalized support character $z_i = e^{\mathrm{i}kia}$; the normalization cancels from the quotient. The material vertex velocity at the vertex with bisector angle $\varphi_i = (i + \frac{1}{2})a$ is exactly

$$e^{\mathrm{i}k\varphi_i}\left\{\frac{\cos(ka/2)}{\cos(a/2)}e_r(\varphi_i) + \mathrm{i}\frac{\sin(ka/2)}{\sin(a/2)}e_\theta(\varphi_i)\right\}.$$

For fixed k its physical side-affine interpolant converges in $H^{1/2}(\mathbb{T};\mathbb{C}^2)$ to the smooth disk field

$$V_k^\circ(\theta) = e^{\mathrm{i}k\theta}\{e_r(\theta) + \mathrm{i}ke_\theta(\theta)\}, \qquad V_k^\circ \cdot e_r = e^{\mathrm{i}k\theta}.$$

We record the fixed-slot Hessian passage rather than citing an unassembled stage. Pull the area-normalized regular polygons to the disk by the normalized near-identity conformal charts. Their stiffness forms and mass densities converge to the disk forms in every fixed L^p, their first eigenfunctions converge strongly in H_0^1, and the uniform convex-domain gradient bound makes $\nabla u_N \otimes \nabla u_N \to \nabla u_\circ \otimes \nabla u_\circ$ in L^2. Polarize the exact volume formula for the normalized eigenvalue Hessian with the fixed smooth extension of V_k° in both slots. Every direct stiffness, mass, area, and product-of-first-jets row then converges by the preceding strong coefficient convergence.

For the Schur row, put the fixed first-variation functional into the reduced resolvent. The spectral gap above the ground state is uniform, the functionals converge in H^{-1}, and the resolvent identity on the moving ground-state complements gives strong H_0^1 convergence of the reduced states. Thus the Schur pairing converges as well. This proves convergence of the complete polygonal normalized Hessian to the disk Hessian on V_k°; no mesh-ratio estimate is used because the slot is fixed and smooth.

At the disk, scale and translation modes have already been projected out and the normalized Hessian is

$$D^2\Phi_\circ[V_k^\circ, V_k^\circ] = 2q_D(e^{\mathrm{i}k\theta}).$$

On the other hand, the exact normal-sine interpolant $f_{N,k} = \mathcal{I}_{\alpha_*}z$ converges to $e^{\mathrm{i}k\theta}$ in $H^{1/2}$, so $g_{N,k} \to q_D(e^{\mathrm{i}k\theta})$. Since $p_{N,k}^{ss}$ is the polygonal half-Hessian coefficient, (16.53) follows. □

Proposition 16.16 (Jointly renormalized regular support sector)**.** *There is N_s such that, for $N \geq N_s$ and every quotient character $2 \leq k \leq N-2$,*

$$\boxed{p_{N,k}^{ss} \geq \frac{11}{20}g_{N,k} = \left(\frac{1}{2} + \frac{1}{20}\right)g_{N,k}.} \tag{16.54}$$

Consequently the pure-support row (16.48) is closed with $\mu_s = 1/20$.

Proof. We keep one normalization until the last step. By complex conjugation assume $2 \leq k \leq N/2$, and put

$$\beta = \frac{2}{N}, \qquad a = \pi\beta, \qquad x = \frac{k}{N} \in (0, \tfrac{1}{2}], \qquad \xi = 2\pi x.$$

Use first the unnormalized complex character $z_i = e^{2\pi\mathrm{i}xi}$. Every quadratic value is then N times its coefficient in the orthonormal complex character. The orthonormal real vector $\sqrt{2/N}\,\mathrm{Re}\,z$ has the same modal coefficient.

Let

$$w_\beta(t) = |2\sin(\pi t)|^\beta, \qquad 0 < t < 1.$$

For the constant disk flux $d_\circ$, the exact corner trace is $d_\circ \sigma_{N,k} w_\beta$. Its only circular frequencies are $k + mN$, and the beta integral gives

(16.55)
$$I_{\beta,m}(x) := \int_0^1 w_\beta(t) e^{-2\pi \mathrm{i}(m+x)t}\, dt = e^{-\pi \mathrm{i}(m+x)} \frac{\Gamma(1+\beta)}{\Gamma(1+\beta/2+m+x)\Gamma(1+\beta/2-m-x)}.$$

Since $\pi d_\circ^2 = \lambda_\circ$ by Rellich's identity, harmonic energy and the area row, divided by $N d_\circ^2$, combine into

$$P_\beta(x) := 2\pi \sum_{m\in\mathbb{Z}} |m+x||I_{\beta,m}(x)|^2 + \frac{\cos(2\pi x) - \cos(\pi\beta)}{\sin(\pi\beta)}. \tag{16.56}$$

The two terms in this formula separately contain the same β^{-1} pole with opposite signs.

We extract the finite part before estimating. For $0 < y < 1$, put

$$T_\beta(y) := \sum_{n\geq 0} (n+y) \left\{ \frac{\Gamma(n+y-\beta/2)}{\Gamma(n+y+1+\beta/2)} \right\}^2.$$

Reflection in (16.55) gives the positive identity

$$\sum_m |m+x||I_{\beta,m}(x)|^2$$

(16.57)
$$= \frac{\Gamma(1+\beta)^2}{\pi^2} \left\{ \sin^2\pi(x-\beta/2) T_\beta(x) + \sin^2\pi(x+\beta/2) T_\beta(1-x) \right\}.$$

Separate the $n = 0$ term. Taylor's formula for $\log\Gamma$, together with $|\psi_{\mathrm{dg}}^{(r)}(u)| \leq C_r(1 + u^{-r-1})$, gives, uniformly for $0 < \beta \leq x/2$ and $x \leq 1/2$,

$$\begin{aligned}
T_\beta(x) &= \zeta(1+2\beta, x) + O\left(\frac{\beta}{x^2} + \frac{\beta|\log x|}{x} + \frac{\beta^2}{x^3} + \beta \right), \\
T_\beta(1-x) &= \zeta(1+2\beta, 1-x) + O(\beta), \\
\zeta(1+2\beta, x) &= \frac{1}{2\beta} - \psi_{\mathrm{dg}}(x) + O\left(\frac{\beta|\log x|}{x} + \frac{\beta^2|\log x|^2}{x} + \beta \right),
\end{aligned}$$

where $\psi_{\mathrm{dg}} = \Gamma'/\Gamma$. Indeed, after comparison with $(n+y)^{-1-2\beta}$ the derivative of the logarithmic quotient is

$$-2\psi_{\mathrm{dg}}(n+y) - (n+y)^{-1} + 2\log(n+y) = O((n+y)^{-2}),$$

so all $n \geq 1$ remainders are absolutely summable; the singular powers come only from $n = 0$.

Expand the sine squares in (16.57), retaining their opposite linear terms until cancellation, and use $\Gamma(1+\beta)^2 = 1 - 2\gamma\beta + O(\beta^2)$. We obtain

$$P_\beta(x) = P(x) + O\left(\beta + \beta x|\log x| + \frac{\beta^2}{x} \right), \tag{16.58}$$

$$P(x) := -\frac{2\sin^2(\pi x)}{\pi}\{\psi_{\mathrm{dg}}(x) + \psi_{\mathrm{dg}}(1-x) + 2\gamma\}. \tag{16.59}$$

The high-character normal-sine disk symbol is

$$Q(x) := \frac{2\sin^4(\pi x)}{\pi^3}\{\zeta(3,x) + \zeta(3,1-x)\}. \tag{16.60}$$

It satisfies $Q(x) \asymp x$ on $(0, 1/2]$. We now prove a strict analytic gap. Set

$$D(x) := -\{\psi_{\mathrm{dg}}(x) + \psi_{\mathrm{dg}}(1-x) + 2\gamma\}, \qquad H_3(x) := \zeta(3,x) + \zeta(3,1-x).$$

The digamma and paired Hurwitz series give

$$D(x) = \frac{1}{x} + 2\sum_{r\geq 1}\zeta(2r+1)x^{2r},$$

and

$$H_3(x) = x^{-3} + \sum_{n\geq 1}\{(n-x)^{-3} + (n+x)^{-3}\} \leq x^{-3} + 14\zeta(3) - 8.$$

The tail increases with x and the last constant is its value at $x = 1/2$. Using $\sin(\pi x) \leq \pi x$ and the integral-tail bound $\zeta(3) < 121/100$, we get

$$\begin{aligned} D(x) - \frac{3}{5}\frac{\sin^2(\pi x)}{\pi^2}H_3(x) &\geq \frac{2}{5x} + \frac{24 - 32\zeta(3)}{5}x^2 \\ &> \frac{2}{5x} - \frac{368}{125}x^2 > 0 \qquad (0 < x \leq \tfrac{1}{2}). \end{aligned}$$

Therefore

$$\boxed{P(x) > \frac{3}{5}Q(x), \qquad 0 < x \leq \frac{1}{2}.} \tag{16.61}$$

Dividing (16.58) by $Q(x) \asymp x$ also shows

$$P_\beta(x) = P(x) + o(Q(x)) \tag{16.62}$$

along every sequence for which $k = 2x/\beta \to \infty$.

It remains to replace the constant flux by the true polygonal flux. The exact area normalization of $F_N'(z) = c_N(1 - z^N)^{-\beta}$ is

$$c_N^{-2} = \sum_{m\geq 0}\frac{(\beta)_m^2}{(m!)^2(mN+1)}.$$

Since $(\beta)_m/m! \leq C\beta m^{\beta-1}$ for $m \geq 1$, $c_N = 1 + O(\beta^3)$. Put $W_N = |F_N'|^2 = c_N^2|1 - z^N|^{-2\beta}$. For every fixed $p > 4$, the change of variables $t = r^N$ gives

$$\|W_N - 1\|_{L^p(\mathbb{D})} \leq C_p\beta^{1+1/p}. \tag{16.63}$$

Indeed, on $t \leq 1/2$ one has $|\log|1 - te^{\mathrm{i}\phi}|| \leq Ct$, while on $t \geq 1/2$ all fixed logarithmic moments remain uniformly integrable against $|1 - te^{\mathrm{i}\phi}|^{-2p\beta}$. Since $r\,dr = (\beta/2)t^{\beta-1}dt$, the p-th power of the norm is $O(\beta^{p+1})$.

The pulled eigenfunction satisfies

$$-\Delta v_N = \lambda_N W_N v_N, \qquad v_N|_{\mathbb{T}} = 0, \qquad \int_{\mathbb{D}} W_N v_N^2 = 1.$$

The disk spectral gap first gives an H^1 estimate with the rate in (16.63). Apply a $W^{2,q}$ estimate with $q > 1$ to obtain L^∞ control, substitute it back into the difference equation, and then apply the $W^{2,p}$ estimate. Thus

(16.64)
$$|\lambda_N - \lambda_\circ| + \|v_N - u_\circ\|_{W^{2,p}(\mathbb{D})} \le C_p\beta^{1+1/p}, \qquad \|q_N - d_\circ\|_{C^\alpha(\mathbb{T})} \le C_p\beta^{1+1/p},$$

where $q_N = -\partial_r v_N|_{\mathbb{T}}$ and $\alpha = 1 - 2/p > 1/2$.

For $r \in C^\alpha$ and $f \in H^{1/2}(\mathbb{T})$, the Gagliardo double integral gives

$$[rf]^2_{H^{1/2}} \le 2\|r\|^2_\infty [f]^2_{H^{1/2}} + C[r]^2_{C^\alpha}\|f\|^2_2.$$

The positive Gamma series (16.57) and $T_\beta(y) \le C(\beta^{-1} + y^{-1})$ imply

$$H(\sigma_{N,k}w_\beta) \le Ck^2, \qquad 2 \le k \le N/2. \tag{16.65}$$

Consequently polarization and (16.64) give, with $\varepsilon_N = C_p\beta^{1+1/p}$,

$$\left|H(\sigma_{N,k}c_N^{-1}w_\beta q_N) - d_\circ^2 H(\sigma_{N,k}w_\beta)\right| \le C\varepsilon_N k^2 = o(k) \tag{16.66}$$

uniformly along every growing-character sequence, because $\varepsilon_N k \le C\beta^{1/p}$. The eigenvalue error in the area row has the same bound.

The sector compatibility $\int_{\partial R_N} z_i p_N \partial_\nu u_N \, ds = 0$ follows from cyclic character orthogonality. The uniform gap on $u_N^\perp$, the Poisson L^2 estimate, and $h_N \int_{\partial R_N} p_N^2 \, ds = 2\lambda_N$ show that reduced Helmholtz energy differs from harmonic extension by $O(1)$ in the unnormalized character. This is also $o(k)$ when $k \to \infty$. It follows from (16.62) and (16.66) that

$$\frac{p^{ss,\mathrm{un}}_{N,k}}{Nd_\circ^2} = P(k/N) + o(Q(k/N)) \tag{16.67}$$

on every growing-character sequence.

Finally, the exact Bessel alias formula for $g_{N,k}$, together with

$$\omega_n = |n| + 1 + O(|n|^{-1}), \qquad (1-n^2)^{-2} = n^{-4}\{1 + O(n^{-2})\},$$

gives

$$\frac{g^{\mathrm{un}}_{N,k}}{Nd_\circ^2} = Q(k/N)\{1 + o(1)\} \tag{16.68}$$

uniformly for $k \to \infty$. Indeed every alias has $|k + mN| \ge k$, and replacing $\cos(ka) - \cos a$ by $\cos(ka) - 1$ costs $O(k^{-2})$ relatively.

If (16.54) failed along a sequence, choose a violating character and conjugate it into $2 \le k_N \le N/2$. If k_N is bounded, pass to a constant subsequence; the retained one-fixed-slot disk limit gives $p^{ss}_{N,k_N}/g_{N,k_N} \to 1$. If $k_N \to \infty$, then (16.61), (16.67), and (16.68) give

$$\liminf_N \frac{p^{ss}_{N,k_N}}{g_{N,k_N}} \ge \frac{3}{5}.$$

Both alternatives contradict a ratio below 11/20. This proves (16.54). $\square$

Lemma 16.17 (Polygonal mixed Schwarz–Christoffel finite part)**.** *Fix* $0 < x < 1/2$*, put*

$$\theta = \pi x, \qquad \rho = e^{2\pi i x}, \qquad D(x) = -\{\psi(x)+\psi(1-x)+2\gamma\}, \qquad T(x) = \psi_1(x)-\psi_1(1-x) = -D'(x).$$

For $\beta = 2/N$*,* $a = \pi\beta$*, and a regular fan character with* $k/N \to x$*, the constant-flux polygonal fan–support coefficient has the phase-realified limit*

(16.69)

$$\boxed{L_{\mathrm{poly}}(x) := \lim_{\beta\downarrow 0}\left[-e^{i\theta}\frac{p_\beta^{fs}}{d_\circ^2 a}\right] = \frac{\cos\theta}{2\pi}D(x) - \frac{\sin\theta}{2\pi^2}T(x) = \frac{\sin\theta}{2\pi^2}\{D'(x) + \pi\cot\theta D(x)\}.}$$

Let

$$F_p(y) := \sum_{n\geq 0}\frac{(-1)^n}{(n+y)^p} = 2^{-p}\{\zeta(p, y/2) - \zeta(p,(y+1)/2)\}$$

and define the exact disk-forcing symbol

$$R_I(x) = -\frac{\sin\theta}{4\pi^3}\{F_3(x) - F_3(1-x)\} + \frac{\cos\theta}{4\pi^2}\{F_2(x) + F_2(1-x)\}. \tag{16.70}$$

Then the genuine polygonal completed-covector finite part exists and is

(16.71)

$$\boxed{R_{b,\mathrm{poly}}(x) = \frac{D'(x)+\pi\cot\theta D(x)}{2\pi^2} - R_I(x), \qquad \frac{A_{N,k}}{d_\circ^2 a} = -e^{-i\theta}\sin\theta R_{b,\mathrm{poly}}(x) + o_x(1).}$$

The formulas extend continuously to $x = 1/2$*, where both* R_I *and* $R_{b,\mathrm{poly}}$ *vanish.*

Proof. On one regular Schwarz–Christoffel side let

$$w_\beta(t) = |2\sin\pi t|^\beta, \qquad J_\beta(t) = \int_{1/2}^{t} w_\beta(u)^{-1}\,du, \qquad 0 < t < 1.$$

For the complex fan character $d_i = \rho^i$, differentiating the turning angles gives $\dot\theta_i = (\rho-1)^{-1}\rho^i$ and $\dot h = 0$. Centred physical arclength satisfies $ds = c_N a w_\beta^{-1} dt$ and $s = c_N a J_\beta(t)$, while the pulled-back flux is $p_N = c_N^{-1} w_\beta q_N$. Hence the exact fan–flux datum is

$$-a(\rho-1)^{-1}\rho^i w_\beta(t) J_\beta(t) q_N(t).$$

For $y = m + x$ put

$$I_\beta(y) = \int_0^1 w_\beta(t)e^{-2\pi i y t}\,dt, \qquad S_\beta(y) = \int_0^1 w_\beta(t)J_\beta(t)e^{-2\pi i y t}, dt,$$

and $M_\beta(x) = \sum_m |m+x| S_\beta(m+x)\overline{I_\beta(m+x)}$. The harmonic mixed row is $-2\pi a(\rho-1)^{-1}M_\beta(x)$ after division by $d_\circ^2$; the area row is

$$\frac{1}{4}h_N\sec^2(a/2)(1+\rho^{-1}), \qquad h_N = \sqrt{\frac{\pi}{N\tan(a/2)}}.$$

It remains to extract the joint finite part. Set

$$C_\beta = \int_0^{1/2} w_\beta^{-1} = 2^{-\beta}\frac{\Gamma((1-\beta)/2)}{2\sqrt{\pi}\,\Gamma(1-\beta/2)}.$$

Then $C_0 = 1/2$, $C_0' = 0$, and direct integration at the two endpoints gives

$$w_\beta(t)J_\beta(t) = -C_\beta(2\pi t)^\beta + \frac{t}{1-\beta} + O(t^{\beta+2}+t^3),$$
$$w_\beta(1-u)J_\beta(1-u) = C_\beta(2\pi u)^\beta - \frac{u}{1-\beta} + O(u^{\beta+2}+u^3).$$

The endpoint expansion can be used here with a summable differentiated remainder, as follows. Choose smooth cutoffs at 0 and 1, subtract from w_β its two endpoint models of type t^β, and subtract from $w_\beta J_\beta$ the four endpoint models displayed above. The remainders and their first β–derivatives have two integrable derivatives; differentiating in β merely inserts a logarithm.

Here is the resulting coefficient ledger, which also fixes the two tail signs. Put

$$a_\beta = \frac{\Gamma(1+\beta)}{2\pi}, \qquad \phi_\beta = \frac{\pi}{2}(1+\beta).$$

For $\sigma = \operatorname{sgn} y$, the two endpoint models give

$$A_+(\beta,x) = a_\beta\{e^{-i\phi_\beta} + e^{-2\pi i x}e^{i\phi_\beta}\}, \quad B_+(\beta,x) = C_\beta a_\beta\{-e^{-i\phi_\beta} + e^{-2\pi i x}e^{i\phi_\beta}\},$$
$$A_-(\beta,x) = a_\beta\{e^{i\phi_\beta} + e^{-2\pi i x}e^{-i\phi_\beta}\}, \quad B_-(\beta,x) = C_\beta a_\beta\{-e^{i\phi_\beta} + e^{-2\pi i x}e^{-i\phi_\beta}\}.$$

Thus, for $j = 0, 1$, fixed $x \in (0, 1/2]$, and $0 \le \beta \le \beta_x < 1$,

$$I_\beta(y) = A_\sigma(\beta,x)|y|^{-1-\beta} + E^I_\beta(y),$$
$$S_\beta(y) = B_\sigma(\beta,x)|y|^{-1-\beta} + E^S_\beta(y),$$

$$|\partial_\beta^j E^I_\beta(y)| + |\partial_\beta^j E^S_\beta(y)| \le C_x \frac{\{1+\log(2+|y|)\}^j}{(1+|y|)^2}. \tag{16.72}$$

The linear endpoint models are included in E^S; their coefficients are $O(|y|^{-2})$. The displayed estimate follows by two integrations by parts on the cutoff remainders. It is uniform down to $\beta = 0$ because $t^\beta \log t$ and the second derivatives of $t^{\beta+2}\log t$ are integrable uniformly on $0 \le \beta \le \beta_x$.

Direct multiplication gives

$$B_+(\beta,x)\overline{A_+(\beta,x)} = 2iC_\beta a_\beta^2 \sin(2\pi x - \pi\beta), \qquad B_-(\beta,x)\overline{A_-(\beta,x)} = 2iC_\beta a_\beta^2 \sin(2\pi x + \pi\beta).$$

If $c_\pm(\beta,x)$ denote these two coefficients and

$$R_m(\beta,x) := |m+x|S_\beta(m+x)\overline{I_\beta(m+x)} - c_{\operatorname{sgn}(m+x)}(\beta,x)|m+x|^{-1-2\beta},$$

then (16.72), followed by the product rule, gives

$$|R_m(\beta,x)| + |\partial_\beta R_m(\beta,x)| \le C_x \frac{1+\log(2+|m|)}{(1+|m|)^2}. \tag{16.73}$$

Indeed the coefficients of every subtracted singular model follow from

$$\int_0^\infty t^\alpha e^{-2\pi iyt}\,dt = \Gamma(1+\alpha)e^{-i\pi(1+\alpha)/2}(2\pi y)^{-1-\alpha}, \qquad y>0,$$

first with an exponential cutoff and then by continuation. This proves the displayed bound rather than assuming a formal asymptotic series. Its majorant is summable, so the subtracted series is C^1 at $\beta=0$ and termwise passage to the finite part is justified. The only non-summable tail coefficients are

$$c_\pm(\beta,x) = 2iC_\beta\left\{\frac{\Gamma(1+\beta)}{2\pi}\right\}^2 \sin(2\pi x\mp\pi\beta),$$

multiplying the corresponding $|m+x|^{-1-2\beta}$ tails. The linear endpoint terms produce summable order $|m|^{-2-\beta}$ terms, so no additional $\beta\zeta(1+\beta)$ finite part is possible.

More explicitly,

$$M_\beta(x) = c_+(\beta,x)\zeta(1+2\beta,x) + c_-(\beta,x)\zeta(1+2\beta,1-x) + \sum_{m\in\mathbb{Z}} R_m(\beta,x).$$

At $\beta=0$ the exact formulas below give

$$R_m(0,x) = -\frac{i\sin^2\theta}{2\pi^3}\frac{\operatorname{sgn}(m+x)}{|m+x|^2}, \qquad \sum_m R_m(0,x) = -\frac{i\sin^2\theta}{2\pi^3}T(x).$$

Also $C_0'=0$ and $\partial_\beta\Gamma(1+\beta)^2|_{\beta=0}=-2\gamma$; the opposite $\mp\pi\cos(2\theta)$ derivatives of the two sine coefficients cancel. These identities account for every finite term in the next display.

At $\beta=0$ one has

$$I_0(y) = e^{-\pi iy}\frac{\sin\pi y}{\pi y}, \qquad S_0(y) = \frac{i}{2}I_0(y)\left(\cot\theta - \frac{1}{\pi y}\right).$$

Using $\zeta(1+2\beta,q) = (2\beta)^{-1} - \psi(q) + O(\beta)$ in the two tails and summing the C^1 remainder from (16.73) yields, locally uniformly in x,

$$M_\beta(x) = \frac{i\sin(2\theta)}{4\pi^2\beta} + i\left\{\frac{\sin(2\theta)}{4\pi^2}D(x) - \frac{\sin^2\theta}{2\pi^3}T(x)\right\} + O_x(\beta).$$

Because $2i(\rho-1)^{-1}\sin(2\theta) = 1+\rho^{-1}$ and $h_N\sec^2(a/2) = 1+O(a^2)$, the pole cancels the area row. Multiplication by $-e^{i\theta}$ proves (16.69). The exact disk alias calculation gives (16.70); adding the corrector contribution $-\sin\theta R_I$ proves (16.71). Reflection about $x=1/2$ gives the stated continuous endpoint values. ☐

Proposition 16.18 (Strict support–double-disk gap)**.** *For every* $0<x\le 1/2$ *one has*

$$\boxed{P(x) > 2Q_{\mathrm{mat}}(x)}, \qquad Q_{\mathrm{mat}}(x) := \frac{\sin^4(\pi x)}{\pi^3}\{\zeta(3,x)+\zeta(3,1-x)\}. \tag{16.74}$$

Proof. Write $H_3(x) = \zeta(3,x) + \zeta(3,1-x)$. The alternating Taylor series for $\sin^2 y$ on $0 \le y \le \pi/2$ gives

$$\sin^2(\pi x) \le \pi^2 x^2 (1 - 2x^2).$$

Indeed the alternating expansion gives $\sin^2 y \le y^2 - y^4/3 + 2y^6/45$. Since $y \le \pi/2$ and $1/3 - 2/\pi^2 - \pi^2/90 > 0$, its right side is at most $y^2 - (2/\pi^2)y^4$. The convergent series for D gives

$$D(x) = \frac{1}{x} + 2x^2 \sum_{n\ge1} \frac{1}{n(n^2-x^2)} \ge \frac{1}{x} + 2\zeta(3)x^2.$$

Moreover $H_3(x) - x^{-3}$ is increasing on $[0,1/2]$, term by term, and therefore

$$H_3(x) \le x^{-3} + H_3(1/2) - 8 = x^{-3} + 14\zeta(3) - 8.$$

Using the elementary bounds $6/5 < \zeta(3) < 121/100$ (sum the first ten terms and bound the decreasing tail between $\int_{11}^{\infty} t^{-3}dt$ and $\int_{10}^{\infty} t^{-3}dt$), it follows that

$$\begin{aligned} D(x) - \frac{\sin^2(\pi x)}{\pi^2} H_3(x) &\ge 2x + (8 - 12\zeta(3))x^2 + (28\zeta(3) - 16)x^4 \\ &\ge x\left(2 - \frac{163}{25}x + \frac{88}{5}x^3\right) > 0. \end{aligned}$$

For the last inequality, the cubic bracket has its minimum at $x_0 = \sqrt{163/1320} < 44/125$, where it equals $2 - (326/75)x_0 > 0$. Since $P(x) > 2Q_{\mathrm{mat}}(x)$ is exactly the last displayed strict inequality multiplied by $2\sin^2(\pi x)/\pi$, the proof is complete. □

Remark 16.19 (Retraction of the old period-three completion)*.* The revision-21 period-three completion used the material support constant E_3. The jointly renormalized Schwarz–Christoffel calculation above shows instead that the true real-paired polygonal support finite part is

$$\frac{1}{2}P(1/3) = \frac{9\log 3}{4\pi} = 0.786823093\ldots,$$

whereas the material computation gives $E_3 = 0.77956467\ldots$. The missing difference is a same-order corner finite part. Therefore the old period-three proposition remains withdrawn. The proofs below use only the genuine polygonal diagonal and mixed finite parts computed above; they do not reinstate the material transfer.

Proposition 16.20 (Certified completed-covector ratio)**.** *For* $0 < x < 1/2$*, put*

$$J(x) := \frac{D'(x) + \pi\cot(\pi x)D(x)}{2\pi^2}.$$

Then

$$\boxed{2R_I(x) < J(x) < R_I(x) < 0.} \tag{16.75}$$

Consequently, for $0 \le \mu \le 1/4$ *and* $Q_{\mathrm{mat}} = Q/2$*, put*

$$D_{\mathrm{poly}}(x;\mu) := P(x) - (1+2\mu)Q_{\mathrm{mat}}(x).$$

Then $D_{\mathrm{poly}}(x;\mu) > 0$ *and*

(16.76)
$$\begin{aligned}\Delta_\mu(x) &:= (1-2\mu)R_I(x)^2 D_{\mathrm{poly}}(x;\mu)\\ &\quad - Q_{\mathrm{mat}}(x)\{R_{b,\mathrm{poly}}(x) - 2\mu R_I(x)\}^2 > 0 \qquad (0<x<1/2).\end{aligned}$$

At $x = 1/2$ *both covectors vanish and* $\Delta_\mu(1/2) = 0$.

Proof. Write $s = \sin(\pi x)$, $c = \cos(\pi x)$ and $A = F_2(x) + F_2(1-x)$. Since $A' = -2\{F_3(x) - F_3(1-x)\}$, one has the exact identities

$$J = \frac{(sD)'}{2\pi^2 s}, \qquad R_I = \frac{(s^2A)'}{8\pi^3 s}. \tag{16.77}$$

Thus it suffices to prove the three strict signs

$$-R_I > 0, \qquad U := R_I - J > 0, \qquad V := J - 2R_I > 0.$$

We give the finite rational certificate, including the endpoint regularizations, so that no sampled floating-point value is used.

At the left endpoint the convergent expansions are

$$D = x^{-1} + 2\sum_{r\ge 1}\zeta(2r+1)x^{2r},$$
$$\pi\cot(\pi x) = x^{-1} - 2\sum_{r\ge 1}\zeta(2r)x^{2r-1},$$
$$F_3(x) - F_3(1-x) = x^{-3} - 2\sum_{r\ge 0}\binom{2r+2}{2}\eta(2r+3)x^{2r},$$
$$F_2(x) + F_2(1-x) = x^{-2} + 4\sum_{r\ge 0}(r+1)\eta(2r+3)x^{2r+1}.$$

The singular terms cancel symbolically and give

$$J(0) = -\frac{1}{6}, \qquad R_I(0) = -\frac{1}{12}, \qquad \lim_{x\downarrow 0}\frac{V(x)}{x} = \frac{3\zeta(3)}{4\pi^2} > 0.$$

Ten retained orders, $q = x^2$, and the elementary bounds $\zeta(r) < 121/100$ for $r \ge 3$ and $\zeta(r) < 5/3$ for $r \ge 2$ bound the omitted $D, D', D/x, \cot$, and divided-cotangent tails by

$$\frac{121}{50}\frac{x^{22}}{1-q}, \quad \frac{121}{25}x^{21}\frac{11-10q}{(1-q)^2}, \quad \frac{121}{50}\frac{x^{21}}{1-q},$$
$$\frac{10}{3}\frac{x^{21}}{1-q}, \qquad \frac{10}{3}\frac{x^{20}}{1-q}.$$

The two alternating-F tails are dominated by

$$\frac{8(12)^2q^{11}}{(1-q)^3}, \qquad \frac{4x(12)q^{11}}{(1-q)^2},$$

with the divided second bound obtained by deleting x. Directed rational evaluation on $0 \le x \le 1/8$ then gives

$$R_I < -0.064086, \qquad U > 0.035346, \qquad V/x > 0.034494. \tag{16.78}$$

For the right endpoint put $y = 1/2 - x$ and $C = \cos(\pi y)$. Then

$$D = 4\log 2 + 2\sum_{n\geq 1}(2^{2n+1} - 1)\zeta(2n+1)y^{2n},$$

$$A = \sum_{n\geq 0}(2n+1)2^{2n+3}\beta_D(2n+2)y^{2n}.$$

If $\mathcal{A} = (CD)_y$ and $\mathcal{B} = (C^2A)_y$, the endpoint zeros factor exactly as

$$-R_I/y = \frac{\mathcal{B}/y}{8\pi^3 C}, \quad U/y = \frac{4\pi\mathcal{A}/y - \mathcal{B}/y}{8\pi^3 C}, \quad V/y = \frac{\mathcal{B}/y - 2\pi\mathcal{A}/y}{4\pi^3 C}.$$

With nine retained orders and $q = 4y^2$, the positive tails for $D, D_y/y, A, A_y/y$ are bounded respectively by

$$\frac{121}{25}\frac{q^{10}}{1-q}, \quad \frac{968}{25}q^9\frac{10-9q}{(1-q)^2}, \quad \frac{8(21)q^{10}}{(1-q)^2}, \quad \frac{128(10)(21)q^9}{(1-q)^3}.$$

Directed rational evaluation on $0 \leq y \leq 1/32$ gives

$$-R_I/y > 0.167528, \qquad U/y > 0.105954, \qquad V/y > 0.001213. \tag{16.79}$$

It remains only the compact interval $[1/8, 15/32]$. The executable certificate `tests/check_jc_ar_ratio_symbol.py` partitions it into the 11,264 exact rational cells

$$[i/32768, (i+1)/32768], \qquad 4096 \leq i \leq 15359.$$

On each cell it evaluates twelve terms of the positive D, D' series and the alternating F_2, F_3 series. Their omitted tails are bounded by

$$\frac{4x^2}{3\cdot 12^2}, \qquad \frac{32x}{9\cdot 12^2}, \qquad (12+z)^{-p},$$

respectively. The rational enclosure

$$\frac{103993}{33102} < \pi < \frac{104348}{33215}$$

comes from Machin's identity; sine and cosine use alternating Taylor series with their signed next terms. Every arithmetic operation is rounded outwards at 64 decimal digits. In particular, sign reversal is exact and integer powers use only directed interval multiplication. The finite minima are

$$-R_I > 0.0056578993130883869, \quad U > 0.0038837669640606282, \quad V > 0.0009703006655867654. \tag{16.80}$$

Equations (16.78)– (16.80) prove (16.75). The script was independently audited against exact rational arithmetic, including a hostile seven-digit ambient decimal context; the same 64-digit enclosures result.

Finally Lemma 16.17 gives $R_{b,\mathrm{poly}} = J - R_I$. Hence $0 < t := R_{b,\mathrm{poly}}/R_I < 1$. For $0 \leq \mu \leq 1/4$, $|t - 2\mu| \leq 1 - 2\mu$. Therefore

$$\Delta_\mu \geq (1-2\mu)R_I^2\{P - 2Q_{\mathrm{mat}}\} > 0$$

by Proposition 16.18. This proves (16.76). $\square$

Lemma 16.21 (Uniform polygonal mixed bridge)**.** *Let $\beta = 2/N$, $a = \pi\beta$, $x = k/N$, and $2 \le k \le N/2$. The constant-flux harmonic-plus-area mixed coefficient $p_\beta^{fs,0}$ satisfies*

$$\left| -e^{i\pi x}\frac{p_\beta^{fs,0}}{d_\circ^2 a} - L_{\mathrm{poly}}(x) \right| \le C\left(\beta + \frac{\beta^2}{x}\right). \tag{16.81}$$

The actual reduced Helmholtz-minus-harmonic mixed modal row satisfies

$$\boxed{|E_{N,k}^{fs}| \le C_p a^2.} \tag{16.82}$$

After replacing the constant flux by the true polygonal flux, the complete mixed and diagonal ledgers have the following consequence. If $k_N \to \infty$, their Schur quotient differs from the limiting proxy determined by (16.76) by $o(a^2)$; if k_N is bounded, the entire mixed Schur quotient is $o(a^2)$.

Proof. For (16.81), keep the $m = 0$ term of $M_\beta(x)$ exact. Reflection gives $S_\beta(0) = 0$, so that term is $x^2 O_{C^2}(1)$. On $|m| \ge 1$, subtract the t^β endpoint models from $I_{\beta,m}$ and the t^β and linear endpoint models from $S_{\beta,m}$. Two integrations by parts give, for $0 \le \ell \le 1$ and $0 \le j \le 2$,

$$|\partial_x^\ell \partial_\beta^j R_m(\beta, x)| \le C\frac{1 + \log^2(2 + |m|)}{(1 + |m|)^2}. \tag{16.83}$$

The removed $|m + x|^{-1-2\beta}$ families are first summed as paired Hurwitz zeta functions. After subtracting their common pole and finite part, their Laurent remainders are jointly $C^{1,2}$ for $0 \le \beta \le x \le 1/2$. Reflection makes the first β coefficient vanish at $x = 0$. Two-variable Taylor's formula therefore gives

$$\left| M_\beta(x) - \frac{i\sin(2\pi x)}{4\pi^2\beta} - i\left\{ \frac{\sin(2\pi x)}{4\pi^2}D(x) - \frac{\sin^2(\pi x)}{2\pi^3}T(x) \right\} \right| \le C(\beta x + \beta^2).$$

The exact area pole cancellation proves (16.81).

We prove (16.82), which is the reduced row omitted in the first bridge attempt. Pull the regular polygon to the disk and write $V_N = \lambda_N |F_N'|^2$. In a nontrivial character sector let $U_N f$ solve $(-\Delta - V_N)U_N f = 0$ with boundary value f, and set

$$\mathcal{K}_N(f, g) = \langle (\Lambda_{V_N} - \Lambda_0)f, g\rangle.$$

The ground state is invariant, so every retained sector is orthogonal to it. The first/second spectral gap is uniform; hence, for fixed $p > 4$ and $q = 2p/(p-1)$,

$$\|U_N f\|_{L^q} + \|Hf\|_{L^q} \le C_p \|f\|_\infty. \tag{16.84}$$

The retained SC weight estimate gives $\|V_N - \lambda_\circ\|_{L^p} \le C_p \beta^{1+1/p}$. Alessandrini's identity and (16.84) yield

$$|\mathcal{K}_N(f, g) - \mathcal{K}_\circ(f, g)| \le C_p \beta^{1+1/p} \|f\|_\infty \|g\|_\infty. \tag{16.85}$$

For the unnormalized constant-flux traces,

$$f^s = d_\circ \sigma_{N,k} w_\beta, \qquad f^f = -d_\circ a(\rho-1)^{-1}\sigma_{N,k} w_\beta J_\beta.$$

Their Fourier coefficients at $n = k + mN$ are, apart from the displayed fixed factors, $I_{\beta,m}$ and $S_{\beta,m}$. Uniformly for $0 < \beta \le x \le 1/2$,

$$|I_{\beta,0}| \le C, \quad |S_{\beta,0}| \le Cx, \qquad |I_{\beta,m}| \le \frac{Cx}{1+|m|}, \quad |S_{\beta,m}| \le \frac{C}{1+|m|} \quad (m \ne 0).$$

For the constant disk potential the reduced-minus-harmonic multiplier is

$$\varkappa_n = j_{0,1}\frac{J'_{|n|}(j_{0,1})}{J_{|n|}(j_{0,1})} - |n| = -j_{0,1}\frac{J_{|n|+1}(j_{0,1})}{J_{|n|}(j_{0,1})}, \qquad |\varkappa_n| \le \frac{C}{1+|n|}.$$

Here $|n| \ge 2$, so there is no ground-state resonance. It follows that

$$|\mathcal{K}_\circ(f^f, f^s)| \le C\frac{a}{x}\left\{\frac{x}{1+Nx} + \sum_{m\ne 0}\frac{x}{N(1+|m|)^3}\right\} \le Ca.$$

The true flux obeys $\|q_N - d_\circ\|_\infty \le C_p a^{1+1/p}$; the Green identity $\mathcal{K}_N(f,g) = -\int V_N U_N f\,\overline{Hg}$ and (16.84) show that replacing either slot changes the preceding unnormalized form by $o(a)$. Finally an unnormalized character is $\sqrt{N}$ times an orthonormal one in each mixed slot. Division by $N = 2\pi/a$ proves (16.82).

For completeness, define the positive side-mass normalization by

$$\bar{q}_N^2 := \frac{\int_0^1 w_\beta q_N^2}{\int_0^1 w_\beta}.$$

The regular-side stationarity identity $\int_{S_i} p_N^2\,ds = (\lambda_N/\pi)L_i$, together with $p_N = c_N^{-1} w_\beta q_N$ and $ds = c_N a w_\beta^{-1} dt$, gives exactly

$$\bar{q}_N^2 = \frac{\lambda_N}{\pi} c_N^2 \frac{\int_0^1 w_\beta^{-1}}{\int_0^1 w_\beta} = \frac{\lambda_N}{\pi}\{1 + O(\beta^3)\}.$$

Here $c_N = 1 + O(\beta^3)$ and the expansions of $\int w_\beta^{\pm 1}$ have the same quadratic coefficient because $\int_0^1 \log|2\sin \pi t|\,dt = 0$. Moreover q_N is cell-periodic and $\bar{q}_N$ lies between its minimum and maximum on a cell. Rescaling a cell of length $O(\beta)$ to $[0,1]$ in (16.64) therefore proves

$$\|q_N - \bar{q}_N\|_{C^\alpha_{\mathrm{cell}}} \le C_p \beta^{2-1/p}, \quad \alpha = 1 - 2/p > 1/2.$$

The endpoint coefficient bounds used above also give

$$\sum_m |m+x||I_{\beta,m}|^2 \le C\frac{x^2}{\beta}, \qquad \sum_m |m+x||S_{\beta,m}|^2 \le \frac{C}{\beta}.$$

Multiplication by a C^α_{cell} function is bounded on this Bloch $H^{1/2}$ space; polarization and $a|\rho-1|^{-1} \le C\beta/x$ therefore show that replacing the constant flux by its cell oscillation changes the harmonic mixed row by at most

$C_p\beta^{2-1/p}$. The disk aliases give uniformly

$$cx \le g_{N,k} \le Cx, \qquad |Z_{N,k}| \le Ca, \qquad |g_{N,k}Z_{N,k}| \le Cax. \tag{16.86}$$

On every growing-character sequence, (16.81), (16.82), the support asymptotic, and the preceding true-flux estimate give, for the limiting proxies denoted by bars,

$$\begin{aligned} |\bar A| &\le Cax, & r, \bar r &\ge cx, \\ |A - \bar A| &\le o(ax) + C_p a^{2-1/p} + Ca^2, & |r - \bar r| &= o(x). \end{aligned}$$

Elementary perturbation of $|A + \mu_s g\overline{Z}|^2/r$ then costs

$$o(a^2x) + C_p a^{3-1/p} + C_p a^{4-2/p}/x + O(a^3 + a^4/x) = o(a^2),$$

because $x \ge \beta \asymp a$ and $p > 4$. If k stays bounded, the fixed-slot limit and (16.86) give $r \ge c_k a$, while the numerator is $O_k(a^2) + O_p(a^{2-1/p})$; its quotient is $O_p(a^{3-2/p}) = o(a^2)$. This proves the last assertion. □

Proposition 16.22 (Closure of the regular completed-action Schur block)**.** *For all sufficiently large N, Contract 16.14 holds with*

$$\boxed{\mu_s = \frac{1}{40}, \qquad \mu_f = \kappa.} \tag{16.87}$$

In particular, the all-character matrix (16.47) *is positive semidefinite.*

Proof. The connection between the certified determinant and the Schur row is division-free. The limiting ledgers give

$$\begin{aligned} \bar g &= 2d_\circ^2 Q_{\mathrm{mat}}, & \bar g\,\overline{\bar Z} &= 2d_\circ^2 a e^{-i\pi x}\sin(\pi x) R_I, \\ \bar A &= -d_\circ^2 a e^{-i\pi x}\sin(\pi x) R_{b,\mathrm{poly}}, & \bar r &= d_\circ^2 D_{\mathrm{poly}}(x;\mu_s). \end{aligned}$$

Consequently the limiting part of (16.46) after deletion of the fan reserve is exactly

$$\left(\frac{1}{2} - \mu_s\right)\bar g|\bar Z|^2 - \frac{|\bar A + \mu_s \bar g\,\overline{\bar Z}|^2}{\bar r} = \frac{d_\circ^2 a^2 \sin^2(\pi x)}{Q_{\mathrm{mat}} D_{\mathrm{poly}}(x;\mu_s)}\,\Delta_{\mu_s}(x). \tag{16.88}$$

Both denominators are positive by Proposition 16.18.

The support theorem gives $p^{ss}_{N,k} \ge (1/2 + 1/20)g_{N,k}$, hence the denominator in (16.46), with $\mu_s = 1/40$, satisfies $r_{N,k} \ge g_{N,k}/40 > 0$. Suppose the claimed Schur row failed along a sequence of characters. After conjugation take $2 \le k_N \le N/2$. If k_N is bounded, Lemma 16.21 makes the negative quotient and the corrector term $o(a^2)$; the untouched fan term is $(3\kappa - \mu_f)a^2 = 2\kappa a^2$, a contradiction.

Otherwise pass to a subsequence with $k_N \to \infty$. The same lemma compares the quotient, including its denominator and corrector, with the limiting proxy up to $o(a^2)$. Proposition 16.20 makes that proxy nonnegative for $\mu_s = 1/40$, including sequences with $x_N \downarrow 0$; at $x_N \to 1/2$ its possible zero is again paid by the same $2\kappa a^2$ fan reserve. Thus the actual row is

$2\kappa a^2 - o(a^2) > 0$, another contradiction. Every character sequence has one of these two alternatives, proving the assertion uniformly. $\square$

Remark 16.23 (Why stationary O–H does not prove JC–AR). On an even regular fan take the alternating support vector $z_i = (-1)^i$. The exact polygonal reconstruction in the CT node gives

$$\|V_z\|^2_{H^{1/2}(\mathbb{T};\mathbb{R}^2)} \asymp Na^{-2}, \qquad \mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(z) \asymp N.$$

Thus the full-vector norm in stationary O–H is larger than the stable normal-sine metric by the sharp factor a^{-2}. Its error can therefore be as large as

$$C\omega_{OH}(a)a^{-2}\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(z).$$

The proved statement $\omega_{OH}(a) \to 0$ supplies neither the rate $O(a^2)$ nor a sufficiently small leading constant. Moreover its radial disk trace is not literally the normal-sine completed trace, and it controls the full Hessian rather than the base-subtracted action root (16.40). Hence O–H cannot be used to assert either (16.48) or (16.47).

Proof contract 16.24 (JC–SP: remaining super-near completed-action modulus and output)**.** Suppose the regular completed-action contract JC–AR holds. Let

$$\mathcal{N}_N(d,w) := a^2\sum_i d_i^2 + \mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w), \qquad \mathcal{N}_N(d,w) \le Ma^5. \tag{16.89}$$

For the exact completed chart of JC–AR also put

$$\mathfrak{E}_N(d,w) := \mathfrak{J}_4(\alpha_* + d) + \mathfrak{G}^{\mathrm{JP}}_{\alpha_*+d}(J_{\alpha_*+d}w). \tag{16.90}$$

The finite-coordinate row needed here is closed by Proposition 16.27: uniformly for fixed M,

$$c_M\mathcal{N}_N(d,w) \le \mathfrak{E}_N(d,w) \le C_M\mathcal{N}_N(d,w) \tag{16.91}$$

whenever either side is at most a fixed multiple of a^5. This row includes the moving metric, the J_α transport, the quotient gauges, and their uniform constants. The exact finite corrector cancels here because $z - z_\alpha^D = J_\alpha w$; its nonlinear variation instead belongs to the completed chart and Hessian modulus below.

The exact fourth-defect identity gives the fan part of this bridge and the radius

$$\frac{\|d\|_{\ell^\infty}}{a} + a^{-2}\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w)^{1/2} \le C_Ma^{1/2}. \tag{16.92}$$

Indeed,

$$\mathfrak{J}_4(\alpha_* + d) = \sum_i d_i^2(6a^2 + 4ad_i + d_i^2) \ge 3a^2\sum_i d_i^2$$

for $d_i \ge -a$ and $\sum_i d_i = 0$; the last coefficient is sharp at $d_i = -a$. The support part of (16.91) is the moving-low-mode transversality content of Proposition 16.27; it is not inferred from a bare interchange of the two meshes. Moreover, applying the closed corrector estimate to α_*+td, dividing by t^2, and letting $t \to 0$ gives

$$\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(Z_N d) \le Ca^2 \sum_i d_i^2. \tag{16.93}$$

Thus the initial physical support velocity $\zeta = w + Z_N d$ also satisfies $\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(\zeta) \le C_M a^5$.

Use the affine path in the exact completed coordinates

$$q_t = (td, tw), \qquad \alpha_t = \alpha_* + td, \qquad z_t = J_{\alpha_t}(\bar{z}^D_{\alpha_t} + tw), \qquad 0 \le t \le 1. \tag{16.94}$$

Proposition 16.27 proves that this path remains in the active convex chart and that all chart maps above are defined uniformly. Its Hessian is taken after composition with the exact scalar $\mathbf{A}_N$; hence the moving fan, support, corrector, gauge, and coordinate-acceleration terms are all part of the Hessian. The earlier proposed affine chord of physical vertices has zero vertex acceleration, but its initial chart tangent need not equal the endpoint difference (d, w). It is therefore not used here without a separate tangent-energy comparison.

For later estimates the exact completed scalar has the following corrector-free normal form. Put $c_\alpha = \bar{z}^D_\alpha$, so that $z^D_\alpha = J_\alpha c_\alpha$. Then

$$\boxed{\begin{aligned}\mathbf{A}_N(d,w) &= \Phi_\alpha(h_\alpha \mathbf{1} + J_\alpha(c_\alpha + w)) - \Phi_\alpha(h_\alpha \mathbf{1}) \\ &\quad + \mathfrak{G}^{\mathrm{JP}}_\alpha(J_\alpha c_\alpha) - \frac{1}{2}\mathfrak{G}^{\mathrm{JP}}_\alpha(J_\alpha w) + \frac{\kappa}{2}\mathfrak{J}_4(\alpha), \qquad \alpha = \alpha_* + d.\end{aligned}} \tag{16.95}$$

Indeed $p_\alpha[\xi] = -2g_\alpha(z^D_\alpha, \xi)$, so the forcing, metric, and completed square combine exactly into $\mathfrak{G}^{\mathrm{JP}}_\alpha(z^D_\alpha) - \frac{1}{2}\mathfrak{G}^{\mathrm{JP}}_\alpha(J_\alpha w)$. Thus (16.95) retains the moving fan, support chart, quotient gauge, corrector, base-value subtraction, moving metric, 1/2 normalization, and fourth-defect row in one scalar.

The only remaining analytic row of JC–SP is a relative modulus for the Hessian of the completed action itself. For every fixed M there must be $\varepsilon_M(a) \to 0$ such that

$$\boxed{\Big|D^2\mathbf{A}_N(q_t)[(d,w),(d,w)] - D^2\mathbf{A}_N(0)[(d,w),(d,w)]\Big| \le \varepsilon_M(a)\mathcal{N}_N(d,w) \quad (0 \le t \le 1).} \tag{16.96}$$

A bound $\varepsilon_M(a) = O_M(a^{1/2})$ would be sufficient. The finite rank scale/translation gauge change, moving corrector, and base-value subtraction are part of the left side and must not be dropped.

One stronger uniform operator formulation, sufficient for (16.96), is the following. Let R_N be the Riesz operator of $\mathcal{N}_N$ on the fixed regular quotient and put $H_N(q) = D^2\mathbf{A}_N(q)$. With

$$r_N(q) = \frac{\|d(q)\|_{\ell^\infty}}{a} + a^{-2}\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w(q))^{1/2},$$

the natural operator statement is

$$\boxed{\Omega_N(r) := \sup_{r_N(q)\le r} \left\| R_N^{-1/2}\{H_N(q) - H_N(0)\}R_N^{-1/2} \right\|_{\mathrm{op}} \longrightarrow 0} \tag{16.97}$$

as $r \downarrow 0$, uniformly in N. A Lipschitz estimate $\Omega_N(r) \le Cr$, together with $r_N(q_t) \le C_M a^{1/2}$ from the chart/energy bridge, would imply the stated $O_M(a^{1/2})$ rate. This operator statement is stronger than the diagonal path estimate and is recorded only as a sufficient formulation. In block form it must simultaneously contain

$$\begin{aligned} |\Delta H_{ff}[d_1, d_2]| &\le Cr\, a^2 \|d_1\|_2 \|d_2\|_2, \\ |\Delta H_{fs}[d, u]| &\le Cr\, a \|d\|_2 \mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(u)^{1/2}, \\ |\Delta H_{ss}[u_1, u_2]| &\le Cr\, \mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(u_1)^{1/2} \mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(u_2)^{1/2}. \end{aligned}$$

The nonzero period-three mixed row is therefore retained rather than estimated as a separate remainder.

The absolute analytic CT estimate is not (16.96). On the alternating support mode it again loses a^{-2}, so a generic full-vector modulus gives only $a^{-2}\omega(C_M a^{1/2})\mathcal{N}_N$. Analytic Lipschitz continuity or $\omega(r) \to 0$ does not make this quantity small. The remaining part of JC–SP therefore is an open natural-operator statement.

There is a scalar reduction which is weaker than proving the full operator modulus. For $q = (d, w)$ define

$$f_q(\zeta) := \mathbf{A}_N(\zeta d, \zeta w), \qquad r_N(q) := \frac{\|d\|_{\ell^\infty}}{a} + a^{-2}\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w)^{1/2}.$$

Suppose there are absolute $c_0, C_0 > 0$ such that every exact completed coordinate in (16.95) extends holomorphically to $|\zeta| r_N(q) \le c_0$ and

$$\boxed{|f_q''(\zeta)| \le C_0 \mathcal{N}_N(d, w)} \tag{16.98}$$

there. If $r_N(q) \le c_0/2$, Cauchy's estimate on the disk of radius $c_0/(2r_N(q))$ about $t \in [0, 1]$ gives

$$|f_q'''(t)| \le \frac{2C_0}{c_0} r_N(q) \mathcal{N}_N(d, w).$$

Since the completed-coordinate path is affine, $f_q''(t) = D^2\mathbf{A}_N(q_t)[q, q]$. Integration therefore proves

$$\boxed{|D^2\mathbf{A}_N(q_t)[q, q] - D^2\mathbf{A}_N(0)[q, q]| \le \frac{2C_0}{c_0} r_N(q) \mathcal{N}_N(d, w).} \tag{16.99}$$

The bridge gives $r_N(q) = O_M(a^{1/2})$, so (16.98) would close the required path modulus with $\varepsilon_M(a) = O_M(a^{1/2})$.

The precise analytic blocker behind (16.98) is a nonstationary quasiuniform resummed completed-jet identity, abbreviated QRCI. It must be obtained by differentiating the exact scalar (16.95); the stage-130 polygon-minus-disk longitudinal ledger is retracted and is not such a derivative. Terminal two-ray reduced-DtN, its full-cell complement, the negative cotangent/area pole, and every endpoint-localized derivative of $h_\alpha, J_\alpha, c_\alpha$, the moving metric, and the base subtraction must be combined before absolute values are taken. The existing full-vector Hessian estimate loses a^{-2} on alternating support data, while the available mesh-free completed estimate is stationary and uses a degenerate radial comparison form. Neither proves QRCI.

Remark 16.25 (The scalar radial-chart shortcut fails at the critical index). One cannot obtain the missing bound by differentiating only the scalar radial graph. If the fixed-normal support numbers move as $h_i(t) = h_i + tz_i$, then on side i $R_t(\varphi) = (h_i + tz_i)/\cos(\varphi - \theta_i)$. At a regular vertex the two one-sided traces of the formal derivative differ by

$$J_i = \frac{z_{i+1} - z_i}{\cos(a/2)}. \tag{16.100}$$

Every nonconstant cyclic character has a nonzero such jump. A jump is not in $H^{1/2}(\mathbb{T})$, because its Slobodetskii integral contains

$$\frac{|J_i|^2}{4} \int_0^{\varepsilon/2} \int_0^{\varepsilon/2} \frac{du\,dv}{(u+v)^2} = +\infty.$$

Thus $t \mapsto R_t$ is not differentiable as an $H^{1/2}$-valued curve, so the scalar CT analytic-Hessian theorem cannot be invoked. Freezing the vertices by a continuous physical affine-side field removes the jump but restores the already recorded a^{-2} loss. This leaves the endpoint-resummed NQCH cancellation as a genuine analytic input.

Remark 16.26 (Exact radius resummation, retraction of DAR, and the live completed third jet). The artificial endpoint radius is not itself an obstruction. If a homogeneous full-cell corner power has amplitude A, exponent β, and the terminal window has relative radius ϑ, then in either the common or jump channel, with U_β and D_β denoting the exact two-ray common and jump coefficients,

$$A^2\vartheta^{2\beta}K_\beta + A^2(1-\vartheta^{2\beta})K_\beta = A^2K_\beta, \qquad K_\beta \in \{U_\beta, D_\beta\}.$$

Hence the apparent $\beta \log \beta$ loss created by stopping at $\vartheta = \beta$ cancels exactly, including after differentiation along the completed ray.

The former amplitude row DAR is retracted as an independent contract. It was written, with

$$K_i(t) = \frac{D_{\beta_i(t)}}{\pi} j_i(q;t)^2,$$

as $\sum_i (A_i^2)' K_i + (\sum_i C_i^{\rm end})'$. But the exact endpoint jet is unchanged by the analytic reallocation

$$\widetilde{A}_i^2 = A_i^2 + \gamma_i, \qquad \widetilde{C}_i^{\rm end} = C_i^{\rm end} - \gamma_i K_i,$$

whereas the displayed DAR expression changes by $-\sum_i \gamma_i K_i'$. Choosing a sufficiently small constant nonzero γ_i and a nonconstant K_i makes this change nonzero while preserving $\widetilde{A}_i^2 > 0$ locally whenever $A_i^2 > 0$. Thus the old row depends on how the full-cell amplitude is separated from its endpoint descendants; it is not a gauge-invariant statement about the scalar (16.95).

The replacement live row is the third derivative of that scalar itself. For $q = (d, w)$ and $q_t = tq$, put $f_q(t) = \mathbf{A}_N(q_t)$. The required real estimate is

$$\boxed{|f_q'''(t)| = |D^3 \mathbf{A}_N(q_t)[q, q, q]| \le C r_N(q) \mathcal{N}_N(q), \qquad 0 \le t \le 1.} \tag{16.101}$$

It is gauge invariant and includes the full-cell flux, both terminal channels, their exact complements, every endpoint and nonendpoint chain-rule descendant, the moving metric, the corrector, the quotient transport, and the base subtraction.

For clarity, it is obtained directly from the normal form. If

$$Y_t = h_{\alpha_t} \mathbf{1} + J_{\alpha_t}(c_{\alpha_t} + tw), \quad B_t = h_{\alpha_t} \mathbf{1}, \quad C_t = J_{\alpha_t} c_{\alpha_t}, \quad W_t = t J_{\alpha_t} w,$$

and $X_t = (\alpha_t, Y_t)$, then for every scalar family F

$$\frac{d^3}{dt^3} F(X_t) = D^3 F(X_t)[X_t', X_t', X_t'] + 3D^2 F(X_t)[X_t', X_t''] + DF(X_t)[X_t'''].$$

Apply this identity to the two Φ terms and the two $\mathfrak{G}^{\rm JP}$ terms in (16.95), with Y_t, B_t, C_t, W_t respectively, and add the exact fourth-defect derivative. In particular

$$\frac{\kappa}{2} \frac{d^3}{dt^3} \mathfrak{I}_4(\alpha_* + td) = 12\kappa \left(a \sum_i d_i^3 + t \sum_i d_i^4 \right),$$

whose absolute value is at most $C r_N(q) a^2 \sum_i d_i^2$. No stage-130 ledger enters this chain rule.

The complex-ray bound (16.98) implies (16.101) by the Cauchy estimate already displayed above. Conversely, (16.101) directly implies the real path modulus (16.96) by integration, but does not by itself assert a holomorphic-tube bound. Either route would suffice for JC–SP. Neither is proved here; renaming the missing estimate does not close QRCI or the main theorem.

Its required output is nevertheless exact. Put $\mu_0 = \min\{\mu_f, \mu_s\}$. Taylor's formula in the completed chart gives

$$\mathbf{A}_N(d, w) = \int_0^1 (1 - t) D^2 \mathbf{A}_N(q_t)[(d, w), (d, w)] \, dt.$$

Thus JC–AR and (16.96) imply, for sufficiently small a depending only on M,

$$\boxed{\mathbf{A}_N(d,w) \geq \left(\mu_0 - \frac{1}{2}\varepsilon_M(a)\right)\mathcal{N}_N(d,w) \geq 0.} \tag{16.102}$$

Together with the energy bridge (16.91), this is precisely the super-near action statement (16.119); it is an output of JC–SP, not a fourth assumed row.

Proposition 16.27 (Canonical finite bridge and completed-path activity)**.** *There is a canonical analytic choice of the scale/translation trivialization J_α near the regular fan for which* (16.91) *holds, with constants independent of N, whenever either side is at most Ma^5. For this choice the whole path* (16.94) *remains in the active convex chart, and J_{α_t}, its inverse, and the exact corrector are defined uniformly for $0 \leq t \leq 1$.*

Proof. Write $\alpha_i = a + d_i$, $\sum_i d_i = 0$, and $\varrho = \|d\|_{\ell^\infty}/a$. Choose normal angles with $\theta_0 = 0$ and define the material map, for $0 \leq s \leq a$, by

$$F_d(ia+s) = \theta_i + \frac{\alpha_i}{a}s.$$

Thus

$$\|F_d' - 1\|_{L^\infty} \leq \varrho, \qquad \|F_d - \mathrm{id}\,\|_{L^\infty} \leq 2\pi\varrho. \tag{16.103}$$

Composition by F_d and F_d^{-1} is consequently bounded on $H^{1/2}(\mathbb{T})$ uniformly when $\varrho \leq 1/2$, as follows directly from the Gagliardo double-integral norm.

Direct integration of the two sine basis functions gives

$$L_\alpha z := \sum_i \left\{\tan\frac{\alpha_{i-1}}{2} + \tan\frac{\alpha_i}{2}\right\} z_i = \int_{\mathbb{T}} \mathcal{I}_\alpha z. \tag{16.104}$$

The side with normal θ_i in the tangential reference polygon has length $h_\alpha\{\tan(\alpha_{i-1}/2) + \tan(\alpha_i/2)\}$, so $L_\alpha z = 0$ is exactly the affine scale row. For $b \in \mathbb{R}^2$ put $\tau_\alpha(b)_i = b \cdot e_r(\theta_i)$. Since first harmonics solve $f'' + f = 0$ on each cell,

$$\mathcal{I}_\alpha \tau_\alpha(b) = b \cdot e_r \tag{16.105}$$

exactly.

Let X_* be the regular nodal complement of characters $0, 1, N-1$. For $w \in \mathsf{X}_*$ define

$$c_\alpha(w) = \frac{L_\alpha w}{L_\alpha \mathbf{1}}, \qquad f_\alpha(w) = \mathcal{I}_\alpha(w - c_\alpha(w)\mathbf{1}),$$

let $\beta_\alpha(w)$ be the L^2 projection coefficient of $f_\alpha(w)$ onto $\{\cos\theta, \sin\theta\}$, and set

$$J_\alpha w = w - c_\alpha(w)\mathbf{1} - \tau_\alpha(\beta_\alpha(w)). \tag{16.106}$$

Equations (16.104) and (16.105) show that $J_\alpha w$ lies in the moving scale/translation quotient and that $J_{\alpha_*} = I$.

We prove uniform transversality. On a material cell, with $r = s/a$, the two pulled-back basis functions are

$$B_0(\alpha_i, r) = \frac{\sin((1-r)\alpha_i)}{\sin\alpha_i}, \qquad B_1(\alpha_i, r) = \frac{\sin(r\alpha_i)}{\sin\alpha_i}.$$

Taylor expansion, cellwise integration, and real interpolation give, for $\|v\|_{\ell^2_a}^2 = a\sum_i |v_i|^2$,

$$\|\mathcal{I}_\alpha v \circ F_d - \mathcal{I}_* v\|_{L^2} \le C\varrho a^2 \|v\|_{\ell^2_a}, \tag{16.107}$$

$$\|\mathcal{I}_\alpha v \circ F_d - \mathcal{I}_* v\|_{H^{1/2}} \le C\varrho a^{3/2} \|v\|_{\ell^2_a}. \tag{16.108}$$

Indeed the pointwise basis error is $O(\varrho a^2)$ and its physical derivative is $O(\varrho a)$.

Cyclic equivariance implies that the continuous Fourier support of the regular sine interpolant of a nodal character k lies in $k + N\mathbb{Z}$. Hence $\mathcal{I}_* w$ is exactly orthogonal to $1, \cos\theta, \sin\theta$, not merely asymptotically so. The regular mass matrix and the disk multiplier give

$$\|w\|_{\ell^2_a} \le C\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w)^{1/2}. \tag{16.109}$$

Since $|\tan(\alpha_i/2) - \tan(a/2)| \le C\varrho a$ and $L_\alpha \mathbf{1} \asymp 1$, weighted Cauchy–Schwarz yields

$$|c_\alpha(w)| \le C\varrho\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w)^{1/2}. \tag{16.110}$$

Changing variables $\theta = F_d(s)$ in the first-harmonic coefficients, subtracting the exactly zero regular coefficient, and using (16.103)–(16.110) gives

$$|\beta_\alpha(w)| \le C\varrho\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w)^{1/2}. \tag{16.111}$$

No inverse estimate enters this step: the multiplier $e_r(F_d)F_d' - e_r$ is $O(\varrho)$ in L^∞.

Put $T_\alpha w = \mathcal{I}_\alpha J_\alpha w$. The functions $(\mathcal{I}_\alpha \mathbf{1}) \circ F_d$ and $e_r \circ F_d$ have uniformly bounded $H^{1/2}$ norms. Therefore (16.108), (16.110), and (16.111) imply

$$\|T_\alpha w \circ F_d - \mathcal{I}_* w\|_{H^{1/2}} \le C\varrho\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w)^{1/2}. \tag{16.112}$$

Now $T_\alpha w$ is exactly orthogonal to the three disk null modes, so $q_D(T_\alpha w) \asymp \|T_\alpha w\|_{H^{1/2}}^2$. Absorbing the right side of (16.112) for a universal $\varrho \le \varrho_0$ proves

$$c\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w) \le \mathfrak{G}^{\mathrm{JP}}_{\alpha}(J_\alpha w) \le C\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(w). \tag{16.113}$$

It also proves that J_α is injective. The two quotient spaces have dimension $N-3$, hence it is onto and has a uniform inverse.

The exact identity

$$\mathfrak{J}_4(\alpha_* + d) = \sum_i d_i^2(6a^2 + 4ad_i + d_i^2)$$

has coefficient at least $3a^2$ for every $d_i \ge -a$. If either side of (16.91) is at most Ma^5, it follows first that $\varrho \le C_M a^{1/2}$, hence $\varrho \le \varrho_0$ for large N. The same coefficient is then bounded above by a universal multiple of a^2.

Combining this fact with (16.113) proves (16.91). The corrector does not occur in this comparison because $z - z_\alpha^D = J_\alpha w$ identically.

It remains to prove activity. If $\Lambda_{\alpha,i}(z)$ denotes the change of side i under a support increment z, the exact support-line formula is

(16.114)

$$\Lambda_{\alpha,i}(z) = \frac{z_{i+1} - z_i \cos\alpha_i}{\sin\alpha_i} + \frac{z_{i-1} - z_i \cos\alpha_{i-1}}{\sin\alpha_{i-1}} = f'(\theta_i+) - f'(\theta_i-), \quad f = \mathcal{I}_\alpha z.$$

After subtracting an exact translation, which changes no side length, the quasiuniform one-dimensional finite-element inverse estimates give

$$\|f\|_{H^1} \le Ca^{-1/2}\|f\|_{H^{1/2}}, \qquad \|f'\|_{L^\infty} \le Ca^{-1/2}\|f'\|_{L^2}.$$

Because $L_\alpha z = 0$ removes the constant mode, the disk multiplier is equivalent to the remaining $H^{1/2}$ norm. Thus

$$\max_i |\Lambda_{\alpha,i}(z)| \le Ca^{-1}\mathfrak{G}_\alpha^{\mathrm{JP}}(z)^{1/2}. \tag{16.115}$$

Along (16.94), exact linearity gives $z_t = z_{\alpha_t}^D + J_{\alpha_t}(tw)$. The closed corrector estimate, the fourth-defect identity, and (16.113) give

$$\mathfrak{G}_{\alpha_t}^{\mathrm{JP}}(z_{\alpha_t}^D) \le C_M a^5, \qquad \mathfrak{G}_{\alpha_t}^{\mathrm{JP}}(J_{\alpha_t}(tw)) \le C_M a^5.$$

Consequently $\mathfrak{G}_{\alpha_t}^{\mathrm{JP}}(z_t)^{1/2} \le C_M a^{5/2}$. The reference side lengths are at least ca, whereas (16.115) bounds every change by $C_M a^{3/2} = o(a)$. Thus every side remains positive. The companion nodal and derivative inverse bounds give normal and tangential vertex displacements $O_M(a^2)$ and $O_M(a^{3/2})$, respectively, so the whole path remains in a fixed smaller radial $W^{1,\infty}$ chart. This proves all assertions. □

Remark 16.28 (Scope of the regular/super-near split). JC–AR and JC–SP isolate the regular completed-action root, the finite energy bridge, and its diagonal propagation. Their exact output is only the super-near action bound (16.102). The full no-ratio Contract 16.34 additionally requires JC–CL to absorb every endpoint outside (16.89). No far implication is assumed in JC–AR or JC–SP.

Proof contract 16.29 (JC–CL: coarse/far action localization)**.** Write $d = d_\alpha = z_\alpha^D$ and $z = d + w$. Exact disk completion first removes every disk covector from the coarse problem:

(16.116)

$$\boxed{\mathcal{A}_\alpha(d+w) = \Phi_\alpha(h_\alpha\mathbf{1} + d + w) - \Phi_\alpha(h_\alpha\mathbf{1}) - \frac{1}{2}\mathfrak{G}_\alpha^{\mathrm{JP}}(w) + \mathfrak{G}_\alpha^{\mathrm{JP}}(d) + \frac{\kappa}{2}\mathfrak{J}_4(\alpha).}$$

Indeed $q_D(B_\alpha + K_\alpha z) - q_D(B_\alpha) = \mathfrak{G}_\alpha^{\mathrm{JP}}(z-d) - \mathfrak{G}_\alpha^{\mathrm{JP}}(d)$, and Proposition 16.3 gives the formula. This identity also fixes the admissible path. The active chart supplies only $h_\alpha\mathbf{1}+[0,1](d+w)$; it does not imply that $h_\alpha\mathbf{1}+d$ or $h_\alpha\mathbf{1}+d+[0,1]w$ is active. In the presence of an arbitrarily short side, the energy bound $\mathfrak{G}_\alpha^{\mathrm{JP}}(d) \lesssim \mathfrak{J}_4(\alpha)$ has no ratio-free inverse side-length consequence.

Thus a coarse proof may not Taylor-expand about the disk corrector without a new activity lemma.

Put

$$E_\alpha(z) := \mathfrak{J}_4(\alpha) + \mathfrak{G}^{\mathrm{JP}}_\alpha(z - d_\alpha). \tag{16.117}$$

A sufficient coarse estimate is the existence of constants $c_F, C_F > 0$ such that, on the whole active no-ratio chart,

$$\boxed{\mathcal{A}_\alpha(z) \geq c_F E_\alpha(z) - C_F a^5.} \tag{16.118}$$

For $E_\alpha(z) \leq Ma^5$, set $d = \alpha - \alpha_*$ and $w = J_\alpha^{-1}(z - d_\alpha)$. The energy bridge and output required in JC–SP then give

$$\mathcal{A}_\alpha(z) \geq 0 \quad \text{whenever} \quad E_\alpha(z) \leq Ma^5 \tag{16.119}$$

for every fixed M. Taking $M \geq C_F/c_F$, Contract 16.34 follows. Indeed, (16.119) handles the first branch, while on $E_\alpha(z) > Ma^5$ equation (16.118) gives

$$\mathcal{A}_\alpha(z) > (c_F M - C_F)a^5 \geq 0.$$

No retained estimate proves (16.118). The existing two-regime estimates for F_4, the normal-sine/Bessel/secant transfers, and JE–V concern pure fan disk endpoints or scalar tails. They do not control arbitrary support amplitudes. In particular, at $\alpha = \alpha_*$ one may have $\mathfrak{J}_4 = 0$ and $\mathfrak{G}^{\mathrm{JP}}_{\alpha_*}(z) \gg a^5$ while remaining in the radial chart. The fixed-fan Hessian identity JP–EH supplies no lower bound for that branch. The nonzero physical-curvature mixed row also cannot be inserted as a vanishing error, by the CT mixed counterexample.

This fixed-fan obstruction has an exact scalar form. Cyclic symmetry at α_* gives $b_* = d_* = 0$, so JC–ALG reduces to

$$\mathcal{A}_*(z) = \Phi_*(h_*\mathbf{1} + z) - \Phi_*(h_*\mathbf{1}) - \frac{1}{2}\mathfrak{G}^{\mathrm{JP}}_*(z).$$

Consequently the restriction of (16.118) to the regular fan requires the following regular-fan Bregman coercivity row, with $c_s = c_F$ and $C = C_F$:

$$\boxed{\Phi_*(h_*\mathbf{1} + z) - \Phi_*(h_*\mathbf{1}) \geq \left(\frac{1}{2} + c_s\right)\mathfrak{G}^{\mathrm{JP}}_*(z) - Ca^5.} \tag{16.120}$$

Equivalently, JP–EH rewrites its left side as the support-time integral of the exact regular-fan reduced Helmholtz DtN, area, and lower rows. The two classical Minkowski concavities for $|P|^{1/2}$ and $\lambda_1(P)^{-1/2}$ do not order their ratio and hence do not prove this inequality. Thus (16.120), followed by the genuinely nonregular coarse localization, is the smallest presently isolated content of JC–CL.

At the regular fan cyclic symmetry gives $\ell_* = d_* = 0$. Define

$$\mathcal{L}_*(z) := \int_0^1 (1-t)\{H_{*,tz} - H_{*,0}\}[z,z], dt. \tag{16.121}$$

Proposition 16.30 (The fixed-reserve RF–IM target is false)**.** *Let $N = 3M \to \infty$, $a = 2\pi/N$, and, on the regular fan, put*

$$d_i = \cos\frac{2\pi i}{3}, \qquad s_* = \frac{4h_*\sin^2(a/2)}{1+2\cos a}, \qquad z_N = (1-a^2)s_*d.$$

The segment $h_*\mathbf{1} + [0,1]z_N$ is strictly active and remains in the radial small chart. Moreover

$$\mathfrak{G}_*^{\mathrm{JP}}(z_N) = \frac{13d_\circ^2\zeta(3)}{4\pi^2}a^3\{1+o(1)\}, \tag{16.122}$$

$$\frac{\Phi_*(h_*\mathbf{1}+z_N)-\Phi_*(h_*\mathbf{1})}{\mathfrak{G}_*^{\mathrm{JP}}(z_N)} \longrightarrow \frac{10}{13}, \tag{16.123}$$

$$\frac{\frac{1}{2}H_{*,0}[z_N,z_N]}{\mathfrak{G}_*^{\mathrm{JP}}(z_N)} \longrightarrow \frac{2\pi^2\log 3}{13\zeta(3)}. \tag{16.124}$$

Consequently

$$\frac{\mathcal{L}_*(z_N)}{\mathfrak{G}_*^{\mathrm{JP}}(z_N)} \longrightarrow \frac{10}{13} - \frac{2\pi^2\log 3}{13\zeta(3)} = -0.6185019068\ldots < -\frac{1}{20}. \tag{16.125}$$

Thus no N-independent C and $\eta < 1/20$ can make the former RF–IM bound $\mathcal{L}_*(z) \geq -\eta\mathfrak{G}_*^{\mathrm{JP}}(z) - Ca^5$ valid on every regular active radial path. This does not disprove RF–BC: on the same family

$$\frac{\mathcal{A}_*(z_N)}{\mathfrak{G}_*^{\mathrm{JP}}(z_N)} \longrightarrow \frac{7}{26} > 0. \tag{16.126}$$

Proof. For a fixed normal fan,

$$L_i(h) = \frac{h_{i-1}+h_{i+1}-2h_i\cos a}{\sin a}.$$

Since $d_{3j} = 1$ and $d_{3j+1} = d_{3j+2} = -1/2$, exactly the $3j$-sides first vanish at $s = s_*$, while all other sides grow. Taking $s_N = (1-a^2)s_*$ makes the entire segment strictly active. Also $s_* = a^2/3 + O(a^4)$, so the support displacement is $O(a^2)$ and its adjacent difference quotient is $O(a)$.

At $s = s_*$, deletion of the zero sides leaves equal support numbers and alternating normal gaps $a, 2a$. Denote the area-normalized disk sources of the regular and alternating fans by B_{reg} and B_{alt}. In the Fourier normalization used here, their leading material profiles satisfy

$$F''_{\mathrm{reg}} = 1 - \sum_{j\in\mathbb{Z}}\delta_{j+1/2}, \qquad F''_{\mathrm{alt}} = 1 - \delta_{1/2} - 2\delta_2$$

with periods 1 and 3, respectively. Hence

$$\|B_{\mathrm{reg}}\|^2_{\dot H^{1/2}} = \frac{\zeta(3)}{4\pi^3}a^3 + O(a^4).$$

For $\varkappa = 2\pi/3$,

$$\widehat{F}_{\mathrm{alt},\ell} = \frac{e^{-i\varkappa\ell/2} + 2e^{-2i\varkappa\ell}}{3(\varkappa\ell)^2}.$$

The numerator has squared modulus 9 for even ℓ and 1 for odd ℓ. The even and odd ℓ^{-3} sums are $\zeta(3)/8$ and $7\zeta(3)/8$, so

$$\|B_{\mathrm{alt}}\|^2_{\dot H^{1/2}} = \frac{3\zeta(3)}{2\pi^3}a^3 + O(a^4).$$

The source-profile Taylor remainder is $O(a^{7/2})$ in $H^{1/2}$. Since the first nonconstant frequencies are of order a^{-1} and the disk multiplier is $\omega_n = |n| + 1 + O(|n|^{-1})$,

$$q_D(B_{\mathrm{alt}}) - q_D(B_{\mathrm{reg}}) = \frac{5d_\circ^2\zeta(3)}{2\pi^2}a^3 + O(a^4). \tag{16.127}$$

The closed disk-chart scalar expansion applies separately to both quasiuniform fans:

$$\Phi_\alpha(h_\alpha \mathbf{1}) = \lambda_\circ + q_D(B_\alpha) + O(a^4).$$

Indeed the complete cubic term is bounded by $\|B\|_{W^{1,\infty}}\|B\|^2_{H^{1/2}} = O(a^4)$, and the tame fourth-and-higher remainder is $O(a^5)$. The support distance from s_N to the collapsed endpoint is $O(a^4)$; two-sided parallel-body inclusions and Dirichlet monotonicity change Φ by $O(a^4)$. Thus (16.127) gives the numerator in (16.123).

For the regular normal-sine metric,

$$Q(x) = \frac{2\sin^4(\pi x)}{\pi^3}\{\zeta(3,x) + \zeta(3,1-x)\}.$$

The real character d has half the energy of the unnormalized complex character. Since

$$\zeta(3,1/3) + \zeta(3,2/3) = 26\zeta(3), \qquad Q(1/3) = \frac{117\zeta(3)}{4\pi^3},$$

and $s_N^2 = a^4/9 + O(a^6)$, the factor $(N/2)d_\circ^2 Q(1/3)$ proves (16.122), and then (16.123).

Finally the true polygonal support symbol gives

$$\frac{\frac{1}{2}H_{*,0}[z_N, z_N]}{\mathfrak{G}_*^{\mathrm{JP}}(z_N)} \longrightarrow \frac{P(1/3)}{Q(1/3)}.$$

The digamma third-point formula yields

$$P(1/3) = \frac{9\log 3}{2\pi}, \qquad \frac{P(1/3)}{Q(1/3)} = \frac{2\pi^2\log 3}{13\zeta(3)},$$

which proves (16.124). The exact Taylor identity

$$\mathcal{L}_*(z_N) = \Phi_*(h_*\mathbf{1} + z_N) - \Phi_*(h_*\mathbf{1}) - \frac{1}{2}H_{*,0}[z_N, z_N]$$

now proves (16.125). The strict comparison with $-1/20$ follows already from $\pi^2 > 9$, $\log 3 > 1$, and $\zeta(3) < 5/4$. Since $\mathfrak{G}_*^{\mathrm{JP}}(z_N) \asymp a^3$, an Ca^5 remainder cannot absorb it. Subtracting $1/2$ from the value ratio gives (16.126). $\square$

Proof contract 16.31 (RF–REL: relative preservation of the full root reserve)**.** There exist $\delta > 0$ and C, independent of N, such that every regular active radial path satisfies

$$\boxed{\mathcal{L}_*(z) \geq -\frac{1-\delta}{2}\{H_{*,0}[z,z] - \mathfrak{G}_*^{\mathrm{JP}}(z)\} - Ca^5.} \tag{16.128}$$

Lemma 16.32 (RF–REL implies RF–BC)**.** *Contract 16.31 implies*

$$\mathcal{A}_*(z) \geq \frac{\delta}{20}\mathfrak{G}_*^{\mathrm{JP}}(z) - Ca^5.$$

Proof. Taylor's identity and the regular action give

$$\mathcal{A}_*(z) = \frac{1}{2}\{H_{*,0}[z,z] - \mathfrak{G}_*^{\mathrm{JP}}(z)\} + \mathcal{L}_*(z).$$

RF–REL and $H_{*,0} \geq (11/10)\mathfrak{G}_*^{\mathrm{JP}}$ imply

$$\mathcal{A}_*(z) \geq \frac{\delta}{2}\{H_{*,0}[z,z] - \mathfrak{G}_*^{\mathrm{JP}}(z)\} - Ca^5 \geq \frac{\delta}{20}\mathfrak{G}_*^{\mathrm{JP}}(z) - Ca^5.$$

□

Thus the regular coarse row remains open, but its current sufficient interface is RF–REL rather than the disproved fixed 1/20-reserve target.

Remark 16.33 (Collapse-word audit and the exact JC–CL cut). Every fixed periodic pattern of deleted regular-fan sides can be encoded by a positive integer gap word $(m_1, \ldots, m_r)$. At its collapse endpoint the support profile on a gap of length m is $u_j = j(m-j)/2$, and the disk-source, metric, and root-reserve coefficients reduce to finite Fourier/Hurwitz-zeta sums. These formulas recover exactly the period-three ratios 10/13 and 7/26 above. The reproducible integer-word precheck `diagnostics/explore_regular_collapse_words.py` enumerates all words of total period at most 16 and several families through period 4096. It finds no negative action and no vanishing relative reserve; the smallest sampled ratios are approximately

$$\mathcal{A}_*/\mathfrak{G}_*^{\mathrm{JP}} = 0.0912513, \qquad \mathcal{A}_*/\mathcal{R}_0 = 0.181689, \qquad \mathcal{R}_0 = (H_{*,0} - \mathfrak{G}_*^{\mathrm{JP}})/2.$$

These numbers are not lower bounds on the full active microscopic class. For $z_i = a^2 u_i$, the exact leading envelope is

$$B_u(x) = \min_i \left\{u_i + \frac{(x-i)^2}{2}\right\}, \quad s_i = i + \frac{1}{2} + u_{i+1} - u_i, \quad g_i = s_i - s_{i-1} \geq 0.$$

At fixed microscopic period, the complete *leading* active class is the gap simplex $g_i \geq 0$, $\sum_i g_i = p$, and along support time $g_i(t) = (1-t) + tg_i$. If $S_p(t)$ is the resulting leading disk-source energy, its exact pair ledger is

$$S_p''(t) = \frac{1}{\pi p}\sum_{i,j}(v_i - v_j)^2 \log\left|2\sin\frac{\pi\{(i-j) + t(v_i - v_j)\}}{p}\right|, \qquad v_i = u_{i+1} - u_i.$$

Here the diagonal terms are defined by $0^2 \log 0 := 0$. If distinct corner atoms collide at $t = 1$, this formula and the support-time Taylor integral are understood as improper identities; the resulting logarithmic singularity is integrable against $1 - t$. Put

$$\widehat{u}_k = \frac{1}{p}\sum_{j=0}^{p-1} u_j e^{-2\pi ikj/p}, \qquad \mathcal{Q}_p[u] = 2\pi \sum_{k=1}^{p-1} |\widehat{u}_k|^2 Q(k/p).$$

Consequently the smallest leading regular target is the universal monotone-transport inequality

$$2\pi\{S_p(1) - S_p(0)\} \geq \left(\frac{1}{2} + c_0\right) \mathcal{Q}_p[u] \tag{16.129}$$

for every period and every point of that simplex. Continuous-simplex searches lower the sampled ratios to about 0.082293 relative to $\mathcal{Q}_p$ and 0.163711 relative to the root reserve at $p = 384$; they find no negative value, but do not prove a uniform constant. Equation (16.129), its finite-N secant/cubic/tail transfer, and the nonregular original-path row all remain open. This is a floating-point necessary-family precheck, not a proof of RF–REL. It also shows that the regular path loss is order one rather than an $o(1)\mathfrak{G}_*^{\mathrm{JP}}$ error. Even a proof of RF–REL would leave the genuinely nonregular original-path inequality below as a second, independent input.

For a genuinely nonregular fan the exact original-path form is, with $z = d + w$,

$$\boxed{\begin{aligned} \mathcal{A}_\alpha(d+w) = \ell_\alpha[d+w] + \int_0^1 (1-t) H_{\alpha,t(d+w)}[d+w, d+w], dt \\ - \frac{1}{2}\mathfrak{G}_\alpha^{\mathrm{JP}}(w) + \mathfrak{G}_\alpha^{\mathrm{JP}}(d) + \frac{\kappa}{2}\mathfrak{J}_4(\alpha). \end{aligned}} \tag{16.130}$$

The nonregular part of JC–CL is precisely a lower bound for this one budget by $c_F\{\mathfrak{J}_4(\alpha) + \mathfrak{G}_\alpha^{\mathrm{JP}}(w)\} - C_F a^5$, uniformly in all gap ratios. Separating its covector from its integrated Hessian would recreate the unavailable JE–FM step.

For a stationary-minimizer bypass one may replace (16.118) by the weaker localization statement: every stationary competitor x with $\Phi_N(x) \leq \Phi_N(0)$ satisfies

$$6a^2 \sum_i d_i^2 + \mathfrak{G}_{\alpha_*}^{\mathrm{JP}}(z - Z_N d) \leq M a^5. \tag{16.131}$$

Together with JC–AR and JC–SP, this would put the competitor in the super-near action-positive branch and exclude a lower value through the exact JP–JC/NR assembly. The present global wrapper gives only qualitative near-disk entry, not (16.131). Thus the stationary route has three open analytic rows, not two.

Proof contract 16.34 (JC–PL: pathwise LCA payment)**.** After decreasing the no-ratio fan and radial endpoint-chart thresholds so that $\varepsilon_V(S) \le \kappa/2$, prove, for every active normalized support vector $z \in \mathsf{X}_\alpha$ in that chart,

$$\boxed{\mathcal{A}_\alpha(z) \ge 0.} \tag{16.132}$$

The proof must use a disjoint, complete pair ledger. The terminal two-ray power and the negative cotangent/area jump row must be joined before any estimate; every unused terminal annulus must remain in the nonterminal ledger; same-cell, adjacent, and first-separation LCA pairs must be counted exactly once. The support-time integral must be taken before a side-ratio supremum, using Lemma 16.4. The true ground-state multiplier, reduced-energy, mixed-flux, area, and fixed-rank rows must retain the same completed covector b_α of (16.12). All constants must be independent of the smallest angle and adjacent ratios. In particular the contract includes endpoints with an arbitrarily short positive side: smallness of an affine-side parametrization relative to the tangential fan polygon is not an assumption. Its admissible chart is exactly the closed JP–NS output

$$h_\alpha \mathbf{1} + [0,1]z \text{ active}, \qquad \sup_{0\le t\le 1} \|\rho_t\|_{W^{1,\infty}} + \mathfrak{G}_\alpha^{\mathrm{JP}}(z)^{1/2} \le C(S+\rho_0),$$

not the false labelled affine-side chart of Proposition JP–NS (radial-not-affine).

Equivalently, by Proposition 16.3, prove

$$\boxed{\mathcal{R}_\alpha(z) \ge -\frac{\kappa}{2}\mathfrak{J}_4(\alpha) - \frac{1}{2}\mathfrak{G}_\alpha^{\mathrm{JP}}(z - z_\alpha^D).} \tag{16.133}$$

The stage-130 moving-disk longitudinal calculation omits material chain-rule terms and is not an admissible proof of either display.

Theorem 16.35 (JC–PL closes the active-chart theorem)**.** *Assume Contract 16.34. Then every normalized polygon in the active near-disk chart satisfies*

$$\boxed{\Phi_\alpha(h_\alpha \mathbf{1} + z) - \Phi_{\alpha_*}(h_* \mathbf{1}) \ge \kappa \mathfrak{J}_4(\alpha) + \frac{1}{2}\mathfrak{G}_\alpha^{\mathrm{JP}}(z - z_\alpha^D).} \tag{16.134}$$

Moreover

$$\mathfrak{G}_\alpha^{\mathrm{JP}}(z_\alpha^D) \le \frac{C_p^2}{4}\mathfrak{J}_4(\alpha). \tag{16.135}$$

Proof. Add and subtract the two disk endpoint values. Equations (16.8), (16.5), and (16.37) give

$$\begin{aligned}\Phi_\alpha(h_\alpha \mathbf{1} + z) - \Phi_{\alpha_*}(h_* \mathbf{1}) \ge{}& P_D^{\mathrm{ns}}(\alpha) - P_D^{\mathrm{ns}}(\alpha_*) + \mathfrak{G}_\alpha^{\mathrm{JP}}(z - z_\alpha^D) \\ &- \varepsilon_V(S)\mathfrak{J}_4(\alpha) + \mathcal{R}_\alpha(z).\end{aligned}$$

Insert (16.35), (16.133), and $\varepsilon_V(S) \le \kappa/2$. The fan coefficient is $2\kappa - \kappa/2 - \kappa/2 = \kappa$, and the support coefficient is $1 - 1/2 = 1/2$. This proves (16.134). Equations (16.36) and (16.7) prove (16.135). ∎

Remark 16.36 (Precise mathematical cut). Contract 16.34 is not a restatement of either discarded separated estimate. At $\alpha = \alpha_*$ it asks only for the coarse one-sided support bound encoded by (16.133); away from the regular fan it gives the forcing access to the complete integrated Hessian surplus before a Schur complement is taken. The exact algebra and the support-time logarithm cancellation are now proved. Revision 24 further resolves the former vague "regular plus path" description into one closed and two open exact subcontracts. The all-character regular completed-action Schur matrix JC–AR is now closed. The two remaining open contracts are the completed-action Hessian modulus JC–SP and the coarse/far localization JC–CL. The former period-three completed-covector claim remains retracted, and the fixed-reserve RF–IM target is disproved. What is still missing is the ratio-free pathwise LCA payment for the true nonstationary ground-state multiplier and all remaining lower rows. No retained node implies either open analytic estimate.

16.4. **DAG disposition.**

item	revision-24 disposition
$D0_{\mathrm{ns}}$, JE–P, JE–V	closed inputs, equations (16.35)–(16.37);
JE–FM	unavailable; the old material falsification is retracted and the row is absent from the active DAG;
JP–SC	retained only as a possible stronger auxiliary estimate; no longer an active prerequisite;
JC–ALG and JC–TL	closed by Proposition 16.3 and Lemmas 16.4–16.5;
pure-support JC–AR row	closed for all characters by Lemma 16.15 and Proposition 16.16;
period-three completed-covector row	retracted by Remark 16.19; its true polygonal mixed finite part is now closed at every fixed Bloch ratio, as is the strict support–double-disk gap; the ratio certificate and uniform all-character bridge are closed by Proposition 16.22;
RF–IM	disproved by Proposition 16.30; RF–REL is the replacement open sufficient interface for RF–BC;
super-near bridge/activity	closed by Proposition 16.27;
terminal-radius pure-power resummation	closed by Remark 16.26; the old DAR row is retracted as noninvariant, while the completed third-jet/QRCI row stays open;

JC–AR	closed all-character regular action Schur contract;
JC–SP, JC–CL	remaining natural Hessian-modulus and coarse/far subcontracts inside JC–PL;
JC–PL	unique open conceptual node, Contract 16.34;
JC–PL → JP–JC → J0	closed algebraically by Theorem 16.35.

17. Node NR: joint endpoint assembly and J0 handoff

Audited source: `NR_assembly.tex`.

17.1. **Incoming joint comparison.** Use the notation of the JP and JC nodes. In particular,

$$z_\alpha^D = -\frac{1}{2}\mathcal{R}_\alpha^{-1}p_\alpha, \qquad \mathfrak{G}_\alpha^{\mathrm{JP}}(z_\alpha^D) \le \frac{C_p^2}{4}\mathfrak{J}_4(\alpha), \tag{17.1}$$

and the exact disk completion is

$$q_D(B_\alpha + K_\alpha z) = P_D^{\mathrm{ns}}(\alpha) + \mathfrak{G}_\alpha^{\mathrm{JP}}(z - z_\alpha^D). \tag{17.2}$$

The closed disk endpoint has margin $2\kappa\mathfrak{J}_4$, and closed value matching has error at most $\varepsilon_V(S)\mathfrak{J}_4$. JC–ALG and JC–TL are closed. The sole unavailable input is JC–PL; its exact algebraic output is the fixed-fan scalar comparison JP–JC,

$$\mathcal{R}_\alpha(z) \ge -\frac{\kappa}{2}\mathfrak{J}_4(\alpha) - \frac{1}{2}\mathfrak{G}_\alpha^{\mathrm{JP}}(z - z_\alpha^D), \qquad \varepsilon_V(S) \le \frac{\kappa}{2}. \tag{17.3}$$

17.2. **Conditional NR/J0 theorem.**

Theorem 17.1 (JP–JC implies J0)**.** *Assume JC–PL, equivalently its JP–JC output* (17.3)*. Then every normalized polygon in the active near-disk chart satisfies*

$$\boxed{\Phi_\alpha(h_\alpha\mathbf{1} + z) - \Phi_{\alpha_*}(h_*\mathbf{1}) \ge \kappa\mathfrak{J}_4(\alpha) + \frac{1}{2}\mathfrak{G}_\alpha^{\mathrm{JP}}(z - z_\alpha^D).} \tag{17.4}$$

The corrector obeys the first estimate in (17.1)*, uniformly in the smallest angle and all adjacent ratios.*

Proof. By the definition of the honest joint residual,

$$\begin{aligned}\Phi_\alpha(h_\alpha\mathbf{1} + z) - \Phi_{\alpha_*}(h_*\mathbf{1}) &= q_D(B_\alpha + K_\alpha z) - q_D(B_{\alpha_*}) \\ &\quad + \mathcal{R}_\alpha(z) + E_{V,\alpha}, \qquad |E_{V,\alpha}| \le \varepsilon_V(S)\mathfrak{J}_4(\alpha).\end{aligned}$$

Insert (17.2), the closed estimate $P_D^{\mathrm{ns}}(\alpha) - P_D^{\mathrm{ns}}(\alpha_*) \ge 2\kappa\mathfrak{J}_4(\alpha)$, and (17.3). Since $\varepsilon_V(S) \le \kappa/2$, the remaining coefficients are exactly κ and $1/2$, which proves (17.4). The corrector estimate is the closed JE–P row followed by exact Hilbert completion. ∎

Corollary 17.2 (Exact downstream handoff)**.** *The implication*

$$\boxed{\text{JP-DE} + \text{JP-JC} \Longrightarrow \text{NR} \Longrightarrow \text{J0}}$$

is proved. JP–DE abbreviates the closed $D0_{\mathrm{ns}}$*, JE–P, and JE–V rows. JC–PL is the only unavailable premise; JP–JC is dependent on it.*

17.3. **DAG certificate.**

edge	disposition
JP–EH and JP–NS	closed exact scalar/Hessian chart and disk metric;
JP–DE	closed disk Schur gap, corrector bound, and value matching;
JP–SC, JE–FM, JP–FS, JP–BR	removed from the active chain; the old period-three material argument for JE–FM is retracted, and all four rows are unnecessary on the joint route;
JC–ALG and JC–TL	closed exact action and support-time cancellation;
JC–PL	unique open pathwise LCA payment;
JP–JC	dependent exact output of JC–PL;
JP–DE+JP–JC → NR → J0	proved by Theorem 17.1.

Part 5. Side activation and the global theorem

18. Foundation F57: stationary side mass and first moment

Audited source: `convex_largeN_polya_szego_stage57_stationary_flux_reduction.tex`.

18.1. **Active stationary polygons.** Let

$$P = \bigcap_{i=1}^{M} \{x \in \mathbb{R}^2 : x \cdot \nu_i < h_i\} \tag{18.1}$$

be a bounded convex active polygon. Write

$$\nu_i = (\cos\theta_i, \sin\theta_i), \qquad \tau_i = (-\sin\theta_i, \cos\theta_i), \tag{18.2}$$

and let S_i be the side on the line $x \cdot \nu_i = h_i$. Every point of S_i has the unique representation

$$x = h_i\nu_i + s\tau_i, \qquad s_i^- < s < s_i^+. \tag{18.3}$$

Thus its length is $\ell_i = s_i^+ - s_i^-$. Let $u > 0$ be the $L^2(P)$-normalized first eigenfunction,

$$-\Delta u = \lambda u \quad \text{in } P, \qquad u = 0 \quad \text{on } \partial P, \qquad \int_P u^2 = 1, \tag{18.4}$$

and put

$$p_i(s) = -\partial_\nu u(h_i\nu_i + s\tau_i) \geq 0. \tag{18.5}$$

Convex polygonal regularity gives $u \in H^2(P)$ and $p_i \in L^2(S_i)$. The distributed Lipschitz-domain derivative, or equivalently approximation inside a fixed active chart, gives the Hadamard formula

$$D\lambda(P)[V] = -\sum_i \int_{S_i} p_i^2 (V \cdot \nu_i)\, ds \tag{18.6}$$

for every vertex-compatible polygonal material field V.

We use the scale-invariant functional

$$\Lambda(P) = \frac{|P|}{\pi}\lambda_1(P). \tag{18.7}$$

Stationarity of Λ is equivalent to stationarity of λ_1 under the constraint $|P| = A$.

18.2. **The two exact sidewise moment identities.**

Theorem 18.1 (Stationary flux equidistribution)**.** *Suppose that P is stationary for Λ with respect to all active vertex perturbations. Then, on every side S_i,*

$$\frac{1}{\ell_i}\int_{s_i^-}^{s_i^+} p_i(s)^2\, ds = \frac{\lambda}{|P|}, \tag{18.8}$$

$$\int_{s_i^-}^{s_i^+} \left(s - \frac{s_i^- + s_i^+}{2}\right) p_i(s)^2\, ds = 0. \tag{18.9}$$

Both identities are independent of the smallest side and of all mesh ratios.

Proof. Work first with the Lagrangian

$$\mathcal{F}(P) = \lambda_1(P) + \mu|P|. \tag{18.10}$$

Fix the normal fan and replace h_i by $h_i + tz_i$. On the open side S_i the induced normal velocity is z_i; on every other open side it is zero. The endpoint motions are tangential to the neighboring sides and do not contribute to the boundary integrals. Consequently

$$D\lambda(P)[z] = -\sum_i z_i \int_{S_i} p_i^2\, ds, \qquad D|P|[z] = \sum_i z_i\ell_i. \tag{18.11}$$

The z_i are independent in an active chart. Stationarity gives

$$\int_{S_i} p_i^2\, ds = \mu\ell_i \qquad (1 \leq i \leq M). \tag{18.12}$$

To identify the multiplier, use the dilation field $V(x) = x$. Scaling gives

$$D\lambda(P)[x] = -2\lambda, \qquad D|P|[x] = 2|P|. \tag{18.13}$$

Hence $-2\lambda + 2\mu|P| = 0$, or

$$\mu = \lambda/|P|. \tag{18.14}$$

Equations (18.12)–(18.14) prove (18.8).

It remains to rotate one supporting line. Replace θ_i by $\theta_i + tq_i$ while keeping h_i fixed. Differentiating

$$x_t \cdot \nu_i(t) = h_i \tag{18.15}$$

at a point $x = h_i\nu_i + s\tau_i$ gives

$$V \cdot \nu_i = -sq_i. \tag{18.16}$$

The normal velocity is zero on every other open side. Therefore

$$D\lambda(P)[q_i] = q_i \int_{s_i^-}^{s_i^+} sp_i(s)^2 \, ds,$$

$$D|P|[q_i] = -q_i \int_{s_i^-}^{s_i^+} s \, ds. \tag{18.17}$$

Stationarity of $\lambda + \mu|P|$ and the sign convention in (18.12) give

$$\int_{s_i^-}^{s_i^+} sp_i(s)^2 \, ds = \mu \int_{s_i^-}^{s_i^+} s \, ds. \tag{18.18}$$

Subtract the midpoint times (18.12) from (18.18). This is (18.9). ☐

Corollary 18.2 (Sidewise probability law)**.** *For every side define*

$$d\mathbb{P}_i(s) = \frac{p_i(s)^2}{(\lambda/|P|)\ell_i} \, ds. \tag{18.19}$$

Then $\mathbb{P}_i$ is a probability measure and

$$\int s \, d\mathbb{P}_i(s) = \frac{s_i^- + s_i^+}{2}. \tag{18.20}$$

In particular, a stationary short side cannot be analyzed using only its total flux: its first flux moment is fixed as well.

Proof. This is exactly (18.8) and (18.9). ☐

Remark 18.3 (Why the moment row is not redundant). For a shallow corner model $p(s)^2 \simeq Cs^{2\varepsilon_-}(\ell - s)^{2\varepsilon_+}$, the normalized first moment is the mean of a beta distribution. Its displacement from $\ell/2$ has the sign of $\varepsilon_- - \varepsilon_+$. Thus (18.9) detects an imbalance of the two endpoint corner exponents which the average identity (18.8) cannot see.

19. Node A0: strict transverse side activation

Audited source: `A0_side_activation.tex`*.*

19.1. **A sharp one-dimensional tent inequality.** For $L > 0$ and $x \in (0, L)$, let

$$K_x(s) = \begin{cases} \dfrac{s(L-x)}{L}, & 0 \le s \le x, \\ \dfrac{x(L-s)}{L}, & x \le s \le L. \end{cases} \tag{19.1}$$

Thus K_x is the Dirichlet Green kernel of $-d^2/ds^2$, viewed as a function of s, and the unit-height triangular tent is

$$h_x(s) = \frac{L}{x(L-x)} K_x(s). \tag{19.2}$$

Lemma 19.1 (Log-concave density dominates its constant rearrangement)**.** *Let $q : (0, L) \to (0, \infty)$ be log-concave and integrable, with*

$$\lim_{s \downarrow 0} q(s) = \lim_{s \uparrow L} q(s) = 0. \tag{19.3}$$

Put

$$\mu = \frac{1}{L} \int_0^L q(s)\, ds. \tag{19.4}$$

If

$$\int_0^L \left(s - \frac{L}{2} \right) q(s)\, ds = 0, \tag{19.5}$$

then, for every $x \in (0, L)$,

$$\boxed{\int_0^L h_x(s)\{q(s) - \mu\}\, ds > 0.} \tag{19.6}$$

Proof. Set $g = q - \mu$ and

$$G(y) = \int_0^y g(s)\, ds, \qquad H(y) = \int_0^y G(r)\, dr. \tag{19.7}$$

Mass and barycenter matching give

$$\int_0^L g = 0, \qquad \int_0^L s g(s)\, ds = 0, \tag{19.8}$$

and therefore

$$H(L) = \int_0^L (L-s) g(s)\, ds = 0. \tag{19.9}$$

Log-concavity says that every superlevel set of q is an interval. In view of (19.3) and $\int q = \mu L$, the set $\{q > \mu\}$ is a nonempty interval $(a, b) \Subset (0, L)$, up to irrelevant closed plateau endpoints. Thus g has the sign pattern

$$-, \quad +, \quad -.$$

Consequently G first decreases, then increases, and finally decreases to $G(L) = 0$. On the last interval it must be nonnegative, because it is decreasing to zero. Hence G changes sign exactly once, from negative to positive. It follows that H first decreases and then increases. Together with $H(0) = H(L) = 0$, this gives

$$H(y) < 0, \qquad 0 < y < L. \tag{19.10}$$

Strictness follows because q cannot equal the positive constant μ while satisfying (19.3).

Now set

$$F(x) = \int_0^L K_x(s)g(s)\,ds.$$

The Green-kernel identity gives

$$-F'' = g, \qquad F(0) = F(L) = 0.$$

Moreover $F'(0) = L^{-1}\int_0^L (L-s)g(s)\,ds = 0$ by (19.8). Hence $F' = -G$ and $F = -H$. Equation (19.10) yields $F(x) > 0$. Multiplication by the positive factor in (19.2) proves (19.6). □

Remark 19.2. The proof is a convex-order statement, but no external convex-order theorem is needed. The only special input is that a log-concave density crosses its average on one interval; the two moment equalities then force the sign of every tent pairing.

19.2. **Log-concavity of the side flux.** Let P be a bounded convex polygon and let $u > 0$ be its first Dirichlet eigenfunction:

$$-\Delta u = \lambda u \quad \text{in } P, \qquad u = 0 \quad \text{on } \partial P. \tag{19.11}$$

Fix an open side S, identify it with $0 < s < L$, and let $\nu_{\rm in}$ be its inward unit normal. Define

$$p(s) = \partial_{\nu_{\rm in}} u(s) > 0. \tag{19.12}$$

Lemma 19.3 (The boundary flux is log-concave on a side)**.** *The function p in (19.12) is log-concave on $(0, L)$. Moreover,*

$$p(s) \longrightarrow 0 \quad \text{as } s \downarrow 0 \text{ or } s \uparrow L. \tag{19.13}$$

Consequently p^2 satisfies the hypotheses of Lemma 19.1.

Proof. The Brascamp–Lieb theorem [1] gives log-concavity of u in every bounded convex domain. Fix $s_0, s_1 \in (0, L)$ and $0 < \theta < 1$. For all sufficiently small $r > 0$, the three points

$$s_j + r\nu_{\rm in} \quad (j = 0, 1), \qquad ((1-\theta)s_0 + \theta s_1) + r\nu_{\rm in}$$

belong to P. Log-concavity of u gives

$$\begin{aligned} &\frac{u(((1-\theta)s_0 + \theta s_1) + r\nu_{\rm in})}{r} \\ &\qquad\qquad \geq \left(\frac{u(s_0 + r\nu_{\rm in})}{r}\right)^{1-\theta} \left(\frac{u(s_1 + r\nu_{\rm in})}{r}\right)^{\theta}. \end{aligned} \tag{19.14}$$

Letting $r \downarrow 0$ proves log-concavity of p.

At a convex vertex of interior angle $\omega < \pi$, the standard corner expansion of a Dirichlet solution starts with $r^{\pi/\omega}\sin(\pi\vartheta/\omega)$. Hence its gradient is $O(r^{\pi/\omega-1})$, which tends to zero. This proves (19.13). $\square$

19.3. **Activating one new side.** Use the scale-invariant functional

$$\Lambda(P) = \frac{|P|}{\pi}\lambda_1(P). \tag{19.15}$$

Suppose P is stationary under all active vertex variations. On each side S of length L, stages 70–72 give the exact identities

$$\frac{1}{L}\int_0^L p(s)^2\,ds = \frac{\lambda}{|P|} =: \mu, \tag{19.16}$$

$$\int_0^L \left(s - \frac{L}{2}\right) p(s)^2\,ds = 0. \tag{19.17}$$

Choose $x \in (0, L)$, insert at that point a collinear marked vertex, and move it a distance $\varepsilon > 0$ in the outward normal direction, keeping the two endpoints of S fixed. For small ε this replaces S by two sides of a convex polygon and adds a triangular cap. At $\varepsilon = 0$, the normal velocity on S is exactly the tent h_x of (19.2).

Theorem 19.4 (Strict side-activation descent)**.** *For every side and every $x \in (0, L)$, the outward splitting path P_ε satisfies*

$$\boxed{\frac{d}{d\varepsilon}\Lambda(P_\varepsilon)\Big|_{\varepsilon=0+} < 0.} \tag{19.18}$$

Thus a stationary convex M-gon is not stationary, and is not a local minimum, in the closed class of convex polygons with at most $M+1$ active sides.

Proof. Hadamard's formula and the area derivative give, for an outward normal velocity h,

$$\begin{aligned} D\Lambda(P)[h] &= \frac{\lambda}{\pi}\int_{\partial P} h\,ds - \frac{|P|}{\pi}\int_{\partial P} p^2 h\,ds \\ &= \frac{|P|}{\pi}\int_{\partial P}\left(\frac{\lambda}{|P|} - p^2\right) h\,ds. \end{aligned} \tag{19.19}$$

For the splitting path, $h = h_x$ on S and is zero on the other sides. Lemma 19.3 applies to $q = p^2$, while (19.16)–(19.17) are precisely its mass and barycenter hypotheses. Lemma 19.1 therefore gives

$$\int_0^L h_x(p^2 - \mu)\,ds > 0.$$

Insert this inequality and $\mu = \lambda/|P|$ into (19.19) to obtain (19.18).

The path has one additional active vertex for every sufficiently small $\varepsilon > 0$. Since its scale-invariant value decreases to first order, the final assertion follows. $\square$

Corollary 19.5 (No minimizing side loss)**.** *Let P minimize Λ in the class of convex polygons with at most N sides. If P has at least three sides and is stationary in its active stratum, then it has exactly N active sides.*

Proof. If P had $M < N$ active sides, Theorem 19.4 would produce a convex $(M+1)$-gon with strictly smaller Λ, contradicting minimality. □

20. Node G0: compact global assembly and equality rigidity

Audited source: `G0_global_wrapper.tex`.

20.1. **Statement and normalization.** For $A > 0$, let $\mathcal{P}_N^{\mathrm{conv}}(A)$ be the class of bounded convex planar polygons of area A with exactly N genuine sides, after consecutive collinear sides are merged. Let $R_N(A)$ be the regular N-gon of area A.

Theorem 20.1 (Global theorem conditional on J0)**.** *Assume the active near-disk coercivity input in 20.3. Then For every $A > 0$ there is N_0 such that, for all $N \geq N_0$ and all $P \in \mathcal{P}_N^{\mathrm{conv}}(A)$,*

$$\lambda_1(P) \geq \lambda_1(R_N(A)). \tag{20.1}$$

Equality holds if and only if P is congruent to $R_N(A)$.

Put

$$\Lambda(\Omega) = \frac{|\Omega|}{\pi}\lambda_1(\Omega). \tag{20.2}$$

It is invariant under dilations. Hence it is enough to work at area π; the general statement follows by the dilation $P \mapsto (\pi/A)^{1/2}P$.

Let $\mathcal{K}_{\leq N}^{\mathrm{conv}}(\pi)$ be the Hausdorff-closed class of area-π convex polygons with at most N active sides. This class is used only to take a global minimum. The theorem itself concerns exactly N sides.

20.2. **The two retained local inputs.** We isolate the only two nonclassical inputs used in the global assembly.

Theorem 20.2 (Strict transverse side activation; stage 132)**.** *Let Q be a stationary convex M-gon for Λ in its active stratum. Insert a marked point in the relative interior of any side and move it outward while keeping the two endpoints fixed. For all sufficiently small positive displacements this gives a convex $(M+1)$-gon Q_ε and*

$$\frac{d}{d\varepsilon}\Lambda(Q_\varepsilon)\bigg|_{\varepsilon=0+} < 0. \tag{20.3}$$

The assertion is independent of all side lengths and mesh ratios.

The proof in stage 132 uses the stationary mass and barycenter identities for the squared ground-state flux on a side, log-concavity of that flux, and Hadamard's formula. In particular, it is a first-variation theorem at the lower-side point; it is not a mountain-pass deformation theorem.

Intermediate obligation 20.3 (Active near-disk value coercivity; current J0 input)**.** There are $\eta_0, S_0, c_0 > 0$ such that the following holds. Let P be an area-π active convex N-gon with positive normal fan

$$\alpha_i > 0, \qquad \sum_i \alpha_i = 2\pi, \qquad a = \frac{2\pi}{N}, \qquad S = a + \max_i \alpha_i \le S_0.$$

Assume, after translation, that its radial graph is within η_0 of the disk in $W^{1,\infty}$. Let h be its support vector in the fixed translation complement of JP. Put $h_0 = h_\alpha \mathbf{1}$, $A_0 = A_\alpha(h_0)$, and $s_\alpha(h) = A'_\alpha[h_0; h]/(2A_0)$. Normalize scale by

$$\widetilde{h} = \frac{h}{s_\alpha(h)}, \qquad s_\alpha(\widetilde{h}) = 1, \qquad \widetilde{h} = h_\alpha \mathbf{1} + z, \qquad z \in \mathsf{X}_\alpha.$$

Shrinking the physical chart if necessary, $s_\alpha(h) > 0$ and this normalization stays in the same uniform near-disk chart. Then

$$\boxed{\lambda_1(P) - \lambda_1(R_N(\pi)) \ge c_0 \left\{ \mathfrak{J}_4(\alpha) + \mathfrak{G}^{\mathrm{JP}}_\alpha (z - z^\sharp_\alpha) \right\},} \tag{20.4}$$

where

$$\mathfrak{J}_4(\alpha) = \sum_i \alpha_i^4 - N a^4. \tag{20.5}$$

The support metric is positive on the fixed affine scale section X_α after the fixed translation normalization. Since Φ_α is scale invariant and the original P has area π, the JP/NR value inequality for $\widetilde{h}$ is exactly (20.4) for $\lambda_1(P)$. Every constant is independent of the smallest physical, polar, and conformal cells and of adjacent ratios.

For clarity, the conditional analytic chain behind input 20.3 is

$$\begin{gathered} \text{TR: exact physical trace, disk endpoint gap, value split} \\ \downarrow \\ \text{CT core} \longrightarrow \{\text{JP–EH}, \text{JP–NS}\} \\ \text{JP–NS+TR+CT core+QB+DF+DM} \longrightarrow \text{JP–DE} \\ \text{JP–EH+JP–NS+JP–DE} \longrightarrow \{\text{JC–ALG}, \text{JC–TL}\} \\ \text{JC–ALG+JC–TL} \longrightarrow \text{JC–PL} \longrightarrow \text{JP–JC} \\ \downarrow \\ \text{JP–DE+JP–JC} \longrightarrow \text{NR} \longrightarrow \text{J0.} \end{gathered} \tag{20.6}$$

No stationary Hessian is integrated on the nonstationary chart.

20.3. **Compact minimization in the closed side class.**

Lemma 20.4 (Compact spectral sublevels)**.** *For fixed N and $L < \infty$, the family*

$$\{P \in \mathcal{K}^{\mathrm{conv}}_{\le N}(\pi) : \lambda_1(P) \le L\} \tag{20.7}$$

is Hausdorff precompact up to translations. Its Hausdorff limits remain in $\mathcal{K}^{\mathrm{conv}}_{\le N}(\pi)$*, and* λ_1 *is continuous along such nondegenerate convex-domain convergence. Consequently the infimum of* λ_1 *over* $\mathcal{K}^{\mathrm{conv}}_{\le N}(\pi)$ *is attained.*

Proof. If $w(P)$ is the minimum width, containment in a strip and the one-dimensional Poincaré inequality give

$$\lambda_1(P) \ge \frac{\pi^2}{w(P)^2}.$$

Thus a spectral sublevel has a positive width bound. For convex bodies, fixed area and a positive minimum-width bound give a diameter bound. To see this without an implicit eccentricity lemma, let $[A,B] \subset P$ be a diameter, $D = |A-B|$, and use it as the horizontal axis. If y_+ and y_- are the extreme signed heights of P above and below that axis, the two triangles with base $[A,B]$ and apices at the corresponding support points have disjoint interiors and lie in P. Hence

$$|P| \ge \frac{D}{2}(y_+ - y_-) \ge \frac{D}{2} w(P), \qquad D \le \frac{2\pi}{w(P)}.$$

After fixing, for example, the Steiner point at the origin, Blaschke compactness then gives a Hausdorff convergent subsequence. Area is continuous for convex bodies, and having at most N genuine sides is closed (parametrize by at most N cyclic vertices and allow adjacent vertices to coalesce). Dirichlet spaces of bounded convex domains converge in the Mosco sense under this Hausdorff convergence, so the first eigenvalue converges. A minimizing sequence may be restricted to the sublevel determined by any one competitor, proving attainment. ☐

Proposition 20.5 (A closed-class minimizer loses no side)**.** *Every minimizer* Q_N *of* λ_1 *over* $\mathcal{K}^{\mathrm{conv}}_{\le N}(\pi)$ *has exactly* N *active sides.*

Proof. After collinear sides are merged, suppose that Q_N has $M < N$ genuine sides. The active M-gon stratum is a smooth finite-dimensional manifold near Q_N after the area and rigid gauges are fixed. Since Q_N is a global minimizer, it is stationary on that stratum. Theorem 20.2 supplies a convex $(M+1)$-gon with strictly smaller Λ, hence with strictly smaller λ_1 at fixed area. It still belongs to $\mathcal{K}^{\mathrm{conv}}_{\le N}(\pi)$, a contradiction. ☐

Remark 20.6 (Why SB disappears)*.* Stage 133 correctly observes that different one-side activation branches need not be connected below the value of their common lower stratum. Proposition 20.5 does not connect them. It selects a global minimizer and needs only one strict feasible descent to contradict minimality. Therefore the side-branch bypass SB is not an open edge of this proof.

20.4. **Qualitative global-to-local entry.** Let B be the area-π disk and $\lambda_B = \lambda_1(B)$.

Lemma 20.7 (Regular polygons converge spectrally to the disk)**.** *The area-π regular polygons satisfy*

$$R_N(\pi) \longrightarrow B \quad \textit{in Hausdorff distance}, \qquad \lambda_1(R_N(\pi)) \longrightarrow \lambda_B. \tag{20.8}$$

Proof. The inradius and circumradius of the area-normalized regular N-gon both tend to one. This proves Hausdorff convergence. Spectral convergence is the convex-domain stability used in Lemma 20.4. □

Lemma 20.8 (Near-disk entry from a competing value)**.** *Let $N_j \to \infty$ and let $P_j \in \mathcal{K}^{\mathrm{conv}}_{\le N_j}(\pi)$ satisfy*

$$\lambda_1(P_j) \le \lambda_1(R_{N_j}(\pi)). \tag{20.9}$$

Then, after translations,

$$P_j \longrightarrow B \quad \textit{in Hausdorff distance}, \qquad \|h_{P_j} - 1\|_{L^\infty(\mathbb{S}^1)} \longrightarrow 0. \tag{20.10}$$

If every P_j is active and $\alpha_{\max}(P_j)$ denotes its largest gap between consecutive outer normals, then

$$\alpha_{\max}(P_j) \longrightarrow 0. \tag{20.11}$$

Finally its radial function $r_j = 1 + \rho_j$ satisfies

$$\|\rho_j\|_{W^{1,\infty}(\mathbb{S}^1)} \longrightarrow 0. \tag{20.12}$$

Proof. Faber–Krahn and (20.9)–(20.8) give

$$\lambda_B \le \lambda_1(P_j) \le \lambda_1(R_{N_j}(\pi)) \longrightarrow \lambda_B.$$

The sequence lies in one fixed spectral sublevel. The proof of Lemma 20.4, with no fixed side bound needed for the convex compactness part, gives Hausdorff precompactness after translation. Every limit has area π and first eigenvalue λ_B. Faber–Krahn rigidity makes the limit the disk. Since every subsequential limit is the disk, the whole sequence converges, and convergence of convex support functions gives (20.10).

Suppose (20.11) failed. Along a subsequence two adjacent supporting normals would have a gap at least $\varepsilon > 0$. Their support numbers tend uniformly to one by (20.10); their intersection vertex in the bisector direction would therefore have distance at least $\sec(\varepsilon/2) + o(1) > 1$ from the origin. This contradicts Hausdorff convergence to B.

For a ray meeting the side with normal angle θ_i and support number h_i,

$$r_j(\theta) = \frac{h_i}{\cos(\theta - \theta_i)}. \tag{20.13}$$

Hausdorff convergence and the inclusions $(1-\varepsilon_j)B \subset P_j \subset (1+\varepsilon_j)B$ give $r_j \to 1$ uniformly, while support convergence gives $h_i \to 1$ uniformly in i. Thus, whenever the ray at θ meets side i,

$$\cos(\theta - \theta_i) = \frac{h_i}{r_j(\theta)} \longrightarrow 1$$

uniformly. In particular $|\theta - \theta_i| \to 0$; this also proves the needed statement at one-sided sector endpoints. Equation (20.13) and its one-sided derivative therefore give (20.12). $\square$

20.5. **Contradiction and equality rigidity.**

Proof of Theorem 20.1. By scale invariance, take $A = \pi$.

Assume first that the inequality fails for arbitrarily large side counts. Then there are $N_j \to \infty$ and exact N_j-gons P_j such that

$$\lambda_1(P_j) < \lambda_1(R_{N_j}(\pi)). \tag{20.14}$$

Let Q_j minimize λ_1 over $\mathcal{K}^{\mathrm{conv}}_{\le N_j}(\pi)$. Lemma 20.4 and Proposition 20.5 show that Q_j exists and has exactly N_j active sides. Moreover

$$\lambda_1(Q_j) \le \lambda_1(P_j) < \lambda_1(R_{N_j}(\pi)). \tag{20.15}$$

Lemma 20.8 gives

$$\|\rho_{Q_j}\|_{W^{1,\infty}} \to 0, \qquad S_j = \frac{2\pi}{N_j} + \alpha_{\max}(Q_j) \to 0.$$

Thus input 20.3 applies for all sufficiently large j and gives

$$\lambda_1(Q_j) - \lambda_1(R_{N_j}(\pi)) \ge c_0\{\mathfrak{J}_4(\alpha^{(j)}) + \mathfrak{G}^{\mathrm{JP}}_{\alpha^{(j)}}(z_j - z^{\sharp}_{\alpha^{(j)}})\} \ge 0,$$

contradicting (20.15). Hence the inequality in (20.1) holds for every sufficiently large N.

Suppose next that nonregular equality cases exist for arbitrarily large N. Choose $N_j \to \infty$ and nonregular exact N_j-gons P_j with

$$\lambda_1(P_j) = \lambda_1(R_{N_j}(\pi)).$$

The inequality just proved shows that this common value is the minimum over the exact N_j-side class. If the minimum over the closed class with at most N_j sides were smaller, its minimizer would have exactly N_j sides by Proposition 20.5, contradicting the proved inequality. Hence each P_j is also a closed-class minimizer. Lemma 20.8 and input 20.3 apply and force

$$\mathfrak{J}_4(\alpha^{(j)}) = 0, \qquad \mathfrak{G}^{\mathrm{JP}}_{\alpha^{(j)}}(z_j - z^{\sharp}_{\alpha^{(j)}}) = 0. \tag{20.16}$$

Strict convexity of x^4 under $\sum_i \alpha_i = 2\pi$ makes the first identity equivalent to $\alpha_i = 2\pi/N_j$ for every i. At the uniform fan cyclic symmetry gives $z^{\sharp}_{\alpha^{(j)}} = 0$ on the exact quotient. Positivity of the support metric then makes the support vector constant modulo translations and scale. Thus P_j is congruent to the regular N_j-gon, a contradiction.

Both a strict failure and a nonregular equality can therefore occur only for finitely many side counts. Increasing N_0 proves the theorem. $\square$

21. Final interface verification

21.1. **Discharge ledger.** The QB targets 8.6 and 8.7 are proved by Corollary 13.12 and Theorem 14.7. The adaptive sampling target 11.5 is Theorem 13.11; the provisional ELCA target is bypassed by the zero-gap source decomposition. The strong ZG–$T1$ square, full-vector reconstruction, separate-curvature estimate, and CT–JM estimate are retracted rather than assumed. The old two-sided vanishing-modulus TR–NR remainder is likewise archived and excluded from this assembly; the current NR node uses only the one-sided fixed-loss output of JC–PL through JP–JC.

The old CT–JP/CT–M4 support Taylor branch is retired. The closed theorem JP–EH, Theorem 15.2, gives the exact nonstationary support Hessian, including reduced energy, mixed flux, and area second variation. Proposition 15.5 closes JP–NS: its sine interpolant reproduces translations exactly, is controlled by the physical vertex displacement without inverse-angle loss, and keeps the whole support segment active. Proposition 15.3 supplies a certified alternating-mesh negative direction for the old cubic candidate. Thus the proof neither assumes nor silently reuses that candidate.

Theorem 15.20 proves the $D0_{\mathrm{ns}}$ endpoint and JE–P principal forcing estimate; Theorem 15.21 proves JE–V. These closed rows form JP–DE. The near-regular period-three material proposition in the JP node is retracted after the polygonal corner finite-part audit and proves neither JE–FM nor its negation. The JC node therefore defines the honest residual 16.8, keeps its forcing and integrated Hessian together in 16.9, and proves the exact completed-action identity and support-time logarithm cancellation, and isolates the sole open pathwise payment JC–PL, Contract 16.34. Its algebraic output is JP–JC, equation 16.133. Theorem 16.35 and its synchronized NR handoff 17.1 prove the exact implication to J0. Hence NR, J0, G0, and the main theorem are dependent only on JC–PL through JP–JC.

21.2. **Conditional composition.** The correct-source gap and no-ratio interpolant give the angular disk endpoint; F129 and TR transfer it to physical traces. The CT core and QB dictionary identify the exact disk cubic rows while retaining curvature. ZG proves DF and PT proves DM. Exact disk Hilbert completion gives $P_D^{\mathrm{ns}}(\alpha) + \mathfrak{G}_\alpha^{\mathrm{JP}}(z - z_\alpha^D)$. JP–JC permits at most half of the closed fan and support margins to be lost by the genuine polygonal-to-disk residual. The assembly in Theorem 17.1 therefore gives J0 without a separate flux estimate. Strict activation and compact minimization then yield Theorem 20.1, which is the conditional conclusion of Theorem 1.1 after scale normalization.

Since JC–PL remains open, this verifies the conditional implication but does not certify the unconditional target theorem.

References

[1] H. J. Brascamp and E. H. Lieb, *On extensions of the Brunn–Minkowski and Prékopa–Leindler theorems, including inequalities for log concave functions, and with an application to the diffusion equation*, J. Functional Analysis **22** (1976), 366–389.